

Elements of the History of Mathematics

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Preface

This work gathers together, without substantial modification, the majority of the historical Notes which have appeared to date in my *Eléments de Mathématique*. Only the flow has been made independent of the *Eléments* to which these Notes were attached; they are therefore, in principle, accessible to every reader who possesses a sound classical mathematical background, of undergraduate standard.

Of course, the separate studies which make up this volume could not in any way pretend to sketch, even in a summary manner, a complete and connected history of the development of Mathematics up to our day. Entire parts of classical mathematics such as differential Geometry, algebraic Geometry, the Calculus of variations, are only mentioned in passing; others, such as the theory of analytic functions, that of differential equations or partial differential equations, are hardly touched on; all the more do these gaps become more numerous and more important as the modern era is reached. It goes without saying that this is not a case of intentional omission; it is simply due to the fact that the corresponding chapters of the *Eléments* have not yet been published.

Finally the reader will find in these Notes practically no bibliographic or anecdotal information about the mathematicians in question; what has been attempted above all, for each theory, is to bring out as clearly as possible what were the guiding ideas, and how these ideas developed and reacted the ones on the others.

The numbers in square brackets refer to the Bibliography at the end of the volume.

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Chapter 1

FOUNDATIONS OF MATHEMATICS; LOGIC; SET THEORY.

The study of what is usually called the “foundations of mathematics”, which has been carried out ceaselessly since the beginning of the 19th century, was impossible to bring to fruition except with the help of a parallel effort to systematise Logic, at least those parts that govern the links between mathematical statements. So it is not possible to separate the history of set theory and the formalisation of mathematics from that of “Mathematical Logic”. But traditional logic, like that of the modern philosophers, covers in principle, an area of applications far greater than Mathematics. Therefore the reader must not expect to find in what follows a history of Logic, even in a very summarised form; we have limited ourselves as far as possible to retracing the evolution of Logic only in so far as it impinged on that of Mathematics. It is because of this that we will say nothing about the non-classical logics (many-valued logics, modal logics); all the more so will we be unable to tackle the history of those controversies which, from the Sophists to the Vienna School, have never stopped dividing philosophers both as to the possibility and the manner of applying Logic to objects in the real world or to concepts of human thought.

That there was a well-developed prehellinic mathematics is not today in any doubt. Not only are the notions (already very abstract) of whole number and of the measurement of quantities commonly used in the most ancient documents which have reached us from Egypt or Chaldea, but Babylonian algebra, because of the elegance and sureness of its methods, should not be

thought of as a simple collection of problems solved by empirical fumbling. And, if nothing is found in the texts which resembles a “proof” in the formal meaning of the word, it is reasonable to believe that the discovery of such methods of solution, whose generality appears through particular numerical applications, was not possible without a minimum of logical links (perhaps not entirely conscious, but rather like those on which a modern algebraist depends when he undertakes a calculation, before “setting down formally” all the details) ([232], pp. 203 ff.).

The essential originality of the Greeks consisted precisely of a conscious effort to order mathematical proofs in a sequence such that passing from one link to the next leaves no room for doubt and constrains universal assent.

That Greek mathematicians made use, in the course of their research, just like modern mathematicians, of “heuristic”, rather than convincing, arguments, is what was proved for example (if there were any need), by the “treatise of method” of Archimedes [153 c]; note also in this, allusions to results “found, but not proven” by earlier mathematicians.¹ But from the first detailed texts that are known to us (and which date from the middle of the Vth century), the ideal “canon” of a mathematical text is properly settled. It will find its highest expression in the great classics, Euclid, Archimedes and Apollonius; the notion of proof, in these authors, differs in no way from ours.

We have no texts allowing us to follow the first steps of this “deductive method”, which seems to us already near perfection at the exact moment when we become aware of its existence. One can only think that it fits fairly naturally into the perpetual search for “explanations” of the world which characterises Greek thought and which is so discernible already amongst the Ionian philosophers of the VIIth century; moreover tradition is unanimous in ascribing the development and refinement of the method to the Pythagorean School, in a period which fits between the end of the VIth century and the middle of the Vth century.

It is on this “deductive” mathematics, fully conscious of its goals and methods, that the philosophical and mathematical thought of subsequent times will be concentrated. We will see on the one hand the establishment little by little of “formal” Logic modelled on mathematics, to conclude with the creation of formalised languages; on the other hand, mainly starting at the beginning of the XIXth century, the basic concepts of mathematics will be queried more

and more and a great effort will be made to clarify their nature, especially after the birth of the Theory of sets.

THE FORMALISATION OF LOGIC.

The general impression which seems to emerge from the (very patchy) texts that we have on Greek philosophical thought of the Vth century, is that it is dominated by an increasingly conscious effort to extend to the whole field of human thought the procedures for conducting discussion put in hand with so much success by contemporary rhetoric and mathematics - in other words, to create Logic in its most general form. The tone of philosophical writings undergoes at this time a sudden change: while in the VIIth and VIth centuries the philosophers affirm or vaticinate (or at most outline vague arguments, based on equally vague analogies), starting with Parmenides and especially Zenon, they argue and try to draw out general principles which can serve as a basis for their dialectic: it is in Parmenides that one finds the first statement of the principle of the excluded middle, and the proofs "by contradiction" of Zenon of Elea remain famous. But Zenon writes in the middle of the Vth century; and, whatever the uncertainties of our documentation,² it is very likely that at this time, mathematicians, in their own sphere, were currently using these principles.

As we said above, it is not for us to retrace the innumerable difficulties which abound at every step in the gestation of this Logic, and the controversies which result, from the Eleatics to Plato and Aristotle, via the Sophists; let us pick up here only the role played in this evolution by the assiduous cultivation of oratorical art and the analysis of language which is one of its corollaries, developments that it is agreed should be attributed mainly to the Sophists of the Vth century. On the other hand, even if the influence of mathematics is not always explicitly recognised, it is none the less manifest, in particular in the writings of Plato and Aristotle. It can be said that Plato was almost obsessed by mathematics; without being himself an inventor in this area, he kept up, after a particular period in his life, with the discoveries by contemporary mathematicians (many of whom were his friends or his pupils), and never after ceased from a most direct kind of interest, even going so far as to suggest new directions of research; so it is that constantly, in his writings, mathematics serves as illustration or model (and sometimes even, as with the

Pythagoreans, feeding his leanings towards mysticism). As for his pupil Aristotle, he could not have avoided receiving the minimum of mathematical foundations required of the pupils of the Academy, and a volume has been produced consisting of extracts of his work that are about mathematics or refer to it [153 d]; but he seems never to have made a great effort to keep in touch with the mathematical movements of his time, and he only quotes in this area results that had been popularised a long time before. This displacement will besides only become more marked with the majority of later philosophers, of whom many, due to the lack of technical preparation, believe themselves in all good faith to be speaking knowledgeably about mathematics, whereas they will only be referring to a stage long since overtaken in its evolution.

The end product of this period, as far as Logic is concerned, is the monumental work of Aristotle [6], whose great merit is to have succeeded in systematising and codifying for the first time procedures of reasoning which had remained vague or unformulated in his predecessors.³ We must above all retain here, as our aim, the general thesis of this work, to know that it is possible to reduce all correct reasoning to the systematic application of a small number of immutable rules, which are independent of the particular nature of the objects in question (an independence which is clearly demonstrated by the representation of the concepts or propositions by means of letters — probably borrowed by Aristotle from the mathematicians). But Aristotle concentrates his attention almost exclusively on a particular type of relations and logical chains, making up what he calls the “syllogism”: it is mainly a case of relations that we would translate nowadays in the form $A \subset B$ or $A \cap B \neq \emptyset$ in the language of the theory of sets,⁴ and with the way of chaining these relations or their negations, by means of the schema

$$(A \subset B \text{ and } B \subset C) \Rightarrow (A \subset C).$$

Aristotle was still sufficiently informed of the mathematics of his era in order to have seen that schemas of this type were not sufficient to take account of all the logical operations used by mathematicians, nor, all the more, of the other applications of Logic ([6], An. Pr., I, 46).⁵ At least the deep study of the different forms of “syllogisms” which he undertakes (and which is almost entirely consecrated to the elucidation of the perpetual difficulties raised by the ambiguity or the obscurity of the objects on which the reasoning bears) gives him among others the chance

to formulate the rules for obtaining the negation of a proposition ([6], An. Pr., I, 46). It is also to Aristotle that is due the credit for having distinguished with great precision the role of “universal” propositions from that of “particular” propositions, the first sketch of quantifiers. ⁶ But it is too well known how the influence of his writings (often interpreted in a narrow and unintelligent way), which remain still very influential until well into the XIXth century, was to encourage philosophers in their neglect of the study of mathematics, and to block the progress of formal Logic. ⁷

However this latter continued to make progress in Antiquity, in the surroundings of the Megaric and Stoic schools, rivals of the Peripatetics. Our information about these tenets are unfortunately all at second hand, often passed on by adversaries or mediocre commentators. The essential progress achieved by these logicians consists, it would appear, of the formation of a “propositional calculus” in the meaning which it has today: instead of being restricted, like Aristotle, to propositions of the particular form $A \subset B$, they state rules about completely indeterminate propositions. Further, they had analysed the logical links between these rules in such a deep way that they knew how to deduce them all from five of them, set down as “unprovable”, by means of procedures very similar to modern methods [23]. Unfortunately their influence was fairly ephemeral, and their results were to fall into oblivion until the day when they were rediscovered by the logicians of the XIXth century. Aristotle remains as the uncontested master in Logic until the XVIIth century; it is known in particular that the scholastic philosophers are entirely under his sway, and if their contribution to formal logic is far from negligible [25], it does not contain any progress at the highest level compared with the achievements of the philosophers of Antiquity.

It is appropriate, however, to note here that it does not seem that the works of Aristotle or his successors had a notable influence on mathematics. The Greek mathematicians conducted their research along the path opened by the Pythagoreans and their successors of the IVth century (Theodorus, Theeta, Eudoxus) without apparently bothering with formal logic in the presentation of their results: a finding that should hardly astonish when one compares the flexibility and precision acquired already by that time by mathematical reasoning, to the very rudimentary state of Aristotelian logic. And when logic will overtake this stage, it is again the new acquisitions

from mathematics that will guide it in its evolution.

With the development of algebra, it was indeed impossible to avoid being struck by the analogy between the rules of formal logic and the rules of algebra, the ones like the others having in common the property of being applicable to objects (propositions or numbers) which are not precisely determined. And when in the XVIIth century algebraic notation took its definitive form in the hands of Viète and Descartes, almost immediately one can see appearing diverse attempts at a symbolic notation intended for the representation of logical operations; but before Leibniz, these tentative efforts, for example like that of Herigone (1644) at writing down the proofs of elementary Geometry, or that of Pell (1659) at writing down those of Arithmetic, remain very superficial and do not lead to any progress in the analysis of mathematical reasoning.

With Leibniz, we are in the presence of a philosopher who is also a mathematician of the first rank, and who will know how to draw out of his mathematical experience the germ of the ideas that will bring formal logic out of the scholastic dead-end.⁸ A universal spirit if ever there were one, inexhaustible source of original and fruitful ideas, Leibniz would be interested even more in Logic as it found itself at the very heart of his grand projects for formalising language and thought, at which he never ceased working throughout his life. Trained in his childhood in scholastic logic, he was seduced by the idea (going back to Raymond Lulle) of a method which would reduce all human concepts to primitive concepts, making up an “Alphabet of human thought”, and would rebuild them in a quasi mechanical way to obtain all true propositions ([198 b], v. VII, p. 185; cf. [69a], chap. II). Still very young, he had also conceived another much more original idea, that of the usefulness of symbolic notations as an “Ariadne’s string” string of thought:⁹ “The true method”, he said, “must provide us with a *filum Ariadnes*, that is to say a kind of sensitive and coarse means that guides the mind, in the same way as lines drawn in geometry and the type of operations that are prescribed to apprentices in Arithmetic. Without that our mind would not know how to go along a long path without straying.” ([198 b], v. VII, p. 22; cf. [69 a], p. 90). Knowing little of the mathematics of his time until about his 25th year, it is at first in the form of a “universal language” that he presents his projects ([69 a], chap. III); but as soon as he comes in contact with Algebra, he adopts it as model for his “universal Characteristic”. By that he means a type of symbolic language, capable of express-

ing without ambiguity all human thought, of reinforcing our powers of reasoning, of avoiding errors by means of an effort of entirely mechanical concentration, finally constructed so that “the chimera that the person himself who presents them does not hear, would be impossible to write down in these characters” ([198 a], v. I, p. 187). In innumerable passages in his writings where Leibniz alludes to this grandiose project and to the progress to which its attainment would lead (cf. [69 a], chap. IV and VI), it is seen with what clarity he had conceived the notion of a formalised language, a pure combination of symbols of which only the chaining is important,¹⁰ in such a way that a machine would be able to produce all the theorems,¹¹ and that all controversies would be resolved by a simple calculation ([198 b], v. VII, p. 198-203). If these hopes might appear excessive, it is none the less true that it is to this constant tendency of Leibniz’s thought that must be tied a good part of his mathematical work, beginning with his work on the symbolism of the infinitesimal Calculus (see p. 190 ff.); he was himself perfectly aware of this, and linked explicitly also to his “Characteristic” his ideas on indicial notation and determinants ([198 a], v. II, p. 204; cf. [69 a], pp. 481-487) and his sketch of “geometrical Calculus” (see pp. 50 and 61 ff.; cf. [69 a], chap IX). But in his mind the essential part had to be symbolic Logic, or, as he puts it a “Calculus ratiocinator”, and though he does not succeed in creating this calculus, at least we see him trying to at least three times. On his first attempt, he has the idea of associating with each “primitive” term a prime number, each term made up of several primitive terms being represented by the product of the corresponding prime numbers;¹² he seeks to translate into this system the usual rules of syllogism, but runs up against major complications caused by negation (that he tries, fairly naturally, to represent by a change of sign) and abandons rapidly this path ([198 c], pp. 42-96; cf. [69 a], pp.326-344). In later attempts, he seeks to give Aristotelian logic a more algebraic form; sometimes he keeps the notation AB for the conjunction of two concepts; sometimes he uses the notation $A + B$;¹³ he notes (in multiplicative notation) the law of idempotence $AA = A$, remarks that one can substitute the proposition “all A is B” by the equality $A = AB$ and that one can recover starting from there most of the rules of Aristotle by a purely algebraic calculation ([198 c], pp. 229-237 and 356-399; cf. [69 a], pp.345-364); he also has the idea of the empty set (“non Ens”), and recognises for instance the equivalence of the propositions “all A is B” and “A.(not B) is not” (loc. cit.). Further, he remarks that his

logical calculus is applicable not only to the logic of concepts, but also to that of propositions ([198 c], p. 377). He appears to be therefore very near to “Boolean calculus”. Unfortunately, it appears that he was not able to free himself completely from the scholastic influence; not only does he set himself as almost the only aim of his calculus the transcription, into his notation, of the rules of syllogism, ¹⁴ but he goes as far as sacrificing his most felicitous ideas to the desire to recover completely the rules of Aristotle, even those that were incompatible with the idea of an empty set. ¹⁵

The work of Leibniz remained for the most part unpublished until the beginning of the XXth century, and had only little direct influence. During the whole of the XVIIIth and the beginning of the XIXth centuries, different authors (de Segner, J. Lambert, Ploucquet, Holland, De Castillon, Gregonne) sketched attempts similar to those of Leibniz, without ever progressing substantially beyond the point at which he had stopped; their work only had a feeble effect, which means that most of them ignored all results due to their predecessors. ¹⁶ It is in any case under the same conditions that G. Boole, who must be considered to be the real creator of modern symbolic logic [29], writes. His key idea consists in putting himself systematically in the position of considering “extension”, so of calculating directly with sets, and writing xy for the intersection of two sets, and $x + y$ for their union when x and y have no element in common. He introduces as well a “universe” denoted 1 (the set of all elements) and the empty set denoted 0, and he writes $1 - x$ for the complement of x . As Leibniz had done, he interprets the relation of inclusion by the relation $xy = x$ (from which he extracts without difficulty the justification of the rules of classical syllogism) and his notation for union and complements gives his system a flexibility that was missing in his forerunners. ¹⁷ Furthermore, by associating with each proposition the set of “cases” in which it holds, he interprets the relation of implication as an inclusion, and his calculus of sets gives him in this way the rules of the “propositional calculus”.

In the second half of the XIXth century, Boole’s system formed the basis for the work of an active school of logicians who improved it and completed it at several points. It is thus that Jevons (1864) enlarged the meaning of the operation of union $x + y$ by extending it to the case in which x and y are arbitrary; A. de Morgan in 1858 and C. S. Peirce in 1867 proved the duality laws

$$(\mathbf{CA}) \cap (\mathbf{CB}) = \mathbf{C}(A \cup B), (\mathbf{CA}) \cup (\mathbf{CB}) = \mathbf{C}(A \cap B);^{18}$$

De Morgan tackles also, in 1860, the study of relations, defining inversion and the composition of binary relations (that is to say the operations corresponding to the operations $\overline{G}^{-1}$ and $G_1 \circ G_2$ on graphs).¹⁹ All this work is systematically expounded and developed in the massive and prolific work of Schröder [277]. But it is fairly interesting to note that the logicians of whom we have been speaking do not seem very interested in the application of their results to mathematics and that, on the contrary, Boole and Schröder especially seem to have as their principal aim to develop “Boolean” algebra by imitating the methods and problems of classical algebra (often in a fairly artificial way). The reasons for this attitude must doubtless be seen in the fact that the Boolean calculus still lacked a facility for transcribing most of mathematical reasoning,²⁰ and only supplied in this way a very partial answer to the great dream of Leibniz. The construction of a formalism better adapted to mathematics — of which the introduction of variables and quantifiers, due independently to Frege [117 a, b, c] and C. S. Peirce [248 b], make up the major stage — was the work of logicians and mathematicians who, unlike the above, had above all as their aim applications to the foundations of mathematics.

Frege’s project [117 b and c] was to create a foundation for arithmetic based on a logic formalised by a “writing of concepts” (Begriffsschrift) and we will come back later (p. 29) to the way in which he defines the natural numbers. His work is characterised by extreme precision and attention to detail in the analysis of concepts; it is because of this tendency that he introduces many a distinction which turns out to be of great importance in modern logic: for instance it is he who first distinguishes between the statement of a proposition and the assertion that this proposition is true, between the relation of belonging and that of inclusion, between an object x and the set $\{x\}$ reduced to this single object, etc.. His formalised logic, which contains not only “variables” with the meaning used in mathematics, but also “propositional variables” representing indeterminate relations, susceptible to quantification, would later (through the work of Russell and Whitehead) supply the fundamental tool for metamathematics. Unfortunately, the symbols which he adopts are hardly inspiring, of a terrifying typographic complexity and far removed from the practice of mathematicians; which had the result of turning away these latter

and reducing considerably the influence of Frege on his contemporaries.

Peano's goal was at the same time much vaster and much more down to earth; it consisted in publishing a "Formulary of mathematics", written entirely in formal language and containing, not only mathematical logic, but all the results of the most important branches of mathematics. The speed with which he managed to complete this ambitious project, helped by a host of enthusiastic collaborators (Vailati, Pieri, Padoa, Vacca, Vivanti, Fano, Burali-Forti) is witness to the excellence of the symbolism which he had adopted: following closely the current practice of mathematicians, and introducing numerous well-chosen abbreviating symbols, his language remains as well fairly easily legible, thanks notably to an ingenious system for replacing brackets by full stop separators [246 f]. Much notation due to Peano is today adopted by the majority of mathematicians: we quote $\in, \supset$ (but, contrary to the present use, with the meaning of "is contained in" or "implies" ²¹), $\cup, \cap, A - B$ (set of differences $a - b$ where $a \in A$ and $b \in B$). On the other hand, it is in the "Formulary" that is found for the first time a thorough analysis of the general notion of function, of those of direct image ²² and reciprocal image, and the remark that a sequence is only a function defined on $\mathbb{N}$. But quantification, with Peano, is subject to hampering restrictions (one can only, in principle, quantify, in his system, relations of the form $A \rightarrow B, A \Leftrightarrow B$ or $A = B$). Further the almost fanatical zeal of some of his disciples laid them wide open to ridicule; criticism, often unjustified, by H. Poincaré in particular, was a heavy blow to the Peano school and became an obstacle to the diffusion of his doctrines in the mathematical world.

With Frege and Peano the essential elements of the formal languages used today were acquired. The most widespread is doubtless that hammered out by Russell and Whitehead in their great work "Principia Mathematica", which happily links the precision of Frege and the convenience of Peano [266]. Most actual formal languages are differentiated from it only by changes of secondary importance, aimed at simplifying its use. Among the most ingenious, we quote the "functional" writing of relations (for instance $\in xy$ instead of $x \in y$, thought up by Lukasiewicz, thanks to which brackets can be completely omitted; but the most interesting is without doubt the introduction by Hilbert of the symbol τ , which allows the consideration of the quantifiers $\exists$ and $\forall$ as abbreviation signs, the avoidance of the introduction of the "universal" functional

symbol ι of Peano and Russell (which is only applied to functional relations), and finally which avoids the need to formulate the axiom of choice in the theory of sets ([163 a], v.III, p. 183).

THE NOTION OF TRUTH IN MATHEMATICS.

Mathematicians have always been sure that they prove “truths” or “true propositions”; such a conviction can obviously only be sentimental or metaphysical, and it is not by getting on to mathematical ground that it can be justified, nor even given a meaning that does not make it a tautology. The history of the concept of truth in mathematics is the concern therefore of the history of philosophy and not of that of mathematics; but the evolution of this concept has had an undeniable influence on that of mathematics, and because of this we can not let it go by in silence.

Let us note first that it is as rare to see a mathematician in possession of a strong philosophical culture as to see a philosopher who has an extensive knowledge of mathematics; the opinions of mathematicians on topics in philosophy, even when these questions are concerned with their field, are most often opinions received at second or third hand, coming from doubtful sources. But, precisely because of this, it is these average opinions which interest the mathematical historian, at least as much as the original views of thinkers such as Descartes or Leibniz (to mention two who were also mathematicians of the first rank), Plato (who at least kept up with the mathematics of his time), Aristotle or Kant (of whom the same could not be said).

The traditional notion of mathematical truth is that which goes back to the Renaissance. In this concept, there is no great difference between those objects which are the concern of mathematicians and those which are the concern of natural science; both are knowable and man has obtained a grasp on them both through intuition and through reason; there are no grounds to doubt either intuition or reason, which are fallible only if used incorrectly. “One must have”, says Pascal, “altogether a wrong spirit to reason badly about principles so great that it is almost impossible for them to escape” ([244], v. XII, p. 9). Descartes, by his fireside, convinced himself that “there have been only Mathematicians who were able to find some proofs, that is to say some sure and certain reasons” ([85 a], v.VI, p. 19) and that (if his tale is to be believed) well before he had built up a metaphysics in which “that very same thing”, he says, “that I previously

took as a rule, namely that the objects that we can visualise very clearly and very distinctly are all true, is only ensured because God is or exists, and that he is a perfect being” ([85 a], v. VI, p. 38). If Leibniz objects to Descartes that it cannot be seen how to recognise that an idea is “clear and distinct”,²³ he also considers axioms as obvious and undisputed consequences of the definitions as soon as the expressions are understood.²⁴ It must not be forgotten either that, in the language of that time, mathematics includes many sciences that we do not recognise anymore as such, and sometimes even as far as the art of the engineer; and in the confidence that they inspire, the surprising success of their applications to “natural philosophy”, to the “mechanical arts”, to navigation, comes in for a large part.

In this way of looking at things, axioms are no more susceptible to being discussed or put into question than the rules of deduction; at most the choice can be left to each person, according to his preferences, to reason “in the way of the ancients” or to give free reign to his intuition. The choice of point of departure is also a matter of individual preference, and one sees appearing numerous “editions” of Euclid where the solid logical framework of the *Elements* becomes a strange travesty; some surveys are given of the infinitesimal calculus and rational mechanics, supposedly deduced from fundamentals which are remarkably badly established; and Spinoza was perhaps giving in good faith his *Ethics* as being proved in the manner of geometers “more geometrico demonstrata”. If it is hard to find in the XVIIIth century two mathematicians who agree on any matter whatsoever, if the polemics occur daily, endless and acrimonious, the notion of truth remains none the less not in question. “Only having one truth about each object”, says Descartes, “whoever finds it knows as much as can be known about it” ([85 a], v. VI, p. 21).

Although no Greek mathematical text of the high period on these questions has survived, it is probable that the point of view of Greek mathematicians on this subject had many more nuances. It is by experience only that the rules of deduction were elaborated to the point of inspiring complete confidence; before they could be considered to be above all discussion, it was necessary to go through mainly fumbings and paralogisms. It would also be to misconstrue the critical spirit of the Greeks, their taste for discussion and sophistry, to imagine that the “axioms” themselves that Pascal judged to be the most obvious (and that, according to a legend spread by his sister, he had, with infallible instinct, discovered himself during his childhood) were not

the subject of long discussions. In an area that was not that of geometry strictly speaking, the paradoxes of the Eleates have preserved for us some traces of these polemics; and Archimedes when he makes the observation ([5 b], v. II, p. 265) that his predecessors made use in many situations of the axiom to which we usually ascribe his name, adds that what is proved by means of this axiom “has been accepted no less than what has been proved without it”, and that it is sufficient that his own results be accepted on the same basis. Plato, in accordance with his metaphysical views, presents mathematics as a means of access to a “truth in itself” and the objects it deals with as having a real existence in the world of ideas; he characterises none the less the mathematical method precisely in a famous extract from the Republic: “Those that are involved with geometry and arithmetic ... assume even and odd, three kinds of angles; they treat them as known objects: once assumed, they esteem that they no longer need to give account of them either to themselves or to others, [considering it] as clear to each one; and starting there, they proceed in sequence, in order to reach by common agreement the goal that their research had suggested” ([250], Book VI, 510 c-e). That which makes up a proof is therefore firstly a point of departure supplying an arbitrary start (even though “clear to everyone”), and beyond which, he says a bit further on, one does not try to dig; and then, a discussion that follows in sequence a series of intermediate stages; finally, at each step, the consent of the interlocutor guaranteeing the correctness of the reasoning. It must be added that once the axioms are stated, no new appeal to intuition is allowed on principle: Proclus, quoting Géminus, recalls that “we have learnt from the pioneers of this science themselves, to take no account of conclusions which are only plausible when it is a case of reasoning which must be part of our geometric doctrine” ([153 e], v.I, p. 203).

Thus it is to experience and the cauldron of criticism that is due the elaboration of the rules of mathematical reasoning; and if it is true, as has been argued in a plausible way [317 d] , that Book VIII of Euclid has kept for us part of the arithmetic of Archytas, it is not surprising to see there the rigidity of the rather pedantic reasoning that does not fail to appear in all mathematical schools where “rigour” is discovered or believed to have been discovered. But, once having entered into the practice of mathematicians, it does not seem that these rules of reasoning have ever been doubted until quite recently: if with Aristotle and the Stoics, some of the rules are

deduced from others by schemes of reasoning, the primitive rules are always assumed to be evident. In the same way, having gone back to the “hypotheses”, “axioms” and “postulates” that appeared to them to supply a solid foundation for the science of their time (such as those for example that they put forward in the first “Elements” that tradition ascribes to Hippocrates of Chio, about 450 B. C.), the Greek mathematicians from the classical period seem to have bent all their efforts to the discovery of new results rather than to a critique of these foundations that, at that time, could not have failed to be sterile; and, putting aside all metaphysical preoccupations, it is from this general accord amongst mathematicians on the bases of their science that the text of Plato quoted above bears testimony.

On the other hand Greek mathematicians do not seem to have believed it possible to explain the “primary notions” that were their points of departure, straight line, surface, ratio of quantities; if they give “definitions” of them, it is obviously a crisis of conscience and without any illusions about their range. It goes without saying that on the other hand, about definitions other than those of “primary notions” (definitions often called “nominal”), the Greek mathematicians and philosophers had perfectly clear ideas. It is in this context that intervenes explicitly, for the first time no doubt, the question of “existence” in mathematics. Aristotle does not miss observing that a definition does not imply the existence of the object defined, and that there must be over and above that either a postulate or a proof. No doubt his observation was derived from the practice of mathematicians; in any case Euclid takes care to postulate the existence of a circle, and to prove that of an equilateral triangle, of parallels, of a square, etc. as he introduces them into his arguments ([153 e], Book I); these proofs are “constructions”; in other words, he exhibits, basing himself on the axioms, mathematical objects that he proves satisfy the definitions that he needs to justify.

Thus we see Greek mathematics in the classical era reaching a kind of empirical certainty (whatever the metaphysical bases for it from some philosopher or other might be); if it cannot be conceived that the rules of reasoning can be questioned, the success of Greek science, and the feeling that exists of the inopportunity of a critical revision, play a great part in the confidence that the axioms themselves inspire, a confidence that is rather of the order of that (almost unlimited, as well) that was put during the last century in the principles of theoretical physics.

It is in any case what is suggested by the motto of the school “*nihil est in intellectu quod non prius fuerit in sensu*”, which is precisely that against which Descartes stands out, as not giving a firm enough basis to that which Descartes hoped to extract from the use of reason.

One must come down to the beginning of the XIXth century to see mathematicians recover from the arrogance of a Descartes (not to say that of a Kant or that of a Hegel, this latter somewhat late, as was fashionable, about the science of his time ²⁵), to a position as well balanced as that of the Greeks. The first blow to the classical concepts is the establishment of non-Euclidean hyperbolic geometry by Gauss, Lobatschevsky and Bolyai at the beginning of the century. We will not undertake to retrace here in detail the origins of this discovery, the outcome of numerous unfruitful attempts to prove the parallel postulate (see [105 a and b]). At the time its effect on the principles of mathematics was perhaps not so deep as is sometimes said. It simply forces the abandonment of the pretensions of the previous century to the “absolute truth” of Euclidean geometry, and all the more, the Leibnizian point of view that definitions imply axioms; these latter no longer appear as at all “obvious”, but rather as hypotheses of which it must be determined whether they are adapted to the mathematical representation of the world of the senses. Gauss and Lobatschevsky believe that the debate between the different possible geometries can be determined by experience ([206], p. 76). It is also the point of view of Riemann, of whom the famous inaugural Lecture “On the hypotheses which form the foundation of geometry” had as aim to furnish a general mathematical framework for the various natural phenomena: “What remains to be resolved” he says, “is the question of knowing to what extent and up to what point these hypotheses are found to be confirmed by experience” ([259 a], p. 284). But that is a problem which visibly has nothing anymore to do with Mathematics; and none of the previous authors seem to put into question that, even if a “geometry” does not correspond to an experimental reality, its theorems remain no less “mathematical truths”. ²⁶

All the same, if it so, it is certainly no longer to an unlimited confidence in classical “geometric intuition” that such a conviction is due; the description that Riemann seeks to give of “multiplicities n times extended”, the object of his work, only relies on “intuitive” ²⁷ considerations to reach a justification for the introduction of “local co-ordinates”; from that moment on he apparently seems to feel himself on solid ground, namely that of Analysis. But this latter is

grounded in the end on the concept of real number, which remained until then of a very intuitive nature; and progress in the theory of functions was leading in this respect to some very worrying results: with the research of Riemann himself on integration, and more so with the examples of curves without tangents, constructed by Bolzano and Weierstrass, it is the whole of the pathology of mathematics that was beginning. For a century we have seen so many monsters of this species that we are a bit blasé, and the most weird teratological characters must be accumulated in order still to astound us. But the effect produced on the majority of mathematicians of the XIXth century went from disgust to consternation: “How” H. Poincaré asks himself, “can intuition deceive us at this point?” ([251 d], p. 19); and Hermite (not without a spark of humour which the commentators of this famous sentence do not all seem to have perceived) declares that he “turns away with fear and horror from this lamentable plague of continuous functions that do not have a derivative” ([160], v. II, p. 318). The worst was that these phenomena, so contrary to common sense, could no longer be laid at the door of ideas badly elucidated, as at the time of the “indivisibles” (see p. 173), since they survived after the reform of Bolzano, Abel and Cauchy, who had established the foundation of the notion of limit in a manner as rigorous as the theory of ratios (see p. 154). It is thus fully to the gross and incomplete character of our geometric intuition that the account must be laid, and it is reasonable that since then it has remained discredited quite justifiably as a means of proof.

This realisation would inevitably react on classical mathematics, starting with geometry. In whatever respect the axiomatic construction of Euclid had been held, more than one imperfection had been noticed, and that already in antiquity. It was the postulate on parallels that had been the object of the greatest number of criticisms and attempts at proof; but the followers and commentators of Euclid had also attempted to prove other postulates (notably that of the equality of right angles) or recognised the insufficiency of certain definitions, such as those of the straight line or plane. In the XVIth century, Clavius, an editor of the *Elements*, notes the absence of a postulate guaranteeing the existence of the fourth proportional; for his part, Leibniz remarks that Euclid uses geometric intuition without mentioning it explicitly, for example when he admits (*Elements*, Book I, prop. 1) that two circles, each of which goes through the centre of the other have a common point ([198 b], v. VII, p. 166). Gauss (who himself did not deny himself the use

of such topological considerations) draws attention to the role played in Euclidean constructions by the notion of a point (or a straight line) being “between” two others, notion that is however not defined ([124 a], v. VIII, p. 222). Finally, the use of displacements — notably in the “case of equality of triangles” — long assumed to be obvious, ²⁸ was soon to appear to the critics of the XIXth century as relying also on unstated axioms. One ends up thus, in the period from 1860 to 1885, with different partial revisions of the beginnings of geometry (Helmholtz, Méray, Houël) tending to remedy some of these gaps. But it is only with M. Pasch [245] that the abandonment of all appeals to intuition is a programme properly formulated and followed with full rigour. The success of his enterprise soon brought him numerous emulators who, principally between 1890 and 1910, gave quite varied statements of the axioms of Euclidean geometry. The most famous of these works were those of Peano, written in his symbolic language [246 d], and especially the “Grundlagen der Geometrie” of Hilbert [163 c], appearing in 1899, a book which, by its lucidity and the depth of its exposition, was to become immediately, with full justification, the charter for modern axiomatics, even to the extent of leading to forgetting its forerunners. It is indeed that, not content with giving there a complete system of axioms for Euclidean geometry, Hilbert classifies these axioms into different groups of different types, and sets himself to determine the exact area of influence of each of these groups of axioms, not only in developing the logical consequences of each of them in isolation, but also in discussing the different “geometries” obtained when one omits or modifies certain of these axioms (geometries amongst which those of Lobatschevsky and of Riemann appear only as particular cases ²⁹); he thus puts clearly in the picture, in an area considered until then as one of those nearest the reality of the senses, the freedom of which the mathematician disposes in his choice of postulates. In spite of the disarray caused in more than one philosopher by these “metageometries” with strange properties, the thesis of the “Grundlagen” was rapidly adopted almost unanimously by mathematicians; H. Poincaré, although hardly guilty of favouring formalism, recognised in 1902 that the axioms of geometry are conventions, for which the notion of “truth”, as it is normally understood, has no more meaning ([251 c], pp. 66-67). “Mathematical truth” resides thus uniquely in logical deduction starting from premises arbitrarily set by axioms. As will be seen later (pp. 35 ff.), the validity of the rules of reasoning used in making these deductions was itself soon to be put in

question, bringing about thus a complete reshaping of the basic conceptions of mathematics.

OBJECTS, MODELS AND STRUCTURES.

A) Objects and structures of mathematics. — From Antiquity to the XIXth century, there is a common understanding as to what are the principal objects of the mathematicians; they are exactly those that are mentioned by Plato in the passage quoted earlier (p. 13); numbers, quantities and figures. If, at first, must be added the objects and phenomena which are the subject of Mechanics, Astronomy, Optics and Music, these “mathematical” disciplines are always clearly separated, amongst the Greeks, from Arithmetic and Geometry, and starting with the Renaissance they fairly quickly attain the rank of independent sciences.

Whatever the philosophical nuances by which the conception of mathematical objects are coloured by such and such a mathematician or philosopher, there is at least one point on which there is unanimity: it is that these objects are given to us and it is not in our power to assign to them arbitrary properties, in the same way as a physicist cannot change a natural phenomenon. Truth to say, psychological reactions doubtless form one part of these views, reactions which are not for us to follow up, but which each mathematician knows when he wears himself out in vain efforts to seize upon a proof which seems to slip away endlessly. From there to assimilate this resistance to the obstacles that the world of senses sets in our path there is only a single step; and even today, more than one, who claims an intransigent formalism, would subscribe voluntarily, in his innermost being, to this admission of Hermite: “I believe that numbers and the functions of Analysis are not the arbitrary result of our minds; I think that they exist outside of us, with the same character of necessity as the things of objective reality, and we meet them or discover them, and study them, as do the physicists, the chemists and the zoologists” ([160], v. II, p. 398).

There is no question in the classical conception of mathematics, of straying away from the study of numbers and figures; but this official doctrine, to which every mathematician believes himself bound to bring his verbal adherence, becomes no less bit by bit an intolerable burden, as new ideas accumulate. The embarrassment of the algebraists up against negative numbers

does not cease until analytical Geometry gives it a convenient “interpretation”; but, well into the XVIIIth century still, d’Alembert, discussing the question in the *Encyclopédie* ([75 a], article NÉGATIF), loses his nerve suddenly after a column of fairly confused explanations, and is content to conclude that “the rules of algebraic operations on negative quantities are generally assumed by everybody and perceived generally as exact, whatever idea is linked elsewhere to these quantities”. As for imaginary numbers, the scandal is far bigger; for if they are “impossible” roots and if (until about 1800) no way of “interpreting” them is seen, how can one, without contradiction, talk of these indefinable beings, and above all why introduce them? D’Alembert keeps here a prudent silence and does not even state these questions, no doubt because he realises that he would not be able to answer other than was done naively by A. Girard a century earlier ([129], f. 22): “It could be said: what use are these impossible solutions? I answer: for three things, for the certainty of the general rule, and that there is no other solution, and for its usefulness.”

In Analysis, the situation, in the XVIIth century, was no better. It is a happy circumstance that analytical Geometry was to appear, as if at a designated point, to give a “representation” in the shape of a geometric figure, of the great creation of the XVIIIth century, the notion of a function, and so assist powerfully (with Fermat, Pascal or Barrow) at the birth of the infinitesimal Calculus (cf. p. 193). But it is known on the other hand, to what philosophico-mathematical controversies the notions of the infinitely small and indivisible were going to give rise. And if d’Alembert is happier here, and recognises that in the “metaphysics” of the infinitesimal Calculus there is nothing other than the notion of limit ([75 a], articles DIFFÉRENTIEL and LIMITE, and [75 b]), he is no more able than his contemporaries, to understand the real meaning of expansion in divergent series, and to explain the paradox of exact results obtained at the end of calculations with expressions deprived of any numerical interpretation. Finally, even in the domain of “geometric certainty”, the Euclidean framework bursts: when Stirling, in 1717, does not shrink from saying that a certain curve has a “double imaginary point at infinity” ([299], p. 93 of the new edition), he would certainly be in trouble trying to link such an “object” to commonly understood notions; and Poncelet who, at the beginning of the XIXth century, gives considerable development to such ideas in founding projective geometry (see p. 131), is content

still to invoke as justification a “principle of continuity” which is altogether metaphysical.

One can conceive that, under these conditions (and at the exact moment when, paradoxically, the “absolute truth” of mathematics is proclaimed with the greatest force), the notion of proof seems to become more and more blurred during the XVIIIth century, since it is out of the question to set down, as did the Greeks, the notions on which reasoning is conducted, and their fundamental properties. The return towards rigour, which starts at the beginning of the XIXth century, brings some improvement to this state of things, but does not stop for all that the stream of new notions: one sees thus appearing in Algebra the imaginaries of Galois ([123], p. 113-127), the ideal numbers of Kummer [188 b], which are followed by vectors and quaternions, n -dimensional spaces, multivectors and tensors (see pp. 61 ff.), not to speak of Boolean algebra. No doubt great progress (which precisely allows the return to rigour, without losing any of the conquests of previous times) occurs with the possibility of giving “models” for these new notions in more classical terms: the ideal numbers or the imaginaries of Galois are interpreted by means of the theory of congruences (see pp. 82 ff.), n -dimensional geometry only appears (if so desired) as a pure language for stating results in algebra “with n variables”; and for the classical imaginary numbers — of which the geometric representation by the points of a plane (see pp. 161 ff.) marks the beginning of this flowering of Algebra — there is soon the choice between the geometric “model” and an interpretation in terms of congruences (cf. p. 82). But mathematicians begin at last to feel sharply that it is to fight against a natural slope along which their work is dragging them, and that it must be allowable, in mathematics, to reason about objects that have no sensible “interpretation”: “It is not of the essence of mathematics”, says Boole in 1854, “to be occupied with the ideas of number and quantity” ([29], v. II, p. 13).³⁰ The same preoccupation leads Grassmann, in his “Ausdehnungslehre”

of 1844, to present his calculus in a form from which the notions of number or of geometric object are, at first, excluded.³¹ And, a bit later, Riemann, in his inaugural Lecture, takes care at the beginning not to speak of “points”, but rather of “determinations” (Bestimmungsweise), in his description of “multiplicities n times extended”, and underlines that, in such a multiplicity, the “metric relations” (Massverhältnisse) “can only be studied for abstract quantities and can only be represented by formulae; under certain conditions, one can however decompose them

into relations of which each taken in isolation is susceptible to a geometric representation, and in that way it is possible to express the results of the calculation in a geometric form” ([259 a], p. 276).

From that moment, the expansion of the axiomatic method is an accomplished fact. If, for a while longer, it is believed useful to control, when possible, the “abstract” results by geometric intuition, at least it is admitted that the “classical” objects are no longer the only ones that the mathematician can legitimately study. It is that — precisely because of the multiple “interpretations” or “models” that are possible — it has been recognised that the “nature” of mathematical objects is in the end secondary, and that it matters little, for example, that a result is presented as a theorem of “pure” geometry, or as a theorem of algebra by the means of analytical geometry. In other words, the essence of mathematics — this elusive notion that could until then only have been expressed by vague names such as “general rule” or “metaphysics” — appears as the study of relationships between objects that are only (voluntarily) known and described by some of their properties, precisely those that are put as axioms at the foundations of their theory. It is this that had already been clearly seen by Boole in 1847, when he wrote that mathematics deals with “operations considered in themselves, independently of the diverse objects to which they can be applied” ([29], v. I, p. 3). Hankel, in 1867, inaugurating the axiomatisation of algebra, defends a mathematics that is “purely intellectual, a pure theory of forms, which has as its purpose, not the combination of quantities, or of their images, the numbers, but objects of thought (“Gedankendinge”) to which may correspond effective objects or relations, even though such a correspondence is not necessary” ([146], p.10). Cantor, in 1883, echoes this claim of a “free mathematics” by proclaiming that “mathematics is entirely free in its development, and its concepts are only linked by the necessity of being consistent, and are co-ordinated with concepts introduced previously by means of precise definitions” ([47], p.182). Finally, the revision of Euclidean geometry succeeds in spreading and popularising these ideas. Pasch himself, although still attached to a certain “reality” of geometric objects, recognises that geometry is in fact independent of their significance, and consists purely in the study of their relations ([245], p.90); a concept that Hilbert pushes to its logical conclusion in underlining that even the names of the basic notions of a mathematical theory can be chosen arbitrarily,³² and that Poincaré expresses

by saying that the axioms are “disguised definitions”, thus completely reversing the scholastic point of view.

It would thus be tempting to say that the modern notion of “structure” is attained in substance around 1900; in fact it will need still another thirty years of apprenticeship before it appears in all its glory. It is no doubt not difficult to recognise structures of the same kind when they have a fairly simple nature; for the group structure, for example, this stage was reached already by the middle of the XIXth century. But at the same time, Hankel can be seen fighting — without quite succeeding — to bring out the general ideas of field and extension, that he does not succeed in expressing except in the form of a “principle of permanence” which is semi metaphysical [146], and which will be only formulated in a definitive form by Steinitz [294 a] 40 years later. Above all it has been fairly difficult, in this matter, to free oneself from the impression that mathematical objects are “given” to us with their structure; only a fairly long use of functional Analysis has been able to familiarise modern mathematicians with the idea that, for example, there are several “natural” topologies on the rational numbers, and several measures on the number line. With this disassociation the transition to general definition of structures was finally realised.

B) Models and isomorphisms. — The intervention of the notion of “model” or “interpretation” of one mathematical theory by means of another will have been noticed on several occasions. That is not a recent idea, and no doubt what can be seen here is a manifestation ceaselessly recurring of a deep feeling for the unity of the various “mathematical sciences”. If one can take as authentic the traditional maxim “All things are numbers” of the first Pythagoreans, that can be considered as the trace of a first attempt to bring back the geometry and algebra of the times to arithmetic. Although the discovery of the irrationals seemed to close for ever that route, the reaction to which it gave birth in Greek mathematics was a second attempt at synthesis this time taking geometry as the basis, and absorbing among others the solution methods for algebraic equations inherited from the Babylonians. ³³ It is known that this conception was to remain until the fundamental reform of R. Bombelli and of Descartes, assimilating every measure of quantity to a measure of length (in other words, to a real number; cf. p. 151). But with the creation of analytical geometry by Descartes and Fermat, the tendency is again reversed, and a much

tighter fusion of geometry and algebra is obtained, but this time to the benefit of algebra. Further, elsewhere, Descartes goes further and conceives of the essential unity of “all these sciences that are commonly called Mathematics ... Even though their subjects are different”, he says “they still do not fail to converge, in that they consider nothing other than the various relations or ratios that are to be found there” ([85 a], v. VI, pp. 19-20).

³⁴ However, this point of view only tended to make Algebra the fundamental mathematical science; a conclusion against which Leibniz protests vigorously, who also himself, as has been seen, conceived of a “universal Mathematics”, but on a much vaster scale and already quite close to modern ideas. Making precise the “accord” of which Descartes spoke, he glimpses, in fact, for the first time, the general notion of isomorphism (which he calls “similitude”), and the possibility of “identifying” relations or operations that are isomorphic; he gives examples addition and multiplication ([69 a], pp. 301-303). But these audacious views remained without echo amongst his contemporaries, and one must await the expansion of Algebra which takes place around the middle of the XIXth century (see pp. 51 ff.) to see the beginnings of the realisation of the Leibnizian dreams. We have already underlined that it is at that moment that the “models” multiply and that it becomes usual to go from one theory to another by a simple change of language; the most striking example of this is perhaps duality in projective geometry (see p. 132), where the practice, frequent at that time, of printing face to face, in two columns, the theorems “dual” one to the other, doubtless plays a great part in the full realisation of the notion of isomorphism. From a more technical point of view, certainly the notion of isomorphic groups for abelian groups is known to Gauss, for groups of permutations to Galois (cf. pp. 51 ff.); it is acquired in a general way for any group whatsoever around the middle of the XIXth century. ³⁵ Following this, with each new axiomatic theory, it is a natural development to define a notion of isomorphism; but it is only with the modern notion of structure that it was finally recognised that every structure carries within itself a notion of isomorphism, and that it is not necessary to give a special definition of it for each type of structure.

C) The arithmetisation of classical mathematics. — The use, more and more widespread, of the notion of “model” was also going to allow the realisation in the XIXth century

of the unification of mathematics dreamed of by the Pythagoreans. At the beginning of the century, whole numbers and continuous quantities seemed as irreconcilable as ever in antiquity; real numbers remain linked to the notion of geometric quantity (at least to that of length), and it is to this latter that appeal had been made for the “models” of negative numbers and imaginary numbers. Even rational numbers were traditionally attached to the idea of the splitting of a quantity into equal parts; only the whole numbers remained apart, as “exclusively the product of our minds” as is said by Gauss in 1832, putting them in opposition to the notion of space ([124 a], v. VIII, p. 201). The first efforts to bring together Arithmetic and Analysis were made at first with rational numbers (positive and negative) and are due to Martin Ohm (1822); they were taken up again around 1860 by several authors, notably Grassmann, Hankel and Weierstrass (in his unpublished lectures); it is to this latter that it appears is due the idea of obtaining a “model” of the positive rational numbers or of whole negative numbers by considering classes of pairs of whole numbers. But the most important step remained to be taken, namely to find a “model” of irrational numbers in the theory of rational numbers; around 1870, it had become an urgent problem, in view of the necessity, after the discovery of “pathological” phenomena in Analysis, to eliminate every trace of geometric intuition in the vague notion of “quantity” in the definition of real numbers. It is known that this problem was solved around this time, almost simultaneously by Cantor, Dedekind, Méray and Weierstrass, and using fairly different methods (see p. 155).

From this moment, whole numbers became the foundation of all classical mathematics. Further, “models” based on Arithmetic acquired still more importance with the extension of the axiomatic method and the conception of mathematical objects as free creations of the mind. There remained in fact one restriction to this freedom claimed by Cantor, the question of “existence” which had already preoccupied the Greeks, and which arose here in a much more immediate way, precisely since all appeals to an intuitive representation had now been abandoned. We will see later (pp. 36 ff.) of what a philosophico-mathematical maelstrom the notion of “existence” was going to be the centre in the first years of the XXth century. But in the XIXth century that had not been reached, and to prove the existence of a mathematical object having given properties,

it was simply, as for Euclid, “constructing” an object with given properties. That was precisely what arithmetic “models” were for: once the real numbers were “interpreted” in terms of whole numbers, complex numbers and Euclidean geometry were also, thanks to analytical Geometry, and it was the same for all new algebraic objects introduced since the beginning of the century; finally — a discovery that had a great effect — Beltrami and Klein had even obtained Euclidean “models” of the non-Euclidean geometries of Lobatschevsky and of Riemann (see p. 134), and in consequence “arithmetised” (and thereby completely justified) these theories which had at first sight aroused such distrust.

D) The axiomatisation of arithmetic.—It was in line with this evolution that a subsequent turn towards the foundations of arithmetic itself was made, and indeed that is what can be seen around 1880. It appears that before the XIXth century no attempt had been made to define addition and multiplication of whole numbers except by a direct appeal to intuition; Leibniz is the only one who, faithful to his principles, warns expressly that “truths” as “obvious” as $2 + 2 = 4$ are no less susceptible to proof if one thinks about the definitions of the numbers involved ([198 b], v. IV, p. 403; cf. [69 a], p. 203); and he did not consider at all that commutativity of addition and multiplication were intrinsic.³⁶ But he does not take his thoughts on this subject any further, and about the middle of the XIXth century, still no progress in this direction had been made: Weierstrass himself, whose lectures contributed considerably to extend the “arithmeticalising” point of view, does not seem to have felt the need for a logical clarification of the theory of whole numbers. The first steps in this direction seem to be due to Grassmann, who, in 1861 ([134], v. II₂, p. 295) gives a definition of addition and multiplication for whole numbers, and proves their fundamental properties (commutativity, associativity, distributivity) using only the operation $x \mapsto x + 1$ and the principle of induction. This latter had been clearly conceived and used for the first time in the XVIIth century by B. Pascal ([244], v. III, p. 456)³⁷ — even though one can find in Antiquity some more or less conscious applications of it — and was currently used by mathematicians since the second half of the XVIIth century. But it is only in 1888 that Dedekind ([79], v. III, pp.359-361) stated a complete system of axioms for arithmetic (a system repeated 3 years later by Peano and usually known by

his name [246 c]), which contained in particular a precise formulation of the principle of induction (that Grassmann uses yet without stating it explicitly).

With this axiomatisation, it seemed that the definitive foundations of mathematics had been reached. In fact, at the very moment when the axioms of arithmetic were being clearly formulated, these same, for many mathematicians (starting with Dedekind and Peano themselves) were already deprived of this role of primordial science, in favour of the latest arrival among the theories of mathematics, the theory of sets; and the controversies that were going to unravel around the notion of whole number cannot be isolated from the great “crisis of the foundations” of the years 1900-1930.

THE THEORY OF SETS.

It can be said that at all times, mathematicians and philosophers have used reasoned arguments from the theory of sets in a more or less conscious way; but in the history of their conceptions of this subject, one must separate sharply all questions linked to the idea of cardinal number (and in particular to the notion of infinity) from those that only introduce the notions of belonging and inclusion. These latter are among the most intuitive and seem never to have raised controversy: it is on them that one can most easily base a theory of syllogism (as Leibniz and Euler were to show us), or axioms such as “the whole is greater than the sum of its parts”, without talking about that which, in geometry, is concerned with intersections of curves and surfaces. Until the end of the XIXth century, no problem arises in talking about the set (or “class” with some authors) of objects possessing such and such a given property;³⁸ and the famous “definition” given by Cantor (“By a set is meant a gathering into one whole of objects which are quite distinct in our intuition or our thought” ([47], p. 282)) will give rise, at the time of its publication, to hardly any objections.³⁹ It is altogether different as soon as the notion of set is mixed with that of number or quantity. The question of the infinite divisibility of the line (doubtless posed already by the first Pythagoreans) was, as is known, to lead to considerable philosophical difficulties: from the Eleates to Bolzano and Cantor, mathematicians and philosophers will throw themselves without success against the paradox of the finite quantity made up of an infinity of points without size. It would be of

no interest to us to retrace, even summarily, the interminable and impassioned polemics that are aroused by this problem, that constituted a ground particularly favourable to metaphysical or theological digressions; let us note only the point of view at which, from Antiquity, most mathematicians stop. It consists essentially in refusing to debate, short of being able to conclude in an irrefutable way — an attitude that we will find again among the modern formalists; in the same way as these latter arrange for the elimination of the intervention of “paradoxical” sets (see below pp. 31 ff.), the classical mathematicians carefully avoid introducing into their reasoning the “actual infinity” (that is to say sets made up of an infinity of objects conceived as existing simultaneously, at least in their thought), and content themselves with “potential infinity”, that is to say the possibility of increasing every given quantity (or of decreasing it if it is a case of a “continuous” quantity).⁴⁰ If this point of view contained a certain dose of hypocrisy,⁴¹ it allowed however the development of the larger part of classical mathematics (including the theory of ratios and later the infinitesimal Calculus);⁴² it even seemed an excellent safeguard, above all after the quarrels caused by the infinitely small, and had become a dogma almost universally subscribed to until well into the XIXth century.

A first glimmer of the general notion of equipotence appears in a remark of Galileo ([122 b], v. VIII, pp. 78-80): he observes that the map $n \mapsto n^2$ establishes a bijective correspondence between the natural numbers and their squares and that it follows that the axiom “the whole is greater than the part”

could not be applied to an infinite set. But far from inaugurating a rational study of infinite sets, this remark seems to have had no other effect than to reinforce mistrust of actual infinity; it is already the conclusion of Galileo himself, and Cauchy, in 1833, only quotes him in order to approve of his attitude.

The necessities of Analysis — and particularly the deep study of functions of real variables, which was taking place during the whole of the XIXth century — are at the origins of what was to become the modern theory of sets. When Bolzano, in 1817, proves the existence of the lower limit of a set bounded below in $\mathbb{R}$ [27 c], he still reasons, like the greater part of his contemporaries, “by comprehension”, speaking not of an arbitrary set of real numbers, but of an arbitrary property of these latter. But when, 30 years later, he writes his “Paradoxien des

Unendlichen” [27 b] (published in 1851, three years after his death), he does not hesitate to claim the right to the existence of the “actual infinity” and to speak of arbitrary sets. He defines, in this work, the general notion of the equipotence of two sets, and proves that two compact intervals in $\mathbb{R}$ are equipotent; he notes also that the characteristic difference between finite sets and infinite sets consists in that an infinite set E is equipotent to a subset distinct from E , but he does not give any convincing proof of this assertion. The general tone of this work is in any case more philosophical than mathematical, and for lack of disassociating sufficiently clearly the notion of the power of a set from that of size or order of infinity, Bolzano fails in his attempts to construct infinite sets of greater and greater power, and lets himself be drawn on this occasion into mixing his reasoning with considerations on divergent series that are totally devoid of sense.

It is due to the genius of G. Cantor that the theory of sets as we know it today was created. He also started with Analysis, and his work on trigonometric series, inspired by those of Riemann, bring him naturally, in 1872, to a first attempt to classify “exceptional” sets which occur in this theory,⁴³ by means of the notion of successive “derived sets”, which he introduces at this point. It is no doubt in connection with these researches, and also with his method for defining the real numbers ([47], pp. 92-97), that Cantor begins to be interested in the problems of equipotence, for in 1873, he remarks that the set of rational numbers (or the set of algebraic numbers) is countable; and in his correspondence with Dedekind, which starts about this time [48], he is seen posing the question of the equipotence between the set of whole numbers and the set of real numbers, which he succeeds in solving negatively a few weeks later. Then, from 1874 on, it is the problem of dimension that preoccupies him, and for three years he seeks in vain to establish the impossibility of a bijective correspondence between $\mathbb{R}$ and $\mathbb{R}^n (n > 1)$, before succeeding, to his great surprise,⁴⁴ in defining such a correspondence. Having such results as new as they were surprising, he devotes himself entirely to the theory of sets. In a series of 6 memoirs published in the *Mathematische Annalen* between 1878 and 1884, he tackles both the problems of equipotence, the theory of totally ordered sets, the topological properties of $\mathbb{R}$ and of $\mathbb{R}^n$ and the problem of measure; and it is admirable to see with what clarity these notions appear in his hands little by little, notions which appear so inextricably tied to the classical conception of the “continuum”. From 1880, he has the idea of iterating “transfinitely” the construction

of “derived sets”; but this idea takes on life only two years later, with the introduction of well ordered sets, one of the most original discoveries of Cantor, thanks to which he can tackle a detailed study of cardinal numbers and formulate the “problem of the continuum” [47].

It was impossible that such bold conceptions, reversing a twice millennial tradition, and leading to results so unexpected and of such a paradoxical appearance, would be accepted without resistance. In fact, amongst the mathematicians who were influential at that time in Germany, Weierstrass was the only one to follow with some favour the work of Cantor (his erstwhile pupil); this latter was to run up on the other hand against the irreducible opposition of Schwarz and especially of Kronecker.⁴⁵ It is, it would appear, as much the constant tension generated by the opposition to his ideas, as his fruitless efforts to prove the continuum hypothesis, which caused in Cantor the first symptoms of a nervous illness from which his mathematical productivity would suffer.⁴⁶ He only really took up again his interest in the theory of sets in 1887, and his last publications date from 1895-1897; he develops there mostly the theory of totally ordered sets and the calculus of ordinals. He had also proved in 1890 the inequality $m < 2^m$; however, not only did the problem of the continuum remain without answer, but there remained in the theory of cardinals a more serious gap, for Cantor had been unable to establish the existence of a well ordering between arbitrary cardinals. This gap was going to be filled, on the one hand by the theorem of F. Bernstein (1897) showing that the relations $a \leq b$ and $b \leq a$ imply $a = b$,⁴⁷ and above all by the theorem of Zermelo [342 a] proving the existence of a well order on every set — a theorem conjectured already in 1883 by Cantor ([47], p. 169).

However Dedekind, from the beginning, had not ceased to follow with a sustained interest the researches of Cantor; but while this latter concentrated his attention on infinite sets and their classification, Dedekind followed his own thoughts on the notion of number (which had already led him to his definition of irrationals by “cuts”). In his opusculum *Was sind und was sollen die Zahlen*, published in 1888, but the essentials dating from 1872-78 ([79], v. III, p. 335), he shows how the notion of natural number (on which, as has been seen, the whole of classical Mathematics had ended up by relying) could itself be derived from fundamental notions in the theory of sets. Developing (no doubt the first explicitly) the elementary properties of arbitrary maps from one set to another (neglected until Cantor, who was only interested in bijective correspondences),

he introduces, for every map f of a set E into itself, the notion of “chain” of an element $a \in E$ relative to f , namely the intersection of the sets $K \subset E$ such that $a \in K$ and $f(K) \subset K$.⁴⁸ He then takes as his definition of an infinite set E the fact that there exists a bijective map ϕ of E into E such that $\phi(E) \neq E$;⁴⁹ if further there exists such a map ϕ and an element $a \notin \phi(E)$ for which E is the chain of a , Dedekind says that E is “simply infinite”, remarks that the “axioms of Peano” are then satisfied and shows (before Peano) how starting from there all the theorems of elementary arithmetic are obtained. The only missing part of his exposé is the axiom of infinity that Dedekind (following Bolzano) believes himself able to prove by considering the “world of thought” of humans (“Gedankenwelt”) as a set.⁵⁰

On another tack, Dedekind had been led by his work on arithmetic (and notably the theory of ideals) to consider the notion of ordered set in a more general setting than Cantor. While this latter restricts himself exclusively to totally ordered sets,⁵¹ Dedekind tackles the general case, and makes in particular a deep study of lattices ([79], v. II, pp. 236-271). This work was hardly noticed at the time; although these results, found again by several authors, have been the object of numerous publications since 1935, their historical importance arises much less from the possibilities for applications, fairly thin no doubt, of this theory, than from the fact that they constituted one of the first examples of careful axiomatic construction. On the other hand, the first results of Cantor on sets which were countable or had the power of the continuum were rapidly to have numerous and important applications right up to the most classical questions of Analysis⁵² (without talking naturally of those parts of the Cantorian work that inaugurated General Topology and measure theory; on that see pp. 141 and 221). Moreover, from the last years of the XIXth century the first uses of the principle of transfinite induction were appearing, having become, especially after the proof of the theorem of Zermelo, an indispensable tool in all parts of modern mathematics. Kuratowski was to give, in 1922, an often more amenable version of this principle, avoiding the use of well ordered sets ([189 a], p. 89); it is in this form, found again later by Zorn [344], that it is mainly used at the present time.⁵³

Towards the end of the XIXth century, the essential concepts of Cantor had thus gained their cause.⁵⁴ We have seen that, at about this time, the formalisation of mathematics is achieved and the use of the axiomatic method is almost universally agreed. But, at the same time, a “crisis

of foundations” of a rare violence was opening up, which was to shake the mathematical world for more than 30 years, and to seem at times to compromise, not only all the recent acquisitions, but even the most classical parts of Mathematics.

THE PARADOXES OF THE THEORY OF SETS AND THE CRISIS OF FOUNDATIONS.

The first “paradoxical” sets appeared in the theory of cardinals and ordinals. In 1897, Burali-Forti remarked that one cannot consider the existence of a set consisting of all the ordinals, because this set would be well ordered, and so isomorphic to one of its segments distinct from the whole, which is absurd [42].⁵⁵ In 1899, Cantor observes (in a letter to Dedekind) that it is no longer possible to speak of the cardinals forming a set, nor to speak of the “set of all sets” without reaching a contradiction (the set of subsets of this latter “set” Ω would be equipotent to a subset of Ω , which is contrary to the inequality $m < 2^m$) ([47], pp. 444-448). In 1905 finally, Russell analysing the proof of this inequality, shows that the reasoning that establishes it proves (without appealing to the theory of cardinals) that the notion of the “set of all sets which are not elements of themselves” is, it also, contradictory [266].⁵⁶

It could be believed that such “antinomies” would only be manifested in the peripheral regions of mathematics, characterised by the consideration of sets of a “size” inaccessible to intuition. But other “paradoxes” were soon to threaten the most classical parts of mathematics. Berry and Russell ([266], v. I, pp.63-64), simplifying an argument of J. Richard [258], observe in effect that the set of whole numbers of which the definition can be expressed in less than sixteen words in English is finite, but that it is however contradictory to define a whole number as “the least whole number which is not definable by less than sixteen words in English”, because this definition consists of fifteen words only.

Although such reasoning, so far from that usually used by mathematicians, may have appeared to many of them as a kind of riddle, they pointed the way none the less to the necessity for a revision of the bases of mathematics, aiming to eliminate “paradoxes” of this nature. But although there was unanimity concerning the urgency of this revision, radical divergences were

soon to appear concerning the manner of realising it. For a first group of mathematicians, be they “idealists”, be they “formalists”,⁵⁷ the situation created by the “paradoxes” of the theory of sets was very analogous to that which resulted, in geometry, from the discovery of non-Euclidean geometries or of “pathological” curves (such as curves without tangents); it must lead to a similar, but more general, conclusion namely that it is hopeless to seek to base an arbitrary mathematical theory on an appeal (explicit or not) to “intuition”. The position can be summed up by the very words of the principal adversary of the formalist school: “The formalist” says Brouwer ([39 a], p. 83), “submits that human reason does not have at its disposal exact pictures of straight lines or numbers greater than ten, for example ... It is true that certain relations between mathematical entities, that we take as axioms, we can deduce from other relations according to fixed rules, with the conviction that in this way we derive truth from other truths by logical reasoning ... [But] for the formalist, mathematical exactitude resides only in the development of the sequence of relations, and is independent of the meaning that one could wish to give to the relations or entities that they link.”

It consists, therefore, for the formalist, in giving to the theory of sets an axiomatic basis quite analogous to that of elementary geometry, where one does not concern oneself with what the “objects” are that are called “sets”, nor what is meant by the relation $x \in y$, but where one enumerates the conditions to be imposed on this latter relation; of course, this must be done in such a way as to include, as far as possible, all the results of the theory of Cantor, all the while making it impossible for “paradoxical” sets to exist. The first example of such an axiomatisation was given by Zermelo in 1908 [342 c]; he avoids sets that are “too large” by the introduction of an “axiom of selection (Aussonderung)”, a property $P(x)$ only determining a set made up of the elements which have it if $P(x)$ already entails a relation of the form $x \in A$.⁵⁸ But the elimination of paradoxes analogous to “Richard’s paradox” could only be done by restricting the meaning attached to the notion of “property”; on that, Zermelo contents himself with describing in very vague terms a type of property that he calls “defined”, and to indicate that one must limit oneself to these latter in the application of the axiom of selection. This point was made precise by Skolem [286 a] and Fraenkel [113 b]; as they observed, its elucidation forces us to place ourselves in a completely formalised system, where the notions of “property” and of

“relation” have lost all “significance” and have become simple designations for the collections formed following explicit rules. That necessitates of course that one must put into the system the rules of Logic that are used, which was not yet the case in the system of Zermelo-Fraenkel.

Other axiomatisations of the theory of sets were proposed following on from this. Let us mention principally that of von Neumann [342 a and c] which, more than the system of Zermelo-Fraenkel, approached the primitive conception of Cantor: this latter, to avoid paradoxical sets, had already proposed ([47], pp. 445-448), in his correspondence with Dedekind, to distinguish between two kinds of sets, the “multiplicities” (“Vielheiten”) and the “sets” (“Mengen”) proper, the second ones being characterised in that they can be thought of as a single object. It is this idea that von Neumann makes precise in distinguishing two types of objects, “sets” and “classes”; in his system (almost completely formalised), classes are distinguished from sets in that they cannot be placed on the left of the sign $\in$. One of the advantages of such a system is that it rehabilitates the notion of “universal class” used by the logicians of the XIXth century (which naturally is not a set); let us point out also that the system of von Neumann avoids (for the theory of sets) the introduction of axiom schemes, replaced by suitable axioms (which makes its logical study easier). Some variants of the system of von Neumann have been given by Bernays and Gödel [130 b].

The elimination of paradoxes seems to have been well carried out by the systems above, but at the cost of restrictions which cannot but seem to be very arbitrary. In favour of the system of Zermelo-Fraenkel, one can say that it restricts itself to formulating restrictions that only sanction the current practice in the applications of the notion of set to the various mathematical theories. The systems of von Neumann and of Gödel are further removed from the usual conceptions; on the other hand, it is not excluded that it is easier to insert certain mathematical theories still in their beginnings in the framework provided by such systems rather than in the narrower framework of the system of Zermelo-Fraenkel.

Certainly one could not affirm that any of these solutions give the impression of being definitive. If they satisfy the formalists, it is that these latter refuse to take into consideration the individual psychological reactions of each mathematician; they assume that a formalised language has completed its task when it can transcribe mathematical reasoning into a form lacking am-

biguity, and so useful as a vehicle for mathematical thought; everyone is free, they would say, to think what he likes about the “nature” of mathematical objects or the “truth” of the theorems that he uses, so long as his reasoning can be transcribed into the common language. ⁵⁹

In other words, from a philosophical point of view, the attitude of the formalists consists in being disinterested in the problem set by the “paradoxes”, in abandoning the platonic position which aimed to attribute to mathematical notions an intellectual “content” common to all mathematicians. Many mathematicians draw back faced with this break with tradition. Russell, for example, seeks to avoid paradoxes by analysing their structure in a deeper way. Taking up again an idea first brought forward by J. Richard (in the article [258] where he displayed his “paradox”) and developed later by H. Poincaré [251 c], Russell and Whitehead observe that the definitions of paradoxical sets all violate the following principle, called the “principle of the vicious circle”: “An element of which the definition involves the totality of the elements of a set can not belong to that set” ([266], v. I, p. 40). Also it is this statement that serves as basis for the Principia, and it is to satisfy it that the “theory of types” is developed in this work. Like that of Frege by which it is inspired, the logic of Russell and Whitehead contains “propositional variables”; the theory of types proceeds to a classification of these different variables, of which the main lines are the following. Starting from a “domain of individuals” which are not specified and which can be qualified as “objects of order 0”, the relationships where the variables (free or tied) are individuals, are said to be “objects of the first order”; and in general, the relations in which the variables are objects of order $\leq n$ (one at least being of order n) are said to be “objects of order $n + 1$ ”. ⁶⁰ A set of objects of order n can then only be defined by a relation of order $n + 1$, a condition that allows the elimination without trouble of the paradoxical sets. ⁶¹ But the principle of the “hierarchy of types” is so restrictive that by adhering to it strictly, one ends up with a mathematics of inextricable complexity. ⁶² To escape this consequence, Russell and Whitehead are constrained to introduce an “axiom of reducibility” affirming the existence, for every relation between “individuals”, of a relation of the first order to which it is equivalent; a condition quite as arbitrary as the axioms of the formalists, and which reduces considerably interest in the construction of the Principia. So the system of Russell and Whitehead has had more success with the logicians than with the mathematicians; it is in any case not entirely for-

malised, ⁶³ and as a result there are numerous obscure details. Various efforts have been made to simplify and clarify this system (Ramsey, Chwistek, Quine, Rosser); tending to use languages that are more and more fully formalised, these authors replace the rules of the Principia (which still had a certain intuitive foundation) with restrictions only taking account of the wording of the aggregations being considered; not only do these rules appear to be as gratuitous as the prohibitions formulated in the systems of Zermelo-Fraenkel or of von Neumann, but, being further removed from the practice of mathematicians, they have, in many cases, led to unacceptable consequences, which the author had not foreseen (such as the paradox of Burali-Forti, or the negation of the axiom of choice).

For mathematicians of the preceding schools, it was above all a case of not renouncing any part of the heritage of the past: “From the paradise that Cantor has created for us”, says Hilbert ([163 c], p. 274), “no one should be able to chase us”. To do this, they are willing to accept limitations on mathematical reasoning, hardly essential because consonant with use, but which do not seem to be imposed by our mental customs and the intuitive idea of the notion of set. Anything seems to them preferable to the intrusion of psychology into the criteria of validity in mathematics; and rather than “putting into account the characteristics of our brains”, as Hadamard says ([12], p. 270), they are resigned to imposing on the mathematical domain limits which are for the most part arbitrary, so long as they include classical Mathematics and do not risk hampering future progress.

It is a very different matter when we see the attitude of mathematicians belonging to the tendency of which it remains for us to speak. If the formalists accept the renunciation of the control of the “eyes of the mind” so far as mathematical reasoning is concerned, the mathematicians that have been called “empiricists”, “realists” or “intuitionists” refuse this abdication; they must have a kind of internal certainty guaranteeing the “existence” of the mathematical objects with which they are concerned. As long as it was only a case of renouncing spatial intuition, there was no serious objection, since the arithmetical “models” allowed retrenchment behind the intuitive notion of whole numbers. But irreducible opposition manifests itself when it is a matter of reducing the notion of whole number to that (much less intuitively precise) of set, then of imposing on the handling of sets barriers without intuitive foundation. The first in time

of these opponents (and the one who, by the authority of his genius, was to exercise the most influence) was H. Poincaré; having admitted, not only the axiomatic point of view for geometry and the arithmetisation of Analysis, but also a good part of the theory of Cantor (which he was one of the first to apply fruitfully in his work), he refuses on the other hand to conceive that arithmetic may be able, it also, to be justified by an axiomatic treatment; the principle of complete induction seems to him in particular a fundamental intuition of our mind, where it is impossible to see a pure convention ⁶⁴ [251 c]. Hostile in principle to formalised languages, of which he contested the usefulness, he confuses constantly the notion of whole number in formalised mathematics, and the use of whole numbers in the theory of proof, which was starting up then, and of which we will speak later; no doubt it was difficult at that time to make as clearly as today — after 50 years of study and discussion — this distinction which however had been clearly felt by a Hilbert or a Russell.

The critics of this kind multiplied after the introduction of the axiom of choice by Zermelo in 1904 [342 a]. Its use in many previous proofs in Analysis or the theory of sets had until then gone by almost unnoticed; ⁶⁵ it is in following an idea suggested by Erhard Schmidt that Zermelo, explicitly formulating this axiom, deduced from it in an ingenious way a satisfactory proof of the existence of a well ordering on every set. Coming at the same time as the “paradoxes”, it appears that this new way of reasoning, by its odd allure, had thrown many a mathematician into confusion; it is enough to see the strange misunderstandings which surged up about this topic, in the following volume of *Mathematische Annalen*, under the signature of mathematicians who were as familiar with the Cantorian methods as Schoenflies and F. Bernstein. The criticisms of E. Borel, published in this same volume, are more substantial and attach themselves clearly to the point of view expressed by H. Poincaré about whole numbers; they are developed and discussed in an exchange of letters between E. Borel, Baire, Hadamard and Lebesgue, who had remained in the classical tradition in French mathematics [12]. Borel starts by denying the validity of the axiom of choice because it contains in general an uncountable infinity of choices, which is inconceivable intuitively. But Hadamard and Lebesgue observe that a countable infinity of arbitrary successive choices is no more intuitive, since it consists of an infinity of operations, that it is impossible to conceive as really taking place. For Lebesgue, who enlarged the debate,

everything reduces to knowing what is meant by saying that a mathematical object “exists”; it is necessary, for him, that one “names” explicitly a property defining it in a unique way (a “functional” property, we would say); for a function such as the one used by Zermelo in his reasoning, that is what Lebesgue calls a “law” of choice; if, he continues, this requirement is not satisfied, and that one limits oneself to “thinking” about this function instead of “naming” it, can one be sure that during the reasoning process one is always thinking of the same thing ([12], p. 267)? In any case, this brings Lebesgue to new doubts; already the question of the choice of a single element in a set seems to him to raise difficulties: one must be sure that such an element “exists”, that is to say that one can “name” at least one of the elements of the set. ⁶⁶ Can one then speak of the “existence” of a set of which one does not know how to “name” each element? Already Baire does not hesitate to deny the “existence” of the set of all subsets of a given infinite set (*loc. cit.*, pp. 263-264); in vain does Hadamard observe that these requirements lead to renouncing even the possibility of speaking of the real numbers: it is to just that conclusion that E. Borel finally agrees. Putting aside the fact that countability seems to have acquired freehold, one has returned just about to the classical position of the adversaries of the “actual infinite”.

All these objections were not very systematic; it was down to Brouwer and his school to undertake a complete remould of mathematics guided by similar, but even more radical, principles. We would not dare to summarise here a doctrine as complex as intuitionism, which consists as much of psychology as mathematics, and we will limit ourselves to indicating some of the most striking aspects, referring for more details to the work of Brouwer himself [39 a and b] and to the exposition by Heyting [162]. For Brouwer, Mathematics is identical with the “exact” part of our thought, based on the first intuition of the sequence of natural numbers, and which it is impossible to translate without mutilation into a formal system. It is in any case only “exact” in the thought of mathematicians, and it is a chimera to hope to develop an instrument of communication between them which is not subject to all the imperfections and ambiguities of language; at most it is possible to hope to awaken in the interlocutor a favourable state of mind by descriptions that are more or less vague ([162], pp. 11-13). Intuitionist mathematics attaches hardly more importance to logic than to language: a proof is not conclusive by virtue of logical

rules fixed once and for all, but because of the “immediate evidence” of each of its steps. This “evidence” must moreover be interpreted in a more restrictive way than by E. Borel and his followers: it is thus that in intuitionist mathematics, it cannot be said that a relation of the form “ R or (not R)” is true (principle of the excluded middle), unless for all system of values given to the variables appearing in R , one can prove that one of the two propositions R , “not R ” is true; for example, from the equation $ab = 0$ between two real numbers, it cannot be concluded that “ $a = 0$ or $b = 0$ ”, for it is easy to create explicit examples of real numbers a , b for which one has $ab = 0$ without knowing, at the actual time, how to prove either of the two propositions $a = 0$, $b = 0$ ([162], p. 21).

It is not astonishing that, starting from such principles, the intuitionist mathematicians finish up with results that are very different from the classical theorems. A whole part of these disappear, for example most of the “existential” theorems in Analysis (such as the theorems of Bolzano and Weierstrass for real functions); if a function of a real variable “exists” in the intuitionist sense, it is ipso facto continuous; a bounded monotone sequence of real numbers does not necessarily have a limit. On the other hand, many classical notions become ramified, for the intuitionist, into several distinct fundamental notions: there are thus two notions of convergence (for a sequence of real numbers) and eight of countability. It goes without saying that transfinite induction and its applications to modern Analysis are (as for the greater part of the theory of Cantor) condemned without appeal.

It is only in this way that, according to Brouwer, mathematical propositions can acquire a “content”; the formalist reasonings that go beyond what the intuitionists allow are judged to be without value, since one can no longer give them a “meaning” to which the intuitive notion of “truth” could be applied. It is clear that such judgements can only rely on a pre-existing notion of “truth”, of a psychological or metaphysical nature. It is to say in practice that they escape all discussion.

It is undoubted that the vigorous attacks that came from the intuitionist camp have sometimes forced not only the mathematical schools in the avant-garde, but even the followers of traditional mathematics to go on the defensive. A famous mathematician has admitted having been impressed by the attacks to the point of having voluntarily restricted his work to branches

of mathematics judged to be “certain”. But such cases can hardly have been frequent. The intuitionist school, of which the memory is no doubt destined to remain only as an historical curiosity, would at least have been of service by having forced its adversaries, that is to say definitely the immense majority of mathematicians, to make their position precise and to take more clearly notice of the reasons (the ones of a logical kind, the others of a sentimental kind) for their confidence in mathematics.

METAMATHEMATICS

The absence of contradiction has at all times been considered as a *sine qua non* condition for all mathematics, and from the time of Aristotle, logic was sufficiently developed so that it was perfectly well known that from an inconsistent theory anything could be deduced. The proofs of “existence”, considered as indispensable since Antiquity, visibly had no other aim than to guarantee that the introduction of a new concept did not risk entailing a contradiction, particularly when this concept was too complicated to fall immediately under “intuition”. We have seen how this requirement became more pressing with the arrival of the axiomatic point of view in the XIXth century and how the construction of arithmetical “models” responded to it. But arithmetic itself, could it be inconsistent? A question that without doubt would not have been thought of before the end of the XIXth century, so much did the whole numbers appear to belong to what was most certain in our intuition; but after the “paradoxes” everything seemed to have been questioned, and one can understand that the feeling of insecurity that they created had brought mathematicians, around 1900, to bear with more care on the problem of the non-contradiction of arithmetic, so as to save at least classical mathematics from shipwreck. So this problem is the second of those that Hilbert enumerated in his famous lecture at the international Congress of 1900 ([163 a], v. III, pp. 229-301). In doing this, he laid down a new principle that was to have great repercussions: whereas in traditional logic the non-contradiction of a concept only made it “possible”, it was equivalent for Hilbert (at least for mathematical concepts defined axiomatically) to the existence of the concept. This implied apparently the necessity of proving a priori the non-contradiction of a mathematical theory before even being able to develop it legitimately; it is just in this way that H. Poincaré understands it when, to make a breach in

formalism, he takes to his account the idea of Hilbert, underlining with a mischievous pleasure the extent to which the formalists were far, at that stage, from being able to achieve it ([251 c], p. 163). We will see later how Hilbert was to pick up the gauntlet; but before that, it must be noted here that, under the influence of his authority and that of Poincaré, the requirements set down by this latter were to be for a long time accepted without reservation as well by the formalists as by their adversaries. One consequence was the belief, very widespread among the formalists, that the theory of proof of Hilbert was an integral part of mathematics, of which it constituted the indispensable prolegomena. This dogma does not seem justified to us,⁶⁷ and we consider that the intervention of metamathematics in the exposition of logic and mathematics can and should be reduced to the very elementary part that is concerned with the handling of abbreviating symbols and criteria of deduction. It is thus not a case, contrary to what Poincaré thinks, of “claiming the freedom from contradiction”, but rather to consider, with Hadamard, that the absence of contradiction, even when it is not proven, is noted ([12], p. 270).

It remains for us to give a brief historical sketch of the efforts of Hilbert and his school, and to retrace summarily, not only the evolution that was finally to lead to the negative result of Gödel and justify a posteriori the scepticism of Hadamard, but also all the progress that followed on from this, concerning the knowledge of the mechanism of mathematical reasoning, and which makes modern metamathematics an autonomous science of incontestable interest.

Already in 1904, in a lecture at the international Congress ([163 c], pp. 247-261), Hilbert attacks the problem of consistency in arithmetic. He observes first that there is no question of proving it by making use of a model,⁶⁸ and outlines the principles of another method: he proposes considering the true propositions of formalised arithmetic as collections of signs without meaning, and to prove that in using the rules governing the formation and linkage of these collections, it is impossible to obtain a collection which is a true proposition and whose negation is also a true proposition. He sketches even a proof of this nature for a formalism less extensive than that of arithmetic; but, as was observed by H. Poincaré soon after ([251 c], p. 185), this proof uses essentially the principle of recurrence, and so appears to rely on a vicious circle. Hilbert did not answer this criticism immediately, and about fifteen years passed by without anyone attempting to develop his ideas; it is in 1917 only that (motivated by the wish to answer

the attacks of the intuitionists) he harnesses himself again to the problem of the foundations of mathematics, which he will not cease from now on to consider until the end of his scientific career. In his work on this subject, which is spread out from 1920 to 1930 approximately, and in which a whole school of young mathematicians take part (Ackermann, Bernays, Herbrand, von Neumann), Hilbert brings out little by little the principles of his “theory of proof” in a more precise way: acknowledging implicitly the validity of the criticism of Poincaré, he admits that in metamathematics, the arithmetical reasoning that is used can only be based on our intuition of the whole numbers (and not on formalised arithmetic); to do this it appears to be essential to restrict this reasoning to “finite procedures” (“finite Prozesse”) of a type allowed by the intuitionists: for example, a proof by contradiction cannot prove the metamathematical existence of a collection or sequence of collections, there must be an explicit law of construction.⁶⁹ On the other hand Hilbert enlarges in two directions his initial programme: not only does he tackle the consistency of arithmetic, but he aspires also to prove the consistency of the theory of real numbers and even that of the theory of sets;⁷⁰ further, to the problems of consistency come to be added those of independence of axioms, of completeness and of decision. We will pass quickly in review these various questions and point out the principal research to which they have given rise.

The proof of the independence of a system of propositions $A_1, A_2, \dots, A_n$ consists in showing that, for each index i , A_i is not a theorem in the theory T_i obtained by taking as axioms the A_j with index $j \neq i$. This point will be established if one knows a consistent theory T'_i in which the $A_j (j \neq i)$ are theorems, as well as “not A_i ”; thus one can consider the problem in two ways depending on whether or not one allows that certain theories (such as arithmetic or the theory of sets) are consistent. In the second case, one has to deal with a problem of “absolute” consistency. On the contrary, the first type of problem reduces to a problem of “relative” consistency, by means of a construction of appropriate “models”, and many proofs of this kind have been thought up, well before mathematics had taken up a completely formalised aspect: it is enough to remember the models of non-Euclidean geometry (see p. 133), the questions of the independence of the axioms of elementary geometry dealt with by Hilbert in the “Grundlagen der Geometrie” [163 c] as well as the work of Steinitz on the axiomatisation of Algebra and those of Hausdorff and

his successors on that of Topology.

A theory T is said to be complete if, for every proposition A of T (not containing any letter other than the constants of T), one of the propositions A , $(\text{not } A)$, is a theorem of T .⁷¹ If one puts on one side certain very rudimentary formalisms, of which the completeness is easily proved ([164], p. 35), the results obtained in this area are essentially negative; the first in time, is due to K. Gödel [130 a] who showed that, if T is consistent and if the axioms of formalised arithmetic are theorems of T , then T is not complete. The fundamental idea of his ingenious method consists in establishing a bijective correspondence (of course, by means of “finite processes”) between the metamathematical statements and certain propositions in formalised arithmetic; we will restrict ourselves to a broad sketch.⁷² For each collection A which is a term or a relation of T , one begins by associating (by a procedure which is an explicit construction, which can be applied quasi mechanically) a whole number $g(A)$, in a bijective way. In the same way, to each proof D of T (considered as a succession of collections) one can associate a whole number $h(D)$ in a bijective way. Finally, one can give a procedure for constructing explicitly a relation $P(x, y, z)$ of T ,⁷³ such that, in T , $P(x, y, z)$ requires that x, y, z are whole numbers, and satisfies the following two conditions:

1° if D is a proof of $A(\lambda)$, where $A(x)$ is a relation of $\mathcal{T}$, and λ is an explicit whole number (that is to say a term of $\mathcal{T}$ which is a whole number), then $P(\lambda, g(A(x)), h(D))$ is a theorem of $\mathcal{T}$;

2° if the explicit whole number μ is not of the form $h(D)$, or if $\mu = h(D)$ and if D is not a proof of $A(\lambda)$, then $(\text{not } P(\lambda, g(A(x)), \mu))$ is a theorem of $\mathcal{T}$.

Then let $S(x)$ be the relation $(\text{not } (\exists z)P(x, y, z))$, and let $\gamma = g(S(x))$, which is a term of $\mathcal{T}$. If $\mathcal{T}$ is consistent, there is no proof of the proposition $S(\gamma)$ in $\mathcal{T}$. In fact, if D were such a proof, $P(\gamma, g(S(x)), h(D))$ would be a theorem of $\mathcal{T}$; but this relation is none other than $P(\gamma, \gamma, h(D))$, and it follows that $(\exists z)(P(\gamma, \gamma, z))$ would also be a theorem of $\mathcal{T}$; as this last relation is equivalent to $(\text{not } S(\gamma))$, $\mathcal{T}$ would be inconsistent. On the other hand what has just been said shows that for every explicit whole number μ , $(\text{not } P(\gamma, \gamma, \mu))$ is a theorem of $\mathcal{T}$. The result is that there is no proof, in $\mathcal{T}$, of $(\text{not } S(\gamma))$, for this last relation is equivalent to $(\exists z)P(\gamma, \gamma, z)$ and the existence of a whole number μ such that $P(\gamma, \gamma, \mu)$ would require that

$\mathcal{T}$ were inconsistent, by virtue of what comes before. ⁷⁴ This metamathematical theorem of Gödel has been subsequently generalised in various directions ([181], chap. XI). ⁷⁵

The relation $S(\gamma)$ of T of which it is shown thus that there is no proof in T either of this relation, or of its negation, is obviously made up for the needs of the cause and does not tie up in any natural way with any mathematical problem. Much more interesting is the fact that if $\mathcal{T}$ denotes the theory of sets (with the system of axioms of von Neumann-Bernays), neither the continuum hypothesis nor its negation are provable in $\mathcal{T}$. This remarkable result was established in two stages: in 1940, Gödel proved that the theory obtained by adjoining to $\mathcal{T}$ the continuum hypothesis $2^{\aleph_0} = \aleph_1$ was not inconsistent [130 b]; then, in 1963, P. Cohen proved the same when one adjoins to $\mathcal{T}$ the relation $2^{\aleph_0} = \aleph_2$ (or $2^{\aleph_0} = \aleph_n$ for an arbitrary whole number $n > 1$) [67].

The decision problem (“Entscheidungsproblem”) is without doubt the most ambitious of all those which are stated in metamathematics: it consists in knowing whether, for a given formal language, one can imagine a “universal procedure” which is quasi mechanical which, when applied to any relation of the formalism in question, indicates in a finite number of operations whether that relation is true or not. ⁷⁶ The solution of this problem was already part of the grand design of Leibniz, and it appears that at one time the school of Hilbert believed its realisation was quite close. It is a fact that one can describe such procedures for formalisms containing few primitive symbols and axioms ([181], pp. 136-141). But the efforts made to make precise the decision problem, by outlining exactly what is meant by “universal procedure” have until now resulted only in negative results ([181], pp.432-439). Besides, the solution of the decision problem for a theory T entails immediately the knowledge of whether T is consistent or not, since it suffices to apply the “universal procedure” to a relation of T and its negation; and we will see that it is impossible to solve the question in this way for the usual mathematical theories.

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It is indeed in the question of the consistency of mathematical theories — the origin and very heart of Metamathematics — that the results have revealed themselves to be the most deceiving. During the years 1920-1930, Hilbert and his school had developed new methods for attacking these problems; after having proved the consistency of partial formalisms, covering

part of arithmetic (cf. [158], [324 d]), they believed they had reached the goal and proved, not only the consistency of arithmetic, but also that of the theory of sets, when Gödel, relying on the incompleteness of arithmetic, deduced from it the impossibility of proving, by the “finite procedures” of Hilbert, the consistency of every theory T containing this latter.⁷⁸

The theorem of Gödel does not however shut the door completely on attempts to prove consistency, so long as one abandons (at least partially) the restrictions of Hilbert concerning “finite procedures”. It is thus that Gentzen in 1936 [127], succeeds in proving the consistency of formalised arithmetic, by using “intuitively” transfinite induction up to the countable ordinal ε_0 .⁷⁹ The value of the “certainty” that one can attach to such reasoning is without doubt less convincing than for that which satisfies the initial requirements of Hilbert, and is essentially a matter of personal psychology for each mathematician; it remains no less true that similar “proofs” using “intuitive” transfinite induction up to a given ordinal, would be considered important progress if they could be applied, for example, to the theory of real numbers or to a substantial part of the theory of sets.

Chapter 2

NOTATION; COMBINATORIAL ANALYSIS.

History and archaeology make us conscious of a large number of “systems of notation”; their initial goal was to attach to each individual whole number (up to a limit which depends on the needs of the application) a name and a written representation, made up of a combination of a restricted number of symbols, accomplished by following more or less regular laws. By far the most frequent procedure consists in splitting up the whole numbers into sums of “successive units” $b_1, b_2, \dots, b_n, \dots$, of which each is a multiple by a whole number of the preceding one; and if in general b_n/b_{n-1} is taken to be equal to the same number b (the “base” of the system, most often 10), many exceptions to this rule are noticed, as with the Babylonians, where b_n/b_{n-1} is sometimes equal to 10, sometimes to 6 [232], and in the chronological system of the Mayas, where b_n/b_{n-1} is equal to 20 except for $n = 2$, and where $b_2/b_1 = 18$ [228]. As for the corresponding written representation, it must indicate the number of “units” b_i of each order i ; in many systems (such as with the Egyptians, the Greeks and the Romans), successive multiples $k.b_i$, (where k varies from 1 to $(b_{i+1}/b_i) - 1$) are represented by symbols which depend at the same time on k and on i . A first and important improvement consisted in representing all the numbers $k.b_i$ (for the same value of k) by the same symbol: it is the principle of the “notation of position”, where the index i is indicated by the fact that the symbol representing $k.b_i$ appears in the i th position in the sequence of “slices” making up the number being represented. The first

system of this kind is found among the Babylonians, who, without doubt already 2000 years before Christ, denote by the same sign all the multiples $k.60^{\pm i}$ corresponding to arbitrary values of the exponent i ([232], pp. 93-109). The inconvenience of such a system resides of course in the ambiguity of the symbols used, so long as nothing indicates that the units of a certain order may be absent, in other words, so long as the system is not completed by the introduction of a “zero”. One sees however the Babylonians making do without such a symbol for the greater part of their history; in fact they only use such a “zero” in the last two centuries before Christ, and still only in the interior of a number; until then, the context alone had to specify the significance of the number under consideration. Only two other systems use a “zero” systematically: that of the Mayas (in use, it would appear, already from the beginning of the Christian era [228]), and our actual decimal system, which (by the agency of the Arabs) is derived from Hindu mathematics, where its use is attested already from the first centuries of our era. It must be noted moreover that the conception of zero as a number (and not as a simple symbol of separation) and its introduction into calculations, also count amongst the original contributions of the Hindus ([78], v. I). Of course, once the principle of the notation of position has been acquired, it was easy to extend it to any base whatever; the discussion concerning the merits of different “bases” proposed since the XVIIth century is a matter of the techniques of numerical Calculation and should not be tackled here. We limit ourselves to remarking that the operation that serves as foundation to these systems, the so-called “Euclidean” division, does not appear before the Greeks, and no doubt goes back to the first Pythagoreans, who made it the essential tool of their theoretical Arithmetic (see p. 85).

The general problems of enumeration, grouped under the name of “combinatorial Analysis”, do not seem to have been tackled before the last centuries of classical Antiquity: only the formula $\binom{n}{2} = n(n-1)/2$ is attested to in the IIIrd century of our era. The Hindu mathematician Bhaskara (XIIth century) knows the general formula for $\binom{n}{p}$. A more systematic study is to be found in a manuscript of Levi ben Gerson, at the beginning of the XIIIth century: he obtains the recurrence formula allowing the calculation of the number V_n^p of arrangements of n objects p at a time, and in particular the number of permutations of n objects; he states also rules equivalent to the relations $\binom{n}{p} = V_n^p/p!$ and $\binom{n}{n-p} = \binom{n}{p}$ ([311], v. VI, pp. 64-65). But this manuscript seems to have been

ignored by his contemporaries, and its results have only been rediscovered little by little by the mathematicians of the following centuries. Among the later advances, let us pick out that Cardan proves that the number of non-empty subsets of a set of n elements is $2^n - 1$; Pascal and Fermat, setting up the calculus of probability, rediscover the expression for $\binom{n}{p}$, and Pascal is the first to observe the relationship between these numbers and the binomial formula: this latter seems to have been already known to the Arabs from the XIIIth century, the Chinese in the XIVth century, and it was rediscovered in the West at the beginning of the XVIth century, as well as the method of calculation by recurrence known as the “arithmetical triangle”, that is normally attributed to Pascal ([311], v. VI, pp. 35-38). Finally Leibniz, around 1676, obtains (without publishing it) the general formula for “multinomial coefficients”, rediscovered independently and published 20 years later by de Moivre.

Chapter 3

THE EVOLUTION OF ALGEBRA.

There are few notions, in Mathematics, that are more primitive than that of a law of composition: it seems inseparable from the first rudiments of calculation with whole numbers and measurable quantities. The most ancient documents that remain from the Mathematics of the Egyptians and Babylonians show us that they are already in possession of a complete system of rules of calculation for whole numbers > 0 , rational numbers > 0 , lengths and areas; even though the Babylonian texts that have reached us only deal with problems in which the data have explicit numerical values, ¹ they leave no doubt as to the generality attributed to the rules that are used, and denote a quite remarkable technical facility in the handling of equations of the first and second degrees ([232], pp. 179 and following). No hint of a justification of the rules used is found in any case, nor even a precise definition of the operations that occur: these like those remain in the domain of empiricism.

A similar concern, on the other hand, shows itself already very clearly among the Greeks of the classical era; one does not meet yet, it is true, an axiomatic treatment of the theory of whole numbers (such an axiomatisation will not appear until the end of the XIXth century; see p. 24); but, already in the Elements of Euclid, many are the passages giving formal proofs of rules of calculation quite as intuitively “obvious” as those of calculation with whole numbers (for example, the commutativity of the product of two rational numbers). The most remarkable proofs of this kind are those that are connected with the theory of quantities, the most original creation of Greek Mathematics (equivalent, as we know, to our theory of real numbers > 0 ; see pp. 148 ff.): Euclid considers there, amongst others, the product of two ratios of quantities,

shows that it is independent of the form in which the two ratios are given (first example of a “quotient” of a law of composition by an equivalence relation), and that it is commutative ([107], Book V, prop. 22-23). ²

We must not hide, though, the fact that this progress towards rigour is accompanied, in Euclid, with a stagnation, and even some points of regression, where the techniques of algebraic calculations are concerned. The crushing preponderance of Geometry (in the context of which the theory of quantities is manifestly conceived) paralyses all autonomous development of an algebraic notation: the elements occurring in the calculations must, at every moment, be “represented” geometrically: over and above, the two laws of composition which occur are not defined on the same set (addition of ratios is not defined in a general way, and the product of two lengths is not a length but an area); there is, as a result, a lack of flexibility which makes it almost impractical to handle algebraic relations of degree greater than the second.

It is only with the decline of classical Greek Mathematics that one sees Diophantus return to the traditions of the “logisticians” or professional calculators, who had continued to apply as they stood the rules inherited from the Egyptians and Babylonians: not burdening themselves any more with geometrical representations for those “numbers” that he considers, he is naturally led to develop rules for abstract algebraic calculation; for example, he gives rules which (in modern language) are equivalent to the formula $x^{m+n} = x^m x^n$ for small values (positive or negative) of m and n ([91 a], v. I, pp. 8-13); a bit further on (pp.12-13) is found the statement of the “rule of signs”, first hint of calculations with negative numbers; finally Diophantus uses, for the first time, a literal symbol to represent an unknown in an equation. On the other hand, he seems barely bothered to connect to general ideas the methods which he applies to the solving of his problems; as to the axiomatic conception of laws of composition, such as was beginning to dawn with Euclid, it appears foreign to the thought of Diophantus as well as to his immediate followers; it will not be found again in Algebra before the beginning of the XIXth century.

It was necessary first, during the intermediary centuries, that on the one hand a system of algebraic notation was developed, adequate for the expression of abstract laws, and that, on the other hand, the notion of “number” received a broadening that allowed, by the observation of sufficiently diversified special cases, an ascent to general concepts. On that point, the axiomatic

theory of ratios of quantities, created by the Greeks, was insufficient, because it only made precise the intuitive notion of a real number > 0 and the operations on these numbers, already known to the Babylonians in a more confused form; it will be a case on the contrary, now, of “numbers” of which the Greeks had not conceived, and of which, at first, no sensible “representation” presented itself: on the one hand, zero and negative numbers, which appeared from the high Middle Ages in Hindu Mathematics; on the other, imaginary numbers, creation of the Italian algebraists in the XVIth century.

If one puts aside the zero, introduced first of all as a symbol of notation before being considered as a number (see p. 45), the common character of these extensions is to be (at least at first) purely “formal”. By that it must be understood that the new “numbers” appear first of all as the results of operations applied in situations where they do not have, according to their strict definition, any meaning (for example, the difference $a - b$ of two whole numbers when $a < b$): whence the name of “false”, “fictitious”, “absurd”, “impossible”, “imaginary”, etc. numbers, that have been attributed to them. For the Greeks of the classical era, taken up above all with clear thinking, such extensions were inconceivable; they could only come from calculators more disposed than were the Greeks to have faith even though somewhat mystical in the power of their methods (“the generality of Analysis”, as would say the XVIIIth century), and to allow themselves to be dragged along by the mechanics of their calculations without checking at each step its soundness; a faith justified besides most often a posteriori, by the exact results to which the extensions, to these new mathematical objects, of rules of calculation valid only, in all rigour, for numbers known previously, led. That explains how there is a gradually increasing boldness in considering for themselves (independently of all applications to concrete calculations), these generalisations of the notion of number, which, at first, only arose as intermediaries in a sequence of operations of which the point of departure and the goal were true “numbers”; once this step was taken, more or less tangible interpretations of these new entities began to be sought, which acquired thus the right to be quoted in Mathematics. ⁴

On this point, the Hindus are already conscious of the interpretation that negative numbers must have in certain cases (a debt in a commercial problem, for instance). In the following centuries, as there is a diffusion into the West (by the intermediary of the Arabs) of the methods

and results of Greek and Hindu mathematics, one becomes more used to the handling of these numbers, and one begins to have other “representations” for them which are geometric or dynamic. There in any case, together with a progressive improvement in algebraic notation, is the only notable progress in Algebra during the end of the Middle Ages.

At the beginning of the XVIth century, Algebra took new flight, thanks to the discovery, by the mathematicians of the Italian school, of the solution “by radicals” of the equation of the 3rd, then of that of the 4th degree (see pp. 72 ff.); it is on this occasion that, in spite of their repugnance, they found themselves so to speak constrained to introduce into their calculations imaginary numbers; little by little, however, confidence in calculation with these “impossible” numbers is born, as with negative numbers, and that although no “representation” of them was thought up for more than two centuries.

On the other hand, algebraic notation received its decisive improvements from Viète and from Descartes; starting with this latter, algebraic notation is already, with a few exceptions, that which we use today.

From the middle of the XVIIth century to the end of the XVIIIth century, it appears that the vast horizons opened up by the creation of the infinitesimal Calculus made to some extent for a neglect of Algebra in general, and in particular mathematical thought on the laws of composition or on the nature of real and complex numbers. ⁵ It is thus that the composition of forces and the composition of velocities, well known in Mechanics already from the end of the XVIIth century, did not exert any repercussion on Algebra, although they contained already in embryo vector calculus. Indeed we must await the movement of ideas which, around 1800, leads to the geometrical representation of complex numbers (see p. 161), to see, in pure Mathematics, the use of vector addition. ⁶

It is at about the same time that, for the first time in Algebra, the notion of a law of composition extends. in two different directions, to elements that no longer come associated with “numbers” (in the widest meaning given up to that point to this word) except by distant analogies. The first of these extensions is due to C. F. Gauss, at the time of his arithmetical researches on quadratic forms $ax^2 + bxy + cy^2$ with integral coefficients. Lagrange had defined, in the set of forms with the same discriminant, an equivalence relation, ⁷ and had, on the other hand,

proved an identity which supplied, in this set, a commutative law of composition (not defined everywhere); starting from these results, Gauss shows that this law is compatible with a slightly different equivalence relation ([124 a], v. I, p. 272): “One sees by that”, he then says, “what should be understood by a class composed from two or more classes”. He proceeds afterwards to the study of the “quotient” law that he has thus just defined, establishes in substance that it is (in modern language) an abelian group law, and this by arguments of which the generality surpasses by far, in the majority of cases, the special case that Gauss considers (for example, the argument by which he proves the uniqueness of the symmetric element can be applied to any law of composition whatever (ibid., p. 273)). But he does not stop there: returning a little later to the question, he recognises the analogy between the composition of classes and the multiplication of integers modulo a prime number ⁸ (ibid. p. 371), but notes also that the group of classes of quadratic forms of given discriminant is not always a cyclic group; the indications that he gives on this matter make us think that he might perhaps have realised, at least in this particular case, the general structure of finite abelian groups ([124 a], v. I, p. 374 and v. II, p. 266).

The other series of researches of which we wish to speak ends, it also, with the notion of group, thus introducing it in a definitive way into Mathematics: it is the “theory of substitutions”, a development of the ideas of Lagrange, Vandermonde and Gauss on the solution of algebraic equations. We will not give here a detailed history of this question (see pp. 75 ff.); we must remember the definition by Ruffini [265], then Cauchy ([56 a], (2), v. I, p. 64), of the “product” of two permutations of a finite set, ⁹ and the first notions about finite groups of permutations: transitivity, primitivity, neutral element, commuting elements, etc.. But these first researches remain, on the whole, fairly superficial, and it is Evariste Galois who must be considered the real initiator of the theory: having, in his memorable work [123], reduced the study of algebraic equations to that of the groups of permutations that he associates with them, he deepens considerably this latter, as much in its concern with general properties of groups (it is Galois who first defines the notion of normal subgroup and recognises its importance), as in the determination of groups possessing particular properties (where the results that he obtains are classed still today amongst the most subtle of the theory). It is also to

Galois that the first idea of the “linear representation of groups” is due, ¹⁰ and this fact

proves clearly that he was in possession of the notion of the isomorphism of two group structures, independently of their “realisations”.

However, if it appears incontestable that the brilliant methods of Gauss and Galois had led them to a very wide conception of the law of composition, they did not have the opportunity to develop particularly their ideas on this point, and their work did not have any immediate action on the evolution of abstract Algebra.¹¹ It is in a third direction that the clearest progress towards abstraction is made: following reflections on the nature of imaginaries (of which the geometrical representation had caused, at the beginning of the XIXth century, fairly substantial work), the algebraists of the English school bring out first, from 1830 to 1850, the abstract notion of a law of composition, and enlarge immediately the field of Algebra by applying this notion to a host of new mathematical objects: the algebra of Logic with Boole (see p. 8), vectors, quaternions and general hypercomplex systems with Hamilton [145 a], matrices and non-associative laws with Cayley ([58], v. I, pp. 127 and 301, and v. II, pp. 185 and 475). A parallel evolution takes place independently on the continent, notably as far as vector Calculus is concerned (Möbius, Bellavitis), linear Algebra and hypercomplex systems (Grassmann) (see pp. 61).¹²

From all this bubbling up of original and fruitful ideas that come to revive Algebra in the first half of the XIXth century, this latter emerges renewed even in its trends. Before, methods and results gravitated around the central problem of the solution of algebraic equations (or diophantine equations in Number Theory): “Algebra”, says Serret in the Introduction to his Higher Algebra Course [284], “is, properly speaking, the Analysis of equations”. After 1850, even if the treatises on Algebra still leave for a long time the preeminence to the theory of equations, new research is no longer dominated by the concern for immediate applications to the solution of numerical equations, and is more and more oriented towards what we would now consider as the essential problem of Algebra, the study of algebraic structures as such.

This work is divided fairly neatly into three streams, which extend respectively the three movements of ideas that we have analysed above, and which are pursued in parallel without mutual measurable influence until the last years of the XIXth century.¹³

It is first the building up, by the German school of the XIXth century (Dirichlet, Kummer, Kronecker, Dedekind, Hilbert) of the theory of algebraic numbers, coming out of the work of

Gauss, to whom is due the first study of this kind, that of numbers $a + bi$ (a and b rational). We will not follow here the evolution of this theory: we must simply pick up, for our purposes, the abstract algebraic notions that emerge there. Already from the first successors of Gauss, the idea of a field (of algebraic numbers) is at the base of all the work in question (as are also the researches of Abel and Galois on algebraic equations); its field of application grows when Dedekind and Weber [80] copy the theory of algebraic functions of one variable onto that of algebraic numbers. Also due to Dedekind ([79], v. III, p. 251) is the introduction of the notion of ideal, which provides a new example of a law of composition between sets of elements; it is to him and Kronecker that the ever greater role played by abelian groups and modules in the theory of algebraic fields goes back; we will return to this later on (see pp. 63 ff. and 93 ff.).

We leave also till later the history of the development of linear Algebra (p. 57) and hyper-complex systems (p. 117), which take place without introducing any new algebraic notion during the end of the XIXth century and the beginning of the XXth century, in England (Sylvester, W. Clifford) and in America (B. and C. S. Peirce, Dickson, Wedderburn) following the route laid out by Hamilton and Cayley, in Germany (Weierstrass, Dedekind, Frobenius, Molien) and in France (Laguerre, E. Cartan) in a manner independent of the Anglo-Saxons and following fairly different methods.

As for the theory of groups, it is mainly under the form of the theory of finite permutation groups that it is developed at first, following the publication of the works of Galois and their diffusion by the works of Serret [284], and above all the great “Treatise on Substitutions” of C. Jordan [174 a]. This latter summarises there, perfecting them greatly, the work of his predecessors on the particular properties of permutation groups (transitivity, primitivity, etc.), obtaining results of which the greater part have hardly been surpassed since; he studies as well, in a deep way, very important particular groups, the linear groups and their subgroups (see p. 138); as well, it is he who introduces the fundamental notion of the representation of one group by another, as well as (a little later) that of quotient group, and who proves part of the theorem known by the name of “Jordan-Hölder Theorem”.¹⁴ It is finally to Jordan that the first study of infinite groups goes back [174 b], that S. Lie on the one hand, F. Klein and H. Poincaré on the other, were to develop considerably, in two different directions, a few years later.

In the meantime, it had gradually become obvious that what is essential in a group is its law of composition and not the nature of the objects making up the group (see for example [58], v. II, pp. 123 and 131 and [79], v. III, p. 439). However, even the researches on finite abstract groups are still for a long time conceived as studies about groups of permutations, and it is only around 1880 that there is a conscious development of the autonomous theory of finite groups. We cannot follow further the history of this theory; we limit ourselves to mentioning two of the tools which still today are amongst the most used in the study of finite groups, and which both go back to the XIXth century: the theorems of Sylow ¹⁵ on p-groups, which date from 1872 [303] and the theory of characters, created in the last years of the century by Frobenius ([119], v. III, pp.1-37). We send the reader wishing to deepen his knowledge of the theory of finite groups and the numerous and difficult problems it gives rise to, to the monographs [44 b], [131], [291] and [341].

It is towards 1880 also that presentations of groups begin to be studied systematically; before that, only presentations of particular groups had been studied, for example of the alternating group $\mathfrak{A}_5$ in the work of Hamilton [145 b], or of the monodromy groups of differential equations and Riemann surfaces (Schwarz, Klein, Schläfli). W. Dyck was the first [97] to define (without giving it a name) the free group generated by a finite number of generators, but he is less interested in it for itself than as a “universal” object allowing the precise definition of what is meant by a group given “by generators and relations”. Developing an idea first put out by Cayley, and visibly influenced elsewhere by the work of the school of Riemann mentioned above, Dyck describes an interpretation of a group with a given presentation, where each generator is represented by a product of two inversions with respect to tangent circles or secants (see also [44 b], chap. XVIII). A little later, after the development of the theory of automorphic functions by Poincaré and his successors, then the introduction by Poincaré of the tools of algebraic Topology, the first studies of fundamental groups will go hand in hand with that of group presentations (Dehn, Tietze), the two theories providing mutual support. It is in any case a topologist, J. Nielsen who, in 1924 introduced the terminology free group in the first deep study of its properties [234]; almost immediately afterwards E. Artin (still in connection with topological questions) introduces the notion of free product of groups, and O. Schreier defines more gener-

ally the free product with amalgamated subgroup. Here again, we cannot tackle the history of subsequent developments in this direction, referring to the work [216] for more details.

It is not either the place to speak of the extraordinary prosperity that has been known, since the end of the XIXth century, by the idea of group (and that of invariant, to which it is intimately linked) in Analysis, in Geometry, in Mechanics and in theoretical Physics. It is by an analogous invasion of this notion, and of the algebraic notions that are related to it (groups with operators, rings, ideals, modules) in those parts of Algebra which until then appeared fairly remote from their proper domain, that the last period of the evolution that we are retracing here is marked, and which ends with the synthesis of the three tendencies that we have followed above. This unification is above all the work of the modern German school: begun with Dedekind and Hilbert in the last years of the XIXth century, the work of axiomatisation of Algebra was vigorously pursued by E. Steinitz [294 a], then, from 1920, under the impulsion of E. Artin, E. Noether and the algebraists of their schools (Hasse, Krull, O. Schreier, van der Waerden). The treatise of van der Waerden [317 a], published in 1930, gathered together for the first time these works in a complete exposition, opening the way and serving as guide to the multiple subsequent researches into abstract Algebra.

Chapter 4

LINEAR ALGEBRA AND MULTILINEAR ALGEBRA.

Linear algebra is at the same time one of the oldest branches of mathematics and one of the newest. On the one hand, problems are found in the origins of mathematics which are solved by a single multiplication or division, that is to say by the calculation of a value of a function $f(x) = ax$, or by the solution of an equation $ax = b$: those are typical problems of linear algebra, and it is not possible to deal with them, nor even to state them correctly, without “thinking linearly”.

On the other hand, not only these questions, but almost everything that touches on equations of the first degree, had for a long time been relegated to elementary teaching, when the modern development of the notions of field, ring, topological vector space, etc. . . ., came to free and give importance to the essential notions of linear algebra (for example duality); it is then that the essentially linear character of almost all modern algebra was realised, of which this “linearisation” is even one of the distinguishing traits, and that the place that is its due was given back to linear algebra. To give its history, from the point of view where we put ourselves, will thus be a task that is as important as it is difficult; and we will have to be content here with fairly summary indications.

From what comes earlier, it follows that linear algebra has without doubt been born in order to supply the needs of practising calculators; it is thus that we see the rule of three ([311], v.

I, pp. 150-155) and the rule of false position, more or less clearly stated, playing an important role in all the manuals of practical arithmetic, from the Rhind papyrus of the Egyptians up to those which are in use in our primary schools, passing through Aryabhata, the Arabs, Leonardo of Pisa and the innumerable “books of calculation” of the Middle Ages and of the Renaissance; but they have perhaps never been anything other than an extract, for the use of practitioners, of more advanced scientific theories.

As for actual mathematicians, the nature of their research on linear algebra is a function of the general structure of their science. Ancient Greek mathematics, as it is set out in the *Elements* of Euclid, has developed two abstract theories of a linear character, on the one hand that of quantities ([107], book V, cf. p. 152), on the other hand that of whole numbers ([107], book VII). With the Babylonians, we find methods that are much closer to our elementary algebra; they know how to solve, and in a very elegant way ([232], pp. 181-183) systems of equations of the first degree. For a very long time, however, progress in linear algebra was mainly tied to that of algebraic calculation, and it is under this aspect, foreign to the present Note, that it is reasonable to consider it; to reduce, indeed, a linear system to an equation of the type $ax = b$, it is enough, if only one unknown is involved, to know the rules (already substantially stated by Diophantus) by which one can move terms from one member to another, and combine similar terms; and, if many unknowns are involved, it is enough to know as well how to eliminate them successively until only one remains. So the treatises on algebra, until the XVIIIth century, were thought to have done everything as far as the first degree is concerned, once they had laid out these rules; as for a system with as many unknowns as equations (they do not consider any others) where the first members would not be linearly independent forms, they content themselves invariably with noting in passing that that indicates a badly posed problem. In the treatises of the XIXth century, and even certain more recent works, this point of view has only been changed by the progress of notation, which allows the writing of systems of n equations in n unknowns, and by the introduction of determinants, which allow the statement of formulae for the explicit solution in the “general case”; this progress, for which the credit would be due to Leibniz ([198 a], v. I, p. 239) if he had published his ideas on the subject, is due principally to mathematicians of the XVIIIth and the beginning of the XIXth century.

But we must first consider several streams of ideas which, much more than the study of systems of linear equations, have contributed to the development of linear algebra in the sense in which we understand it. Inspired by the study of Apollonius, Fermat ([109], v. I, pp. 91-110; French translation, v. III, pp. 85-101), conceives, even before Descartes ([85 a], v. VI), the principles of analytic geometry, has the idea of the classification of plane curves by means of their degree (which, having become more and more familiar to all mathematicians, can be considered as definitely acquired towards the end of the XVIIth century), and states the fundamental principle of an equation of the first degree, in the plane, representing a straight line, and an equation of the second degree a conic: a principle from which he deduces straightaway some “very pretty” consequences concerning geometric loci. At the same time, he announces ([109], v. I, pp. 184-188; French translation, v. III, pp. 159-163) the classification of problems as problems solved, problems which reduce to an equation in two unknowns, to an equation in three unknowns, etc.; and he adds: the first consist in determining a point, the second a line or plane locus, those following a surface; etc. (“...such a problem does not seek only a point or a line, but a whole surface proper to the question; it is from that that are born the superficial loci, and so on for those following”, loc. cit. p. 186; there can be found already the embryo of geometry in n dimensions). This piece of writing, stating the principle of dimension in algebra and in geometric algebra, indicates a fusion of algebra and geometry, absolutely in line with modern ideas, but which, as has been seen already, took more than two centuries to penetrate minds.

At least these ideas ended up soon in the flowering of analytic geometry, which acquires its whole range in the XVIIIth century with Clairault, Euler, Cramer, Lagrange and many others. The linear character of the formulae for the transformation of co-ordinates in the plane and in space, which cannot fail to have been noticed already by Fermat, is put into prominence for example by Euler ([108 a], (1), v. IX, chap. II-III, and Append. chap. IV), who builds on this the classification of planar lines, and that of surfaces, according to their degree (invariant precisely because of the linearity of these formulae); it is also he (loc. cit., chap. XVIII) who introduces the word “affinity” to indicate the relation between curves that can be deduced from one another by a transformation $x' = ax, y' = by$ (but without noticing anything of a geometric

invariant in this definition which remains attached to a particular choice of axes). A bit later we see Lagrange ([191], v. III, pp. 661-692) devoting a whole memoir, which will remain justly famous for a long time, to typically linear and multilinear problems of analytical geometry in three dimensions. It is at about the same time, in connection with the linear problem consisting of the search for a plane curve passing through given points, that the notion of determinant takes shape, first in a somewhat empirical manner, with Cramer [72] and Bézout [21]; developed afterwards by several authors, this notion and its essential properties are definitively brought out by Cauchy ([56 a], (2), v. I, pp. 91-169) and Jacobi ([171], v. III, pp. 355-392).

On the other hand, while mathematicians had some slight tendency to hold first degree equations in some disdain, the solution of differential equations constituted on the contrary a major problem; it was natural that, amongst these equations, one picked out at an early stage the linear equations, with constant coefficients or not, and that their study contributed to enhance the value of linearity and that which goes with it. That is just what is seen with d'Alembert [75 c], Lagrange ([191], v. I, pp. 471-668) and Euler ([108 a], (1), v. XII); but only the first of these judges it useful to say that the general solution of a non homogeneous equation is the sum of a particular solution and of the general solution of the corresponding homogeneous equation; moreover, when these authors state that the general solution of the linear homogeneous equation of order n is a linear combination of n particular solutions, they do not add that these must be linearly independent, and make no effort to render explicit this latter notion; on this point as on so many others, it is only, it appears, the teaching of Cauchy at the École Polytechnique that will throw some light ([56 b], pp. 573-574). But already Lagrange (*loc. cit.*) introduces also (solely for calculations, it is true, and without giving it a name) the adjoint equation $L^*(y) = 0$ of a linear differential equation $L(y) = 0$, a typical example of duality by virtue of the relation

$$\int zL(y)dx = \int L^*(z)ydx$$

valid when y and z are zero at the ends of the interval of integration; more precisely, and 30 years before Gauss had explicitly defined the transpose of a linear substitution in 3 variables, we see there without doubt the first example of a “functional operator” L^* the transpose or “adjoint” of an operator L given by means of a bilinear function (here the integral $\int yzdx$).

At the same time, and with Lagrange also ([191], v. III, pp. 695-795), linear substitutions, in 2 or 3 variables at first, were on the way to conquer arithmetic. It is clear that the set of values of a function $F(x, y)$, when x and y take all integer values, does not change when an arbitrary linear substitution, with integer coefficients, of determinant 1, is performed on x, y ; it is on this fundamental observation that Lagrange bases the theory of the representation of numbers by forms, and that of the reduction of forms; and Gauss, by means of a step of which it has become difficult for us to appreciate all the boldness, gets out of it the notion of equivalence and that of a class of forms (cf. p. 50); on this point, he recognises the necessity of some elementary principles relative to linear substitutions, and introduces in particular the notion of transpose or adjoint ([124 a], v. I, p. 304). From that moment on, the arithmetic study and the algebraic study of quadratic forms in 2, 3 and later n variables, that of bilinear forms that are closely linked to them, and more recently the generalisation of these notions to an infinity of variables, was, up to our time, to form one of the most fruitful sources of progress for linear algebra (cf. pp. 136 ff.).

But, a still more decisive progress perhaps, Gauss, in these same *Disquisitiones*, created (cf. p. 51) the theory of finite abelian groups, which arises in four different ways, by means of the additive group of whole numbers modulo m (for m an integer), by means of the multiplicative group of numbers prime to m modulo m , by means of the group of classes of binary quadratic forms, and finally by means of the multiplicative group of m th roots of unity; and, as we have already noted, it is clearly as abelian groups, or to put it better as modules over $\mathbb{Z}$, that Gauss treats all these groups, studies their structure, their isomorphism relations, etc.. In the module of “complex integers” $a + bi$, he is seen later studying an infinite module over $\mathbb{Z}$, of which he has no doubt not missed seeing the isomorphism with the module of periods (discovered by him in the complex domain) of elliptic functions; in any case this idea appears already clearly with Jacobi, for example in the famous proof of the impossibility of a function with 3 periods and in his views on the problem of inversion of abelian integrals ([171], v. II, pp. 25-50), to end up soon with the theorems of Kronecker ([186 a], v. III₁, pp. 49-109).

Here, in streams of which we have sought to follow the thread and sometimes the meanderings, another comes to join in, which for a long time had remained underground. As it will be

brought out elsewhere in more detail (see p. 130), “pure” geometry in the sense in which it had been understood in the previous century, that is to say essentially the projective geometry of the plane and of space without the use of co-ordinates, had been created in the XVIIth century by Desargues [84], whose ideas, appreciated at their true worth by a Fermat, put into use by a Pascal, had subsequently fallen into disuse, eclipsed by the brilliant progress of analytical geometry; it is brought back to pride of place towards the end of the XVIIIth century, by Monge, then Poncelet and his emulators, Brianchon, Chasles, sometimes completely and voluntarily separated from analytical methods, sometimes (especially in Germany) mixed intimately with them. Yet projective transformations, from whatever point of view they are considered (synthetic or analytical), are none other of course than linear substitutions on the projective or “barycentric” co-ordinates; conics (in the XVIIth century), and later quadrics, of which the projective theory has for a long time been the principal object of study of this school, are none other than quadratic forms, of which we have already brought out above the close connection with linear algebra. To these notions is joined that of polarity: created, it also, by Desargues, the theory of poles and polars becomes in the hands of Monge and his successors, and soon under the name of principle of duality, a powerful instrument of transformation for geometric theorems; if one does not dare affirm that its links with adjoint differential equations were noticed only belatedly (they are indicated by Pincherle at the end of the century), what has none the less not been missed, Chasles bears testimony to it ([60 b], p. 55), is to notice its kinship with the notion of reciprocal spherical triangle, introduced into spherical trigonometry by Viète ([319], p. 418) and Snellius already in the XVIth century. But duality in projective geometry is only one aspect of the duality of vector spaces, taking account of the modifications that are imposed by passing from an affine space to a projective space (which is a quotient space of it, by the relation “scalar multiplication”).

The XIXth century, more than any period in our history, was rich in mathematicians of the first rank; and it is difficult in a few pages, even limiting oneself to the most outstanding aspects, to describe all that the fusion of these movements of ideas produces in their hands. Between the purely synthetic methods on the one hand, a kind of Procrustean bed where their orthodox adepts torture themselves, and the analytic methods linked to a system of co-ordinates arbitrarily inflicted on space, one feels soon the need for a kind of geometrical calculus, dreamed of but

not created by Leibniz, imperfectly outlined by Carnot [51]: it is first the addition of vectors that appears, implicit in Gauss in his geometrical representation of imaginary numbers and the application that he makes of it to elementary geometry (cf. p. 162), developed by Bellavitis under the name of “method of equipollence”, and that takes its definitive shape in Grassmann, Möbius, Hamilton; at the same time, under the name of “barycentric calculus”, Möbius gives a version of it adapted to the needs of projective geometry ([223], v. I).

At the same time, and amongst the same men, the passage takes place, so natural (as soon as this path has been entered) that we have seen it already announced by Fermat, from the plane and “ordinary” space to space of n dimensions; a passage that is even inevitable, since algebraic phenomena, which are interpreted for 2 or 3 variables as themselves in geometric terms, subsist without change for any number of variables; to impose therefore, in the use of geometric language, a limitation to 2 or 3 dimensions, would be for the modern mathematician a yoke as inconvenient as that which continually prevented the Greeks from extending the notion of number to ratios of incommensurable quantities. Also the language and ideas relating to space of n dimensions appear almost simultaneously on all sides, obscurely in Gauss, clearly among the mathematicians of the subsequent generation; and their more or less great measure of boldness in using it took less account of their mathematical leanings than of their philosophical views or even just practical considerations. In any case, Cayley and Grassmann, in about 1846, manipulate these concepts with the greatest of ease (and that, says Cayley differing from Grassmann ([58], v. I, p. 321), “without recourse to any metaphysical notion”); in Cayley, one remains constantly very near to the analytic interpretation and co-ordinates, whereas in Grassmann it is from the first, with the addition of vectors in space of n dimensions, the geometric aspect that takes the lead, to end up with developments of which we shall speak in a moment.

However, the impulsion that came from Gauss was pushing mathematicians, in two different ways, towards the study of algebras or hypercomplex systems. On the one hand, one could not avoid trying to extend the domain of real numbers other than by the introduction of the “imaginary unit” $i = \sqrt{-1}$, and to open up thus perhaps domains vaster and more fruitful than that of the complex numbers. Gauss convinced himself ([124 a], v. II, p. 178) of the impossibility of such an extension, at least if one seeks to keep the principal properties of the complex numbers,

that is to say in modern language, those that make it a commutative field; and, either because of his influence, or independently, his contemporaries seem to have shared this conviction, which will only be justified much later by Weierstrass ([329 a], v. II, pp. 311-332) in a precise theorem. But, as soon as one interprets the multiplication of complex numbers as rotations in the plane, one is led, if intending to extend this idea to space, to envisage (since rotations in space form a non abelian group) non commutative multiplications; it is that which is one of the ideas that guides Hamilton ¹ in his discovery of quaternions [145 a], the first example of a non commutative field. The singularity of this example (the only one, as Frobenius was to prove later, that can be constructed on the field of reals) restricted somewhat its scope, in spite or perhaps even because of the formation of a school of fanatical “quaternionists”: a strange phenomenon, which repeats itself later around the work of Grassmann, then the popularisers who draw out of Hamilton and Grassmann what has been called the “vector calculus”. The abandonment a little later of associativity, by Graves and Cayley who construct the “Cayley numbers”, does not open up an interesting vein. But, after Sylvester had introduced matrices, and (without giving it a name) had clearly defined rank ([304], v. I, pp. 145-151), it is again Cayley ([58], v. II, pp. 475-496) who creates its calculus, not without noting (an essential fact, often lost sight of later on) that a matrix is only an abbreviated notation for a linear substitution, just as in fact Gauss denoted by (a, b, c) the form $aX^2 + 2bXY + cY^2$. What is there is besides only one of the aspects, the most interesting for us no doubt, of the abundant production by Sylvester and Cayley on determinants and all that is linked with them, a production bristling with ingenious identities and impressive calculations.

It is also (among other things) an algebra over the reals that Grassmann discovers, the exterior algebra to which his name remains linked. His work, prior even to that of Hamilton ([134], v I₁), created in a moral solitude which was almost complete, remained for a long time badly known, because no doubt of its originality, because also of the philosophical fog by which it begins to be shrouded, and which for example turns Möbius away at first. Furnished with concerns analogous to those of Hamilton but with a wider range (and which, as he realises soon, were those themselves of Leibniz), Grassmann constructs a vast algebraico-geometric edifice, resting on a geometric or “intrinsic” conception (already just about axiomatised) of the vector space of

n dimensions; amongst the most elementary of the results which he ends up with, let us quote for example the definition of linear independence of vectors, that of dimension and the fundamental relation $\dim V + \dim W = \dim (V + W) + \dim (V \cap W)$ (loc. cit., p. 209; cf. [134], v. I, p. 21). But it is above all the exterior multiplication, then the interior multiplication, of multivectors, that provide him with the tools by means of which he deals easily with, first the problems of linear algebra proper, then those which are concerned with the Euclidean structure, that is to say with the orthogonality of vectors (where he finds the equivalent of the duality which he is missing).

The other path opened up by Gauss, in the study of hypercomplex systems, is that which starts with the complex integers $a + bi$; from these, one carries on quite naturally to more general algebras or hypercomplex systems, over the ring $\mathbb{Z}$ of integers or over the field $\mathbb{Q}$ of rationals, and first of all to those foreseen already by Gauss which are generated by the roots of unity, then to the fields of algebraic numbers and to modules of algebraic integers: those are the principal objects of the work of Kummer, the study of the former is tackled by Dirichlet, Hermite, Kronecker, Dedekind. Here, contrary to what happens with algebras over the reals, it is not necessary to give up any of the characteristic properties of commutative fields, and it is to these that, during the whole of the XIXth century, work is limited. But the linear properties, and for example the search for a basis for the integers of the field (indispensable for a general definition of the discriminant) play in many respects an essential role; and, in Dedekind in any case, the methods are destined to become typically “hypercomplex”; besides Dedekind himself, without addressing in a general way the problem of algebras, is aware of this aspect of his work, and of that which ties it for example to the results of Weierstrass on hypercomplex systems over the reals ([79], in particular vol. II, pp. 1-19). At the same time the determination of the structure of the multiplicative group of units in a field of algebraic numbers, carried out by Dirichlet in his famous notes ([92], v. I, pp. 619-644) and almost at the same time by Hermite ([159], v. I, pp. 159-160), was eminently suitable to clarify the ideas about modules over $\mathbb{Z}$, their systems of generators, and, when they have them, their bases. Then the notion of ideal, that Dedekind defines in algebraic number fields (as a module over the ring of integers of the field) ([79], v. III, p. 251), whereas Kronecker introduces in polynomial rings (under the name of “system of

modules”) an equivalent notion ([186 a], v. II, pp. 334-342), gives the first examples of modules over rings more general than $\mathbb{Z}$; and in the same authors, then in Hilbert, is released little by little in particular cases the notion of a group with operators, and the possibility of always constructing, starting with such a group, a module over a suitably defined ring.

At the same time, the arithmetico-algebraic study of quadratic and bilinear forms, and their “reduction” (or, what amounts to the same, that of matrices and their “invariants”) leads to the discovery of general principles concerning the solution of systems of linear equations, principles that, for lack of the notion of rank, had escaped Jacobi.² The problem of the solution in integers of systems of linear equations with integral coefficients is tackled and solved, at first in a special case by Hermite, then in all its generality by H. J. Smith ([287], v. I, pp. 367-409); the results of this latter are found again, only in 1878, by Frobenius, within the framework of a vast programme of research instituted by Kronecker, and in which Weierstrass also participates; it is incidentally Kronecker, in the course of his work, who gives the definitive form to the theorems on linear systems with real (or complex) coefficients, which is also elucidated in an obscure manual, with the scrupulous care which characterises him, by the famous author of *Alice in Wonderland*; as for Kronecker, he disdains publishing his results, and leaves it to his colleagues and disciples; even the word rank is only introduced by Frobenius. It is also in their teaching at the university of Berlin that Kronecker [186 b] and Weierstrass introduce the “axiomatic” definition of determinants (as a multilinear alternating function of n vectors in space of n dimensions, normalised so as to take the value 1 for the identity matrix), a definition equivalent to that which can be deduced from the calculations of Grassmann; it is in his lectures that Kronecker still, without feeling the need to give it a name, and in a still not intrinsic form, introduces the tensor product of spaces and the “Kroneckerien” product of matrices (a linear substitution induced, on a tensor product, by given linear substitutions applied to the factors).

These researches should not be separated either from invariant theory, created by Cayley, Hermite and Sylvester (the “invariant trinity” of whom later Hermite speaks in his letters) and which, from a modern point of view, is above all a representation theory of linear groups. There emerges, as an algebraic equivalent of the duality of projective geometry, the distinction between series of cogredient and contragredient variables, that is to say vectors in a space and vectors in

the dual space; and after attention had been drawn first to forms of low degree, then of arbitrary degree, in 2 or 3 variables, little time is lost in examining bilinear, then multilinear forms, in several series of “cogredient” or “contragredient” variables, which is equivalent to the introduction of tensors; this becomes conscious and is popularised when, under the inspiration of invariant theory, Ricci and Levi-Civita, in 1900, introduce the “tensor calculus” into differential geometry [257], which later was very much in fashion following its use by “relativist” physicists. It is still the progressive interpenetration of invariant theory, of differential geometry, and of the theory of partial differential equations (especially the problem of Pfaff so called, and its generalisations) which brings little by little geometers to consider, first of all bilinear alternating forms of differentials, in particular the “bilinear covariant” of a form of degree 1 (introduced in 1870 by Lipschitz, then studied by Frobenius), to finish with the creation, by E. Cartan ([52 a], v. II, pp. 303-396) and Poincaré ([251 b], v. III, chap. XXII) of the calculus of exterior differential forms. Poincaré introduces these, in the context of the formation of his integral invariants, as being expressions which appear in multiple integrals, whereas Cartan, guided no doubt by his researches on algebras, introduces them in a less formal way, but not without observing also that the algebraic part of their calculus is identical to the exterior multiplication of Grassmann (hence the name he adopts), thereby putting definitively into its true place the latter’s work. The translation into the notation of tensor calculus, of exterior differential forms, shows besides immediately their link with asymmetric tensors, which will, as soon as one returns to a purely algebraic point of view, make it to be seen that they are to alternating multilinear forms what covariant tensors are to arbitrary multilinear forms; this aspect becomes clearer still with the modern theory of representations of linear groups; and by that is recognised, for example, the substantial identity between the definition of determinants given by Weierstrass and Kronecker, and that which results from the calculus of Grassmann.

We reach thus the modern era, where the axiomatic method and the notion of structure (felt at first, defined only recently), allow us to separate concepts that up till then had been inextricably mixed, to formulate that which was vague or subconscious, to prove in the generality which is appropriate the theorems which had only been known in particular cases. Peano, one of the creators of the axiomatic method, and one of the first mathematicians also to appreciate at its

true worth the work of Grassmann, gives already in 1888 ([246 b], chap. IX) the axiomatic definition of vector spaces (whether of finite dimension or not) over the field of reals, and, with a fully modern notation, linear maps from such a space to another; a little later, Pincherle seeks to develop applications of linear algebra, thus conceived, to the theory of functions, in a direction, it is true, which has hardly shown itself to be fruitful; at least his point of view will allow him to recognise in the “Lagrange adjoint” a particular case of the transposition of linear maps: which appears soon, still more clearly, and for partial differential equations as well as for differential equations, in the course of the memorable work of Hilbert and his school on Hilbert space and its applications to analysis. It is in connection with these latter researches that Toeplitz [309 a], also introducing (but by means of co-ordinates) the most general vector space over the reals, makes the fundamental observation that the theory of determinants is useless for the proof of the main theorems of linear algebra, which allows the extension of these without difficulty to infinite dimensional spaces; and he indicates also that the linear algebra thus understood is applicable of course to any commutative base field.

On the other hand, with the introduction by Banach, in 1922, of the spaces which carry his name, ³ one meets, it is true in a problem which is topological as much as algebraic, spaces which are not isomorphic to their dual (see pp. 214 ff.). Already between a vector space and its dual, there is no “canonical” isomorphism, that is to say determined by the structure, which had been reflected for a long time by the distinction between cogredient and contragredient. It seems nonetheless that there is no doubt that the distinction between a space and its dual was only definitively established following the work of Banach and his school; it is this work too which raised an interest in the notion of codimension. As for the duality or “orthogonality” between vector subspaces of a space and those of its dual, the way in which it is formulated today demonstrates an analogy which is not just coincidental with the modern formulation of the main theorem of Galois theory, and with the duality of Pontrjagin so called in locally compact abelian groups; this goes back to Weber, who, in the course of arithmetic researches, stated the bases of it in 1866 for finite groups [327 d]; in Galois theory, the “duality” between subgroups and subfields takes shape with Dedekind and Hilbert; and the orthogonality between vector subspaces is visibly derived, first of all from the duality between linear varieties in projective geometry, and

also from the notion and properties of completely orthogonal varieties in a Euclidean space or a Hilbert space (whence its name). All these threads gather together in the contemporary era, in the hands of algebraists such as E. Noether, Artin and Hasse, of topologists such as Pontrjagin and Whitney, not without mutual influence being exerted between one group and the other.

At the same time a critical examination is being undertaken, which is meant to eliminate, at each point, the hypotheses which are not truly indispensable, and especially those which would prevent certain applications. Thus is realised the possibility of substituting rings for fields in the notion of vector space, and, creating the general notion of module, treating at one and the same time these spaces, abelian groups, particular modules already studied by Kronecker, Weierstrass, Dedekind, Steinitz, and even groups with operators, and for example to apply to them the Jordan-Hölder theorem; at the same time the transition to non commutativity is made, by means of the distinction between right modules and left modules, to which the modern development of the theory of algebras by the American school (Wedderburn, Dickson) and especially the German school (E. Noether, Artin) was leading.

Finally there has emerged, quite recently, the last of the tendencies of which we must speak here: contained in embryo in the theorem of Dedekind ([79], vol. III, p. 29) according to which arbitrary automorphisms of a commutative field are always linearly independent, the linearisation of Galois theory is carried out with Artin [7 a], then it is soon generalised to arbitrary extensions of commutative fields, and even to extensions of non commutative fields ([172 d], chap. VII).

Chapter 5

POLYNOMIALS AND COMMUTATIVE FIELDS.

The theory of commutative fields — and the theory of polynomials, which is closely linked to it — derive directly from what constituted, until the middle of the XIXth century the principal object of classical Algebra: the solution of algebraic equations, and geometric construction problems which are equivalent to that.

As soon as the solution of an algebraic equation of degree > 1 is sought, one finds oneself up against calculation difficulties of a totally new kind, the determination of an unknown being impossible to accomplish by means of “rational” calculations starting from given quantities. This difficulty must have been seen at a very early stage: and it must be counted as one of the most important contributions of the Babylonians that they knew how to reduce the solution of quadratic and biquadratic equations to a single new algebraic operation, the extraction of square roots (as is proved by the numerous numerical equations that are found solved in this way in the texts that have reached us ([232], pp. 183-193)). As far as formal calculation is concerned, Antiquity will never reach further than this point in the problem of solving algebraic equations; the Greeks of the classical era limit themselves indeed to finding again the Babylonian formulae in geometrical terms, and their use in an algebraic form is not attested to before Hero (100 after J. C.) and Diophantus.

It is in a totally different direction that the Greeks make decisive progress. We have poor

information on the way in which the Babylonians conceived of and calculated the square roots of non square integers; ¹ in the rare texts which have reached us on this question, they seem to be satisfied with fairly crude approximation methods ([232], pp. 33-38). The Pythagorean school which had determined with rigour the concept of commensurable quantities, and attached to it a quasi religious character, could not hold this point of view; and it is possible that it was the failure of repeated attempts to express $\sqrt{2}$ as a rational that led them finally to prove that this number is irrational. ²

We will say elsewhere (cf. p. 148) how this discovery, which marks a major turning point in the history of Mathematics, reacted profoundly on the concept of “number” among the Greeks, and led them to create an algebra of exclusively geometric character, so as to find in it a way of representing (or perhaps a proof of “existence” for) the incommensurable ratios, that they refused to consider as numbers. Most often, an algebraic problem is reduced by them to the intersection of two auxiliary plane curves, suitably chosen, or to several successive determinations of such intersections. Late traditions with little authority make the introduction of a first classification of these constructions, destined for a long and brilliant career, go back to Plato: for reasons that are more philosophical than mathematical, it appears, he would have put aside the constructions “by ruler and compass” so called, that is to say those where there are no auxiliary curves except for straight lines and circles. ³ In any case, Euclid, in his Elements [107], limits himself exclusively to treating problems soluble in this way (without however characterising them by a particular name); a circumstance which doubtless would contribute not a little to fastening the attention of mathematicians of the following centuries on these problems. But we know today ⁴ that algebraic equations that can be solved “by ruler and compass” are of a very special type; in particular, an irreducible equation of the 3rd degree (over the field of rationals) can not be solved in this way, and the Greeks had met very early on problems of the 3rd degree which have remained famous, such as the duplication of the cube (solution of $x^3 = 2$) and the trisection of an angle; the quadrature of the circle put them on the other hand in presence of a transcendental problem. We see them, in order to solve these problems, introducing numerous algebraic (conics, cissoid of Diocles, conchoid of Nicomedes) or transcendental (quadratic of Dinostratus, spiral of Archimedes) curves; which could not have failed to lead them to an

autonomous study of these various curves, thus preparing the way for future developments in analytic Geometry, in algebraic Geometry and in infinitesimal Calculus. But these methods do not help to make any progress in the solution of algebraic equations, ⁵ and the only work of Antiquity which made a worthy contribution to this question and exerted a lasting influence on the algebraists of the Middle Ages and the Renaissance is Book X of the Elements of Euclid [107]; in this Book (to which some historians would make the principal results of Theetetus go back), he considers the ex-

pressions obtained by the combination of several radicals, such as $\sqrt{\sqrt{a} \pm \sqrt{b}}$ (a and b rational), gives conditions under which these expressions are irrational, classes them into numerous categories (which he shows are distinct), and studies the algebraic relations between these various irrationals, such as those which we would write today as

$$\sqrt{\sqrt{p} + \sqrt{q}} = \sqrt{1/2(\sqrt{p} + \sqrt{p-q})} + \sqrt{1/2(\sqrt{p} - \sqrt{p-q})};$$

the whole expressed in the usual geometric language of the Elements, which makes the exposition particularly heavy and awkward.

After the decline of classical Greek mathematics, the concepts concerning algebraic equations are modified. There is no doubt that, during the whole of the classical period, the Greeks were in possession of methods of indefinite approximation for square roots, about which we are unhappily badly informed. ⁶ With the Hindus, then the Arabs and their western emulators in the Middle Ages, the extraction of roots of all orders becomes an operation that tends to be considered as fundamental on the same basis as the rational operations of algebra, and to be denoted, like these latter, by symbols which are more and more manageable in calculations. ⁷ The theory of the equation of the second degree, which is perfected during the whole of the Middle Ages (number of roots, negative roots, cases of impossibility, double roots), and that of biquadratic equations, give models for formulae for the solution of equations “by radicals”, from which algebraists are going to try for centuries to copy analogous formulae for the solution of equations of higher degree, and in the first instance for equations of the 3rd degree. Leonardo of Pisa, the principal introducer of Arab science to the West in the XIIIth century, recognises in any case that the irrationals classified by Euclid in his Xth book can not be of use to this end (a new proof

of impossibility, in a theory that contains so many), and we see him already trying to do similar calculations with cube roots, obtaining relations such as

$$\sqrt[3]{16} + \sqrt[3]{54} = \sqrt[3]{250},$$

analogous to Euclid's formulae for square roots (and of which besides previous examples are found among the Arabs). But three centuries of unfruitful efforts will still go by before Scipio del Ferro, at the beginning of the XVIth century, finally succeeds, for the equation $x^3 + ax = b$, in obtaining the solution formula ⁸

$$x = \sqrt[3]{\frac{b}{2} + \sqrt{\left(\frac{b}{2}\right)^2 + \left(\frac{a}{3}\right)^3}} + \sqrt[3]{\frac{b}{2} - \sqrt{\left(\frac{b}{2}\right)^2 + \left(\frac{a}{3}\right)^3}}. \quad (5.1)$$

We cannot describe here the picturesque side of this sensational discovery — the quarrels that it provoked between Tartaglia on the one hand, Cardan and his school on the other — nor the figures, often appealing, of the scholars who were the protagonists. But we must note the decisive progress that ensued for the theory of equations, in the hands of Cardan and his pupils. Cardan, who had less repugnance than most of his contemporaries at using negative numbers, observes thus that equations of the third degree may have three roots, and the biquadratic equations four ([50], v. IV, p. 259), and he remarks that the sum of the three roots of $x^3 + bx = ax^2 + c$ (an equation in which he knows how to eliminate the term in x^2) is always equal to a (ibid.). No doubt guided by this relation and the intuition of his general character, he has the initial idea of the notion of multiplicity of a root; above all he is emboldened (not without oratorical preparation) to calculate formally with expressions containing square roots of negative numbers. It is likely that he was brought to this by the fact that such expressions arise naturally in the use of the formula (1) when $\left(\frac{b}{2}\right)^2 + \left(\frac{a}{3}\right)^3 < 0$ (a case — called “irreducible” — where Cardan had recognised the existence of three real roots); it is what appears in any case with clarity in his disciple R. Bombelli, who, in his *Algebra* ([28 a], p. 293) proves the relation

$$\sqrt[3]{2 + \sqrt{-121}} = 2 + \sqrt{-1}$$

and takes care to give explicitly the rules for the calculation of complex numbers in a form

which is very close to modern expositions.⁹ Finally, already in 1545, another pupil of Cardan, L. Ferrari, succeeds in solving the general equation of the 4th degree, with the help of an auxiliary equation of the 3rd degree.¹⁰

After such a rapid advance, the following period, until the middle of the XVIIIth century, does little more than develop the new ideas introduced by the Italian school. Thanks to the essential progress that it brings to algebraic notation, Viète can express in a general manner the relations between the coefficients and roots of an algebraic equation, at least when the roots are all positive¹¹ ([319], p. 158). Bolder, A. Girard [129] does not hesitate to affirm (of course without proof) that an equation of degree n has exactly n roots, on condition that “impossible roots” are counted, each with its degree of multiplicity, and that these roots satisfy the relations given by Viète; he obtains also, for the first time, an expression for the sums of powers of roots up to exponent 4.

But the spirit of the XVIIth century is turned towards other directions and it is only as an indirect consequence that Algebra profits somewhat from the new discoveries of analytic Geometry and the infinitesimal Calculus. So, to the method of Descartes for obtaining the tangents to algebraic curves (cf. p. 178) is linked the criterion for the multiplicity of a root of an algebraic equation, which is stated by his disciple Hudde ([85 b], v. I, pp. 433 and 507-509). It is no doubt also to the influence of Descartes that one must go back for the distinction between algebraic functions and transcendental functions, parallel to that which he introduces in his *Géométrie* between “geometric” and “mechanical” curves (cf. pp. 176 and 193.) In any case, this distinction is perfectly clear in J. Gregory who, in 1667, tries even to prove that the area of a circular sector can not be an algebraic function of the chord and radius ([136 a]; cf. [155]). The expression “transcendental” is from Leibniz, who does not cease to be interested in these questions of classification for the whole of his career, and who, in about 1682, discovers a simple proof of the result sought by Gregory, by proving that $\sin x$ can not be an algebraic function of x ([198 a], v. V, pp. 97-98).¹² With his friend Tschirnhaus, he is besides one of the only mathematicians of his time who is still interested in the problem of the solution “by radicals” of algebraic equations. In his early stages, we see him studying the “irreducible case” of the equation of the 3rd degree, and convincing himself (in fact without sufficient proof) that it is impossible in this case to get

rid of imaginary quantities from the formulae of the solution ([198 d], pp. 547-564). At about the same time, he attacks also without success the solution by radicals of the equation of the 5th degree; and when later, Tschirnhaus pretends to have solved the problem by making all the terms of the equation disappear except for the two extremes by means of a transformation of the form $y = P(x)$, where P is a polynomial of the 4th degree, suitably chosen, Leibniz realises straightaway that the equations which determine the coefficients of $P(x)$ are of degree > 5 , and judges that the method is destined for failure ([198 d], pp. 402-403).

It appears that it is the needs of the new Analysis that had little by little revived interest in algebra. The integration of rational functions, accomplished by Leibniz and Johann Bernoulli, and the question of imaginary logarithms which is closely linked to it, are the occasion for deepening the calculus of imaginary numbers, and for answering the question of the decomposition of a polynomial into factors of the first degree (“fundamental theorem of algebra”).¹³ From the beginning of the XVIIIth century, the solution of the binomial equation $x^n - 1 = 0$ is reduced by Cotes and de Moivre to the division of the circle into n equal parts; in order to obtain the expression of its roots “by radicals”, it is sufficient therefore to know how to do it when n is an odd prime, and de Moivre remarks that the substitution $y = x + \frac{1}{x}$ reduces the problem then to the solution “by radicals” of an equation of degree $(n - 1)/2$. As far as the “fundamental theorem” is concerned, after the repeated setbacks in the general solution “by radicals” (including several attempts by Euler ([108 a], (1), v. VI, pp. 1-19 and 170-196)), proofs begin to be sought a priori, not using explicit formulae for the solution. Without going into the detail of the proposed methods (which would end up with the proofs of Lagrange and Gauss; cf. p. 91 and 161), it is fitting to note here the point of view from which the problem is seen in the middle of the XVIIIth century: it is assumed (without any justification other than a vague feeling of generality, coming no doubt, as with A. Girard, from the existence of relations between the coefficients and roots) that an equation of degree n always has n “ideal” roots, with which one can calculate as with numbers, without knowing if they are numbers (real or complex); and what it is necessary to prove is (by using if necessary calculations on ideal roots) that at least one of these roots is an ordinary complex number.¹⁴ In this defective form, it is recognised that here is the first embryo of the general idea of a “formal adjunction” which, in spite of the objections of Gauss ([124 a],

v. III, p. 1), would become the basis of the modern theory of commutative fields.

With the fundamental memoirs of Lagrange ([191], v. III, pp. 205-421) and of Vandermonde [315], the year 1770 sees a new and decisive period in the history of the theory of algebraic equations opening up. To the empiricism of the more or less fortunate attempts at solution formulae, which until then had reigned without dispute, a systematic analysis of the problems as stated and of the methods susceptible to solve them is going to follow, an analysis which in sixty years will lead to the definitive results of Galois. Lagrange and Vandermonde both start from the ambiguity which multiple determinations of radicals introduce in the formulae for the solution of equations of degree ≤ 4 ; this fact had attracted the attention of Euler ([108 a], (1), v. VI, pp. 1-19) who had showed among other things how, in the formula of del Ferro, one must associate the determinations of the radicals which appear there so as to obtain 3 roots and not 9.

¹⁵ Lagrange remarks that each of the cubic radicals in the formula of del Ferro can be written in the form $\frac{1}{3}(x_1 + \omega x_2 + \omega^2 x_3)$, where ω is a cube root of one, x_1, x_2, x_3 are the three roots of the equation in question, taken in a certain order, and he makes the fundamental observation that the function $(x_1 + \omega x_2 + \omega^2 x_3)$ of the three roots can only take two distinct values for all permutations of the three roots, which explains a priori the success of the methods for the solution of this equation. A similar analysis of the solution methods for the equation of the 4th degree leads him to the function $x_1 x_2 + x_3 x_4$ of the four roots, which only takes three distinct values for all permutations of the roots, and is as a consequence a root of an equation of the third degree with coefficients that are rational functions of those of the given equation; ¹⁶ these facts constitute, says Lagrange, “the true principles, and, so to say, the metaphysic ¹⁷ of the solution of equations of the 3rd and of the 4th degree” ([191], v. III, p. 357). Relying on these examples, he proposes studying in general, for an equation of degree n , the number ν of values ¹⁸ that a rational function V of the n roots can take when they are arbitrarily permuted; in reality he inaugurates thus (with this terminology still tightly adapted to the theory of equations) the theory of groups and that of fields, of which he obtains already several fundamental results by use of the same principles as those that are used today. For example, he shows that the number ν is a divisor of $n!$ by means of the argument that is used today for proving that the order of a subgroup of a finite group divides the order of the group. More remarkable still is the theorem

where he shows that, if V_1 and V_2 are two rational functions of the roots such that V_1 and V_2 are invariant under the same permutations, then each of them is a rational function of the other and the coefficients of the equation (a particular case of the theorem of Galois characterising a subextension of a Galois extension as the field of invariants of its Galois group): “This problem”, he says, “appears to me one of the most important in the theory of equations, and the general solution that we will give for it will be the means of throwing new light on this part of Algebra” ([191], v. III, p. 374).

All these researches are naturally, in the mind of Lagrange, preliminaries to the analysis of possible methods for the solution of algebraic equations by a successive reduction to equations of lesser degree, such a method being linked, as he shows, to the formation of rational functions of the roots taking less than n values under permutations of the roots. Guided no doubt by his results on the equation of the 3rd degree, he introduces in general the “Lagrange resolvents” $y_k = \sum_{h=1}^n \omega_k^h x_h$, where ω_k is an n th root of unity ($1 \leq k \leq n$), shows clearly how the knowledge of these n numbers results in that of the roots x_k , and looks in general for the degree of the equation which is satisfied by the y_k ; he shows for example that, if n is prime, the y_k are roots of an equation of degree $n-1$, whose coefficients are rational functions of a root of an equation of degree $(n-2)!$ with coefficients that are expressed rationally in terms of the coefficients of the given equation. “Here are, if I am not mistaken”, he concludes, “the real principles for the solution of equations, and the most correct analysis for getting there; everything reduces, as can be seen, to a kind of calculation of combinations, by which one finds a priori the results which should be expected” ([191], v. III, p. 403).

As for the memoir of Vandermonde, independent of that of Lagrange, it meets this latter at many points, particularly so far as the idea of seeking rational functions of the roots taking as few distinct values as possible under permutations of the roots is concerned,¹⁹ and the study of the “Lagrange resolvents” that he also introduces for this purpose. His work is far from having the clarity and generality of that of Lagrange; on one point however it goes clearly further, in applying the same ideas to the equation for the division of the circle $x^n - 1 = 0$ for n an odd prime. While Lagrange is content to remind readers that this equation reduces to an equation of degree $m = (n-1)/2$, with rational coefficients, without seeking to solve it when $n \geq 11$,

Vandermonde states that the m th powers of the Lagrange resolvents of this equation are rational, because of the relations between the various roots of $x^n - 1 = 0$; but he limits himself to checking the soundness of this assertion in the case $n=11$, without justifying it in a general way.

It is only 30 years later that the result announced by Vandermonde was fully proved by C. F. Gauss.²⁰ His decisive results on the equation $x^n - 1 = 0$ (n an odd prime) fit into the general programme of his memorable research on arithmetic ([124 a], v. I, pp. 413 ff.), and illustrate in a very special way his mastery in the handling of what we would now call the theory of cyclic groups. After having proved that the polynomial $\Phi_n(x) = (x^n - 1)/(x - 1)$ is irreducible for n an odd prime,²¹ he has the idea of writing its $n - 1$ roots in the form $\zeta^{g^k} = \zeta_k$ ($0 \leq k \leq n - 2$), where g is a primitive root of the congruence $z^{n-1} \equiv 1 \pmod{n}$ (which amounts, in modern language, to bringing into the open the fact that the group Γ of the equation $\Phi_n(x) = 0$ is cyclic). To every divisor e of $n - 1$, he puts in correspondence the $f = (n - 1)/e$ “periods” $\eta_\nu = \zeta_\nu + \zeta_{\nu+f} + \zeta_{\nu+2f} + \cdots + \zeta_{\nu+(e-1)f}$ ($1 \leq \nu \leq f$) and shows substantially that the linear combinations with rational coefficients of the η_ν make up a field, generated by any one of the f periods η_ν and of degree f over the field of rational numbers (this field corresponds naturally to the subgroup of Γ of order e). We cannot go into the details of his analysis here, and of the important arithmetical consequences that follow from it; let us notice only that it gives him in particular the famous theorem on the possibility of constructing “by ruler and compass” the polygons having a number of sides equal to a prime number of the form $2^{2^k} + 1$. As for the solution by radicals of the equation $\Phi_n(x) = 0$, it follows easily from the theory of periods, applied to the f th power of a Lagrange resolvent $\sum_{\nu=0}^{f-1} \omega^\nu \eta_\nu$ (where $\omega^f = 1$).

It is directly to Lagrange that the research of his compatriot Ruffini [265] is linked, contemporaneous with the *Disquisitiones*; taking up again the question at the point at which Lagrange had left it, he proposes as goal the proof of the impossibility of the solution “by radicals” of the “general” equation²⁴ of the 5th degree. The proof of Ruffini, prolix and obscure, remains incomplete although reworked several times; but it is already very close to the proof (correct in principle) that Abel would obtain later.²⁵ His main interest resides especially in the introduction of calculations with substitutions and the first notions of the theory of groups, that Ruffini develops to show that there does not exist a function of the 5 roots of the equation taking more

than 2 and less than 5 values when the roots are arbitrarily permuted.

We have already said (see p. 51) how this first sketch of the theory of groups of permutations was developed and systematised by Cauchy a few years later. But, if the notions necessary for the development of the ideas of Lagrange are thus clarified little by little as far as substitutions are concerned, it was still necessary to state in as clear a way the first principles of the theory of fields. That was what was missing for Ruffini, and it is that which Abel and Galois are going to do, in the last phase of the problem of the solution of algebraic equations.

During the whole of his short life, Abel does not fail to be concerned with this problem. While almost still a child, he believed that he had obtained a formula for the solution by radicals of the general equation of the 5th degree. Having realised later his mistake, he did not rest until he had succeeded in proving that such a formula did not exist ([1], v. I, p. 66). But he does not stop there. While his emulator Jacobi develops the theory of elliptic functions as an analyst, it is the algebraic point of view that dominates the work of Abel on this question, centred on the theory of the equations of division of elliptic functions ([1], v. I, pp. 265, 377 and *passim*). He obtains thus new types of equation soluble by radicals by a method copied from that of Gauss for the solution of equations of division of the circle ([1], v. I, pp. 310 and 358); ²⁶ a result from which he rises to the conception of “abelian” equations, for which he proves, in a famous memoir, the solubility by radicals ([1], v. I, p. 478); it is on this occasion that he defines in a precise way the notion of a polynomial irreducible over a given field (the field generated by the coefficients of the equation that he is studying). ²⁷ Finally death laid him low in 1829, just as he attacks the general problem of the characterisation of all the equations soluble by radicals, and having just communicated to Crelle and Legendre results already very near to those of Galois ([1], v. II, pp. 219-243, 269-270 and 279).

It was to him that was reserved, three years later, the crowning of the structure [123]. Like Abel, but in a still clearer way, he starts by defining (up to terminology) the membership by a number of a field generated by given numbers, the notion of adjunction, and irreducible polynomials over a given field. Given an equation $F(x) = 0$, without multiple roots, with coefficients in a given field K , he shows successively that “one can always construct a function V of the roots, such that none of the values that are obtained by permuting in this function the roots in all

possible ways, are equal”, that this function “enjoys the property that all the roots of the proposed equation can be expressed rationally as a function of V ”, and that, $V, V', V'', \dots$ being the roots of the irreducible equation which V satisfies, “if $a = f(V)$ is a root of the proposed equation, $f(V')$ as well will be a root of the proposed equation” ([123], pp. 47-51); in modern language, he thus proves that V , as well as any one of its conjugates over K , generates the field N of the roots of F . He then defines the group Γ of F as the set of permutations of the roots x_i that are obtained by substituting for V , in the rational expression for each of the x_i as a function of V , any one of the conjugates of V ; and he obtains straightaway the fundamental characterisation of the elements of K by the property of being invariant under every permutation of Γ ([123], p. 51). He proves next that, if N contains the field of roots L of another polynomial, the group of N over L is a normal subgroup of Γ (a notion that he introduces at this point) ([123], p. 175). From there he deduces finally the criterion of solubility of an equation by radicals, by means of an argument of which this is the essential part: assuming that the base field K contains all the roots of unity, there must exist by hypothesis an increasing sequence $(K_i)_{0 \leq i \leq m}$ of fields intermediate between K and N , with $K_0 = K, K_m = N$, the field K_{i+1} being obtained by adjoining to K_i all the roots of a binomial equation $x^{n_i} - a_i = 0$ (with $a_i \in K_i$). There exists therefore in Γ a decreasing sequence (Γ_i) of subgroups such that $\Gamma_0 = \Gamma, \Gamma_m = \varepsilon$ (neutral element), Γ_{i+1} being normal in Γ_i and the quotient group Γ_i/Γ_{i+1} being cyclic (a case in which one says that the group Γ is soluble). Conversely, if this is so, the use of a Lagrange resolvent shows that K_{i+1} is obtained by adjoining to K_i all the roots of a binomial equation, and hence the equation $F(x) = 0$ is soluble by radicals.²⁸ The impossibility of the solution by radicals of the “general” equation of degree $n > 4$ is then a consequence of the fact that the group Γ of this equation, isomorphic to the symmetric group S_n is not soluble.

Starting with the middle of the XIXth century, the algebraists, as we have already remarked (cf. p. 53), enlarge considerably the field of their investigations, until then almost entirely confined to the study of equations. In the light of the discoveries of Galois, it appears that the problem of the solution “by radicals” is only a particular case, fairly artificial, of the general problem of the classification of irrationals. It is this latter that, during the whole of the end of the XIXth century, is going to be attacked on several sides, and many disparate results will

accumulate little by little, preparing the way for the synthesis of Steinitz.

First as far as algebraic irrationals are concerned, a fundamental principle of the classification was supplied by Galois theory, reducing the study of an algebraic equation to that of its group. In fact, it is mainly the theory of permutation groups, of which we do not have to talk here (cf. p. 53), that, in pure Algebra, is the principal object of research at that time. Other progress in the theory of algebraic fields comes from the development, at the same time, of Number Theory and of Algebraic Geometry. This progress is concerned mainly with the way of expounding the theory, and is, for the most part, due to Dedekind [79], who introduces the notions of field and ring,²⁹ and (linked to his research on hypercomplex systems) develops systematically the linear aspect of extension theory ([79], v. III; pp. 33 ff.). It is also he who considers the Galois group as made up of the automorphisms of the extension under consideration, and not any longer only as a group of permutations of the roots of an equation; and he proves (for the field of numbers) the fundamental theorem on the linear independence of automorphisms ([79], v. III, p. 29) as well as the existence of normal bases for a Galois extension ([79], v. II, p. 433). Finally, he tackles the problem of algebraic extensions of infinite degree, and notices that Galois theory can not be applied to it as it stands (an arbitrary subgroup of a Galois group not being always identical with the group of an extension with respect to a subextension); and by a bold intuition, he already begins to think of the Galois group as a topological group³⁰ —an idea that will come to fruition only with the theory of Galois extensions of infinite degree, developed by Krull in 1928 [187 d].

Parallel to this evolution the notion of an element transcendental over a field becomes precise. The existence of transcendental numbers is proved for the first time by Liouville in 1844, by means of an explicit construction, based on the theory of diophantine approximations [204 c]; Cantor, in 1874, gives another “non constructive” proof using simple considerations about the power of sets [47]; finally, Hermite proves in 1873 the transcendence of e , and Lindemann in 1882 that of π by a method which is analogous to that of Hermite, putting thus a final seal on the ancient problem of the squaring of the circle.³¹

As for the role of transcendental numbers in algebraic calculations, Kronecker observes in 1882 that, if x is transcendental over a field K , the field $K(x)$ is isomorphic to the field of rational

fractions $K(X)$ ([186 a], v. II, p. 253). He makes besides the adjunction of indeterminates to a field the keystone of his exposition of the theory of algebraic numbers ([186 a], v. II, pp. 245-387). On the other hand, Dedekind and Weber show in the same year [80] how arithmetical methods can help to set up the theory of algebraic curves. One sees thus appearing in several directions analogies between Arithmetic and Algebraic Geometry, which will turn out to be extremely fruitful for the one and the other.

In all this research, the fields that arise consist of “concrete” elements in the sense of classical mathematics — (complex) numbers or functions of complex variables. ³² But already Kronecker, in 1882, takes full account of the fact (obscurely sensed by Gauss and Galois) that “indeterminates” only play the role in his theory of the basis elements of an algebra, and not that of variables in the sense of Analysis ([186 a], v. II, pp. 339-340); and, in 1887, he develops this idea, linked to a vast programme that aims at nothing less than recasting all mathematics by rejecting all that cannot be reduced to algebraic operations on the integers. It is on this occasion that, taking up an idea of Cauchy ([56 a], (1), v. X, pp. 312 and 351) who had defined the field $\mathbb{C}$ of complex numbers as the field of remainders $\mathbb{R}[X]/(X^2 + 1)$, Kronecker shows how the theory of algebraic numbers is completely independent of the “fundamental theorem of algebra” and even of the theory of real numbers, all fields of algebraic numbers (of finite degree) being isomorphic to a field of remainders $\mathbb{Q}[X]/(f)$ (f an irreducible polynomial over $\mathbb{Q}$) ([186 a], v. III₁, pp. 211-240). As is remarked a few years later by H. Weber [327 a], developing a first sketch of the axiomatic theory of fields, this method of Kronecker is applicable in fact to all base fields K . Weber indicates in particular that one can take as K a field $\mathbb{Z}/(p)$ (p a prime number), thus making the calculus of congruences “modulo p ” return to field theory; this latter had been born in the second half of the XVIIIth century, with Euler, Lagrange, Legendre and Gauss, and the analogy that it showed up with the theory of algebraic equations was not missed by observers; developing this analogy, Galois (with his research on group theory in mind) had not hesitated to introduce “ideal roots” of an irreducible congruence modulo p , ³³ and had indicated its principle properties ([123], pp. 113-127). ³⁴ When Kronecker’s method is applied to $\mathbb{Z}/(p)$, what is found again however (up to terminology) is the presentation that had already been given by Serret and Dedekind ([79], v. I, p. 40) of the theory of these “Galois imaginaries”.

To all these examples of “abstract fields” there come to be added, at the turn of the century, some fields of a new very different type, the fields of formal power series introduced by Veronese [318], and above all the p -adic fields of Hensel [157 f]. It is the discovery of these latter that led Steinitz (as he says explicitly) to bring out the abstract notions common to all these theories, in a fundamental work [294 a] which can be considered as having given birth to the actual concept of Algebra. Developing systematically the consequences of the axioms of commutative fields, he introduces thus the notion of prime field, of separable (algebraic) elements, of perfect fields, defines the degree of transcendence of an extension, and proves finally the existence of algebraically closed extensions of arbitrary fields.

In recent times, the theory of Steinitz has been completed in several important areas. On the one hand, the work of Artin has brought into prominence the linear character of Galois theory [7 a]. On the other hand, the general notion of derivation (copied from the formal properties of classical differential Calculus) sensed by Dedekind ([79], v. II, p. 412), introduced by Steinitz in the special case of a field of rational fractions ([294 a], pp. 209-212), has been used with success in the study (essential for modern Algebraic Geometry) of transcendental extensions, and notably in the generalisation to these latter of the notion of separability [330 d].

Chapter 6

DIVISIBILITY; ORDERED FIELDS.

The elementary arithmetical operations, and especially the calculus of fractions, could not but lead to numerous empirical verifications of the divisibility of whole numbers. But neither the Babylonians (however so expert at Algebra), nor the Egyptians (in spite of their acrobatic calculations with fractions) seem to have known any general rules governing these properties, and it is to the Greeks that the initiative returns here. Their arithmetic work, of which a masterly exposition is found in Books VII and IX of Euclid [107], does not yield pride of place to any of their most beautiful discoveries in the other branches of mathematics. The existence of the H.C.D. of two integers is proved already at the beginning of Book VII by the procedure known under the name of “Euclid’s algorithm”;¹ it serves as a foundation for all ulterior developments (properties of prime numbers, existence and calculation of the L.C.M., etc.); and the crowning of the structure is formed by the two remarkable theorems proving the existence of an infinity of prime numbers (Book IX, prop. 20) and giving a procedure for constructing perfect even numbers starting with certain prime numbers (this procedure gives in fact all the even perfect numbers, as Euler was to prove). Only the existence and uniqueness of the decomposition into prime factors are not proved in a general form; all the same Euclid proves explicitly that every integer is divisible by a prime number (Book VII, prop. 31), as well as the two following propositions (Book IX, props. 13 and 14):

“If, starting from a unit, as many numbers as desired are in a progression of constant ratio [i. e. geometric], and that the one after the unit is prime, the largest will not be divisible by any except those that occur in the progression” (in other words, a power p^n of a prime number can

not be divided except by the powers of p with exponent $\leq n$).

“If a number is the smallest that is divisible by [given] prime numbers, it will not be divisible by any other prime number with the exception of those initially [given as] dividing it” (in other words, a product of distinct prime numbers $p_1 \dots p_k$ has no other prime factors than $p_1, \dots, p_k$).

It seems therefore that if Euclid does not state the general theorem it is only due to a terminology and notation adequate only for arbitrary powers of an integer. ²

Although a careful study makes it likely that there exists, in the text of Euclid, several successive layers, each corresponding to a stage in the development of Arithmetic, ³ it seems that this evolution was entirely completed between the beginning of the Vth century and the middle of the IVth, and one can only admire the shrewdness and the sure footedness of the logic that are manifested there: we must wait two millennia to take part in comparable progress in Arithmetic.

It is so-called “indeterminate” or “diophantine” problems which are at the source of the ulterior developments of Number Theory. The term “diophantine equations”, as it is used today, is not, historically, completely justified; one means in general by that algebraic equations (or systems of equations) with integral coefficients, for which one is looking only for the solutions in integers: a problem that is normally impossible if the equations are “determined”, that is to say have only a finite number of solutions (in real or complex numbers), but that, on the contrary, often allows solutions when there are more unknowns than equations. Yet, if Diophantus seems clearly to have been the first to have considered “indeterminate” problems, he only exceptionally looks for integral solutions, and is content most of the time with obtaining a single solution in rational numbers [91 a]. This is a type of problem that he can solve most of the time by algebraic calculations where the arithmetic nature of the unknowns does not arise; ⁴ also divisibility theory only plays a very retiring role (the word prime number is only pronounced a single time ([91 a], Book V, problem 9, v. I, pp. 334-335), and the notion of relatively prime numbers is only brought up in connection with the theorem affirming that the quotient of two relatively prime numbers can not be a square unless each of them is a square). ⁵

The study of integral solutions of indeterminate equations does not really start until the Chinese and Hindu mathematicians of the high Middle Ages. The first of these seem to have been led to speculations of this kind by practical problems in the making of calendars (where the deter-

mination of periods common to several cycles of astronomical phenomena constituted precisely a “diophantine” problem of the first degree); to them is owed in any case (no doubt between the IVth and the VIIth century of our era) a rule for the solution of simultaneous linear congruences. As for the Hindus, of whom the mathematics knew its full flowering from the Vth to the XIIIth century, not only did they know how to treat methodically (by the application of Euclid’s algorithm) systems of linear diophantine equations with an arbitrary number of unknowns, ⁶ but they are the first to tackle and solve problems of the second degree, including certain particular cases of “Fermat’s equation” $Nx^2 + 1 = y^2$ ([78], vol. II, pp. 87-307).

We do not need to pursue here the history of the theory of diophantine equations of degree > 1 , which, following on from the work of Fermat, Euler, Lagrange and Gauss, was to end up in the XIXth century with the theory of algebraic integers (cf. pp. 93 ff.). As we have already noted (cf. p. 58), the study of linear systems, which seems no longer to present problems worthy of interest, is somewhat neglected during this period: in particular, general necessary conditions for solutions of an arbitrary system are not sought, nor is the description of the set of solutions. However, towards the middle of the XIXth century, Hermite is led to use, in connection with his research in Number Theory, several lemmas about linear Diophantine equations, and notably a “reduced form” for a linear substitution with integral coefficients ([159], v. I, pp. 164 and 265); finally, after Heger had given in 1858 the necessary condition for a system whose rank is equal to the number of equations, H. J. Smith, in 1861, defines the invariant factors of a matrix with integral entries in a “canonical form” ([287], v. I, pp. 367-409).

But in the interval the notion of an abelian group was gradually becoming more precise, following its introduction by Gauss (cf. p. 60), and so was the importance taken by this notion in the ulterior development of Number Theory. In the particularly deep study, expounded in the *Disquisitiones*, of the finite abelian group of classes of quadratic forms of a given discriminant, Gauss rapidly realised that certain of these groups were not cyclic: “in this case”, he says, “a basis [that is to say a generator] is not sufficient, two, or a greater number, must be taken, which, by multiplication and composition, ⁷ will be able to produce all the classes” ([124 a], v. I, pp. 374-375). It is not certain that, with these words, Gauss wished to describe the decomposition of a group as a direct product of cyclic groups; however, in the same page in the *Disquisitiones*, he

proves that there exists an element in the group whose order is the L.C.M. of the orders of all the elements — in other words, he obtains the existence of the largest invariant factor of the group ([124 a], v. I, p. 373) —; and on the other hand, the notion of direct product was known to him, for in a manuscript dating from 1801 but not published during his lifetime, he sketches a general proof of the decomposition of a finite abelian group as a direct product of p -groups⁸ ([124 a], v. II, p. 266). In any case, in 1868, Schering, the editor of Gauss's work, inspired by these results (and notably by this manuscript that he had just found again) proves (still for the group of classes of quadratic forms) the general decomposition theorem ([272], v. I, pp. 135-148) by a method which, taken up again two years later in abstract terms by Kronecker ([186 a], v. I, pp. 273-282), is essentially that which is still used today. As for torsion-free abelian groups, we have already said (cf. pp. 64 ff.) how the theory of elliptic functions and abelian integrals, developed by Gauss, Abel and Jacobi, led little by little to take note of their structure; the first and most famous example of the decomposition of an infinite group as a direct sum of monogenic groups is given by Dirichlet in 1846 in his memoir on the units of an algebraic number field ([92], v. I, pp. 619-644). But it is only in 1879 that the link between the theory of abelian groups of finite type and the theorem of Smith is recognised and used explicitly by Frobenius and Stickelberger ([120], §10).

At about the same time the theory of similarity for matrices (with real or complex coefficients) was being completed. The notion of eigenvalue of a linear substitution appears explicitly in the theory of systems of linear differential equations with constant coefficients, applied by Lagrange ([191], v. I, p. 520) to the theory of small movements, by Lagrange ([191], v. VI, pp. 655-666) and Laplace ([193], v. VIII, pp. 325-366) to the “secular” inequalities of the planets. It is implicit in many other problems tackled about the middle of the XVIIIth century, such as the determination of the axes of a conic or a quadric (accomplished first by Euler ([108 a], (1), v. IX, p. 384)), or the study (developed also by Euler ([108 a], (2), v. III, pp. 200-201)) of the principal axes of inertia of a solid body (discovered by De Segner in 1755); we know today that it is she also who (in a much more hidden form) took part in the beginnings of the theory of partial differential equations, and in particular in the equation of vibrating strings. But (without talking of this latter case) the common ancestry between these various problems is hardly recog-

nised before Cauchy ([56 a], (2), v. V, p. 248 and v. IX, p. 174). Further, as the majority bring in symmetric matrices, it is the eigenvalues of these latter which are mostly studied at first; let us note here that from 1826, Cauchy proves the invariance under similarity of the eigenvalues of these matrices, and proves that they are real for a symmetric matrix of order 3 ([56 a], (2), v. V, p. 248), a result that he generalises to arbitrary real symmetric matrices three years later ([56 a], (2), v. IX, p. 174).⁹ The general notion of projectivity, introduced by Möbius in 1827 ([223], v. I, p. 217), arrives quickly at the problem of classification of these transformations (in 2 and 3 dimensions first of all), which is none other than the problem of the similarity of the corresponding matrices; but for a long time this question is only treated by the “synthetic” methods which are in vogue in the middle of the XIXth century, and its progress (in any case fairly slow) does not seem to have had any influence on the theory of eigenvalues. It is not the same with another problem from geometry, the classification of pencils of conics or quadrics, which from the modern point of view, reduces to the study of elementary divisors of the matrix $U + \lambda V$ where U and V are two symmetric matrices; it is exactly in this vein that Sylvester, in 1851, tackles this problem, examining with care (with the aim of finding “canonical forms” for the pencil under consideration) what becomes of the minors of the matrix $U + \lambda V$ when a value is substituted for λ which annihilates its determinant ([304], v. I, pp. 219-240). The purely algebraic aspect of the theory of eigenvalues progresses simultaneously; it is thus that several authors (among whom Sylvester himself) prove around 1850 that the eigenvalues of U^n are the n th powers of the eigenvalues of U , whereas in 1858 Cayley, in the memoir in which he sets up the calculus of matrices ([58], v. II, pp. 475-496), states the “Cayley-Hamilton theorem” for a square matrix of arbitrary order,¹⁰ contenting himself with proving it by direct calculation for matrices of order 2 and 3. Finally, in 1868, Weierstrass, taking up again the method of Sylvester, obtains “canonical forms” for a “pencil” $U + \lambda V$, where, this time, U and V are square matrices which are not necessarily symmetric, subject only to the condition that $\det(U + \gamma V)$

is not identically null; he deduces from this the definition of the elementary divisors of an arbitrary square matrix (with complex entries), and proves that they characterise this latter up to similarity ([329 a], v. II, pp. 19-44); these results are besides found again partially (and apparently independently) by Jordan two years later¹¹ ([174 a], pp. 114-125). Here again, it is

Frobenius who, in 1879, shows that the theorem of Weierstrass can be easily deduced from the theory of Smith, extended to polynomials ([119], v. I, pp. 482-544, §13).

We have just alluded to the theory of divisibility of polynomials in one variable; the question of the division of polynomials was naturally to be stated from the beginning of algebra, as the inverse of the operation of multiplication (this latter being already known to Diophantus, at least for polynomials of small degree); but it can be conceived that it is hardly possible to tackle the problem in a general way before a coherent notation for the various powers of the variable was imposed. In fact, examples of the process of “Euclidean” division of polynomials, such as we know it, are hard to find before the middle of the XVIIth century; ¹² and S. Stévin (who essentially uses exponent notation) seems to be the first to have the idea of deducing from it the extension of “Euclid’s algorithm” for the determination of the G.C.D. of two polynomials ([295], v. I, pp. 54-56). Taking that into account, the notion of divisibility remained peculiar to rational integers until the middle of the XVIIIth century. It is Euler who, in 1770, opens a new chapter of Arithmetic by extending, not without temerity, the notion of divisibility to integers in a quadratic extension: seeking to determine the divisors of a number of the form $x^2 + cy^2$ (x, y, c rational integers), he writes $x + y\sqrt{-c} = (p + q\sqrt{-c})(r + s\sqrt{-c})$ (p, q, r, s rational integers) and taking the norms of both members, he does not hesitate to affirm that he obtains in this way all the divisors of $x^2 + cy^2$ in the form $p^2 + cq^2$ ([108 a], (1), v. I, p. 422). In other words, Euler argues as if the ring $\mathbb{Z}[\sqrt{-c}]$ were principal; a little later, he uses an analogous argument to apply the method of “infinite descent” to the equation $x^3 + y^3 = z^3$ (it reduces to writing that $p^2 + 3q^2$ is a cube, which he does by stating $p + q\sqrt{-3} = (r + s\sqrt{-3})^3$). But from 1773, Lagrange proves ([191], v. III, pp. 695-795) that the divisors of numbers of the form $x^2 + cy^2$ are not always of this form, the first example of the fundamental difficulty which was to occur with much greater clarity in the studies, taken up by Gauss and his successors, on divisibility in the fields of roots of unity; ¹³ it is not possible in general to extend directly to these fields the essential divisibility properties of rational integers, the existence of the G.C.D. and the uniqueness of the decomposition into prime factors. It is not the place here to describe in detail how Kummer for the field of roots of unity [188 b], ¹⁴ then Dedekind and Kronecker for arbitrary fields of algebraic numbers, succeed in surmounting this formidable obstacle by the creation of the theory of ideals, one of

the most decisive advances of modern algebra (cf. pp. 93 ff.). But Dedekind, always curious about the foundations of the various mathematical theories, is not content with this success; and, analysing the mechanism of the relations of divisibility, he sets up the basis of the modern theory of ordered groups, in a memoir (without stirring his contemporaries, and forgotten for 30 years) which is without doubt one of the first by date of the works on axiomatic algebra ([79, v. II, pp. 103-147).

From the middle of XVIIIth century, the search for a proof of the “fundamental theorem of algebra” is the order of the day (cf. p. 74). We do not need to remember here the attempt by d’Alembert, who inaugurated the series of proofs using the infinitesimal calculus (cf. p. 161). But, in 1749, Euler tackles the problem in another way ([108 a], (1), v. VI, pp. 78-147): for every polynomial f with real coefficients, he seeks to prove the existence of a decomposition $f = f_1 f_2$ into two polynomials (not constants) with real coefficients, which would give him a proof of the “fundamental theorem” by induction on the degree of f . It is sufficient even, as he remarks, to stop with the first factor of odd degree, and in consequence all the difficulties reduce to considering the case where the degree n of f is even. Euler limits himself then to the study of the case where the desired factors are both of degree $n/2$, and he indicates that, by an elimination calculation suitably undertaken, the unknown coefficients of f_1 and f_2 can be expressed rationally in terms of a root of an equation with real coefficients of which the extreme terms are of opposite signs and which as a consequence has at least one real root. But Euler’s proof is only a sketch, where a number of essential points are passed over in silence; and it is only in 1772 that Lagrange succeeds in solving the difficulties raised by this proof ([191], v. III, pp. 479-516) by means of a very long and very meticulous analysis, where he proves himself capable of a remarkable virtuosity in the use of “Galois” methods (so to speak) newly created by him (cf. pp. 75 ff.).

All the same Lagrange, like Euler and all his contemporaries, does not hesitate to argue formally in a “field of roots” of a polynomial (that is to say, in his language, to consider “imaginary roots” of this polynomial); the Mathematics of his time had not supplied any justification for this type of argument. Also Gauss, deliberately hostile, from its beginnings, to the frantic formalism of the XVIIIth century, raises himself powerfully, in his dissertation, against this abuse ([124

a], v. III, p. 3). But he would not have been himself if he had not felt that it was a case there of an argument with a defective presentation that was intrinsically correct. Also we see him, a few years later ([124 a], v. III, p. 33; cf. also [124 b]) taking up again a simpler variant of Euler's argument, suggested already in 1759 by de Foncenex (but that this latter was not able to bring to a satisfactory end), and deducing from it a new proof of the "fundamental theorem", where he avoids carefully all use of "imaginary" roots: these last being replaced by clever adjunctions and specialisations of indeterminates.

The role of Topology in the "fundamental theorem" is thus reduced to the one theorem according to which a polynomial with real coefficients cannot change sign in an interval without becoming zero (Bolzano's theorem for polynomials). This theorem is also at the bottom of all the separation criteria for real roots of a polynomial (with real coefficients), which constitute one of the favourite subjects of Algebra during the XIXth century.¹⁵ During this research, one could not fail to notice that it is the order structure of $\mathbb{R}$, even more than its topology, that plays there the essential role;¹⁶ for example, Bolzano's theorem for polynomials is still true for the field of all real algebraic numbers. This movement of ideas found its goal in the abstract theory of ordered fields, created by E. Artin and O. Schreier ([7 b] and [8 a and b]); one of its more remarkable results is no doubt the discovery that the existence of an order relation on a field is linked to purely algebraic properties of the field.

Chapter 7

COMMUTATIVE ALGEBRA.

ALGEBRAIC NUMBER THEORY.

“Abstract” commutative algebra is a recent creation, but its development can not be understood except in relation to that of algebraic number theory and algebraic geometry which gave it birth.

It was possible to conjecture without too much improbability that the famous “proof” that Fermat claimed to have of the impossibility of the equation $x^p + y^p = z^p$ for p an odd prime number and x, y, z integers $\neq 0$, was based on the decomposition

$$(x + y)(x + \zeta y) \dots (x + \zeta^{p-1}y) = z^p$$

in the ring $\mathbf{Z}[\zeta]$ (where $\zeta \neq 1$ is a p th root of unity), and on an argument about divisibility in this ring, supposing it to be principal. It is found in any case that an analogous argument is outlined by Lagrange ([191], v. II, p. 531); it is by arguments of this kind, with different variants (notably alterations of the variables with the purpose of lowering the degree of the equation) that Euler ([108 a], v. I, p. 488)¹ and Gauss ([124 a], v. II, p.387) prove Fermat’s theorem for $p = 3$, Gauss (loc. cit. and Dirichlet ([92], v. I, p. 42) for $p = 5$, and Dirichlet the impossibility of the equation $x^{14} + y^{14} = z^{14}$ ([92], v. I, p. 190). Finally, in his first research on the theory of numbers, Kummer had believed that he had obtained in this way a general proof, and it is no doubt this error (which was pointed out to him by Dirichlet) that led him to his studies on the arithmetic of cyclotomic fields, from where he was at last to succeed in deducing a correct

version of his proof for prime numbers $p < 100$ [188 d].

From another angle, the famous memoir of Gauss of 1831 on biquadratic residues, whose results are deduced from a detailed study of divisibility in the ring $\mathbb{Z}[i]$ of “Gaussian integers” ([124 a], v. II, p. 109) showed what advantage there could be for the classical problems of number theory by the extension of the notion of divisibility to algebraic numbers; ² so it is not surprising that between 1830 and 1850 this theory was the object of the work of many

German mathematicians, Jacobi, Dirichlet and Eisenstein first, then, a little later, Kummer and his pupil and friend Kronecker. We should not speak here of the theory of units, too restricted to number theory, where progress was rapid, Eisenstein obtaining the structure of the groups of units for the cubic fields, Kronecker for the cyclotomic fields, shortly before Dirichlet, in 1846 ([92], v. I, p. 640) proves the general theorem, which Hermite had almost reached on his own account ([159], v. I, p. 159). Much more difficult would appear to be the question (central to the whole theory) of the decomposition into prime factors. Since Lagrange had given examples of numbers of the form $x^2 + Dy^2$ (x, y, D integers) having divisors that are not of the form $m^2 + Dn^2$ ([191], v. II, p. 465), it was known substantially that it was not to be expected that in general the rings $\mathbb{Z}[\sqrt{-D}]$ should be principal ideal rings, and to the temerity of Euler a great circumspection had succeeded; when Dirichlet, for example, proves that the relation $p^2 - 5q^2 = r^5$ (p, q, r integers) is equivalent to $p + q\sqrt{5} = (x + y\sqrt{5})^5$ for x, y integers, he limits himself to indicating in a note that “there are analogous theorems for many other prime numbers [other than 5]” ([92], v. I, p. 31). With the memoir of Gauss of 1831 and the work of Eisenstein on cubic residues [102 a], one had, it is true, some advanced studies of the arithmetic of the principal ideal rings $\mathbb{Z}[i]$ and $\mathbb{Z}[\rho]$ ($\rho = (-1 + i\sqrt{3})/2$, a cube root of unity) in perfect analogy with the theory of rational integers, and with these examples at least, the tight link between arithmetic in quadratic fields and the theory of binary quadratic forms developed by Gauss was very apparent; but for the general case a “dictionary” was missing which would have allowed the treatment of a quadratic field by simple translation of the theory of Gauss. ³

In fact, it is not for quadratic fields, but indeed for cyclotomic fields (and for reasons that would not appear clearly until much later (cf. p. 98)) that the enigma would first be resolved. From 1837, Kummer, an analyst to begin with, turns to the arithmetic of cyclotomic fields,

which is not going to cease from keeping him busy almost exclusively for 25 years. Like his predecessors, he studies divisibility in the rings $\mathbf{Z}[\zeta]$, where ζ is a p th root of unity $\neq 1$ (p an odd prime); he soon sees that, there also, rings that are not principal ideal rings occur, blocking all progress in extending the laws of arithmetic [188 a], and it is only in 1845, at the end of 8 years of effort, that the light finally dawns, thanks to his definition of “ideal numbers” [188 c and d].

What Kummer is doing reduces, in modern language, to defining valuations on a field $\mathbf{Q}[\zeta]$: they are in bijective correspondence with his “prime ideal numbers”, the “exponent” with which such a factor appears in the “decomposition” of a number $x \in \mathbf{Z}[\zeta]$ being none other than the value of x in the corresponding valuation. As the conjugates of x also belong to $\mathbf{Z}[\zeta]$, and that their product $N(x)$ (the “norm” of x) is a rational integer, the “prime ideal factors” to be defined had also to be “factors” of rational prime numbers, and to give it its definition, one could limit oneself to saying that they were the “prime ideal divisors” of a prime number $q \in \mathbf{Z}$. For $q = p$, Kummer had already substantially proved [188 a] that the principal ideal $(1 - \zeta)$ was prime and that its $(p - 1)$ th power was the principal ideal (p) ; this case did not therefore raise any new problems. For $q \neq p$, the idea that seems to have guided Kummer is to replace the cyclotomic equation $\Phi_p(z) = 0$ by the congruence $\Phi_p(u) \equiv 0 \pmod{q}$, in other words to decompose the cyclotomic polynomial $\Phi_p(X)$ over the field $\mathbf{F}_q$, and to associate with each irreducible factor of this polynomial a “prime ideal factor”. A simple case (explicitly quoted in the Note [188 b] where Kummer announces his results without proof) is that where $q \equiv 1 \pmod{p}$; if $q = mp + 1$ and if $\gamma \in \mathbf{F}_q$ is a $(q - 1)$ st primitive root of 1, we have, in $\mathbf{F}_q[X]$,

$$\Phi_p(X) = \prod_{k=1}^{p-1} (X - \gamma^{km})$$

since $\gamma^{pm} = 1$. Associating then with each factor $X - \gamma^{km}$ a “prime ideal factor” $\mathfrak{q}_k$ of q , Kummer says that an element $x \in \mathbf{Z}[\zeta]$, of which P is the minimal polynomial over $\mathbf{Q}$, is divisible by $\mathfrak{q}_k$ if in $\mathbf{F}_q$ we have $P(\gamma^{km}) = 0$; to sum up, in modern language, he writes the quotient ring $\mathbf{Z}[\zeta]/q\mathbf{Z}[\zeta]$ as a direct product of fields isomorphic to $\mathbf{F}_q$. For $q \not\equiv 1 \pmod{p}$, the irreducible factors of $\Phi_p(X)$ in $\mathbf{F}_q[X]$ are no longer of the first degree, and it will therefore be necessary to substitute for X in $P(X)$ “Galois imaginary” roots of the factors of Φ_p in $\mathbf{F}_q[X]$. Kummer avoids this difficulty by passing on, as we would say today, to the decomposition field

K of q : if f is the smallest integer such that $q^f \equiv 1 \pmod{p}$, and if one puts $p - 1 = ef$, K is none other than the subfield of $\mathbf{Q}(\zeta)$ constructed from the invariants of the subgroup of order f of the Galois group (cyclic of order $p - 1$) of $\mathbf{Q}(\zeta)$ over $\mathbf{Q}$; in other words it is the unique subfield of $\mathbf{Q}(\zeta)$ which is of degree e over $\mathbf{Q}$; it was very well known since the Disquisitiones of Gauss, being generated by the “periods”

$$\eta_k = \zeta_k + \zeta_{k+e} + \zeta_{k+2e} + \cdots + \zeta_{k+(f-1)e}$$

($0 \leq k \leq e - 1$, $\zeta_v = \zeta^{g^v}$ where g is a primitive root of the congruence $z^{p-1} \equiv 1 \pmod{p}$), which make up a normal basis for it. If $R(X)$ is the minimal polynomial (unitary and with rational integer coefficients) of any one of these “periods” η , Kummer, basing himself on the formulae of Gauss, proves that, over the field $\mathbf{F}_q$, $R(X)$ is still decomposed into distinct factors of the first degree $X - u_j$ ($1 \leq j \leq e$), and it is to each of these u_j that he associates this time a “prime ideal factor” q_j . In order to define “divisibility by q_j ”, Kummer writes every $x \in \mathbb{Z}[\zeta]$ in the form $x = \sum_{k=0}^{f-1} \zeta^k y_k$, where each $y_k \in K$ is itself written in a unique way as a polynomial of degree $\leq e - 1$ in η , with rational integer coefficients; he says that x is divisible by q_j if and only if, when u_j is substituted for η in each of the y_k , the elements of $\mathbf{F}_q$ obtained are all zero. But it was still necessary to define the “exponent” of q_j in x . For that, Kummer introduces what we would now call a uniformiser for q_j , that is to say an element $\rho_j \in K$ such that $N(\rho_j) \equiv 0 \pmod{q}$, $N(\rho_j) \not\equiv 0 \pmod{q^2}$, and finally such that ρ_j is divisible by q_j (in the sense defined above) but by none other of the ideal factors $\neq q_j$ of q . The existence of such a ρ_j had been substantially proved by Kronecker in his dissertation the year before ([186 a], v. I, p. 23); then setting $\rho'_j = N(\rho_j) / \rho_j$, Kummer says that the exponent of q_j in x is equal to h if one has $x\rho_j'^h \equiv 0 \pmod{q^h}$, but $x\rho_j'^{h+1} \not\equiv 0 \pmod{q^{h+1}}$; he starts of course by proving that the relation $x\rho_j' \equiv 0 \pmod{q}$ is equivalent to the fact that x is divisible by q_j (in the previous sense). Once these definitions are set up, the extension to $\mathbb{Z}[\zeta]$ of the usual laws of divisibility for the “ideal numbers” did not offer any serious problems; and from his first memoir [188 a] Kummer was even able, by using the “method of drawers” of Dirichlet, to prove that the “classes” of “ideal factors” were finite in number. ⁵

We will not take up the history of the subsequent work of Kummer on cyclotomic fields, as

far as the determination of the number of classes and the application of the proof of Fermat's theorem in various cases is concerned. Let us only mention the way in which, in 1859, he extends his method in order to obtain (at least partially) the “prime ideal numbers” in a “Kummer field” $\mathbf{Q}(\zeta, \mu)$, where μ is a root of an irreducible polynomial $P(X) = X^p - \alpha$, with $\alpha \in \mathbf{Z}[\zeta]$ [188 e]. It is interesting that Kummer sees this problem precisely as considering $\mathbf{Q}(\zeta, \mu)$ as a cyclic extension of the field $\mathbf{Q}(\zeta)$ taken as “base field”: ⁶ he starts with a “prime ideal number” q of $\mathbf{Z}[\zeta]$, that he assumes divides neither p nor α , and this time, he examines (in modern terms) the polynomial $\bar{P}(X) = X^p - \bar{\alpha}$ in the residual field k of the valuation of $\mathbf{Q}(\zeta)$ corresponding to q ($\bar{\alpha}$ being the canonical image of α in k). As $\mathbf{Q}(\zeta)$ is the field of p th roots of unity, $\bar{P}$ is, either irreducible over k , or a product of factors of the first degree. In the first case, Kummer says that q remains prime in $\mathbf{Z}[\zeta, \mu]$; in the second, he introduces elements w_i ($1 \leq i \leq p$) of $\mathbf{Z}[\zeta]$ whose images in k are the roots of $\bar{P}$, and he associates with each index i a prime factor ideal $\mathfrak{r}_i$ of q ; setting next $W_i(X) = \prod_{j \neq i} (X - w_j)$, he says that, for a polynomial f with coefficients in $\mathbf{Z}[\zeta]$, $f(\mu)$ contains the factor ideal $\mathfrak{r}_i$; m times if one has

$$f(w_i)W_i^m(w_i) \equiv 0 \pmod{q^m}$$

but

$$f(w_i)W_i^{m+1}(w_i) \not\equiv 0 \pmod{q^{m+1}}.$$

To sum up, he obtains in this way the valuations of $\mathbf{Q}(\zeta, \mu)$ unramified over $\mathbf{Q}$, which is sufficient for the applications which he has in mind.

Kummer had the luck to meet, in his study of the particular fields to which his research on Fermat's theorem had first led him, a number of fortuitous circumstances which rendered the study much more feasible. The extension of Kummer's results to the general case presented formidable difficulties and was to cost years of efforts.

After Kronecker and Dedekind, who hold the lead roles in it, the history of the theory of algebraic numbers, during the 40 years that follow the discovery of Kummer, is not without a reminder (but happily without the same acrimonious character) of the rivalry of Newton and

Leibniz 180 years earlier, over the invention of the infinitesimal Calculus. Student and soon a colleague of Kummer at Berlin, Kronecker (whose thesis, as we have seen, was useful to Kummer at an essential point of his theory) was very closely interested in “ideal numbers” with the purpose of applying them to his own research; and we admire his astonishing insight when we see him, already in 1853 ([186 a], v. IV, p. 10), stating the general theorem on the structure of abelian extensions of $\mathbb{Q}$ and, what is perhaps even more remarkable, creating, in the following years, the theory of complex multiplication and discovering the first shoots of the theory of class fields ([186 a], v. IV, pp. 177-183 and 207-217). A letter from Kronecker to Dirichlet; in 1857 ([186 a], v. V, pp. 418-421), shows him already at that time, in possession of a generalisation of Kummer’s theory, which besides is confirmed by Kummer himself in one of his own works ([188 e], p. 57), and Kronecker will often refer to this theory in his memoirs between 1860 and 1880. ⁷

But although at this time none of the mathematicians of the German school of number Theory were ignorant of this work of Kronecker, this latter seems only to have communicated the principles of his methods to a restricted circle of friends and pupils, and when he finally decides to publish them, in his memoir of 1881 on the discriminant ([186 a], v. II, pp. 193-236), and especially in his great “Festschrift” of 1882 ([186 a], v. II, pp. 237-287), Dedekind can not prevent himself from expressing his surprise ([79], v. III, p. 427), having imagined quite different procedures from the echoes which he had heard ([79], v. III, p. 287). Besides Kronecker was far from having to the same degree the remarkable gifts of exposition and clarity of Dedekind, and it is therefore not surprising that it is mostly the methods of this latter, published already in 1871, which formed the framework of algebraic number theory; as interesting as it is, the method of “adjunction of indeterminates” of Kronecker, as far as Number Theory is concerned, is no more to our eyes than a variant of that of Dedekind, and it is mainly in another direction, aimed towards Algebraic Geometry, that Kronecker’s ideas acquired all their importance for the history of Commutative Algebra, as we will see later.

For reasons that could only appear clearly much later, a first preamble to any attempt at a general theory was of course the clarification of the notion of algebraic integer. This was acquired in about 1845-1850, although it is fairly difficult to date its apparition in a precise way;

it appears likely that it is the idea of a system stable under addition and multiplication (or, more precisely, what we now call a $\mathbb{Z}$ -algebra of finite rank) that, more or less consciously, had led to the general definition of algebraic integers: this definition is inevitable when one forces a $\mathbb{Z}$ -algebra of the type $\mathbb{Z}[\theta]$ to be of finite rank, by analogy with the ring $\mathbb{Z}[\zeta]$ generated by a root of unity, which was at the centre of the preoccupations of the arithmeticians of the time. So it is that when, independently, Dirichlet ([92], v. I, p. 640), Hermite ([159], v. I, pp. 115 and 146) and Eisenstein ([102 c], p. 236) introduce the notion of algebraic integer, they do not seem to consider that it consists of a new idea nor to believe that it is useful to make a detailed study thereof; only Eisenstein proves substantially (*loc. cit.*) that the sum and the product of two algebraic integers are algebraic integers, besides without claiming that this result is original.

A point which was much better hidden was the determination of the rings in which one could hope to generalise Kummer's theory. This latter, in his first note [188 b], does not hesitate to affirm that he can find by his methods the theory of binary quadratic forms of Gauss by considering the rings $\mathbb{Z}[\sqrt{D}]$ (D an integer); he never developed this idea, but it certainly appears that neither he nor anyone before Dedekind was able to see that the unique decomposition into prime "ideal" factors is not possible in rings of the form $\mathbb{Z}[\sqrt{D}]$ when $D \equiv 1 \pmod{4}$ (although the example of the cube roots of unity showed that the ring $\mathbb{Z}[\rho]$ considered since Gauss is distinct from $\mathbb{Z}[\sqrt{-3}]$).⁸ Before Dedekind and Kronecker, the only rings studied are still of the type $\mathbb{Z}[\theta]$ or sometimes certain particular rings of the type $\mathbb{Z}[\theta, \theta']$.⁹ As far as Kronecker is concerned, it is possible that the idea of considering the ring of all integers of an algebraic extension was first suggested to him by the study of fields of algebraic functions, where this ring appears in a natural way as the set of functions that are "finite at a finite distance": he insists in any case in his memoir of 1881 on the discriminant (written and announced to the Berlin Academy already in 1862) on this characterisation of the "integers" in these fields ([186 a], v. II, pp. 193-236). Dedekind does not give any indication as to the origin of his own ideas on this point, but from his first publications on number fields in 1871, the ring of all integers of such a field plays a fundamental role in his theory; it is also Dedekind who clarifies the relation between such a ring and its subrings having the same field of fractions, by the introduction of the notion of conductor ([79], v. I, pp. 105-157).

But that was not the only difficulty. To generalise the ideas of Kummer, it was first necessary to avoid going through the decomposition field which naturally could not have any analogue in the case of a non abelian field. This detour seems anyway at first sight very surprising and artificial, for if one starts from the irreducible polynomial $\Phi_p(X)$ of $\mathbf{Z}[X]$, one asks oneself why Kummer does not follow to the bitter end the logical consequences of his ideas, and what it is that stops him using the theory of “Galois imaginaries”, well known at that time. The obstacle appears more clearly in the light of an unhappy attempt to generalise made already in 1865 by Selling, a pupil of Dedekind: given an irreducible polynomial $P \in \mathbf{Z}[X]$, Selling decomposes the corresponding polynomial $\bar{P}(X)$ into irreducible factors in $\mathbf{F}_q[X]$; the roots of this polynomial belong therefore to a finite extension $\mathbf{F}_r$ of $\mathbf{F}_q$; but Selling, in order to define in the same way as Kummer the exponent of a “prime ideal factor” of q in an integer of the field of roots of $P(X)$, does not hesitate to speak in the field $\mathbf{F}_r$ of congruences modulo a power of q ([282], p. 26); and a little later, when he tries to tackle the question of ramification, he “adjoins” to $\mathbf{F}_r$ “imaginary roots” of an equation of the form $x^h = q$ ([282], p. 34). It is obvious that these bold steps (which would be justified by replacing the finite field $\mathbf{F}_q$ by the q -adic field) could not at that time end up with anything except nonsense. Fortunately Dedekind had just in 1857 ([79], v. I, pp. 40-66), under the name of “theory of superior congruences”, taken up under another form the theory of finite fields: ¹⁰ he interprets the elements of these latter as “remainders” of polynomials in $\mathbf{Z}[X]$ following a “double module” consisting of linear combinations, with coefficients in $\mathbf{Z}[X]$, of a prime number p and a unitary irreducible polynomial $P \in \mathbf{Z}[X]$ (which is no doubt for him, as for Kronecker, at the beginnings of the general idea of module at which they will independently end up a little later). By his own testimony ([79], v. I, p. 218) it appears that Dedekind had begun to attack the problem of “ideal factors” of p in a field $\mathcal{Q}(\xi)$, where $P \in \mathbf{Z}[X]$ is the minimal polynomial of ξ , in the following way (at least in the “unramified” case), that is to say when in $\mathbf{F}_p[X]$ the polynomial $\bar{P}$ corresponding to P does not have multiple roots): we write, in $\mathbf{Z}[X]$,

$$P = P_1 P_2 \dots P_h + pG$$

where the $\bar{P}_i$ are irreducible and distinct in $\mathbf{F}_p[X]$; it can be assumed that G is not divisible

(in $\mathbf{Z}[X]$) by any of the P_i and for all i , we set $W_i = \prod_{j \neq i} P_j$; then, if $f \in \mathbf{Z}[X]$, we will say that $f(\xi)$ contains the “ideal factor” $\mathfrak{p}_i$ of p corresponding to P_i k times if we have

$$fW_i^k \equiv 0 \pmod{p^k, P}$$

and

$$fW_i^{k+1} \not\equiv 0 \pmod{p^{k+1}, P}.$$

The relationship with the method followed by Kummer for the “Kummer fields” is manifest here, and one can equally, in this way, rejoin easily the initial definition of Kummer for the cyclotomic fields (see for example the work of Zolotareff [343] who, at first independently of Dedekind, develops these ideas a little later).

However neither Dedekind, nor Kronecker who seems also to have made similar attempts, were to continue further along this path, both stopped by the difficulties presented by ramification ([79], v. I, p. 218 and [186 a], v. II, p. 325).¹¹ If the ring of integers A of a number field K which is considered as admitting a basis (over $\mathbf{Z}$) consisting of the powers of the same integer θ , it is not difficult to generalise the preceding method for the ramified prime numbers in $\mathbf{Z}[\theta]$ (as Zolotareff indicates (loc. cit.)). But there are fields K where no basis of this type exists in the ring A ; and Dedekind even ends up by discovering that there are cases where certain prime numbers p (the “extraordinary factors of the discriminant” of the field K) are such that, whatever the $\theta \in A$, application of the preceding method to the minimal polynomial of θ over $\mathbf{Q}$ would lead to attributing to p multiple ideal factors whereas in fact p is not ramified in A ;¹² he admits to having been stopped for a long time by this unforeseen difficulty, before succeeding in surmounting it by creating in its entirety the theory of modules and ideals, expounded in a masterly way (and already totally modern, in contrast with the discursive style of his contemporaries) in what is no doubt his masterpiece, the famous “XIth supplement” to the book of Dirichlet on number Theory ([79], v. III, pp. 1-222). This work will know three successive versions, but from the first (published as the “Xth supplement” to the second edition of the book of Dirichlet in 1871) the essentials of the method are already in place, and almost in a single step the theory

of algebraic numbers passes from sketches and previous fumblings to a discipline in full maturity and already in possession of its essential tools: from the beginning, the ring of all integers of a number field is placed at the centre of the theory; Dedekind proves the existence of a basis of this ring over $\mathbf{Z}$, and deduces from that the definition of the discriminant of a field, as the square of the determinant consisting of the elements of a basis of the ring of integers and their conjugates; he gives however in the XIth supplement a characterisation of ramified prime numbers (as prime factors of the discriminant) only for quadratic fields ([79] v. III, p. 202), whereas he was in possession of the general theorem since 1871.¹³ The central result of the work is the theorem on the existence and uniqueness of the decomposition of ideals into prime factors, for which Dedekind starts by developing an elementary theory of “modules”; in fact, in the XIth supplement, he reserves this name for the $\mathbf{Z}$ -submodules of a number field, but the conception of it that he builds and the results that he proves are already set out in a way that is immediately applicable to the most general modules;¹⁴ we must note among other things, from 1871, the introduction of the notion of “Noether quotient” which plays an important role (as well besides as the “ascending chain condition”) in the first proof of unique factorisation. In the following two editions, Dedekind was to give as well two other proofs of this theorem, which he considered justifiably to be the keystone of this theory. It must be noted here that it is in the third proof that fractional ideals occur (already introduced by Kummer from 1859 for cyclotomic fields) and the fact that they form a group; we will return later to the second proof (p. 107).

All these results (up to the language used) were without doubt already known to Kronecker in about 1860 as particular cases of his more general concepts of which we will speak later (whereas Dedekind admits not having overcome the last difficulties of the theory until 1869-1870 ([79], v. I, p. 351));¹⁵ as far as number fields are concerned, we must underline in particular that, from this time, Kronecker knew that the whole theory is applicable without any essential change when one departs from a “base field” k which is itself a number field (other than $\mathbf{Q}$), a point of view to which the theory of complex multiplication naturally led; he had also recognised, for certain fields k , the existence of algebraic extensions $K \neq k$ unramified over k ([186 a], v. II, p. 269) which can not happen for $k = \mathbf{Q}$ (as follows from the output of Hermite and Minkowski for the discriminant). Dedekind was never to develop this last point of view (although he indicates its

possibility in his memoir of 1882 on the different), and the first systematic exposition of the theory of “relative fields” is due to Hilbert ([163 a], v. I, pp. 63-363).

Finally, in 1882 ([79], v. I, pp. 359-396), Dedekind completes the theory by introducing the different, which gives him a new definition of the discriminant and allows him to specify the exponents of the prime ideal factors in the decomposition of the latter. It is also at about this time that he becomes interested in the particular points brought out by Galois extensions, introducing the notions of a decomposition group and an inertia group (in a memoir ([79], v. II, pp. 43-48) that was only published in 1894), and even (in papers unpublished during his lifetime ([79], v. II, pp. 410-411)) a sketch of ramification groups, that Hilbert (independently of Dedekind) would develop a little later ([163 a], v. I, pp. 13-23 and 63-363).

So, in about 1895, Algebraic Number theory had finished the first stage of its development; the tools forged during this period of formation are going to allow it to tackle almost straightaway the next stage, the general theory of class fields (or, what amounts to the same thing, the theory of abelian extensions of number fields) which is carried on to our day and which we will not need to describe here. From the point of view of Commutative Algebra, it can be said that at the same time the theory of Dedekind rings is practically completed, apart from their axiomatic characterisation, as well as the structure of modules of finite type over these rings (which, in the case of number fields, will only be clarified in substance by Steinitz in 1912 [294 b]).¹⁶

The subsequent progress of Commutative Algebra is going to arise above all from quite different problems, issuing from Algebraic Geometry (which, besides will influence in a direct way Number Theory, even before the “abstract” developments of the contemporary era).

We are not going to do the detailed history of Algebraic Geometry here, which, until the death of Riemann, hardly touches on our subject. It is sufficient to remember that it had above all as its goal the study of algebraic curves in the complex projective plane, tackled most often by the methods of projective geometry (with or without the use of co-ordinates). In parallel there was developing, with Abel, Jacobi, Weierstrass and Riemann, the theory of “algebraic functions” of one complex variable and their integrals; the link between this theory and the geometry of plane algebraic curves was evidently in their minds, and it was known how on that occasion to “apply Analysis to Geometry”; but the methods used for the study of algebraic functions were mainly

of a “transcendental” nature, even before Riemann; ¹⁷ this characteristic is more accentuated in the work of this latter, with the introduction of “Riemann surfaces” and of arbitrary analytic functions defined on such a surface. Almost immediately after the death of Riemann, Roch and especially Clebsch, recognised the possibility of drawing out from the deep results obtained by the transcendental methods of Riemann numerous and striking applications to the projective geometry of curves, which was naturally going to encourage contemporary geometers to give these results purely “geometric” proofs; this programme, incompletely followed by Clebsch and Gordan, was accomplished by Brill and M. Noether a few years later [37], by means of the study of systems of variable points on a given curve and of auxiliary curves (the “adjoints”) going through such systems of points. But even for the contemporaries, the transcendental methods of Riemann (and notably his use of topological notions and the “Dirichlet principle”) seem to rest on uncertain foundations; and even though Brill and Noether were more careful than the majority of contemporary “synthetic” geometers (see later p. 106), their geometrico-analytic arguments are not safe from all blame. It is essentially in order to give to the theory of plane algebraic curves a solid basis that Dedekind and Weber publish in 1882 their great memoir on this subject [80]: “The researches published herewith”, they say, “have as their goal to state the foundations of the theory of algebraic functions of one variable, one of the main creations of Riemann, in a way that is at the same time simple, rigorous and entirely general. In later researches on this subject, restrictive hypotheses on the singularities of the functions being considered are often made, and the supposed exceptional cases are, either mentioned in passing as limit cases, or entirely neglected. In the same way, certain fundamental theorems on continuity or analyticity are assumed, for which the “evidence” leans on geometric intuitions of a varied nature” ([80], p. 181). ¹⁸

The essential idea of their work is to copy the theory of algebraic functions of one variable onto the theory of algebraic numbers such as it had just been developed by Dedekind; to do this, they must first put themselves in an “affine” point of view (over against their contemporaries who invariably considered algebraic curves as embedded in complex projective space): they start therefore from a finite algebraic extension K of the field $\mathbb{C}(X)$ of rational fractions, and from the ring A of “integral algebraic functions” in K , i. e. from the elements of this field which

are integers over the ring $\mathbb{C}[X]$ of polynomials; their fundamental result, that they obtain without using any topological consideration, ¹⁹ is that A is a Dedekind ring, to which are applied mutatis mutandis (and even, as Dedekind and Weber remark without having seen clearly the reason yet ([79], v. I, p. 268), in a simpler way) all the results of the “XIth supplement”. That done, they prove that their theorems are in fact birationally invariant (in other words, only depending on the field K) and in particular do not depend on the choice of the “line at infinity” made at the beginning. What is no doubt even more interesting for us, is that, wanting to define the points of the “Riemann surface” corresponding to K (and in particular the “points at infinity”, which could not correspond to ideals of A), they are led to introduce the notion of a “place” of the field K : they find themselves faced with the situation that will be found again by Gelfand in 1940 while setting up the theory of normed algebras, namely a set K of elements which are not given in advance as functions, but that nonetheless one wants to be able to consider as such; and, in order to obtain the set of definition of these hypothetical functions, they have for the first time the idea (that Gelfand will take up again, and which has become dull through having been used at every turn in modern mathematics) of associating with a point x of a set E and with a set F of maps of E into a set G the map $f \mapsto f(x)$ of $\mathcal{F}$ into G , in other words to consider, in the expression $f(x)$, f as variable and x as fixed, contrary to the classical tradition. Finally, they have no trouble, starting from the notion of place, in defining the “positive divisors” (“Polygone” in their terminology) that refine the notion of ideal of A and correspond to the “systems of points” of Brill and Noether; but, although they write the principal divisors and the divisors of differentials as “quotients” of positive divisors, they do not give the general definition of divisors, and it is only in 1902 that Hensel and Landsberg will introduce, in analogy with fractional ideals, this notion that will always embarrass the holders of purely “geometric” methods (obliged in spite of themselves to define them under the name of “virtual systems”, but hindered from being able to give them a “concrete” interpretation).

The same year 1882 also sees the appearance of the great memoir of Kronecker awaited for more than 20 years ([186 a], v. II, pp. 237-387). Much more ambitious than the work of Dedekind-Weber, it is unhappily also much more vague and obscure. His central theme is (in modern language) the study of the ideals of a finite integral algebra over one of the rings of

polynomials $\mathbf{C}[X_1, \dots, X_n]$ or $\mathbf{Z}[X_1, \dots, X_n]$; Kronecker limits himself a priori to those of these ideals that are of finite type (the fact that they are all of this kind was only to be proved (for the ideals of $\mathbf{C}[X_1, \dots, X_n]$ several years later by Hilbert during his work on invariants ([163 a], v. II, pp. 199-257)). As far as $\mathbf{C}[X_1, \dots, X_n]$ or $\mathbf{Z}[X_1, \dots, X_n]$ are concerned one was naturally led to associate to each ideal of one of these rings the “algebraic variety” consisting of the common zeros of all the elements of the ideal; and the geometrical studies in 2 or 3 dimensions done during the course of the XIXth century would intuitively lead to the idea that every variety is the union of “irreducible” varieties, finite in number, whose “dimensions” are not necessarily the same. It appears that the proof of this fact was the goal that Kronecker was setting himself, although he does not state it explicitly anywhere, and that nowhere in his memoir can any definition of “irreducible variety” nor of “dimension” be found. In fact he limits himself to indicating summarily how a general method of elimination gives, starting with a system of generators of the ideal under consideration, a finite number of algebraic varieties, for each of which, in a suitable system of co-ordinates, a certain number of the co-ordinates are arbitrary and the others are “algebraic functions”.²¹ But, if it is really the decomposition into irreducible varieties that Kronecker is aiming at, it has to be recognised that he only succeeds in the elementary case of a principal ideal, where he proves substantially, by extending a classical lemma of Gauss on $\mathbf{Z}[X]$ ([124 a], v. I, p. 34), that the rings $\mathbf{C}[X_1, \dots, X_n]$ and $\mathbf{Z}[X_1, \dots, X_n]$ are unique factorisation domains; and, in the general case, it can even be asked if Kronecker was at that time in possession of the notion of prime ideal (what he calls “Primmodulsystem” is an ideal indecomposable as the product of two others ([186 a], v. II, p. 336); that is all the more astonishing as the definition given since 1871 by Dedekind was perfectly general). A little later, Kronecker was however to correct this inadvertence.

It must be said however that Kronecker’s method of elimination, suitably applied, leads exactly to the decomposition of an algebraic variety into its irreducible components: that is what is clearly established by E. Lasker at the beginning of his great memoir of 1905 on ideals of polynomials [194]; he defines correctly the notion of irreducible variety (in $\mathbf{C}^n$) as an algebraic variety V such that the product of two polynomials can not be zero in the whole variety V unless one of them is zero on it, and there is also a definition of dimension that is independent of

the axes chosen. In the interesting historical considerations that he inserts in this work, Lasker indicates on the other hand that he links up, not only with the purely algebraic tendency of Kronecker and Dedekind, but also with the problems raised by the geometric methods of the school of Clebsch and M. Noether, and notably with the famous theorem proved by the latter in 1873 [239]. It consists essentially, as we would say today, of the determination of the ideal α of those polynomials of $\mathbb{C}[X_1, \dots, X_n]$ which are zero on the points of a given set M in $\mathbb{C}^n$; most often M was the “algebraic variety” of the zeros common to a finite number of polynomials f_i , and for a long time it appears that it was assumed (of course without justification) that, at least for $n = 2$ or $n = 3$, the ideal α was quite simply generated by the f_i ;²² M. Noether had shown that already for $n = 2$ and for two polynomials f_1, f_2 , this is in general inexact, and he had given sufficient conditions for α to be generated by f_1 and f_2 . Ten years later, Netto proves that, without any hypothesis on f_1 and f_2 , a power of α is in any case contained in the ideal generated by f_1 and f_2 [231], a theorem that Hilbert generalises in 1893 in his famous “zeros theorem” ([163 a], v. II, pp. 287-344). It is without doubt that, inspired by this result Lasker, in his memoir, introduces the general notion of primary ideal²³ in the rings $\mathbb{C}[X_1, \dots, X_n]$ and $\mathbb{Z}[X_1, \dots, X_n]$ (after having given in these rings the definition of prime ideals, by transcribing the definition of Dedekind), and proves²⁴ the existence of a primary decomposition for every ideal in these rings.²⁵ He does not seem to have been concerned with questions of uniqueness of this decomposition; it is Macaulay who, a little later [211] introduces the distinction between “immersed” and “non-immersed” primary ideals and shows that the latter are determined in a unique way, but not the earlier. It must finally be noted that Lasker also extends his results to the ring of entire series convergent in the neighbourhood of a point, relying on the “preparation theorem” of Weierstrass. This part of his memoir is doubtless the first place where this ring had been considered from a purely algebraic point of view, and the methods that Lasker develops for this purpose were to influence Krull strongly when in 1938 he created the general theory of local rings (cf. [187 h], pp. 204 and *passim*).

The movement of ideas that will end up with modern Commutative Algebra starts to take shape around 1910. If the general notion of a field is acquired already at the beginning of the XXth century, on the other hand the first work where the general notion of a ring is defined is

without doubt that of Fraenkel in 1914 [113 a]. At that time, there were already as examples of rings, not only the integral rings of Number Theory and of Algebraic

Geometry, but also rings of series (formal or convergent), and finally algebras (commutative or not) over a base field. However, as much for the theory of rings as for that of fields, the catalytic role seems to have been played by the theory of p -adic numbers of Hensel, that Fraenkel as well as Steinitz [294 a] mention quite particularly as the point of departure for their research.

The first publication of Hensel on this subject goes back to 1897; in it he starts from the analogy brought to light by Dedekind and Weber between the points of a Riemann surface of a field of algebraic functions K and the prime ideals of a number field k ; he proposes to transport into Number Theory the “Puisseux expansions” (classical since the middle of the XIXth century) which, in the neighbourhood of an arbitrary point of a Riemann surface of K , allow us to express every element $x \in K$ in the form of a convergent series of powers of the “uniformiser” at the point under consideration (a series having only a finite number of terms with negative exponents). Hensel shows in the same vein that if p is a prime ideal of k above a prime number p , one can associate to each $x \in k$ a “ p -adic series” of the form $\sum_i \alpha_i p^i$ (or $\sum_i \alpha_i p^{i/e}$) when p is ramified above p ; the α_i being taken from a system of given representatives of the quotient field of the ideal p ; but his great originality is to have the idea of considering such “expansions” even when they do not correspond to any element of k , by analogy with the expansions in entire series of transcendental functions on a Riemann surface [157 a].

During the whole of his subsequent career, Hensel will devote himself to polishing and perfecting little by little his new calculus; and if his approach can seem to us hesitant or heavy, it must not be forgotten that at least in the beginning he still does not have available any of the topological or algebraic tools of present day mathematics which would have facilitated his task. In his first publications he besides hardly ever speaks of topological notions, and to sum up, for him the ring of p -adic numbers (p a prime ideal of the ring of integers A of a number field k), is, in modern terms, the projective limit of the rings A/p^n for n tending to infinity, with a purely algebraic meaning; and to establish the properties of this ring and of its field of fractions, he must at each step use more or less painfully ad hoc arguments (for example to prove that the p -adic numbers form an integral domain). The idea of introducing topological notions in a p -adic field

hardly appears with Hensel before 1905 [157 d]; and it is only in 1907, after having finished writing the book where he restates the theory of algebraic numbers following his ideas [157 f], that he reaches the definition and essential properties of absolute p -adic values [157 e], starting from which he will be able to develop, by copying the theory of Cauchy, a complete “ p -adic analysis” that he will know how to apply fruitfully in number Theory (notably with the use of the p -adic exponential and logarithm), and whose importance has not ceased to grow since then.

Hensel had well seen, from the beginning, the simplifications that his theory was bringing to classical statements, by allowing the “localisation” of problems and working in a field where not only the properties of divisibility are trivial, but also where, thanks to a fundamental lemma that he got out already in 1902 [157 c], the study of polynomials of which the polynomial “reduced” mod p has no multiple root is reduced to the study of polynomials over a finite field. He had given from 1897 [157 b] striking examples of these simplifications, notably in questions referring to the discriminant (in particular, a short proof of the criterion given by him some years previously for the existence of “extraordinary divisors”). But for a long time it seems as if the p -adic numbers inspired a great distrust in contemporary mathematicians; an attitude current without doubt towards ideas that were too “abstract”, but that the slightly excessive enthusiasm of their author (so frequent in mathematics amongst the zealots of new theories) was not without justifying in part. Not content in fact with applying his theory fruitfully to algebraic numbers, Hensel, impressed like all his contemporaries by the proofs of transcendence of e and π , and perhaps deluded by the qualification of “transcendent” applied at the same time to numbers and to functions, had reached the point of thinking that there existed a link between his p -adic numbers and real transcendent numbers, and he had believed for a moment that he could obtain thus a simple proof of the transcendence of e and even of e^e ([157 d], p. 556).²⁶

Soon after 1910, the situation changes, with the rising of the following generation, influenced by the ideas of Fréchet and of F. Riesz on topology, by those of Steinitz on algebra, and conquered by “abstraction” from the beginning; it will know how to make assimilable and put in its place the work of Hensel. From 1913, Kürschak [190] defines in a general way the notion of valuation, recognises the importance of the archimedean valuations (of which the p -adic valuation gave an example), proves (by copying the proof in the case of real numbers) the existence of the

completion of a field with respect to a valuation, and above all proves in a general way the possibility of the extension of a valuation to an algebraic extension of a given field. But he had not seen that the non-archimedean character of a valuation was already hidden in the prime field; that point was established by Ostrowski, to whom is also due the determination of all the valuations on the field $\mathbb{Q}$, and the fundamental theorem characterising fields furnished with a non archimedean valuation as subfields of $\mathbb{C}$ [242]. In the years that go from 1920 to 1935, the theory will complete itself with a more detailed study of valuations which are not necessarily discrete, including among others the examination of the different circumstances: which arise when one goes to an algebraic or transcendental extension (Ostrowski, Deuring, F. K. Schmidt); on the other hand, in 1931, Krull introduces and studies the general notion of valuation [187 f] which will be much used in the years that follow by Zariski and his school of algebraic Geometry.²⁷ We must also mention here, even though it is outside our remit, the deeper studies on the structure of complete valuation fields and complete local rings, which come from the same period (Hasse-Schmidt, Witt, Teichmüller, I. Cohen).

The work of Fraenkel mentioned above (p. 107) dealt only with a very special kind of ring (artinian rings having only a single prime ideal, which is, as well, assumed to be principal). If one excludes the work of Steinitz on fields [294 a], the first important work on the study of general commutative rings are the two great memoirs of E. Noether on the theory of ideals: that of 1921 [236 a], dedicated to the primary decomposition, which takes up again on the most general level and completes on many points the results of Lasker and Macaulay; and that of 1927 characterising Dedekind rings axiomatically [236 b]. In the same way that Steinitz had shown it for fields, it is seen in these memoirs how a small number of abstract ideas, such as the notion of irreducible ideal, chain conditions and the idea of an integrally closed ring (the last two, as we have seen, already brought out by Dedekind) can on their own lead to general results which seem inextricably linked to results of pure calculation in the cases where they were known beforehand.

With these memoirs of E. Noether, joined to the slightly later work of Artin-van der Waerden on divisor ideals ([317 a], v. II, pp. 105-109) and of Krull linking these ideals to essential valuations [187 f] is thus completed the long study of the decomposition of ideals started a

century before,²⁸ at the same time as modern Commutative Algebra is inaugurated.

The countless later researches on Commutative Algebra are grouped most easily following some major directing tendencies:

A) Local and topological rings. Although contained in embryo in all the previous work in Number Theory and Algebraic Geometry, the general idea of localisation frees itself very slowly. The general notion of a ring of fractions is only defined in 1926 by H. Grell, a pupil of E. Noether, and only for integral domains [137]; its extension to more general rings will only be given in 1944 by C. Chevalley for noetherian rings and in 1948 by Uzkov in the general case. Until about 1940, Krull and his school are practically alone in using in general arguments consideration of the local rings A_p of an integral domain;

these rings will only begin to appear explicitly in Algebraic Geometry with the work of Chevalley and Zariski starting in 1940.²⁹

The general study of local rings themselves only starts in 1938 with the great memoir of Krull [187 h]. The most important results of this work concern dimension theory and regular rings, of which we should not speak here; but it is there also that appears for the first time the completion of an arbitrary local noetherian ring, as well as a still imperfect form of the graded ring associated with a local ring;³⁰ this latter will only be defined in about 1948 by P. Samuel [270] and independently by the research in algebraic Topology of Leray and H. Cartan. Krull, in the previously quoted work, hardly uses topological language; but from 1928 [187 e], he had proved that, in a noetherian ring A , the intersection of powers of a single ideal a is the set of $x \in A$ such that $x(1 - a) = 0$ for an $a \in a$; it is easily deduced from that that for every ideal m of A , the m -adic topology on A induces on an ideal a the m -adic topology of a ; in his memoir of 1938, Krull completes this result by proving that in a local noetherian ring, every ideal is closed. These theorems were soon after extended by Chevalley to semi-local noetherian rings, then by Zariski to the rings which bear his name [340 b]; it is also to Chevalley that the introduction of “linear compactness” in topological rings goes back, as well as the determination of the structure of semi-local complete rings [62 c].

B) Passing from local to global. Since Weierstrass, the habit has existed of associating an

analytic function of one variable (and in particular an algebraic function) with the set of its “expansions” at all the points of the Riemann surface where it is defined. In the introduction to his book on Number Theory ([157 f], p. V), Hensel associates in the same way with every element of an algebraic number field k the set of elements which correspond to it in the completions of k for all valuations ³¹ on k . It can be said that it is this point of view that, in modern Commutative Algebra, has replaced the decomposition formula for an ideal as a product of prime ideals (extending in a certain sense the initial point of view of Kummer). The remark of Hensel reduces implicitly to embedding k in the product of all its completions; that is what is done explicitly by Chevalley in 1936 with his theory of “idèles” [62 b], which perfects the previous analogous ideas of Prüfer and von Neumann (these latter limit themselves to embedding k in the product of its p -adic completions). ³² Although this is a bit outside our remit, it is important to mention here that, thanks to an appropriate topology on the group of idèles, all the techniques of locally compact groups (including Haar measure) can thus be applied to Number Theory in a very effective way.

In a more general setting, the theorem of Krull [187 f], characterising an integrally closed ring as the intersection of valuation rings (which comes back to embedding the ring under consideration in the product of valuation rings) often eases the study of these rings, although the method is not really manageable except for the essential valuations of Krull rings. Examples (fairly elementary) are besides often found in Krull [187 i] of the method of “passing from the local to the global” consisting of proving a property of an integral domain A by reducing it to verifying it for the “localisations” A_p of A at all its prime ideals; ³³ more recently, Serre has noticed that this method is valid for arbitrary commutative rings A , that it is applicable also to A -modules and to their homomorphisms and that it is sufficient often to “localise” at the maximal ideals of A : a point of view that is tied closely to ideas on “spectra” and on sheaves defined on these spectra (see below, p. 115).

C) Integers and integral closure. We have seen that the notion of algebraic integer, first introduced for number fields, had already been extended by Kronecker and Dedekind to algebraic function fields, although in this case it could appear fairly artificial (not corresponding to a projective notion). The memoir of E. Noether of 1927, followed by the

work of Krull starting in 1931, was to show the interest that these notions were to offer for the most general rings.³⁴ It is due to Krull in particular that we owe the lifting theorems for prime ideals in entire algebras [187 g], as well as the extension of the theory of decomposition and inertia groups of Dedekind-Hilbert [187 f]. As for E. Noether, to her is owed the general formulation of the normalisation lemma³⁵ (from which follows among others the zeros theorem of Hilbert) as well as the first general criterion (transcription of classical arguments of Kronecker and Dedekind) allowing the affirmation that the integral closure of an integral domain is finite over the ring.

Finally, it must be pointed out here that one of the reasons for the modern importance of the notion of integrally closed ring is due to the studies of Zariski on algebraic varieties; he has discovered in effect that the “normal” varieties (that is to say those whose local rings are integrally closed) stand out by means of particularly pleasant properties, notably the fact they have no “singularity of codimension 1”; and it was noticed afterwards that analogous phenomena take place for “analytical spaces”. Also “normalisation” (that is to say the operation that, for local rings of a variety, consists in taking their integral closure) has itself become a powerful tool in the arsenal of modern Algebraic Geometry.

D) The study of modules and the influence of Homological Algebra. One of the distinguishing characteristics of the work of E. Noether and W. Krull in Algebra is the tendency to “linearisation”, extending the analogous direction imprinted on field theory by Dedekind and Steinitz; in other words, it is as modules that ideals are considered above all else, and one is therefore led to apply to them all the constructions of linear Algebra (quotient, product, and more recently tensor product and forming homomorphism modules) resulting in general in modules that are no longer ideals. It appears thus fairly quickly that in many questions (whether it is a matter of commutative or non commutative rings), it is not in our interest to limit ourselves to the study of the ideals of a ring A , but that it is necessary, on the contrary to state the theorems more generally for A -modules (eventually subject to certain finiteness conditions).

The intervention of Homological Algebra only reinforced the previous tendency, since this

branch of Algebra is essentially concerned with questions of a linear nature. We should not retrace its history here; but it is interesting to point out that several of the fundamental notions of Homological Algebra (such as those of projective module and that of the functor Tor) were born as a result of the close study of the behaviour of modules over a Dedekind ring relative to the tensor product, a study undertaken by H. Cartan in 1948.

Conversely, it could be foreseen that the new classes of modules introduced in a natural way by Homological Algebra as “universal annihilators” of the functors Ext (projective modules and injective modules) and of the functors Tor (flat modules) would throw a new light on Commutative Algebra. It turns out that it is especially the projective modules and even more the flat modules that have proved to be useful: the importance of these latter arises above all from the remark, first made by Serre [283 b], that localisation and completion naturally introduce flat modules, “explaining” thus in a much more satisfactory way the properties of these two operations that were already known and making their use much easier. It is appropriate to mention besides that the applications of Homological Algebra are far from being limited to that, and that it plays a deeper and deeper role in Algebraic Geometry.

E) The notion of spectrum. The last by date of the new notions of Commutative Algebra has a complex history. The spectral theorem of Hilbert introduced ordered sets of orthogonal projectors of a Hilbert space, making up a “Boolean algebra” (or better a Boolean lattice),³⁶ in bijective correspondence with a Boolean lattice of classes of measurable subsets (for an appropriate measure) of R . It is no doubt his earlier work on operators in Hilbert spaces that, around 1935, led M. H. Stone to study in a general way Boolean lattices, and notably to look for “representations” of them by subsets of a set (or classes of subsets under a certain equivalence relation). He observes that a Boolean lattice becomes a commutative ring (of a very special type besides), when multiplication is defined by $xy = \inf(x, y)$ and addition by $x + y = \sup(\inf(x, y'), \inf(x', y))$. In the particular case where one leaves the Boolean lattice $\mathfrak{P}(X)$ of all subsets of a finite set X , it is seen immediately that the elements of X are in a natural bijective correspondence with the maximal ideals of the corresponding “Boolean” ring; and Stone obtains exactly his general theorem about the representation of a Boolean lattice on the set of maximal ideals that contain it [301 a].

On the other hand, it was known that a classical example of a Boolean lattice was the set of subsets which are both open and closed in a topological space. In a second work [301 b], Stone showed that in fact every Boolean lattice is also isomorphic to a Boolean lattice of this kind. For that it was naturally necessary to define a topology on the set of maximal ideals of a “Boolean” ring; which is done very simply by taking as closed sets, for each ideal α , the set of maximal ideals containing α .

We should not speak here of the influence of these ideas in Functional Analysis, where they played an important role in the birth of the theory of normed algebras developed by I. Gelfand and his school. But in 1945, Jacobson observes [172 c] that the procedure of defining a topology, thought up by Stone, can in fact be applied to every ring A (commutative or not) so long as one takes as set of ideals not the set of maximal ideals, but the set of two sided “primitive” ideals (i. e. the two sided ideals b such that A/b is a primitive ring); for a commutative ring, one obtains of course the maximal ideals. From his side, Zariski in 1944 [340 a], uses an analogous method for defining a topology on the set of places of an algebraic function field. However these topologies remained simple curiosities for the majority of algebraists, because of the fact that they are in general not Hausdorff, and that there was a fairly understandable reluctance to work with such awkward objects. This mistrust was only dissipated when A. Weil showed, in 1952, that every algebraic variety can be furnished in a natural way with a topology of the preceding type and that this topology allows the definition, in a perfect analogy with the case of differential or analytic varieties, of the notion of a fibred space [330 e]; soon after Serre had the idea of extending to these varieties thus topologised the theory of coherent sheaves, thanks to which the topology supplies in the case of “abstract” varieties the same services as the usual topology when the base field is $\mathbb{C}$, notably in what concerns the application of the methods of Algebraic Topology [283 a and b].

From then on it was natural to use this geometric language in all Commutative Algebra. It was rapidly observed that the consideration of maximal ideals is normally insufficient for obtaining convenient statements,³⁷ and that the adequate notion is that of the set of prime ideals of the ring, topologised in the same way. With the introduction of the notion of spectrum, there is at our disposal now a dictionary allowing every theorem of Commutative Algebra to

be expressed in a geometric language very close to that of the Algebraic Geometry of the Weil-Zariski period; which besides has led immediately to a considerable widening of the framework of this last, in such a way that Commutative Algebra is now hardly more from this point of view than the most elementary part [138 a].

Chapter 8

NON COMMUTATIVE ALGEBRA.

We have seen (p. 62) how the first non commutative algebras made their appearance in 1843-44, in the work of Hamilton [145 a] and of Grassmann ([134], v. I₂). Hamilton, in introducing the quaternions, already has a very clear conception of arbitrary algebras of finite rank over the field of real numbers ([145 a], Preface, pp. (26)-(31)).¹ In developing his theory, he has, a little later, the idea of considering what he calls “biquaternions”, that is to say the algebra over the complex numbers having the same multiplication table as the field of quaternions; and he observes on this occasion that this extension has the effect of provoking the apparition of divisors of zero ([145 a], p. 650). The point of view of Grassmann is somewhat different, and for a long time his “exterior algebra” will remain more or less apart from the theory of algebras,² but often in his language, which still lacks precision, the first idea of an algebra (of finite dimension or not) defined by a system of generators and relations cannot miss being recognised ([134], v. II₁, pp. 199-217).

New examples of algebras are introduced in the years 1850-1860, in a more or less explicit way: if Cayley, developing the theory of matrices ([58], v. II, pp. 475-496), does not consider yet that square matrices make up an algebra (a point of view that will not be expressed clearly until the Peirces around 1870 [248 c]), at least he notes already, on this occasion, the existence of a system of matrices of order 2 which satisfy the multiplication table of the quaternions, a remark that can be considered as the first example of a linear representation of an algebra.³ On the other hand, in the memoir where he defines the abstract notion of a finite group, he also gives in passing the definition of the algebra of such a group, without besides getting anything out of this definition ([58], v. II, p. 129).

There is no other notable progress to point out before 1870; but at that time research on the general structure of algebras of finite dimension (over the field of real or complex numbers) starts. It is B. Peirce who takes the first steps along this way; he introduces the notions of nilpotent element, of idempotent element, proves that an algebra (with or without identity) of which at least one element is not nilpotent contains an idempotent $\neq 0$, writes down the famous decomposition

$$x = exe + (xe - exe) + (ex - exe) + (x - xe - ex + exe)$$

(e idempotent, x an arbitrary element), and has the idea (still a bit imprecise) of the decomposition of an idempotent into a sum of “primitive” idempotents orthogonal in pairs [247]. As well, according to Clifford ([65], p.274), ⁴ it is to B. Peirce that must be attributed the notion of the tensor product of two algebras, that Clifford himself applies implicitly to a generalisation of the “biquaternions” of Hamilton ([65], pp. 181-200), and explicitly to the study of the algebras that bear his name, a few years later ([65], pp. 397-401 and 266-276). These new notions are used by B. Peirce for the classification of algebras of small dimension (over the complex numbers), a problem which is also attacked, around 1880, by other mathematicians of the anglo-American school, Cayley and Sylvester in the lead. The great variety of possible structures thus becomes rapidly obvious, and it is no doubt this fact that, during the following period, is going to direct research towards obtaining classes of algebras with more particular properties.

On the continent, where the evolution of ideas is fairly different, such research appears already before 1880. In 1878, Frobenius proves that the quaternions constitute the only example of a non commutative field (of finite dimension) over the field of real numbers ([119], v. I, pp. 343-405) — a result published independently two years later by C. S. Peirce [248 d]. From 1861, Weierstrass, stating accurately a remark of Gauss, had, in his lectures, characterised commutative algebras without nilpotent elements ⁵ over $\mathbb{R}$ or $\mathbb{C}$ as direct sums of fields (isomorphic to $\mathbb{R}$ or $\mathbb{C}$); Dedekind had on his side arrived at the same conclusions around 1870, in connection with his “hypercomplex” conception of the theory of commutative fields; their proofs were published in 1884-85 ([329 a], v. II, pp. 311-332 and [79], v. II, pp. 1-19). It is in 1884 also that H. Poincaré, in a short very cryptic note ([251 a], v. V, pp. 77-79), draws attention to the possibility

of considering the equations $z_i = \phi_i(x_1, \dots, x_n, y_1, \dots, y_n)$ which express the multiplicative law $(\sum_i x_i e_i)(\sum_i y_i e_i) = \sum_i z_i e_i$ in an algebra, as defining (locally of course) a Lie group. This remark seems to have made a big impression on Lie and his disciples (Study, Scheffers, F. Schur and a little later Molien and E. Cartan), busy precisely at this time with the development of the theory of “continuous” groups, and notably the problems of classification (see in particular [271], p. 387); during the period 1885-1905, it guides the mathematicians of this school to apply to the study of the structure of algebras methods of the same kind as those used by them in the study of Lie groups and Lie algebras. These methods rely above all on the consideration of the characteristic polynomial of an element of the algebra relative to its regular representation (a polynomial already met in the work of Weierstrass and Dedekind quoted earlier) and on the decomposition of this polynomial into irreducible factors; a decomposition where, as Frobenius will discover a bit later, is reflected the decomposition of the regular representation into irreducible components.

During the research of the school of Lie on algebras the “intrinsic” notions of the theory gradually emerge. The notion of radical appears in a particular case (that where the quotient by the radical is a direct summand of the field) with G. Scheffers in 1891 [271], more clearly with Molien [224 a] and Cartan ([52 a], v. II₁, pp. 7-105), who study the general case (the word “radical” itself is from Frobenius ([119] v. III, pp. 284-329)). Study and Scheffers [271] put into prominence the concept of an algebra as a direct sum of several others (already foreseen by B. Peirce ([247], p. 221)). Finally with Molien [224 a] quotient algebras of an algebra are introduced, a notion that is essentially equivalent to that of a twosided ideal (defined for the first time by Cartan ([52 a], v. II₁, pp. 7-105)) or of homomorphism (a name also due to Frobenius); the analogy with groups is very sharp here, and a little later in 1904, Epstein and Wedderburn consider composition sequences of twosided ideals and extend to them the Jordan-Hölder theorem. The most important results of this period are those of T. Molien [224 a]: guided by the notion of a simple group, he defines simple algebras (over C) and proves that these are matrix algebras, then proves that the structure of an arbitrary algebra of finite rank over

C reduces essentially to the case (already studied by Scheffers) where the quotient by the radical is a direct sum of fields. These results are soon after found again and established in a

clearer and more rigorous way by E. Cartan ([52 a], v. II₁, pp. 7-105), who introduces on this occasion the notion of semi-simple algebra, and brings out numerical invariants (the “Cartan integers”) linked with an arbitrary algebra over the field C – thus bringing the theory of these algebras to a point beyond which we have hardly progressed since; ⁶ finally he extends the results of Molien and his own to algebras over R .

Around 1900 a movement of ideas grows up which leads to the abandonment of all restrictions on the field of scalars in everything touching on linear algebra; in particular we must bring out the vigorous impulsion given to the study of finite fields by the American school, around E. H. Moore and L. E. Dickson; the most striking result of this research is Wedderburn’s theorem [328 a] proving that every finite field is commutative. In 1907, Wedderburn picks up again Cartan’s results and extends them to an arbitrary base field [328 b]; doing this he abandons completely the methods of his forebears (which become inapplicable as soon as the base field is no longer algebraically closed or totally ordered), and returns, by perfecting it, to B. Peirce’s technique of idempotents, which allows him to put into definitive form the theorem on the structure of semi-simple algebras, whose study is reduced to that of non commutative fields. Further, the problem of the extension of the field of scalars is stated naturally in the perspective which he adopts, and he proves that every semi-simple algebra remains semi-simple after a separable extension of the base field, ⁷ and becomes a direct sum of central algebras of matrices if this extension is taken large enough ([328 b], p. 102). ⁸ A little later, Dickson, for $n = 3$ [88 a] and Wedderburn himself for arbitrary n [328 c] give the first examples of non commutative fields of rank n^2 over their centre, ⁹ inaugurating thus in a particular case the theory of “crossed products” and of “factor sets” that would be developed later by R. Brauer [34 a] and E. Noether [236 c]. Finally, in 1921, Wedderburn proves a particular case of the commutation theorem [328 d].

In between times, from 1896 to 1910, there was developing, in the hands of Frobenius, Burnside and I. Schur, a neighbouring theory to that of algebras, the theory of the linear representation of groups (limited at first to representations of finite groups). It takes its origins from remarks of Dedekind: this latter (even before the publication of his work on algebras) had, around 1880, met during his research on normal bases of Galois extensions, the “Gruppen-determinant” $\det(x_{st-1})$, where $(x_s)_{s \in G}$ is a sequence of indeterminates for which the set of indices is a finite group G

(in other words, the norm of the generic element of the group algebra G relative to its regular representation); and he had observed that when G is abelian, this polynomial decomposes into linear factors (which generalised an identity proved a long time before for “circulant” determinants, which correspond to cyclic groups G). During his very interesting correspondence with Frobenius ([79], v. II, pp. 414-442), Dedekind, in 1896, draws his attention to this property, its link with the theory of characters of abelian groups [327 c] and some analogous results on particular non commutative groups, that he had obtained in 1886. A few months later, Frobenius solved completely the problem of the decomposition of the “Gruppen-determinant” into irreducible factors ([119], v. III, pp. 38-77), thanks to his brilliant generalisation of the notion of character ([119], v. III, pp. 1-37), of which we should not speak here. But it must be noted that in the later development of this theory,¹⁰ Frobenius always remains aware of its relation to the theory of algebras (on which Dedekind had besides never ceased insisting in his letters); and after having introduced for groups the notion of irreducible representation and of completely reducible representation ([119], v. III, pp. 82-103), and shown that the regular representation contains all the irreducible representations, it is by analogous methods that he proposed, in 1903, to take up again the theory of Molien-Cartan ([119], v. III, pp. 284-329). With Burnside [44 a] and I. Schur [279 c], the “hypercomplex” aspect of the theory does not occur explicitly; but it is with them that the fundamental properties of irreducible representations, Schur’s lemma and Burnside’s theorem come to light. Finally, it must be noted for our objective that it is in this theory that appear for the first time two particular cases of the commutation theorem: in the thesis of I. Schur [279 a] who links up (precisely by commutation in the ring of endomorphisms of a tensor space) the representations of the linear group and those of the symmetric group, and in his work of 1905 [279 c], where he shows that matrices permutable with all the matrices of an irreducible representation over the field $\mathbb{C}$ are scalar multiples of I (a result that flows out also from Burnside’s theorem).

It remained to disentangle clearly the common substrate of these theories: it was the work of the German school around E. Noether and E. Artin, in the period 1921-1933 which sees the creation of modern algebra. Already, in 1903, in a memoir on the algebraic integration of linear differential equations ([251 a], v. III, pp. 140-149), H. Poincaré had defined, in an algebra, left

ideals and right ideals and the notion of minimal ideal; he had also remarked that in a semi-simple algebra, every left ideal is the direct sum of its intersections with the simple components, and that in the algebra of matrices of order n , the minimal ideals are of dimension n ; but his work remained unnoticed by algebraists.¹¹ In 1907, Wedderburn defines anew left and right ideals of an algebra and proves some properties (notably that the radical is the largest nilpotent left ideal ([328 b], pp. 113-114)). But we must wait until 1927 for these notions to be used in an essential way in the theory of algebras.¹² Putting into general form procedures of proof appearing previously here and there,¹³ W. Krull in 1925 [187 a] and E. Noether in 1926 [236 b] introduce and use systematically the maximal and minimal conditions; the first is used to extend to abelian groups with operators (which he defines on this occasion) Remak's theorem on the decomposition of a finite group into a direct product of indecomposable groups, whereas the second brings these conditions into the characterisation of Dedekind rings. In 1927, E. Artin [7 c], applying the same idea to non commutative rings, shows how, by a systematic study of minimal ideals, Wedderburn's theorems can be extended to all rings whose left ideals satisfy both the maximal and minimal conditions.¹⁴

On the other hand, Krull, in 1926 [187 b], makes the link between the notion of abelian groups with operators and that of linear representations of groups; a point of view generalised to algebras and developed in detail by E. Noether in a fundamental work of 1929 [236 c] which, by the importance of the ideas introduced and the clarity of the exposition, deserves to stand alongside the memoir of Steinitz on commutative fields as one of the pillars of modern linear algebra.¹⁵

Finally, in a series of works which start in 1927 ([237], [34 a], [236 d]), E. Noether and R. Brauer (who is joined starting in 1929-31 by A. Albert and H. Hasse) take up again the study of left fields at the point where Wedderburn and Dickson had left it. If the most important part of their results consists in a deep study of Brauer groups (in particular over the field of algebraic numbers) and so goes beyond the framework of this note, let us point out in any case that it is in the course of this work that the commutation theorems are stated precisely, as well as the notion of a field neutralising a simple algebra and its relations with the maximal commutative subfields; finally, in 1927, Skolem characterises the automorphisms of simple rings [286 b], a

theorem found again a few years later by E. Noether [236 b] and R. Brauer [34 a].

So, in 1934, the elementary theory of simple and semi-simple rings has just about reached its final aspect (for an exposé of the state of the theory at this stage, see [87]); since then, it has been developing in two different directions, that we limit ourselves to mentioning briefly. On the one hand, the theory of “factor sets” of R. Brauer and E. Noether has recently received new impulsion, following its incorporation into modern Homological Algebra.¹⁶ On the other hand, there have been many attempts, with more or less success, to extend — at least in part — the results of the classical theory to rings without minimal condition¹⁷ or to rings without identity. But to date these extensions have had hardly any repercussions on other branches of mathematics; for more details on this work, we refer you to the recent exposé of N. Jacobson [172 b].

Chapter 9

QUADRATIC FORMS; ELEMENTARY GEOMETRY.

The theory of quadratic forms, in its modern aspect, hardly goes back before the second half of the XVIIIth century, and, as we will see, it was developed mainly to respond to the needs of Arithmetic, Analysis and Mechanics. But the fundamental notions of this theory have in reality appeared from the beginnings of “Euclidean” geometry, of which they formed the frame. For this reason, its history can not be retraced without speaking, at least in a summary way, of the development of “elementary geometry” since antiquity. Of course, we will only be able to link up with the evolution of a few general ideas, and the reader must not expect to find here precise information on the history of such or such a particular theorem, about which it will suffice to refer back to specialised historical or didactic works. ¹ It goes without saying also, when we speak below of various possible interpretations of the same theorem in various algebraic or geometric languages, that we do not intend at all to say that these “translations” have at all times been as familiar as they are today; quite on the contrary, it is the principal goal of this Note to show how, very gradually, mathematicians have become conscious of these relationships between questions often of very different aspect; we will also have to show how, doing this, they have been led to put some coherence in the mass of theorems from geometry bequeathed by the ancients, and finally to try to delimit exactly what should be understood by “geometry”.

If one puts aside the discovery, by the Babylonians, of the formula for the solution of equa-

tions of the second degree ([232], pp. 183-189) it is therefore under their geometric disguise that the birth of the principal concepts of the theory of quadratic forms must be noted. These appear first as squares of distances (in the plane or space of three dimensions) and the corresponding notion of “orthogonality” is introduced by means of the right angle, defined by Euclid as half the straight angle (Elements, Book I, Def. 10); the notions of distance and right angle being linked by the theorem of Pythagoras, the keystone of the Euclidean edifice. ² The idea of angle seems to have been introduced very early in Greek mathematics (which no doubt had received it from the Babylonians, broken in to the use of angles by their long astronomical experience). It is known that in the classical period, only angles less than 2 right angles are defined (the “definition” of Euclid is besides as vague and unusable as the ones he gives of straight line and plane); the notion of direction is not brought out, although Euclid uses (without axiom or definition) the fact that a straight line divides the plane into two regions, which he distinguishes carefully when it is necessary. ³ At this stage, the idea of the group of plane rotations only comes to light in a very imperfect way, by addition (introduced, it also, without explanation by Euclid) of non directed angles of half-lines, which is only defined, in principle, when the sum is at most equal to two right angles. ⁴ As for trigonometry, it is disdained by geometers and abandoned to surveyors and astronomers; it is these latter (Aristarchus, Hipparchus, Ptolemy especially [255]) who establish the fundamental relations between the sides and angles of a right angled triangle (plane or spherical) and draw up the first tables (they consist of tables giving the chord of the arc cut out by an angle $\theta < \pi$ on a circle of radius r , in other words the number $2r \sin \frac{\theta}{2}$; the introduction of the sine, more easily handled, is due to Hindu mathematicians of the Middle Ages); in the calculation of these tables, the formula for the addition of arcs, unknown at this time, is replaced by the equivalent use of the theorem of Ptolemy (going back perhaps to Hipparchus)

on quadrilaterals inscribed in a circle. It must be noted also that Euclid and Hero give propositions equivalent to the formula

$$a^2 = b^2 + c^2 - 2bc \cos A$$

between the sides and angles of an arbitrary plane triangle; but one can hardly see there the first appearance of the notion of a bilinear form associated with a metric form, for lack of the

idea of a vector calculus which will only emerge in the XIXth century.

Displacements (or movements, the distinction between the two notions not being clear in antiquity — nor even much later) are known to Euclid; but, for reasons of which we are ignorant, he seems to feel a clear repugnance to use it (for example in the “case of equality of triangles”, where the impression is gained that he only uses the notion of displacement because he has not been able to formulate an appropriate axiom ([153 e], v. I, pp. 225-227 and 249)); however, it is to the notion of displacement (rotation about an axis) that he has recourse for the definition of cones of revolution and of spheres (Elements, Book XI, def. 14 and 18), as well as Archimedes for that of quadrics of revolution. But the general idea of a transformation, applied to the whole of space, is more or less a stranger to mathematical thought before the end of the XVIIIth century;⁵ and before the XVIIth century, no trace is found either of the notion of composition of movements, nor a fortiori of the composition of displacements. That does not mean to say, of course, that the Greeks were not particularly aware of the “regularities” and “symmetries” of figures, which we now attach to the notion of groups of displacements: their theory of regular polygons, and even more that of regular polyhedra — one of the most remarkable chapters of all their mathematics — is there to prove the opposite.⁶

Finally, the last of the essential contributions of Greek mathematics, in the area which concerns us, is the theory of conics (as far as quadrics is concerned, the Greeks only know certain quadrics of revolution, and do not carry their study of them very far, except for the sphere). It is interesting to note here that, although the Greeks never had the idea of the fundamental principle of analytic geometry (essentially because of the lack of a manageable algebra), they commonly use, for the study of particular “figures”, “ordinates” relative to two (or even more than two) axes in the plane (linked closely to the figure, which is one of the fundamental points where their method differs from that of Fermat and Descartes, whose axes are fixed independently of the figure under consideration). In particular, the first examples of conics (other than the circle) which come up in connection with the problem of the duplication of the cube, are the curves given by the equations $y^2 = ax$, $y = bx^2$, $xy = c$ (Menechme, a pupil of Eudoxus, middle of the IVth century); ⁷ and it is the equation of conics (normally with reference to two oblique axes made up of a diameter and of the tangent to one of its points of contact with the curve) that is

most often used in the study of problems relative to these curves (whereas the “focal” properties only play a very effacing role, contrary to what scholarly traditions going back only to the XIXth century would make us believe). From this vast theory, we must above all remember here the notion of conjugate diameters (already known by Archimedes), and the property that today serves as the definition of the polar of a point, given by Apollonius [153 b] when the point is exterior to the conic (the polar being therefore for him the straight line joining the points of contact of the tangents issuing from this point); from our point of view, these are two examples of “orthogonality” with respect to a quadratic form distinct from the metric form, but it is well understood that the link between these notions and the classical notion of perpendiculars could not possibly be conceived of at that time.

There is hardly any other progress to point out before Descartes and Fermat; but from the beginnings of analytic geometry, the algebraic theory of quadratic forms starts to emerge from its geometric crust: Fermat knows that an equation of the second degree in the plane represents a conic ([109], v. I, pp. 100-102; French translation, v. III, pp. 84-101) and sketches analogous ideas about quadrics ([109], v. I, pp. 111-117; French translation, v. III, pp. 102-108). With the development of analytic geometry in 2 and 3 dimensions during the course of the XVIIIth century two of the central problems of the theory appear (especially in connection with conics and quadrics): the reduction of a quadratic form to a sum of squares and the search for its “axes” with respect to the metric form. For conics, these two problems are too elementary to give rise to important algebraic progress; for an arbitrary number of variables, the first is solved by Lagrange in 1759, in connection with the maxima of functions of several variables ([191], v. I, pp. 3-20). But this problem is almost immediately eclipsed by that of the search for axes, even before invariance of rank had been formulated;⁸ as for the law of inertia, it is only discovered around 1850 by Jacobi ([171], v. III, pp. 593-598), who proves it by the same argument as nowadays, and Sylvester ([304], v. I, pp. 378-381) who limits himself to stating it as quasi-obvious.⁹

The problem of the reduction of a quadric to its axes already presents algebraic difficulties substantially greater than the analogous problem for conics; and Euler, who is the first to tackle it, is not in a position to prove the real character of the eigenvalues, which he admits after an attempt at justification without conviction ([108 a], (1), v. IX, pp. 379-392).¹⁰ Given that

this point is correctly established in about 1800 [140], we must await Cauchy for the proof of the corresponding theorem for forms in an arbitrary number n of variables ([56 a], (2), v. IX, pp. 174-195). It is also Cauchy who, at about the same time, proves that the characteristic equation giving the eigenvalues is invariant under all orthogonal changes of axes ([56 a], (2), v. V, p. 252); ¹¹ but for $n = 2$ or $n = 3$, this invariance was intuitively “obvious” because of the geometric interpretation of eigenvalues by means of axes of the corresponding conic or quadric. Besides, during the course of research on this subject, the elementary symmetric functions of the eigenvalues have also occurred naturally (with various geometric interpretations, in connection notably with the theorems of Apollonius on conjugate diameters), and in particular the discriminant, which (known for a long time for $n = 2$ in connection with the theory of the equation of the second degree) appears for the first time for $n = 3$ with Euler ([108 a], (1), v. IX, p. 382); this latter meets it in connection with the classification of quadrics (in expressing the condition for a quadric not to have a point at infinity) and does not mention its invariance with respect to orthogonal axis changes. But a bit later, with the beginnings of the arithmetical theory of quadratic forms with integral coefficients, Lagrange notes (for $n = 2$) a particular case of the invariance of the discriminant under linear but not orthogonal change of variables ([191], v. III, p. 699), and Gauss establishes, for $n = 3$, the “covariance” of the discriminant for all linear transformations ([124 a], v. I, pp. 301-302). ¹² Once the general formula for the multiplication of determinants was proved by Cauchy and Binet, the extension of the formula of Gauss to an arbitrary number of variables was immediate; it is that which, around 1845, will give the first impulsion to the general theory of invariants.

To the two notions that, amongst the Greeks, took the place of the theory of displacements — that of movement and that of “symmetry” of a figure — has just been added a third in the XVIIth and XVIIIth centuries with the problem of the changing of rectangular axes, which is substantially equivalent to this theory. Euler devotes several works to this question, attaching importance above all to obtaining parametric representations which are manageable for the formulae for changing axes. It is known what use Mechanics was going to make of the three angles which he introduces for this purpose for $n = 3$ ([108 a], (1), v. IX, pp. 371-378). But he does not limit himself to that, considers in 1770 the general problem of orthogonal transformations for

arbitrary n , remarks that the goal is reached here by introducing $n(n-1)/2$ angles as parameters, and finally, for $n=3$ and $n=4$, gives for rotations rational representations (as a function, respectively, of 4 homogeneous parameters and of 8 homogeneous parameters linked by a relation), which are none other than those obtained later by means of the theory of quaternions, and of which he does not indicate the origin ([108 a], (1), v. VI, pp. 287-315).¹³

On the other hand, Euler indicates also how to translate analytically the search for the “symmetries” of plane figures, and it is for this purpose that he is led to prove, substantially, that a plane displacement is a rotation, or a translation, or a translation followed by a symmetry ([108 a], (1), v. IX, pp. 197-199). The rise of Mechanics at this time leads besides to the general study of displacements: but first of all it is only a question of “infinitely small” displacements tangent to continuous movements: they are apparently the only ones that occur in the researches of Torricelli, Roberval and Descartes on the composition of movements and the instantaneous centre of rotation for plane movements (cf. p. 177). This latter is defined in a general way by Johann Bernoulli; d’Alembert in 1749, Euler the following year, extend this notion by proving the existence of an instantaneous axis of rotation for movements which leave a point fixed. The analogous theorem for finite displacements is not stated until 1775 by Euler [108 b], in a memoir where he discovers at the same time that the determinant of a rotation is equal to 1; the following year, he proves the existence of a fixed point for plane similarities ([108 a], (1), v. XXVI, pp. 276-285). But the work of Chasles must be awaited, starting in 1830 [60 a], in order to obtain finally a coherent theory of finite and infinitely small displacements.

We arrive thus at what can be called the golden age of geometry, which is inserted grosso modo between the dates of the publication of the *Géométrie descriptive* of Monge (1795) [225] and of the “Erlangen programme” of F. Klein (1872) ([182], v. I, pp. 460-497). The essential progress that we owe to this brusque renewal of geometry are the following:

- A) The notion of the element at infinity (point, straight line or plane), introduced by Desargues in the XVIIth century [84], but which is hardly expressed during the XVIIIth century except by abuse of language, is rehabilitated and systematically used by Poncelet [252] who thus makes of projective space the general framework of all geometrical phenomena.
- B) At the same time, with Monge and especially Poncelet, the passage to complex projective

geometry is effected. The notion of imaginary point, sporadically used during the XVIIIth century, is here exploited (concurrently with that of the point at infinity) to give statements independent of the “case of the figure” of real affine geometry. If at first the justifications brought in to support these innovations remained very awkward (especially those from the members of the school of “synthetic” geometry, where the use of coordinates ends up as being seen as a disgrace), what cannot be missed is recognising there, under the name of “principle of contingent relations” in Monge, or “principle of continuity” in Poncelet, the first germ of the idea of “specialisation” from modern algebraic geometry. ¹⁴

One of the first results following on from these concepts is the remark that, in complex projective space, all the conics (respectively quadrics) which are non degenerate are of the same kind; which brings Poncelet to the discovery of “isotropic” elements: “Circles placed arbitrarily on a plane”, he says, “are therefore not totally independent among themselves, as could be thought at first sight, they have ideally two imaginary points in common at infinity” ([252], v. I, p. 48). Later, he introduces in the same way the “umbilical”, an imaginary conic at infinity common to all spheres ([252], v. I, p. 370); and if he does not speak particularly of the isotropic generators of the sphere, at least he underlines explicitly the existence of rectilinear generators, real or imaginary, for all quadrics (ibid., p. 371); ¹⁵ notions that his followers (notably

Plücker and Chasles), even more than he, make great use of, in particular in the study of “focal” properties of conics and quadrics.

C) The notions of point transformation and of composition of transformations are, they also, formulated in a general way and introduced systematically as means of proof. Apart from displacements and projections, only certain particular transformations were known up to that point: certain plane projective transformations of the type $x' = a/x$, $y' = y/x$, used by La Hire and Newton, the “affinity” $x' = ax$, $y' = by$ of Clairaut and Euler ([108 a], (1), v. IX, chap. XVIII), and finally a few particular quadratic transformations, with Newton again, Maclaurin and Braikenridge. Monge, in his *Géométrie descriptive*, shows all the use that can be made of plane projections in 3 dimensional geometry. With Poncelet, one of the systematic procedures for proofs, used to surfeit, consists in reducing by projection properties of the conic to those of the circle (a method already used on

occasion by Descartes and Pascal); and in order to go from a quadric to a sphere, he invents the first example of a projective transformation of space, the “homology” ([252], v. I, p. 357); finally it is he also who introduces the first examples of birational transformations of a curve into itself. In 1827, Möbius ([223], v. I, p. 217) (and independently Chasles in 1830 ([60 b], p. 695)), define the most general linear projective transformations; at the same time appear inversion and other types of quadratic transformations, whose study will inaugurate the theory of birational transformations, which will develop in the second half of the XIXth century.

- D) The notion of duality appears fully in the open and is found consciously linked to the theory of bilinear forms. The theory of poles and polars with respect to conics, which, since Apollonius, had only made some progress with Desargues and La Hire, is extended to quadrics by Monge, who, as well as his pupils, sees the possibility of transforming by this means known theorems into new results.¹⁶ But it is again due to Poncelet that the credit returns for having built up these remarks into a general method in the theory of transformations “by reciprocal polars”, and of having made it into a particularly efficacious tool of discovery. A little later, notably with Gergonne, Plücker, Möbius and Chasles, the general notion of duality frees itself from the link with quadratic forms, still too close with Poncelet. In particular, Möbius, while examining the various possibilities for duality in space of 3 dimensions (defined by a bilinear form), discovers in 1833 duality with respect to an alternating bilinear form ([223], v. I, pp. 489-515),¹⁷ especially studied, in the XIXth century, under the form of the theory of “linear complexes” and developed in relation to the “geometry of straight lines” and the “Plücker co-ordinates” introduced by Cayley, Grassmann and Plücker around 1860.
- E) From the beginnings of projective geometry, the intensive study of the properties of classical geometry in their relationships with projective space had rapidly led to dividing them into “projective properties” and “metric properties”; and it is no doubt not exaggerated to see in this separation one of the clearest manifestations, at this time, of what was to become the modern notion of structure. But Poncelet, who is the first to introduce this

distinction and this terminology, is already conscious of what links these two types of properties; and, tackling in the *Traité* the problems concerning angles, whose properties “seem not to be part of those that we have called projective ..., they follow on nonetheless in such a simple way”, he says, “from the principles that form the basis [of this work] ..., that I do not believe that any other geometrical theory could lead to it in a manner which is both more direct and simpler. It will not be astonishing, if one considers that projective properties of figures are necessarily the most general among those that can belong to them; in such a way that they must encompass, as simple corollaries, all the other properties or particular relations of the area” ([252], v. I, p. 248). Truth to tell, after this declaration, it is a little surprising to see him tackle questions about angles in a very roundabout way, by linking them to focal properties of conics, instead of bringing in directly the cyclic points; and in fact, it is only 30 years later that Laguerre (still a pupil at the *École Polytechnique*) gave the expression for a right angle by means of the cross-ratio of these straight lines and isotropic straight lines with the same origin ([192], v. II, p. 13). Finally, with Cayley ([58], v. II, pp. 561-592) the fundamental idea that “metric” properties of a plane figure are none other than the “projective” properties of the figure extended by the cyclic points becomes clearly expressed — a decisive way station towards the “Erlangen programme”.

F) Non-Euclidean hyperbolic geometry, which will come to light around 1830, remains at first a little apart from the movement of which we are tracing the main features. Arising out of preoccupations of an essentially logical order touching on the foundations of classical geometry, this new geometry is presented by its inventors ¹⁸ in the same axiomatic and “synthetic” form as Euclid’s geometry, and without any link to projective geometry (whose introduction following the classical model seems even to be excluded a priori, since the notion of unique parallel disappears in this geometry); it is no doubt because of that that it hardly attracts, for a long time, any interest from the French, German or English schools of projective geometry. Also, when Cayley, in the fundamental memoir quoted earlier ([58], v. II, pp. 561-

592) has the idea of replacing cyclic points (considered as a “tangentially degenerate” conic) by an arbitrary conic (that he calls “absolute”), he is not thinking at all of linking this idea

with the geometry of Lobatschevsky-Bolyai, even though he indicates how his conception leads to new expressions for the “distance” between two points, and that he mentions its links with spherical geometry. The situation changes around 1870, when the non-Euclidean geometries, following the diffusion of the works of Lobatschevsky, and the publication of the works of Gauss and the inaugural lecture of Riemann, come to the fore of mathematical news. Following the path traced by Riemann, Beltrami, without knowing of the work of Cayley, finds again in 1868 expressions for distance as given by this latter, but in quite another context, by considering the interior of a circle as an image of a surface with constant curvature, in which the geodesics are represented by straight lines [18 a]; it is Klein, who two years later, makes (independently of Beltrami) a synthesis of these various points of view, which he completes by the discovery of elliptic non-Euclidean space ([182], v. I, pp. 254-305). ¹⁹

- G) In the second half of the period which we are considering here, a period of critical reflection is inserted, during which the supporters of “synthetic” geometry, not content with having banished co-ordinates from their proofs, pretend to do without real numbers even in their axioms for geometry. The principal representative of this school is von Staudt, who succeeds essentially in bringing about this tour de force [325], very admired in his time and even well into the XXth century; and if today the same importance is not attributed to ideas of this order, whose possibilities for fruitful application are revealed to be fairly thin, it must nonetheless be recognised that the efforts of von Staudt and his disciples contributed to clarifying the role of real or complex “scalars” in classical geometry, and to introducing in that very way the modern conception of geometries over an arbitrary base field.

Around 1860, “synthetic” geometry was at its peak, but the end of its reign was approaching at great speed. Having remained heavy and ungracious during the whole of the XVIIIth century, analytic geometry, in the hands of Lamé, Bobillier, Cauchy, Plücker and Möbius, acquires at last the elegance and conciseness that will allow it to fight on equal terms with its rival. Especially, after about 1850, the ideas of group and invariant, formulated at last in a precise way, invade little by little the scene, and it is seen that the theorems of classical geometry are none other than the

expression of identical relations between invariants or covariants of the group of similarities,²⁰ in the same way that those of projective geometry express the identities (or “syzygies”) between covariants of the projective group. It is this thesis that is magisterially stated by F. Klein in the famous “Erlangen programme” ([182], v. I, pp. 460-497), where he advocates the abandonment of sterile controversies between the “synthetic” tendency and the “analytical” tendency; if, he says, the accusation levelled against this latter of giving a privileged role to an arbitrary system of axes “was most often justified in what concerns the defective way in which the method of co-ordinates was used previously, it founders when it is a case of a rational application of this method . . . The domain of spatial intuition is not forbidden by the analytic method . . .”, and he underlines that “one must not underestimate the advantage that a well adapted formalism brings to subsequent research, in that it goes in advance so to say of thought” (loc. cit., pp. 488-490).

One concludes thus with a rational and “structural” classification of the theorems of “geometry” according to the group from which they arise: the linear group for projective geometry, the orthogonal group for metric questions, the symplectic group for “linear complex” geometry. But under this merciless clarity, classical geometry — with the exception of algebraic geometry and differential geometry,²¹ henceforth constituted as autonomous sciences — flares suddenly and loses all its brightness. Already the generalisation of methods based on the use of transformations had made the development of new theorems somewhat mechanical: “Today,” says Chasles in 1837 in his *Aperçu historique*, “every one can put himself forward, take an arbitrary known truth, and submit it to various general transformation principles; he will draw out other truths, different or more general; and these will be susceptible to similar operations; so that one will be able to multiply, almost to infinity, the number of new truths deduced from the first . . . So anyone can who wishes, in the present state of the science, generalise and create in Geometry; genius is no longer indispensable in order to add a stone to the structure” ([60 b], pp. 268-269). But the situation becomes much clearer with progress in the theory of invariants, which succeeds at last (at least for the “classical” groups) in formulating general methods allowing in principle the writing down of all the algebraic covariants and all their “syzygies” in a purely automatic way; a victory that, at the same time, signals the death, as a field of research, of the classical theory of invariants itself, and of “elementary” geometry,²² which has become in practice a

simple dictionary for it. No doubt, nothing allows the foresight a priori, among the infinity of “theorems” that can thus be rolled out at will, as to which will be those whose statement, in the appropriate geometric language, will have a simplicity and an elegance comparable to those of the classical results, and there remains there a restricted domain where numerous amateurs continue to work with pleasure (the geometry of the triangle, the tetrahedron, of algebraic curves and surfaces of low degree, etc.). But for the professional mathematician, the mine is worked out, since there are no longer there any structural problems, susceptible of reverberating on other parts of mathematics; and this chapter of the theory of groups and invariants can be considered as closed until further notice. ²³

Thus, after the Erlangen programme, Euclidean and noneuclidean geometries, from a purely algebraic point of view, became simple languages, more or less convenient, for expressing the results of the theory of bilinear forms, whose progress went in tandem with that of the theory of invariants. ²⁴ All that concerns the notion of rank of a bilinear form and the relationships between these forms and linear transformations is definitively clarified by the works of Frobenius ([119], v. I, pp. 343-405). It is also due to Frobenius that we have the canonical expression of an alternating form on a free $\mathbb{Z}$ -module ([119], v. I, pp. 482-544); however, the left symmetric determinant had already appeared with Pfaff, at the beginning of the century, in connection with the reduction of differential forms to a normal form; Jacobi, who, in 1827, takes up this problem again ([171], v. IV, pp. 17-29), knows that a left symmetric determinant of odd order is null, and it is he who constructs the expression for the pfaffian and shows that it is a factor of the left symmetric determinant of even order; but he had not realised that this latter is the square of the pfaffian, and this point was only established by Cayley in 1849 ([58], v. I, pp. 410-413). The notion of the symmetric bilinear form associated with a quadratic form is the most elementary case of the process of “polarisation”, one of the fundamental tools of the theory of invariants. Under the name of “scalar product”, this notion will know an immense prosperity, first with the vulgarisers of the “vector calculus”, then, starting in the XXth century, thanks to the unsuspected generalisation that is brought to it by the theory of Hilbert spaces (see p. 212). It is also this latter theory that will bring to light the notion of the adjoint of an operator (which beforehand had hardly come out except in the theory of linear differential equations, and, in tensor

calculus, by the waltz of the co- and contravariant indices under the baton of the metric tensor); it is that which finally will give all its character to the notion of hermitian form introduced first by Hermite in 1853 in connection with his arithmetic research ([159], v. I, p. 237), but having remained on the edge of the great mathematical streams until around 1925 and the applications of complex Hilbert spaces to quantum theories.

The study of the orthogonal group and the group of similarities — clearly conceived and treated as such since the middle of the XIXth century, and having become the heart of the theory of quadratic forms — as well as that of the other “classical” groups (linear group, symplectic group and unitary group), takes up on the other hand a greater and greater importance. We can only mention here the essential role played by these groups, in the theory of Lie groups and differential geometry on the one hand, the arithmetic theory of quadratic forms (see for example [285] and [100]) on the other; ²⁵ it is to this circumstance, as well as to the extension of the concept of duality to the most diverse questions that is due the fact that there is hardly any modern mathematical theory where bilinear forms do not crop up in some form or other. We must in any case note that it is the study of rotation groups (in three dimensions) that led Hamilton to the discovery of quaternions [145 a]; this discovery is generalised by W. Clifford who, in 1876, introduces the algebras that bear his name, and proves that they are tensor products of quaternion algebras, or quaternion algebras and a quadratic extension ([65], pp. 266-276). Found again four years later by Lipschitz [205 b], who uses them to give a parametric representation of orthogonal transformations in n variables (generalising those that Cayley had obtained for $n = 3$ ([58], v. I, pp. 123-126) and $n = 4$ (v. II, pp. 202-215) by the theory of quaternions), these algebras, and the notion of “spinor” that derives from them ([52 b] and [62 a]), would also know a great vogue in modern times by virtue of their use in quantum theories.

It remains for us finally to say a word about the evolution of ideas which led to the almost total abandonment of all restrictions on the ring of scalars in the theory of sesquilinear forms — a common tendency in all modern algebra, but which has perhaps manifested itself here earlier than elsewhere. We have already pointed out the fruitful introduction of geometry over the field of complex numbers (which, besides, during the whole of the XIXth century, did not go without a perpetual and sometimes dangerous confusion between this geometry and real geometry); the

clarity here comes especially from the axiomatic studies of the end of the XIXth century on the foundations of geometry [163 c]. In the course of this research, Hilbert and his emulators, notably, on examining the relationships between the various axioms, were led to construct appropriate counterexamples, where the “base field” (commutative or not) possessed more or less pathological properties, and they thus accustomed mathematicians to “geometries” of a quite new type. From an analytical point of view, Galois had already considered linear transformations where coefficients and variables took values in a finite prime field ([123], p. 145); in developing these ideas, Jordan [174 a] is led in a natural way to consider the classical groups over these fields, groups whose intervention manifests itself in varied mathematical domains. Dickson, around 1900, extends the research of Jordan to all finite fields, and more recently, it has become apparent that a large part of the theory of Jordan-Dickson can be extended to an absolutely arbitrary “base field”; this is due essentially to general properties of isotropic vectors and to Witt’s theorem, which, trivial in the classical cases, were only established for an arbitrary base field in 1936 [337 a].²⁶

But in thus pushing the study of sesquilinear forms towards an “abstraction” which is always greater, it has turned out to be extremely suggestive to keep as is the terminology which, in the case of space of 2 or 3 dimensions, came out of classical geometry, and to extend it to the n-dimensional case and even to infinite dimensional spaces. Passed over in its role as an autonomous and living science, classical geometry is thus transfigured into a universal language of contemporary mathematics, with a suppleness and a usefulness that are incomparable.

Chapter 10

TOPOLOGICAL SPACES.

The notions of limit and continuity go back to antiquity; it would be impossible to make a complete history of them without studying systematically from this point of view, not only the mathematicians, but also the Greek philosophers and in particular Aristotle, nor also without following the evolution of these ideas across the mathematics of the Renaissance and the beginnings of the differential and integral Calculus. Such a study, which it would certainly be very interesting to undertake, would go far beyond the framework of this note.

It is Riemann who must be considered as the creator of topology, as of so many other branches of modern mathematics: it is in fact he who, first, sought to disengage the notion of topological space, conceived the idea of an autonomous theory of these spaces, defined the invariants (the “Betti numbers”) which were to play the greatest role in the later development of topology, and gave its first applications to analysis (periods of abelian integrals). But the movement of ideas in the first half of the XIXth century had not happened without preparing the way for Riemann in more than one way. Indeed, the desire to settle mathematics on a firm base, which was the cause of so much important research during the whole of the XIXth century and up to our day, had led to the correct definition of the notion of a convergent series and of a sequence of numbers tending to a limit (Cauchy, Abel) and that of a continuous function (Bolzano, Cauchy). On the other hand, the geometric representation (by points in the plane) of complex numbers, or, as had been said until then, “imaginary numbers” (qualified sometimes also, in the XVIIIth century, as “impossible” numbers), a representation due to Argand and Gauss (see p. 161), had become familiar to the majority of mathematicians: it constituted progress of the

same order as in our days the adoption of geometric language in the study of Hilbert space, and contained in embryo the possibility of a geometric representation of every object which was susceptible to continuous variation; Gauss, who in any case was naturally led to such conceptions by his research on the foundations of geometry, on non-Euclidean geometry, on curved surfaces, seems already to have had this possibility in view, for he uses the words “size twice extended” in defining (independently of Argand and the French mathematicians) the geometric representation of imaginary numbers ([124 a], v. II, pp. 101-103 and pp. 175-178).

It is on the one hand his research on algebraic functions and their integrals, on the other hand his thoughts (largely inspired by his study of the works of Gauss) on the foundations of geometry, that led Riemann to formulate a programme of study which is exactly that of modern topology, and to give this programme a start towards its realisation. Here for example is how he expresses himself in his Theory of abelian functions ([259 a], p. 91):

“In the study of functions that are obtained by the integration of exact differentials, some theorems of analysis situs are almost indispensable. Under this name, which was used by Leibniz, although perhaps with a somewhat different meaning, it is in order to include that part of the theory of continuous quantities which studies these quantities, not as independent of their position and measurable the ones by means of the others, but in abstracting every idea of measurement and studying solely their relationships of position and inclusion. I reserve the right to treat this object later, in a way that is completely independent of all measurement ...”.

And in his famous inaugural Lecture “On the hypotheses that form the foundation of geometry” ([259 a], p. 272):

“...The general notion of quantity several times extended, ¹ which contains that of spatial quantity as a particular case, has remained completely unexplored ... (p. 272)”.

“...The notion of quantity presupposes an element susceptible of different determinations. Following on whether or not one can pass from one determination to another by continuous transitions, these determinations make up a continuous multiplicity (of which they will be called the points) or a discrete multiplicity (p. 273)”.

“...The measurement consists of a superposition of quantities to compare; to measure, there must therefore be a means of bringing one quantity onto another. In the absence of such a means,

it is not possible to compare two quantities except when one is part of the other... The studies that can then be made about this subject constitute part of the theory of quantities, independent of the measurement, and where the quantities are not considered as having an existence independent of their position, nor as being expressible by means of a unit of measure, but as parts of a multiplicity. Such studies have become a necessity in many parts of mathematics, and in particular for the theory of analytic multiform functions ... (p. 274)".

"...The determination of the position in a given multiplicity is thus reduced, each time that it is possible, to a finite number of numerical determinations. There are, it is true, some multiplicities, in which the determination of the position requires, not a finite number, but an infinite sequence or a continuous multiplicity of determinations of quantities. Such multiplicities are made up for example by the possible determinations of a function in a given domain, the position of a figure in space, etc." (p. 276).

In this last sentence will be noticed the first idea of a study of functional spaces; besides, already in Riemann's Dissertation the same idea is to be found expressed "The set of these functions", he says in connection with the minimum problem known under the name of Dirichlet's principle, "makes up a connected domain, closed in itself" ([259 a], p. 30), which is, in an imperfect form, nonetheless the embryo of the proof that Hilbert would give later of Dirichlet's principle, and even of the majority of the applications of functional spaces to the calculus of variations.

As we have said, Riemann gave a start to the carrying out of this grandiose programme, by defining the "Betti numbers", first of a surface ([259 a], pp. 92-93), then (ibid., pp. 479-482; cf. also [259 c]) of a multiplicity of an arbitrary number of dimensions, and in applying this definition to the theory of integrals; results which inaugurate algebraic Topology, a branch of mathematics whose development has not stopped accelerating since the beginning of the XXth century, and should not be traced here.

As for the general theory of topological spaces, such as was foreseen by Riemann, it was necessary in order for it to develop, that the theory of real numbers, of sets of numbers, of sets of points on the straight line, in the plane and in space, should first be studied more systematically than had been the case in Riemann's time: this study was linked, on the other hand, to researches

(semi philosophical with Bolzano, essentially mathematical with Dedekind) on the nature of irrational numbers, as well as progress in the theory of functions of a real variable (to which Riemann himself brought an important contribution by his definition of the integral and his theory of trigonometric series, and which was the object, among others, of the work of du Bois-Reymond, Dini, Weierstrass); it was the work of the second half of the XIXth century and most particularly Cantor, who was the first to define (first on a straight line, then in Euclidean space of n dimensions) the notions of accumulation point, closed and perfect set, and who obtained the essential results on the structure of these sets on the straight line (cf. p. 155): one should consult on this, not only the Works of Cantor [47], but also his very interesting correspondence with Dedekind [48], where will be found clearly expressed also the idea of the number of dimensions considered as a topological invariant.

Further progress of the theory is stated for example, in a semi-historical, semi-systematic form in the book of Schoenflies [275 a]: by far the most important was the Borel-Lebesgue theorem (proved first by Borel for a closed interval on the straight line and a countable family of open intervals covering it).

The ideas of Cantor had first met a fairly lively opposition (cf. p. 28). At least his theory of sets of points on the straight line and in the plane was soon used and widely spread by the French and German schools of the theory of functions (Jordan, Poincaré, Klein, Mittag-Leffler, then Hadamard, Borel, Baire, Lebesgue, etc.): the first volumes of the Borel collection, in particular, each contain an elementary statement of this theory (see for example [32 a]). As these ideas were spread, people began to think in different ways about their possible application to sets, not of points anymore, but of curves or functions: an idea that saw the light of day, for example, already in 1884, in the title “On the limit curves of a family of curves” of a memoir of Ascoli [10], and which expresses itself in a communication of Hadamard to the congress of mathematicians at Zurich in 1896 [141]; it is tightly linked also to the introduction of “line functions” by Volterra in 1887, and to the creation of the “functional calculus”, or theory of functions where the argument is a function (on that, the work of Volterra on functional Analysis [322 b] can be consulted). On the other hand, in the famous memoir ([163 a], v. III, pp. 10-37) where Hilbert, taking up again on this point the ideas of Riemann, proved the existence of the minimum in the

Dirichlet principle and inaugurated the “direct method” of the calculus of variations, one sees clearly appearing the interest that there is in considering sets of functions where the Bolzano-Weierstrass principle is valid, that is to say every sequence contains a convergent sub sequence; such sets would soon indeed play an important role, not only in the calculus of variations, but in the theory of functions of a real variable (Ascoli, Arzelà) and in that of functions of a complex variable (Vitali, Carathéodory, Montel). Finally the study of functional equations, and most particularly the solution by Fredholm of the type of equation that bears his name [116], made familiar the consideration of a function as an argument, and of a set of functions as the analogue of a set of points, in connection with which it was quite as natural to use geometric language as for the points of a Euclidean space of n dimensions (a space which, also, escapes “intuition”, and for that reason, remained for a long time an object of mistrust for many mathematicians). In particular, the memorable works of Hilbert on integral equations [163 b] end with the definition and the geometric study of Hilbert space by Erhard Schmidt [274 b], in complete analogy with Euclidean geometry (see p. 212).

Meanwhile the notion of axiomatic theory had taken an increasingly greater importance, thanks above all to numerous works on the foundations of geometry, among which those of Hilbert [163 c] exercised a particularly decisive influence; during this very same work, Hilbert had been led to state exactly, from 1902 ([163 c], p. 180), a first axiomatic definition of the “multiplicity twice extended” in the sense of Riemann, a definition which constituted, he said, “the foundation of a rigorous axiomatic treatment of analysis situs”, and already used neighbourhoods (in a meaning restricted by the exigencies of the problem to which Hilbert was then limiting himself).

The first attempts to bring out that which was in common to the properties of sets of points and of functions (without the intervention of a notion of “distance”), were made by Fréchet [115 a] and F. Riesz [260 b]; but the first, starting from the notion of countable limit, does not succeed in constructing, for non metrizable spaces, an axiom system which is convenient and fruitful; at least he recognised the relationship between the Bolzano-Weierstrass principle and the Borel-Lebesgue theorem; it is in this connection that he introduced the word “compact”, even though with a meaning somewhat different from that given to it today. As for F. Riesz, who started from

the notion of point of accumulation (or rather, which comes to the same thing, of “derived” set), his theory was still incomplete, and remained besides as only a sketch.

With Hausdorff ([152 a], chap. 7-8-9) general topology as it is understood today starts. Taking up again the notion of neighbourhood, he knew how to choose, among the axioms of Hilbert on neighbourhoods in the plane, those that could give to his theory at the same time all the precision and the generality that were desirable. The chapter where he develops its consequences remains a model of axiomatic theory, abstract but adapted to applications from the beginning. That was, quite naturally, the point of departure for subsequent research on general topology, and principally the work of the school of Moscow, directed for the greater part towards the problem of metrisation (cf. p. 166): we must remember from this especially here the definition, by Alexandroff and Urysohn, of compact spaces (under the name of “bcompact spaces”), then the proof by Tychonoff [313] of the compactness of products of compact spaces. Finally the introduction of filters by H. Cartan [53], which while supplying a very valuable instrument in view of all kinds of applications (where it is an advantageous substitute for the notion of “Moore-Smith convergence” [227]), came, thanks to the theorem on ultrafilters, to achieve a clarification and simplification of the theory.

Chapter 11

UNIFORM SPACES.

The principal notions and propositions relating to uniform spaces came out little by little from the theory of real variables, and were the object of systematic study only recently. Cauchy, looking to set up rigorously the theory of series (cf. p. 153), took as departure point a principle that he seems to have considered as obvious, according to which a necessary and sufficient condition for the convergence of a sequence (a_n) is that $|a_{n+p} - a_n|$ should be arbitrarily small whenever n is sufficiently large (see for example ([56 a], (2), v. VII, p. 267)). With Bolzano [27 c], he was without doubt one of the first to state explicitly this principle, and to recognise its importance: whence the name “Cauchy sequence” given to sequences of real numbers that satisfy the condition in question, and by extension to sequences (x_n) of points in a metric space such that the distance between x_{n+p} and x_n is arbitrarily small whenever n is sufficiently large; from there finally the name of “Cauchy filter” given to the generalisation of Cauchy sequences in uniform spaces.

When later on the intuitive notion of real number was no longer satisfactory, and that it was sought, in order to give Analysis a solid foundation, to define real numbers starting from rational numbers, it was precisely the Cauchy principle that furnished the most fruitful among the definitions suggested in the second half of the XIXth century; it is Cantor’s definition ([47], pp. 93-96) (developed also, among others, from the ideas of Cantor, by Heine [154 b], and independently, by Méray), according to which one makes a real number correspond to every Cauchy sequence (“fundamental sequence” in the terminology of Cantor) of rational numbers; the same real number will correspond to two Cauchy sequences of rational numbers (a_n) and

(b_n) if $|a_n - b_n|$ tends to zero, and in this case only. The essential idea here is that, from a certain point of view, the set $\mathbf{Q}$ of rational numbers is “incomplete”, and that the set of real numbers is the “complete” set that is deduced from $\mathbf{Q}$ by “completing” it.

On the other hand, Heine, in work mainly inspired by the ideas of Weierstrass and Cantor, was the first to define uniform continuity for numerical functions of one or more real variables [154 a], and proved that every numerical function, continuous on a closed bounded interval of $\mathbf{R}$, is uniformly continuous there [154 b]: it is “Heine’s theorem”. This result is linked to the compactness of a closed bounded interval in $\mathbf{R}$ (“Borel-Lebesgue theorem”, cf. p. 141), and the proof given by Heine of his theorem can also be used, with a few modifications, to prove the Borel-Lebesgue theorem (which seemed to several authors a sufficient reason for giving this theorem the name “Heine-Borel theorem”).

The extension of these ideas to more general spaces was made when, first some special cases were studied, and then in general, metric spaces, where a distance (numerical function of pairs of points, satisfying certain axioms) is given and defines at the same time a topology and a uniform structure. Fréchet, who was the first to state the general definition of these spaces, recognised the importance of the Cauchy principle [115 a], and introduced also for all metric spaces, the notion of pre compact (or “totally bounded” [115 a and b]) space. Hausdorff, who, in his “Mengenlehre” [152 a and b] developed a lot of the theory of metric spaces, recognised in particular that one can apply to these spaces the Cantor construction which was considered above, and so deduce, for every non “complete” metric space (that is to say where the Cauchy principle is not valid), a “complete” metric space.

Metric spaces are “uniform spaces” of a particular kind; uniform spaces have only been defined in a general way recently, by A. Weil [330 b]. Beforehand it was only known how to use the notions and results referring to “uniform structure” when metric spaces were being dealt with: which explains the important role played in a lot of modern work on topology by metric or metrizable spaces (and in particular by compact metrizable spaces) in questions where distance is not of any real use. Once the definition of uniform spaces is stated, there is no difficulty (especially when the notion of filter is also available) in extending to these spaces nearly all the theory of metric spaces, as it is stated for example by Hausdorff (and to extend in the same way,

for example, to all compact spaces, the results set out for compact metric spaces in the Topology of Alexandroff-Hopf [4]). In particular, the completion theorem for uniform spaces is none other than the transposition, without any essential modification, of Cantor's construction for the real numbers.

Chapter 12

REAL NUMBERS.

Every measure of size implies a diffuse notion of real numbers. From the point of view of mathematics, the origins of the theory of real numbers must go back to the progressive formation, in Babylonian science, of a system of numbering capable (in principle) of denoting values as near as required to every real number [232]. The possession of such a system, and the confidence in numerical calculations which can not but follow from this, terminated inevitably, in fact, with a “naive” notion of real numbers, which is hardly different from that which one finds again today (linked to the decimal system of numbering) in elementary teaching or with physicists and engineers; this notion does not allow itself to be defined with exactness, but it can be expressed by saying that a number is considered as defined by the possibility of obtaining approximate values and introducing these into the calculations: which, besides, implies necessarily a certain degree of confusion between the measures of size given by experience, which are naturally not susceptible to indefinite approximation, and such “numbers” as $\sqrt{2}$ (supposing that there exists an algorithm for the indefinite approximation of this number).

A similar “pragmatic” point of view reappears therefore in all the schools of mathematics where calculating skills take over from the desire for rigour and theoretical preoccupations. It is these latter, on the contrary, which dominate Greek mathematics: so we owe to it the first rigorous and coherent theory of ratios of quantities, that is to say, essentially the real numbers; it is the end result of a series of discoveries on ratios and in particular on incommensurable ratios, of which it is difficult to exaggerate the importance in the history of Greek thought, but of which, in the absence of precise texts, we can only just discern the major lines. Greek mathematics in

its beginnings is inseparably linked to speculations, partly scientific, partly philosophical and mystical, on proportions, similitudes and ratios, in particular the “simple ratios” (expressible by fractions with small numerator and denominator); and it was one of the characteristic tendencies of the Pythagorean school to claim to explain all by integers and the ratio of integers. But it was the Pythagorean school, precisely, that discovered the incommensurability of the side of a square with its diagonal (the irrationality of $\sqrt{2}$): the first example, no doubt, of a proof of impossibility in mathematics; just the fact of stating this question implies a clear distinction between a ratio and its approximate values, and suffices to indicate the immense gap which separates Greek mathematicians from their predecessors. ¹

We are badly informed about the movement of ideas that accompanied and followed this important discovery. ² We will limit ourselves to indicating summarily the principal ideas that are at the basis of the theory of ratios of quantities, a theory which, codified by the great mathematician Eudoxus (contemporary and friend of Plato), was definitively adopted by classical Greek mathematics, and is known to us through the Elements of Euclid [107], where it is given a masterly exposition (in Book V of these Elements):

- 1) The word and the idea of number are strictly reserved for integers > 1 (1 is the monad and not a number properly speaking), to the exclusion, not only of our irrational numbers, but even of those that we call rational numbers, these being, for the Greek mathematicians of the classical era, ratios of numbers. There is there much more than a simple question of terminology, the word number being linked for the Greeks (and for the moderns until recent times) to the idea of a system with a double law of composition (addition and multiplication): the ratios of integers are conceived by the classical Greek mathematicians as operators, defined on the set of integers or on a part of this set (the ratio of p to q is the operator which, to N , makes correspond, if N is a multiple of q , the integer $p.(N/q)$), and making up a multiplicative group, but not a system with a double law of composition. In this, the Greek mathematicians were separating themselves voluntarily from the “logisticians” or professional calculators, who had, like their Egyptian and Babylonian predecessors, no scruples at treating as numbers fractions or the sum of an integer and a fraction. It seems besides that they imposed on themselves this restriction on the idea

of number for motives more philosophical than mathematical, and following on from the thoughts of the first Greek thinkers on the unit and the multiple, the unit not being capable of subdivision (in this system of thought) without losing by that selfsame action its character of unit. ³

- 2) The theory of quantities is axiomatically based, and at the same time for all kinds of quantities (one finds allusions to previous theories which, from what appears, treated separately lengths, areas, volumes, time, etc.). The quantities of one kind are characterised by the fact that they are capable of being compared (that is to say that a definition of equality, which is properly speaking an equivalence, and the relations $>$ and $<$ are assumed), of being added and subtracted $A + B$ is defined, and $A - B$ if $A > B$), and satisfy the axiom of “Archimedes” so-called; this is clearly conceived, from the beginning, as a keystone of the edifice (it is in fact indispensable for all axiomatic characterisations of real numbers); it is a pure accident that it has been given the name of Archimedes, and this latter insists, in the introduction to his “Quadrature of the Parabola” ([5 b], v. II, p. 265), on the fact that this axiom has been used by his predecessors, that it plays an essential role in the work of Eudoxus, and that its consequences are no less sure than the determinations of areas and volumes made without its help. ⁴

It is easy to see that, from this axiomatic foundation, follows necessarily the theory of real numbers. It must be noted that, for Eudoxus, the quantities of a given kind form a system with one internal law of composition (addition), but that this system has an external law of composition with as operators the ratios of quantities, these being conceived of as forming an abelian multiplicative group. A and A' being quantities of the same kind, and similarly B and B' , the ratios of A and A' and of B and B' are defined as equal if, whatever the integers m and m' , $mA < m'A'$ forces $mB < m'B'$ and $mA > m'A'$ forces $mB > m'B'$; inequalities between ratios are defined in an analogous manner. That these ratios form a domain of operators for every kind of quantity is equivalent to the axiom (not made explicit but often used in the drafting of Euclid) of the existence of a fourth proportional: a ratio A/A' being given and B' being given, there exists a B , of the same kind as B' , such that $B/B' = A/A'$. Thus the highly original idea of Eudoxus allowed identification between the domains of operators defined for all kinds of quantities;⁵ in

an analogous manner, the set of integers (see above) can be identified with a subset of the set of ratios of quantities, namely with the set of rational ratios (ratios of commensurable quantities); however, because of the fact that these ratios, in as much as they are operators on the integers, are (in general) defined only for a subset of the set of integers, it was still necessary to develop their theory separately (Book VII of Euclid).

The domain of universal operators thus constructed was therefore for Greek mathematicians the equivalent of what is for us the set of real numbers; it is clear besides that with the addition of quantities and the multiplication of ratios of quantities, they possessed the equivalent of what for us is the field of real numbers, even though in a much less handy form. ⁶ It is possible, on the other hand, to ask oneself if they had conceived of these sets (set of quantities of a given kind, or sets of ratios of quantities) as complete in our sense; it is not easy to see, otherwise, why they would have assumed (without even feeling the need to turn it into an axiom) the existence of a fourth proportional; furthermore, certain texts seem to refer to ideas of this kind; finally they certainly assume as obvious that a curve, capable of being described by a continuous movement, can not pass from one side to the other of a straight line without cutting it, a principle that they used for example in their research on the duplication of the cube (construction of $\sqrt[3]{2}$ by the intersection of curves) and which is essentially equivalent to the property under consideration; however, the texts that we possess do not allow us to know with complete precision their ideas on this point.

Thus therefore is the state of the theory of real numbers in the classical era of Greek mathematics. As admirable as the construction of Eudoxus was, and not leaving anything to be desired from the point of view of coherence and rigour, it must be admitted that it lacked in suppleness, and was hardly favourable for the development of numerical calculations and especially of algebraic calculations. What is more, its logical necessity could only be appreciated by minds taken with rigour and exercised in abstraction; it is therefore natural that with the decline of Greek mathematics, one sees reappearing little by little the “naive” point of view which had been kept by means of the tradition of the logisticians; it is that which dominates for example with Diophantus [91 a], a true follower of this tradition rather than official Greek science; this latter, while reproducing for form’s sake the Euclidean definition of number, understands really by the

word “number”, the unknown of the algebraic problems whose solution is, either an integer, or a fractional number, or even an irrational. ⁷ Although this change in attitude, about the subject of number, was linked to one of the most important advances in the history of mathematics, namely the development of Algebra (see p. 48), it does not constitute, be it well understood, progress in itself, but rather regress.

It is not possible for us to follow here all the vicissitudes of the idea of number through Hindu, Arab and Occidental mathematics until the end of the Middle Ages: it is the “naive” notion of number that dominates; and, even though the Elements of Euclid were used as basis for the teaching of mathematics during this period, it is likely that the doctrine of Eudoxus remained generally not understood because the necessity for it did not arise any more. The “ratios” of Euclid were most often qualified as “numbers”; the rules of calculation of integers were applied to them, obtaining thus exact results, without seeking to analyse deeply the reasons for the success of these methods.

We see however already R. Bombelli, in the middle of the XVIth century, writing on this subject, in his Algebra [28 b], ⁸ a point of view that (on condition that we assume acquired the results of book V of Euclid) is essentially correct; having recognised that once the unit of length is chosen there is a bijective correspondence between the lengths and the ratios of quantities, he defines the various algebraic operations (assuming the unit fixed, of course) on the lengths, and, representing numbers by lengths, obtains the geometric definition of the field of real numbers (a point of view for which the credit is most often given to Descartes), and gives thus to his Algebra a solid geometrical basis. ⁹

But the Algebra of Bombelli, even though singularly advanced for his times, did not go beyond the extraction of radicals and the solution by radicals of equations of the 2nd, 3rd and 4th degrees; of course the possibility of extraction of radicals is assumed by him without discussion. Simon Stévin [295], also, adopts an analogous point of view on the subject of number, which is for him that which marks a measure of quantity, and which he considers as essentially “continuous” (without making precise the meaning he gives to this word); if he distinguishes between the “geometrical numbers” and the “arithmetical numbers”, it is only from the accident of their mode of definition, without there being for him there a difference of kind; here besides is his last

word on the subject: “We conclude therefore that there are no absurd, irrational, irregular, inexplicable or deaf numbers; but that there is in them such excellence and agreement that we have subject matter for meditation night and day in their admirable perfection” ([295], p. 10). On the other hand, having been the first to have turned the tool of decimal fractions into a method of calculation, and proposed for them a notation already close to ours, he conceived clearly that these fractions furnish an indefinite approximation algorithm for every real number, as he brings out in his *Appendice algebratique* of 1594 “containing a general rule for all Equations” (a booklet of which the only known example was burnt at Louvain in 1914; but see [295], p. 88). Such an equation having been put into the form $P(x) = Q(x)$ (where P is a polynomial of degree greater than that of the polynomial Q , and $P(0) < Q(0)$), we substitute for x the numbers 10, 100, 1000, ... until we find $P(x) > Q(x)$, which, he says, determines the number of digits of the root; then (if for example the root is found to have two digits) we substitute 10, 20, ... which determines the digit of tens; and then the same for the following digit, then for the successive decimal digits: “And proceeding thus infinitely”, he says, “one reaches infinitely near to that required” ([295], p. 88). As can be seen, Stévin had (no doubt the first) a clear idea of Bolzano’s theorem and had recognised in this theorem the essential tool for the systematic solution of numerical equations; an intuitive conception of the numerical continuum is recognised there, at the same time, so clear that there remained little to do to make it definitively precise.

However, in the two centuries that followed, the definitive establishment of correct methods was twice slowed down by the development of two theories of which we do not have to give the history here: the infinitesimal calculus, and the theory of series. Across the discussions that they brought up, we recognise, as at all periods of the history of mathematics, the perpetual balancing between the seekers occupied with going forward, at the cost of some insecurity, persuaded that there will always be time later to consolidate the conquered territory, and the critical spirits, who (without necessarily giving an inch to the first as far as intuitive faculties and inventive talents go) do not believe they are wasting their efforts in devoting some effort to precise expression and to rigorous justification of their concepts. In the XVIIth century, the principal object of debate is the notion of the infinitely small, which, justified a posteriori by the results which it allows us to reach, seemed in open opposition to the axiom of Archimedes; and we see the most

open spirits of the time finish by adopting a point of view little different from that of Bombelli, and which is distinguished from it mainly by the greater attention paid to the rigorous methods of the ancients; Isaac Barrow (the master of Newton, and who himself played an important part in the creation of the infinitesimal calculus) gives a brilliant exposé of it in his *Lessons in Mathematics* delivered at Cambridge in 1664-65-66 [16 a and b]; recognising the necessity, in order to find again for the subject of numbers the proverbial “geometrical certainty”, of returning to the theory of Eudoxus, he presents at length, and very judiciously, the defence of this latter (who, as he witnessed, appeared intelligible to many of his contemporaries) against those who tackled him with obscurity or even absurdity. On the other hand, defining numbers as symbols which denote the ratio of quantities, and susceptible of being combined amongst themselves by the operations of arithmetic, he obtains the field of real numbers, in terms taken up again after him by Newton in his *Arithmetic* and of which his successors until Dedekind and Cantor would not change anything.

But it is at about this time that the method of expansion in series was introduced, which soon, in the hands of impenitent algebraists, takes on an exclusively formal character and turns away the attention of mathematicians from the questions of convergence that the sane use of series raises in the domain of real numbers. Newton, the principal creator of this method, was still conscious of the necessity of considering these questions: and, if he had not sufficiently elucidated them, he had recognised at least that the power series that he was introducing converged “most of the time” at least as well as a geometric series (whose convergence was already known to the ancients) for small values of the variable ([233 a], v. I, pp. 3-26); at about the same time, Leibniz had observed that an alternating series, with terms decreasing in absolute value and tending to 0, is convergent; in the following century, d’Alembert, in 1768, expresses doubts on the use of non-convergent series. But the authority of Bernoulli and especially Euler means that such doubts are exceptional at this time.

It is clear that the mathematicians who would have had the habit of using series in numerical calculation would never have neglected thus the notion of convergence; and it is not chance that the first who, in this domain as in many others, had brought about a return to correct methods, was a mathematician who, already from the prime of youth, had a love of numerical calculation:

C. F. Gauss, who, almost a child, had used the algorithm of the arithmetico-geometric mean,¹⁰ could not avoid forming for himself a clear idea of limits; and we see him, in a fragment that dates from 1800 (but was only published in our time) ([124 a], v X₁, p. 390), defining precisely, on the one hand the upper bound and the lower bound, on the other hand the upper limit and the lower limit of a sequence of real numbers; the existence of the first (for a bounded sequence) seeming to be assumed as obvious, and the latter having been correctly defined as limits for n tending to $+\infty$, of $\sup_{p \geq 0} u_{n+p}$, $\inf_{p \geq 0} u_{n+p}$. Gauss, as well, gives also, in his memoir of 1812 on the hypergeometric series ([124 a], v. III, p. 139) the first model of a discussion about convergence undertaken, as he says, “in all rigour, and done to satisfy those whose preferences go to the rigorous methods of the ancient geometers”: it is true that this discussion, forming a secondary point in the memoir, does not go back to first principles in the theory of series; it is Cauchy who establishes these first, in his *Cours d’Analyse* of 1821 ([56 a], (2), v. III), in a manner correct at all points, starting from the Cauchy criterion clearly stated, and assumed to be obvious; as, for the definition of number, he keeps to the point of view of Barrow and Newton, it can thus be said that for him the real numbers are defined by the axioms of quantities and the criterion of Cauchy: which indeed is sufficient to define them.

It is at the same time that another important aspect of the theory of real numbers is definitively clarified. As we have said, it has always been assumed as geometrically obvious that two continuous curves can not cross without intersecting; a principle that (made suitably precise) would be equivalent, it also, to the property of the straight line being a complete space. This principle is still part of the basis of the “rigorous” proof given by Gauss in 1799 of d’Alembert’s theorem, according to which every polynomial with real coefficients admits a real or complex root ([124 a], v. III, p. 1); the proof of the same theorem, given by Gauss in 1815 ([124 a], v. III, p. 31), relies, in the same way as a previous attempt by Lagrange, on the principle, analogous but simpler, according to which a polynomial can not change sign without becoming zero (a principle that we have already seen used by Stévin). In 1817, Bolzano gives, starting with the criterion of Cauchy, a complete proof of this latter principle, which he obtains as a particular case of an analogous theorem about continuous numerical functions of a numerical variable [27 c]. Stating clearly (before Cauchy) the “Cauchy criterion”, he seeks to justify it by an argument

that, in the absence of all arithmetical definitions of a real number, was only and could only be a vicious circle; but, once this point is admitted, his work is entirely correct and quite remarkable, as containing, not only the modern definition of a continuous function (given here for the first time), with a proof of the continuity of polynomials, but even the proof of the existence of the lower bound of an arbitrary bounded set of real numbers (he does not speak of sets, but, which amounts to the same thing, of properties of real numbers). On the other hand, Cauchy, in his *Cours d'Analyse* ([56 a], (2), v. III), defining, he also, continuous functions of one or more numerical variables, proves equally that a continuous function of one variable can not change sign without becoming zero, and this by the same argument as Simon Stévin, which naturally becomes correct, once continuity is defined, as soon as the Cauchy criterion is used (or as soon as the equivalent principle of “nested intervals” so-called is assumed, as Cauchy does at this point, of which the convergence of indefinite decimal fractions is of course only a particular case).

Once this point is reached, it only remained for mathematicians to make precise and develop the results acquired, by correcting a few errors and filling a few gaps. Cauchy, for example, had believed for a time that a convergent series, whose terms are continuous functions of a variable, has as sum a continuous function: the correction of this point by Abel, during the course of his important work on series ([1], v. I, p. 219; cf. also v. II, p. 257 and *passim*), ended finally by the elucidation by Weierstrass, in his lectures (unpublished, but which had a considerable influence), of the notion of uniform convergence (see p. 205). On the other hand, Cauchy had, without sufficient justification, assumed the existence of a minimum of a continuous function in one of the proofs given by him for the existence of roots of a polynomial; it is again

Weierstrass who brought clarity to questions of this kind by proving in his lectures the existence of a minimum for functions of numerical variables, defined on closed bounded intervals; it is following on from his critique of the unjustified application of this theorem to sets of functions (of which the “Dirichlet principle” is the best known example) that the movement of ideas begins which ends (see p. 143) with the general definition of compact spaces and the modern statement of the theorem.

At the same time, Weierstrass, in his lectures, had recognised the logical interest that there is

in completely separating the idea of real numbers from the theory of quantities: to use this, in effect, reduces to defining axiomatically the set of points of the straight line (therefore definitively the set of real numbers) and assuming the existence of such a set; although this way of doing things is essentially correct, it is evidently preferable to start only with rational numbers, and to deduce the real numbers from there by completion.¹¹ This is what was done, by different methods, and independently the one from the other, by Weierstrass, Dedekind, Méray and Cantor; whereas the procedure of “cuts” so-called, proposed by Dedekind ([79], v. II, pp. 315-334) was much closer to the definitions of Eudoxus, the other proposed methods are nearer to the method used since Hausdorff to complete a metric space. It is at this moment also that Cantor starts to develop the theory of sets of real numbers, of which Dedekind had had the first idea [48], obtaining thus the principal elementary results on the topology of the straight line, the structure of open sets, of closed sets, the notions of derived sets, perfect totally discontinuous sets, etc..

With Cantor, the theory of real numbers took, with few exceptions, its definitive form; let us indicate briefly in what way it has been extended. Apart from the work in general topology (see p. 143), and applications to Integration (see p. 221), it is mainly a case of research on the structure and classification of sets of points on the straight line and of numerical functions of real variables; they have their origin in the works of Borel [32 a], aiming mainly at measure theory, but which ended up among other things at the definition of “Borel sets”: these are sets belonging to the smallest family of subsets of $\mathbb{R}$, containing the intervals, and closed with respect to union and countable intersection and the operation C . Intimately linked to these sets are the so-called “Baire” functions, that is to say those that can be obtained starting with continuous functions by the operation of limits, repeated “transfinitely”; they were defined by Baire in the course of important work where he abandons entirely the measure point of view in order to tackle systematically the qualitative and “topological” aspect of these questions [11 a]: it is on this occasion that he defines and studies for the first time semi-continuous functions, and with a view to characterise limit functions of continuous functions, he introduces the important notion of a thin set (“first category” set in the terminology of Baire) (see p. 165). As for the numerous works that have followed those of Baire, and which are due mainly to the Russian and especially Polish schools, we can only draw attention to their existence here (see p. 166).

Chapter 13

EXPONENTIALS AND LOGARITHMS.

The history of the multiplicative group R_+^* of real numbers >0 is closely linked to that of the development of the notion of powers of a number >0 , and of the notations employed to indicate them. The conception of a “geometric progression” made up of the successive powers of the same number goes back to the Egyptians and the Babylonians; it was familiar to Greek mathematicians, and one finds already in Euclid (Elements, Book IX, prop. 11) a general statement equivalent to the rule $a^m a^n = a^{m+n}$ for integer exponents >0 . In the Middle Ages, the French mathematician N. Oresme (XIVth century) finds this rule again; it is also with him that appears for the first time the notion of fractional exponent >0 , with a notation already close to ours and some rules for calculation (stated in general form) concerning them (for example the two rules that we now write $(ab)^{1/n} = a^{1/n} b^{1/n}$, $(a^m)^{p/q} = (a^{mp})^{1/q}$ [74]. But the ideas of Oresme were too much in advance of the Mathematics of his time to exercise an influence on his contemporaries and his treatise sunk rapidly into forgetfulness. A century later, N. Chuquet stated again Euclid’s rule; he introduces as well an exponential notation for the powers of unknowns in his equations, and does not hesitate to make use of the exponent 0 and integer exponents <0 .¹ This time (and although the work of Chuquet remained in manuscript and does not seem to have been circulated much), the idea of isomorphism between the “arithmetic progression” of the exponents, and the “geometric progression” of the powers, will no longer be lost sight of; extended to negative exponents and to fractional exponents by Stifel ([298], fol. 35 and 249-250), it ends up finally with the definition of logarithms and the construction of the first tables, undertaken independently by the Scot J. Napier, in 1614-1620, and the Swiss J. Bürgi (whose work did not

appear until 1620, although his conception went back to the first years of the XVIIth century). With Bürgi, the continuity of the isomorphism established between $\mathbb{R}$ and $\mathbb{R}_+^*$ is implicitly assumed by the use of interpolation in the handling of the tables; it is on the contrary explicitly formulated in the definition of

Napier (as explicitly at least as was allowed by the fairly vague concept of continuity that obtained at that time). ²

We do not need to dwell here on the services rendered by logarithms in the numerical Calculus; from the theoretical point of view, their importance dates especially from the beginnings of the infinitesimal Calculus, with the discovery of series expansions of $\log(1 + x)$ and of e^x , and of the differential properties of these functions (see pp. 172 ff.). As far as the definition of exponentials and logarithms is concerned, it will be limited, until the middle of the XIXth century, to assuming intuitively the possibility of extending by continuity to the set of real numbers the function a^x defined for all rational x ; and it is only once the notion of real number is definitively made precise and deduced from that of rational number, that there will be thought of giving a rigorous justification to this extension.

Chapter 14

N DIMENSIONAL SPACES.

We have already had the opportunity of saying how the development of the analytic Geometry of the plane and space led mathematicians to introduce the notion of n dimensional space, which supplied them with a geometric language which was extremely convenient for expressing in a concise and simple way theorems of algebra concerning equations with an arbitrary number of variables, and notably all the general results of linear Algebra (see p. 61). But if, towards the middle of the XIXth century, this language had become current among numerous geometers, it remained purely conventional, and the absence of an “intuitive” representation of spaces of more than three dimensions seemed to forbid in these latter arguments “by continuity” that were allowed in the plane or in space based exclusively on “intuition”. It is Riemann who was the first, in his research on Analysis situs and on the foundations of Geometry, who was bold to argue in that way by analogy with the case of three dimensional space (see p. 176); ¹ following him, numerous mathematicians set out to use, with great success, arguments of this nature, notably in the theory of algebraic functions of several complex variables. But the control of spatial intuition being very limited at that time, one could remain with good reason very sceptical as to the worth in proofs of such considerations, and only allow them purely heuristically, in as much as they made very plausible the exactness of certain theorems. It is thus that H. Poincaré, in his memoir of 1887 on the residues of double integrals of functions of two complex variables, avoids as far as possible all recourse to intuition in four dimensional space: “As this hypergeometric language is repugnant still to many good minds”, he says, “I will only make infrequent use of it”; the “artifices” that he employs to this end allow him to get back to topological arguments in

three dimensional space, where he then hesitates no longer to appeal to intuition ([251 a], v. III, pp. 443 ff.).

Elsewhere, the discoveries of Cantor, and notably the famous theorem establishing that $\mathbb{R}$ and $\mathbb{R}^n$ are equipotent (which seemed to put into question the very notion of dimension)² showed that it was indispensable, in order to provide a solid basis for the arguments of Geometry and Topology, to free them entirely from all recourse to intuition. We have already said (cf. p. 139) that this need is at the origin of the modern conception of general Topology; but even before the creation of this latter, a rigorous study had begun of the topology of number spaces and of their most immediate generalisations (the “ n dimensional varieties”) by methods arising mainly from the branch of Topology called “combinatorial Topology” or better “algebraic Topology”, of which we can not speak here.

Chapter 15

COMPLEX NUMBERS; MEASUREMENT OF ANGLES.

We will not take up again here the complete exposé of the historical development of the theory of complex numbers or that of quaternions, these theories being essentially in the domain of Algebra (see pp. 72 and 62); but we will say a few words about the geometric representation of imaginary numbers which in many respects constitutes decisive progress in the history of Mathematics.

It is to C. F. Gauss that is due without dispute the first clear conception of the bijective correspondence between complex numbers and points of the plane, ¹ and especially the merit of having been the first to have known how to apply this idea to the theory of complex numbers, and to have foreseen clearly the advantage that the analysts of the XIXth century would make of it. During the course of the XVIIth and XVIIIth centuries, mathematicians had little by little reached the conviction that imaginary numbers, which allowed the solution of equations of the 2nd degree, also allowed the solution of algebraic equations of arbitrary degree. Numerous attempts at the proof of this theorem were published during the XVIIIth century; but, even without mentioning those that rely only on a vicious circle, there were none that were not open to serious objections. Gauss, after a detailed examination of these attempts and a tight critique of the gaps, proposed, in his inaugural Dissertation (written in 1797, appeared in 1799), to give finally a rigorous proof; taking up an idea given out in passing by d'Alembert (in the published

proof by this latter in 1746, ² he remarks that the points (a, b) of the plane such that $a + ib$ is a root of the polynomial $P(x + iy) = X(x, y) + iY(x, y)$, are the intersections of the curves $X=0$ and $Y=0$; by a qualitative study of these curves, he shows that a continuous arc of one of them joins points of two distinct regions bounded by the other, and concludes from this that the curves meet ([124 a], v. III, p. 3; see also [124 b]): a proof that, by its clarity and originality, constitutes a considerable progress on previous attempts, and is without doubt one of the first examples of an argument from pure Topology applied to a problem of Algebra. ³

In his Dissertation, Gauss does not define explicitly the correspondence between points of the plane and imaginary numbers; as far as these latter are concerned, and the questions about “existence” that they raised during two centuries, he even adopts a fairly reserved position, intentionally presenting all his arguments in a form in which only real numbers occur. But the march of ideas in his proof would be entirely unintelligible if they did not presuppose a fully conscious identification of the points of the plane and complex numbers; and his contemporaneous research on the theory of numbers and elliptic functions, where complex numbers also occur, only reinforces this hypothesis. At what point the geometric conception of imaginary numbers became familiar to him, and to what results it would lead in his hands, is what is shown clearly in the notes (published only in our day) where he applies complex numbers to the solution of problems of elementary Geometry ([124 a], v. IV, p. 396 and v. VIII, p. 307). Still more explicit is the letter to Bessel of 1811 ([124 a], v. VIII, pp. 90- 91), where he sketches the essentials of the theory of integration of functions of complex variables: “In the same way”, he says, “that one can represent the whole domain of real quantities by means of an indefinite straight line, so one can picture (“sinnlich machen”) the complete domain of all the quantities, the reals and the imaginaries, by means of an indefinite plane, where each point, determined by its abscissa a and its ordinate b , represents at the same time the quantity $a + ib$. The continuous movement of a value of x to another in consequence is made following a line, and so can be accomplished in an infinity of ways ...”

But it is only in 1831 that Gauss (in connection with the introduction of the “Gaussian integers” $a + ib$, where a and b are integers) displayed publicly his ideas on this point in such a clear a manner ([124 a], v. II, *Theoria Residuorum Biquadraticorum*, *Commentatio secunda*, art.

38, p. 109, and *Anzeige*, pp. 174 ff.). In the interval, the idea of the geometric representation of imaginary numbers had been found again independently by two modest seekers, both amateur mathematicians, more or less self-taught, and for whom it was the only contribution to science, both also without much contact with the scientific environment of their time. Because of this, their work was in grave risk of being completely overlooked; it is precisely what happened for the first in time, the Dane C. Wessel, whose pamphlet, which appeared in 1798, very clearly conceived and drawn up, was only brought out of limbo a century later; and the same misadventure almost happened to the second, the Swiss J. Argand, who only by chance saw, in 1813, the exhumation of his work that he had published seven years earlier.⁴ This work provoked an active discussion in the *Annales de Gergonne*, and the matter was the object, in France and in England, of several publications (due to fairly obscure authors) between 1820 and 1830; but it lacked the authority of a big name to put an end to these controversies and to rally mathematicians to the new point of view; and it was necessary to wait until the middle of the century before the geometric representation of imaginary numbers was finally universally adopted, following the publications of Gauss (quoted above) in Germany, the work of Hamilton and Cayley on hypercomplex systems in England, and finally, in France the adherence of Cauchy,⁵ a few years only before Riemann, by a brilliant extension, came also to widen the role of Geometry in the theory of analytic functions, and create Topology at the same time.

The measurement of angles, by the arcs that they cut out on the circle, is as ancient as the notion of angle itself, and is already known to the Babylonians, from whom we have kept the unit of angle, the degree; it is besides only a matter for them of the measurement of angles between 0 and 360° , which sufficed for them, since their angles were mainly used by them to identify the position of celestial objects at predetermined points of their apparent trajectories, and to draw up tables for use in scientific and astrological situations.

With the Greek geometers of the classical era, the definition of angle (Eucl. El., I, def. 8 and 9) is still more restricted, since it only applies to angles less than two right angles; and as on the other hand their theory of ratios and of measurement relied on the comparison of arbitrarily large multiples of the quantities measured, the angles could not be for them a measurable quantity, although one finds naturally with them the notions of equal angles, of angles greater or less one

than the other, and of the sum of two angles when this sum does not exceed two right angles. In the same way as for the addition of fractions, the measurement of angles must therefore have been in their eyes an empirical procedure without scientific value. This point of view is well illustrated by the admirable memoir of Archimedes on spirals ([5 b], v. II, pp. 1-12), where, in default of being able to define these by the proportionality of the radius vector to the angle, he gives a cinematic description of them (def 1, p. 44; cf the statement of prop. 12, p. 46) from which he manages to draw out, as is shown by the rest of his work, all that the general notion of the measurement of angles would have given him if were to have had it. As for the Greek astronomers, they seem, on this point as on many others, to have been content to follow their Babylonian predecessors.

Here also, as in the evolution of the concept of real numbers (cf. p. 150), the loosening of the spirit of rigour, during the decadence of Greek science, brought the return of the “naïve” point of view, which in certain respects, is closer to our own than the rigid Euclidean conception. It is thus that an ill-advised interpolator inserts in Euclid the famous proposition (Eucl. El., VI, 33): “The angles are proportional to the arcs that they cut out on the circle”,⁶ and an anonymous scholastic who comments on the “proof” of this proposition does not hesitate to introduce, without any justification of course, arcs equal to arbitrarily large multiples of a circumference, and the angles corresponding to these arcs ([107], v. V, p. 357). But Vietus even, in the XVIth century, while appearing to touch on our modern conception of angle when he discovers that the equation $\sin nx = \sin \alpha$ has several roots, only obtains the roots that correspond to angles less than 2 right angles ([319], pp. 305-313). It is only in the XVIIth century that this point of view is overtaken in a definitive manner; and, after the discovery by Newton of the expansions in series of $\sin x$ and $\cos x$ had supplied expressions for these functions, valid for all values of the variables, one finds with Euler, in connection with logarithms of “imaginary” numbers, the precise conception of the notion of the measurement of an arbitrary angle ([108 a], (1), v. XVII, p. 220).

Of course, the classical definition of the measurement of an angle by the length of the arc of a circle is not only intuitive, but essentially correct; however, it requires, in order to be made rigorous, the notion of the length of a curve, that is to say the integral Calculus. From the point of view of the structures that come into play, that is a very roundabout procedure, and it is possible to

use no other methods than those of the theory of topological groups; the real exponential and the complex exponential appear thus as deriving from the same origin, the theorem characterising “groups with one parameter”.

Chapter 16

METRIC SPACES.

As we said (see p. 146), the notion of metric space was introduced in 1906 by M. Fréchet, and developed a few years later by F. Hausdorff in his “Mengenlehre”. It acquired great importance after 1920, on the one hand following the fundamental work of S. Banach and his school on normed spaces and their applications to functional Analysis (see pp. 216 ff.), on the other by reason of the interest that was raised by the notion of valuation in Arithmetic and in Algebraic Geometry (where notably completion with respect to a valuation shows itself to be very fruitful).

From the period 1920-1930 are dated a whole series of studies undertaken by the Moscow school on the properties of the topology of a metric space, work which aimed in particular at obtaining necessary and sufficient conditions for a given topology to be metrizable. It is this movement of ideas that caused the appearance of interest in the notion of a normal space, defined in 1923 by Tietze, but whose important role was only recognised following the work of Urysohn [314] on the extension of continuous numerical functions. Apart from the trivial case of functions of a real variable, the problem of extension to the whole space of a continuous numerical function defined on a closed set, was treated for the first time (for the case of the plane) by H. Lebesgue [196 d]; before the definitive result of Urysohn, it had been solved for metric spaces by H. Tietze [307]. The extension of this problem to the case of functions with values in an arbitrary topological space acquired in these last years a considerable importance in algebraic Topology. Recent works have as well brought into the open that, in this type of question, the notion of normal space is hard to handle, because it still offers too many “pathological” possibilities; most of the time it is necessary to substitute for it the more restrictive notion of a

paracompact space, introduced by J. Dieudonné in 1944 [90 a]; in this theory, the most remarkable result is the theorem due to A. H. Stone [300], according to which every metrizable space is paracompact. ¹

We have already focused attention (see p. 155) on the important works of the end of the XIXth century and of the beginning of the XXth century (E. Borel, Baire, Lebesgue, Osgood, W. H. Young) on the classification of sets of points in the spaces R^n , and on the classification and characterisation of numerical functions obtained by iterating, starting with continuous functions, the process of passing to the limit (for sequences of functions). It rapidly became apparent that metric spaces supplied a natural framework for research of this kind, whose development after 1910 is mainly due to the Russian and Polish schools. It is these schools that have between them brought to light the fundamental role played in modern Analysis by the notion of thin sets, and by the theorem on the countable intersection of open sets everywhere dense in a complete metric space, proved first (independently) by Osgood [240] for the numerical straight line and by Baire [11 b] for the spaces R^n .

On the other hand, in 1917, Souslin [289], correcting an error of Lebesgue, showed that the continuous image of a Borel set is not necessarily Borel, which led him to the definition and study of the larger category of sets, since called “analytic” or “Souslin”; after the premature death of Souslin this study was mainly undertaken by N. Lusin (whose ideas had inspired the work of Souslin) and by Polish mathematicians (see [209] and [189 b]). The actual importance of these sets relies mainly on their applications in the theory of integration (where, thanks to their special properties, they allow constructions that would be impossible on arbitrary measurable sets), and in modern potential theory, where the fundamental theorem on the capacity of Souslin sets, proved recently by G. Choquet [63], has already shown itself to be rich in varied applications.

Chapter 17

INFINITESIMAL CALCULUS.

In 1604, at the apogee of his scientific career, Galileo, believes he has proved that, in rectilinear movement where the speed grows proportional to the distance covered, the law of movement will be just the one ($x = ct^2$) that he had discovered in the falling of weights ([122 b], v. X, pp. 115-116). Between 1695 and 1700, there is no volume of *Acta Eruditorum* published monthly at Leipzig, where there do not appear memoirs from Leibniz, the Bernoulli brothers, the marquis de l'Hôpital, dealing with, to all intents and purposes with the notation of which we still make use, the most varied problems of differential calculus, integral calculus and the calculus of variations. It is thus almost exactly during the interval of a century that infinitesimal calculus was wrought or, as was eventually said by the English, Calculus at its best ("calculus"); and nearly three centuries of constant use have not completely blunted this incomparable instrument.

The Greeks neither had nor imagined anything like it. If they knew, no doubt, if only to refuse to use it, an algebraic calculus, that of the Babylonians, of which part of their Geometry is perhaps just a transcription, it is strictly in the domain of geometric invention that what is perhaps their most brilliant mathematical creation is registered, their method for treating problems that for us would tie in to the integral calculus. Eudoxus, treating the volume of the cone and the pyramid, gave the first models of it, which Euclid has more or less faithfully transmitted to us ([107], book XII, prop. 7 and 10). But above all, it is to these problems that most of the work of Archimedes was devoted [5 b and c]; and by a remarkable piece of fortune, we are in a position to read still in their original text, in the sonorous dorian dialect in which he had so carefully composed them, most of his writings, and up to the one, found again recently, where he sets

out the “heuristic” procedures by which he was led to some of his most beautiful results ([5 b], v. II, pp. 425-507). For it is there that there is one of the weaknesses of the “exhaustion” of Eudoxus: an irreproachable method of proof (having assumed certain postulates), it is not a method of discovery; its application relies necessarily on advance knowledge of the result to be proved; also, says Archimedes, “of the results whose proofs Eudoxus was the first to discover, concerning the cone and the pyramid ..., not a small part is due to Democritus, who was the first to state them without proof” (loc. cit., p. 430). This circumstance makes it particularly difficult to analyse in detail the work of Archimedes, an analysis which, truth to tell, seems not to have been undertaken by any modern historian; for because of this fact we are ignorant to what point he was conscious of the lines of relationship which unite the different problems which he treats (links that we will express by saying that the same integral recurs in many places, under varied geometric aspects), and what importance he might have attributed to them. For example, let us consider the following problems, the first solved by Eudoxus, the others by Archimedes: the volume of the pyramid, the area of a segment of the parabola, the centre of gravity of the triangle, and the area of the spiral of Archimedes so-called ($\rho = c\omega$ in polar co-ordinates); they all depend on the integral $\int x^2 dx$, and, without in any way distancing ourselves from the spirit of the method of exhaustion, they can all be reduced to the calculation of “Riemann sums” of the form $\sum_n an^2$. It is in this way in fact, that Archimedes deals with the spiral ([5 b], v. II, pp. 1- 121), by means of a lemma that reduces to writing

$$N^3 < 3 \sum_{n=1}^N n^2 = N^3 + N^2 + \sum_{n=1}^N n < (N + 1)^3.$$

As for the centre of gravity of the triangle, he proves (by exhaustion, by means of a decomposition into parallel strips) that it can be found on each median, hence at their meeting point ([5 b], v. II, pp. 261-315). For the parabola he gives three procedures: one, heuristic, meant only to “give some likelihood to the result”, reduces the problem to the centre of gravity of the triangle, by an argument from statics during which he does not hesitate to consider the segment of the parabola as the sum to infinity of segments of straight lines parallel to the axis ([5 b], v. II, pp. 435-439); another method relies on an analogous principle, but is set out in full rigour by exhaustion ([5 b], v. II, pp. 261-315); a last proof, extraordinarily ingenious but of less scope,

gives the area sought as a sum of a geometric series by means of particular properties of the parabola. Nothing indicates a relationship between these problems and the volume of the pyramid; it is even specified ([5 b], v. II, p. 8) that the problems relating to the spiral have “nothing in common” with certain others relative to the sphere and the paraboloid of revolution, of which Archimedes had the opportunity of speaking in the same introduction and amongst which there is to be found one (the volume of the paraboloid) which reduces to the integral $\int x dx$.

As can be seen from these examples, and apart from the use of particular contrivances, the principle of exhaustion is the following: by a decomposition into “Riemann sums”, upper and lower bounds are obtained for the quantity under consideration, bounds that are compared directly with the expression given for the quantity, or else with corresponding bounds for an analogous problem which has already been solved. The comparison (which, because of the inability to use negative numbers is necessarily done in two parts) is introduced by the sacramental words: “if not, in fact, it would be either greater or less; assume, if possible, that it be greater, etc.; assume, if possible, that it be smaller, etc.” whence the name of the method “apagogic” or “by reduction to the absurd” (“ἀπαγωγή εἰς ἀδύνατον”) that was given to it by the scholars of the XVIIth century. It is in an analogous form that the determination of the tangent to the spiral is drawn up by Archimedes ([5 b], v. II, pp. 62-76), an isolated result, and the only one that we are able to quote as an ancient source for the “differential calculus” apart from the relatively easy determination of tangents to conics, and a few problems of maxima and minima. If in fact, in what concerns “integration”, an immense field of research was offered to Greek mathematicians, not only by the theory of areas and volumes, but also by statics and hydrostatics, they hardly had, due to the lack of dynamics, the opportunity of seriously tackling differentiation. It is true that Archimedes gives a dynamic definition of his spiral; and, short of knowing how he could have been led to the knowledge of its tangent, we have the right to ask ourselves if he did not have some idea of the composition of movements. But in that case would he not have applied such a powerful method to other problems of the same kind? It is more likely that he must have used some heuristic procedure of passing to the limit that the results known to him concerning conics could have suggested; these, of course, are of an essentially simpler nature, since one can construct the points of intersection of a straight line and a conic, and in consequence determine the

condition for coincidence of these points. As to the definition of the tangent, this is conceived of as a straight line which, in the neighbourhood of a certain point of the curve, leaves the curve entirely on the same side; the existence of it is assumed, and it is also assumed that every curve is made up of convex arcs; under these conditions, in order to show that a straight line is tangent to a curve, certain inequalities must be proved, which of course is done with the most complete accuracy.

From the point of view of rigour, the methods of Archimedes leave nothing to be desired; and, in the XVIIth century, still, when the most scrupulous mathematicians wish to put a result which is adjudged to be particularly delicate entirely beyond doubt, it is an “apagogic” proof that they give of it ([109], v. I, pp. 211-254; French translation, v. III, pp. 181-215 and [244], v. VIII, pp. 249-282). As to their fruitfulness, the work of Archimedes is sufficient testimony. But in order to have the right to see there an “integral calculus”, it would be necessary to give in evidence, through the multiplicity of the geometric appearances, some sketch of the classification of the problems according to the nature of the underlying “integral”. In the XVIIth century, as we will see, the search for such a classification becomes little by little one of the principal concerns of the geometers; if no trace of this is found in Archimedes, is it not a sign that such speculations would have appeared to him as exaggeratedly “abstract”, and that he voluntarily, on the contrary, at each opportunity, kept as near as possible to the specific properties of the figure whose study he was pursuing? And must we not conclude that this admirable work, from which the integral calculus, as admitted by its creators, is entirely drawn, is in some sense the opposite of the integral calculus?

It is not with impunity, besides, that one can, in mathematics, allow a ditch to be dug between discovery and proof. In favourable times, the mathematician, without abandoning rigour, only has to write down his ideas almost as he conceives of them; sometimes even he can hope to do it so that it would be thus, at the cost of a felicitous change in the language and notations assumed. But often he must resign himself to choosing between incorrect and possibly fruitful methods of exposition, and correct methods that no longer allow him to express his thought except by distorting it and at the cost of a tiring effort. Neither one nor the other way is exempt from dangers. The Greeks followed the second, and it is perhaps there, even more than in the

sterilising effect of the Roman conquest, that the reason for the surprising halt in their mathematics almost immediately after its most brilliant flowering must be sought. It has been suggested, without it being unlikely, that the oral teaching of the successors of Archimedes and Apollonius could have contained many a new result without their having thought it necessary to inflict on themselves the extraordinary effort required for a publication which conformed to the received canons. It is no longer such scruples in any case that stopped the mathematicians of the XVIIth century, when, faced with the new problems which were being stated in droves, they sought in assiduous study of the writings of Archimedes the means to bypass it.

While the great classics of the Greek literature and philosophy were all printed in Italy, by Alde Manuce and his followers, and almost all before 1520, it is only in 1544, and with Hervagius at Basle, that the earliest edition of Archimedes appears, both Greek and Latin [5 a], without any previous Latin publication having appeared to prepare its way; and, far from the mathematicians of the time (absorbed as they were by their algebraic researches) having immediately felt its influence, we must await Galileo and Kepler, both of them astronomers and physicists rather than mathematicians, for this influence to become manifest. From this moment, and unceasingly until about 1670, there is no name, in the writings of the founders of the infinitesimal Calculus, which recurs more often than that of Archimedes. Many translate and comment on him; all from Fermat to Barrow, quote him with envy; all declare finding there at the same time a model and a source of inspiration.

It is true that these declarations, as we will see, must not all be taken to the letter; there is to be found one of the difficulties that we come up against in trying to get a true interpretation of these writings. The historian must take account also of the organisation of the scientific world of the time, very defective still at the beginning of the XVIIth century, whereas at the end of the same century, by means of the creation of scholarly societies and scientific periodicals, by means of the consolidation and development of the universities, it ends up by resembling strongly what we know today. Deprived of all periodicals until 1655, mathematicians did not have the choice, in order to make their work known, of anything other than by way of letters, and the printing of a book, most often at their own cost, or at the cost of a patron if one could be found. The editors and printers capable of work of this sort were rare, often rather uncertain. After the long delays

and the innumerable troubles that a publication of this kind implied, the author had most often to face up to interminable controversies, provoked by adversaries who were not always in good faith, and carried on sometimes in a surprising bitterness of tone: for, in the general uncertainty in which even the principles of the subject of the infinitesimal calculus was to be found, it was not difficult for each person to find weak points, or at least obscure and contestable, in the arguments of his rivals. It is understandable that under these conditions many scholars keen on peace contented themselves with communicating to a few chosen friends their methods and their results. Some, and especially certain science amateurs, such as Mersenne in Paris, and later Collins in London, kept up a vast correspondence in all countries, in which they communicated extracts here and there, not without mixing in with these extracts stupidities of their own vintage. Possessors of “methods” that, for lack of notions and general definitions, they could not draw up in the form of theorems nor even formulate with some accuracy, mathematicians were thus reduced to making the attempt by means of large numbers of particular cases, and believed that it was not possible to do better, in order to measure its power, than to throw out challenges to their colleagues, accompanied sometimes by publication of their own results in code. The studious youth journeyed, and more perhaps than today; and the ideas of such a scholar were spread sometimes better as a result of the journeys of his pupils than by his own publications, but not without having there another cause of misunderstandings. Finally, the same problems occurring necessarily to a large number of mathematicians, among whom many very distinguished ones, who had only an imperfect knowledge of the results of the one of the others, claims for priority could not fail to arise ceaselessly, and it was not rare that accusations of plagiarism would be added on.

It is therefore in the letters and private papers of the scholars of the time, as much or even more than in their publications proper, that the historian must seek his documents. But, whereas those of Huygens for example have been kept and have been the object of exemplary publication [169 b] , those of Leibniz have only been published in a fragmentary manner so far, and many others are lost without hope. At least the most recent research, based on the analysis of manuscripts, has brought out, in a way that seems irrefutable, a point that the partisan quarrels had somewhat obscured: it is that, each time that one of the great mathematicians of that time

gave an account of his own work, of the evolution of his thought, of the influences that he came under and those that he did not come under, he has done it in an honest and sincere manner, and in all good faith; ¹ these precious testimonies, of which we possess a fairly great number, can thus be used in all confidence, and the historian does not need in their case to transform himself into an examining magistrate. For the remainder, most of the questions of priority that have arisen are totally lacking in sense. It is true that Leibniz, when he adopted the notation dx for the “differential”, was ignorant of the fact that Newton, for about ten years, had been using $\dot{x}$ for the “fluxion”: but what would it have mattered if he had known? To take a more instructive example, who is the author of the theorem $\log x = \int \frac{dx}{x}$, and what is the date thereof? The formula, as we have just written it, is from Leibniz since both one and the other members are written in his notation. Leibniz himself, and Wallis, attribute it to Grégoire de Saint-Vincent. This latter, in his *Opus Geometricum* [135] (appeared in 1647, but drawn up, he says, long before), proves only the equivalent of what follows: if $f(a, b)$ denotes the area of the hyperbolic segment $a \leq x \leq b, 0 \leq y \leq A/x$, the relation $b'/a' = (b/a)^n$ implies $f(a'/b') = n.f(a, b)$; to which his pupil and commentator Sarasa adds almost immediately [269] the remark that the areas $f(a, b)$ can therefore “take the place of logarithms”. If he says no more, and if Grégoire himself had said nothing about it, is it not because, for the majority of mathematicians of the time, logarithms were “calculation aids” with no right to be quoted in mathematics? It is true that Torricelli, in a letter of 1644 [208], speaks of his research on a curve that we would write as $y = ae^{-cx}, x \geq 0$, while adding that where Napier (whom he covers with praise besides) “was only after the arithmetical uses”, he himself “drew out of it a speculation on geometry”; and he had left a manuscript about this curve evidently prepared for publication, but which remained unpublished until 1900 ([310], v. I, pp. 335-347). Descartes had besides encountered the same curve already in 1639 in connection with the “problem of Debeune”, and had described it without speaking of logarithms ([85 a], v. II, pp. 514-517). Whatever, J. Gregory, in 1667, gives, without quoting anybody whatever, ([136 a], reproduced in [169 a], pp. 407-462), a rule for calculating the areas of hyperbolic segments by means of (decimal) logarithms: which implies at the same time a theoretical knowledge of the link between the quadrature of the hyperbola and logarithms, and the numerical knowledge of the link between “natural” and “decimal” logarithms.

Is it about this last point only that the claim of Huygens applies, who contests immediately the novelty of the result of Gregory ([169 b], v. VI, pp. 228-230)? It is that which is no clearer to us than to his contemporaries; these in any case had the clear impression that the existence of a link between logarithms and the quadrature of the hyperbola was something that had been known for a long time, without their being able on this to refer to anything other than epistolary allusions or to the book of Grégoire de Saint-

Vincent. In 1668, when Brouncker gives (with a polished proof of convergence, by comparison with a geometric series) series for $\log 2$ and $\log (5/4)$ ([38]: reproduced in [218], v. I, pp. 213-218), he presents them as expressions of the corresponding segments of a hyperbola, and adds that the numerical values that he obtains are “in the same ratio as the logarithms” of 2 and of 5/4. But in the same year, with Mercator ([220], reproduced in [218], v. I, pp. 167-196) or more exactly in the exposition given immediately by Wallis of the work of Mercator ([326 b], reproduced in [218], v. I, pp. 219-226), the language changes: since the segments of a hyperbola are proportional to logarithms, and that it is well known that logarithms are not defined by their characteristic properties except up to a constant factor, nothing prevents considering the segments of a hyperbola as logarithms, qualified as “natural” (in opposition to the “artificial” or “decimal” logarithms), or hyperbolic; this last step taken (to which the series for $\log(1+x)$ contributes, given by Mercator), the theorem $\log x = \int dx/x$ is obtained, up to notation, or rather it became a definition. What to conclude, if not that it is by almost imperceptible transitions that this discovery was made, and that a dispute about priorities on this matter resembles strongly a quarrel between the violin and the trombone as to the exact moment when a particular motif appears in a symphony? And truth to tell, while at the same time other mathematical creations, the arithmetic of Fermat, the dynamics of Newton, carried a strongly individual cachet, it is very much of the gradual and inevitable development of a symphony, where the “Zeitgeist”, at the same time composer and conductor, holds the baton, that we are reminded by the development of the infinitesimal calculus in the XVIIth century: each undertakes his part with characteristic timbre, but no one is master of the themes that he is creating for the listener, themes that a scholarly counterpoint has almost inextricably entwined. It is thus under the form of a thematic analysis that the history of this must be written; we will be content here with a summary sketch,

and which does not pretend to a meticulous exactitude.² Here in any case are the principal themes that a superficial examination brings out:

- A) The theme of mathematical rigour, contrasting with that of the infinitely small, indivisibles or differentials. It has been seen that both hold an important place with Archimedes, the first in all his work, the second only in the treatise on his Method, that the XVIIth century did not know, so that, if it had been transmitted and not reinvented, it could only have been by philosophical tradition. The principle of the infinitely small appears besides in two distinct forms depending on whether it is a case of “differentiation” or of “integration”. As for this one, let it be first a case of calculating a plane area: it will be divided into an infinity of parallel strips infinitely small, by means of an infinity of equidistant parallels; and each of these strips is a rectangle (even though none of the finite strips that are obtained by means of two parallels at a finite distance apart would be a rectangle). In the same way, a solid of revolution would be decomposed into an infinity of cylinders of the same infinitely small height, by planes perpendicular to the axis; ³ analogous ways of speaking could be used when it is a case of decomposing an area into triangles by means of concurrent straight lines, or to argue about the length of an arc of a curve as if it were the case of a polygon with an infinity of sides, etc.. It is certain that the rare mathematicians who possessed fully the ability to handle the methods of Archimedes, such as Fermat, Pascal, Huygens, Barrow, could, in each particular case, have found no difficulties in replacing the use of this language by rigorous proofs; also they often remark that this language is only there for conciseness. “It would be easy”, says Fermat, “to give proofs in the manner of Archimedes ... ; this being said, it is sufficient to have been warned once for all in order to avoid continual repetitions ...” ([109], v. I, p. 257); in the same way Pascal: “thus one of these methods only differs from another in the way of stating it” ([244], v. VIII, p. 352); ⁴ and Barrow, with his mocking conciseness: “longior discursus apagogicus adhiberi possit, sed quorsum?” (it would be possible to lengthen this by an apagogic discourse, but to what end?) ([16 b], p. 251). Fermat is very careful even, it would appear, not to bring forward anything that he could not justify thus, and condemns himself by that to announcing no general result except by allusion or in the form of a “method”; Barrow, although so careful,

is somewhat less scrupulous. As for the majority of their contemporaries, it can be said at the very least that rigour is not their main concern, and that the name of Archimedes is most often only a dissembling meant to cloak merchandise of great price no doubt, but for which Archimedes would certainly not have assumed responsibility. All the more so is it the case where differentiation is concerned. If the curve, when it is a case of its rectification, is assimilated to a polygon with an infinity of sides, it is a case here of an “infinitely small” arc of the curve which is assimilated to an “infinitely small” segment of a straight line, either the chord, or a segment of the tangent whose existence is assumed; or again it is an “infinitely small” time interval that is being considered, during which (so long as it is a case only of the speed) the movement “is” uniform; still bolder, Descartes, wishing to determine the tangent of a cycloid which does not come under his general rule, assimilates curves rolling one on the other to polygons, so as to conclude that “in the infinitely small” the movement can be assimilated to a rotation about a point of contact ([85 a], v. II, pp. 307-338). Here again, a Fermat, who bases his rules for tangents and for maxima and minima on such infinitesimal considerations, is able to justify them in each particular case ([109], v. I, pp. 133-179; French translation, v. III, pp. 121-156); cf. also ([109], v. II, *passim*, in particular pp. 154-162, and *Supplément aux Oeuvres* pp. 72-86); Barrow gives precise proofs for a large part of his theorems, in the manner of the ancients, starting with simple hypotheses of monotonicity and convexity. But the time was already over for pouring new wine into old wineskins. In all that, as we know today, it was the notion of limits that was being developed; and, if it is possible to extract from Pascal, Newton and from others as well, statements that seem very close to our modern definitions, it is only necessary to put them back in their context to realise the irremovable obstacles that were in the way of a rigorous exposition. When starting in the XVIIIth century some mathematicians caught up by the need for clarity wished to put some order into the confused heap of their riches, such indications, encountered in the writings of their predecessors, were precious to them; when d’Alembert for example explains that there is nothing else in differentiation other than the notion of limit, and defines this with accuracy [75 b], it is possible to believe that he was guided by the considerations of Newton on the “first and last reasons of vanishing

quantities” [233 b]. But, so long as it is a case of the XVIIth century, it must be realised that the way was not open for modern analysis until Newton and Leibniz, turning their back on the past, accept that they must seek provisionally the justification for their new methods, not in rigorous proofs, but in the fruitfulness and coherence of the results.

- B) The dynamic. Already Archimedes, we have seen, gave a dynamic definition of his spiral; and in the Middle Ages a rudimentary theory of the variation of quantities as a function of time is developing (but proof to the contrary lacking, without infinitesimal considerations), and of their graphical representation, which must perhaps have had its origins in Babylonian astronomy. But it is of the greatest importance for the mathematics of the XVIIth century that, from the beginning, the problems of differentiation arose, not only in connection with tangents, but in connection with velocities. Galileo however ([122 a and b]), seeking the law of velocities in the falling of weights (after having obtained the law of distances $x = at^2$, by means of an inclined plane), does not proceed by using differentiation: he makes various hypotheses about the velocity, first $v = \frac{dx}{dt} = cx$ ([122 b], v. VIII, p. 203), then later $v = ct$ (*id.*, p. 208), and seeks to find again the law of distances by reasoning, in a fairly obscure way, about the graph of the velocity as a function of time; Descartes (in 1618) argues in the same way, about the law $v = ct$, but as a true mathematician and with as much clarity as the language of indivisibles can muster⁵ ([85 a], v. X, pp. 75-78); with both, the graph of velocity (in a straight line) plays the main role, and there is reason to ask to what point they were aware of the proportionality between the distances covered and the areas bounded by the axis of time and the curve of the velocity; but it is difficult to say anything on this point, although the language of Descartes seems to imply knowledge of the fact in question (that certain historians wish to put back to the Middle Ages [335]), whereas Galileo does not make any clear remark about it. Barrow states it explicitly in 1670 ([16 b], p. 171); perhaps at that time it was no longer a novelty for anyone, and Barrow does not present it as such; but, no more for this result as for any other, is it appropriate to give a date with too much accuracy. As for the hypothesis $v = cx$, considered also by Galileo, he is happy (*loc. cit.*) to prove that it is not sustainable (or, in modern language, that the equation $\frac{dx}{dt} = cx$ has no solution $\neq 0$ that is zero for $t = 0$

), by an obscure argument that Fermat later ([109], v. II, pp. 267-276) takes the trouble of developing (and which amounts just about to saying that, $2x$ being a solution as well as x , $x \neq 0$ would be contrary to the obvious physical uniqueness of the solution). But this same law $\frac{dx}{dt} = cx$ which, in 1614, was used by Napier to introduce his logarithms, of which he gives a dynamic definition [230 a], which, in our notation, would be written as follows: if, on two straight lines, two bodies are in motion following the laws $\frac{dx}{dt} = a$, $\frac{dy}{dt} = -ay/r$, $x_0 = 0$, $y_0 = r$, then one says that x is the “logarithm” of y (in modern notation, we have $x = r \log(r/y)$). It has been seen that the curve which is a solution of $\frac{dy}{dx} = \frac{c}{x}$ appeared in 1639 with Descartes, who described it dynamically ([85 a], v. II, pp. 514-517); it is true that he qualified fairly disdainfully as “mechanics” all the non algebraic curves, and claimed to exclude them from geometry; but happily this taboo, against which Leibniz, much later, believes he still has to protest vigorously, was not observed by his contemporaries, nor by Descartes himself. The cycloid, the logarithmic spiral appear and are studied with eagerness, and their study is a powerful help in the interlacing of geometric and dynamic methods. The principle of composition of movements, and more precisely of velocities, was at the basis of the theory of movement of projectiles stated by Galileo in the chef-d’oeuvre of his old age, the Discorsi of 1638 ([122 b], v. VIII, pp. 268-313) a theory that contains thus implicitly a new determination of the tangent to the parabola; if Galileo does not expressly state it, Torricelli on the contrary ([310], v. III, pp. 103-159) insists on this point, and even bases on this same principle a general method of determination of tangents for curves which are capable of a dynamic definition. It is true that he was forestalled in this, by several years, by Roberval ([263], pp. 3-67) who says that he was led to this method by the study of the cycloid; this same problem of the tangent to the cycloid gives besides the opportunity to Fermat of proving the power of his method of differentiation ([109], v. I, pp. 162-165), whereas Descartes, incapable of applying to it his algebraic method, invents for the occasion the instantaneous centre of rotation ([85 a], v. II, pp. 307-338).

But, as the infinitesimal calculus is developed, dynamics ceases to be a separate science. It is noticed more and more that in spite of Descartes, algebraic curves and functions do not

have, from the “local” point of view which is that of the infinitesimal calculus, anything that distinguishes them from much more general functions; the functions and curves with a dynamic definition are functions and curves like the others, accessible to the same methods; and the variable “time” is no more than a parameter, whose temporal aspect is purely a matter of language. So with Huygens, even when it is a matter of mechanics, it is geometry that dominates ([169 b], v. XVIII); and Leibniz does not give to time any privileged place in his calculations. On the contrary Barrow conceives of making, of the simultaneous variation of several quantities as functions of a universal independent variable conceived of as “time”, the foundation of an infinitesimal calculus with a geometric tendency. This idea, which must have come to him as he sought to find again the method of composition of movements that he knew existed only by hear-say, is stated in detail, in very clear and very general terms, in the first three of his *Lectiones Geometricae* [16 b]; and for example he proves there with care that, if a moving point has as projections, on two rectangular axes AY , AZ , moving points such that one is moving with a constant speed a and the other with a speed v that grows with time, the trajectory has a tangent with slope equal to v/a , and is concave in the direction of increasing Z . In the following parts of the *Lectiones*, he pursues these ideas very far, and although he puts a sort of affectation in drawing them up almost end to end in a form that is as geometric and as sparingly algebraic as possible, it is permissible to see there, with Jakob Bernoulli ([19 a], v. I, pp. 431 and 453), the equivalent of a good part of the infinitesimal calculus of Newton and Leibniz. Exactly the same ideas form the point of departure for Newton ([233 a], v. I, pp. 201-244 and [233 b]): his “fluents” are the various quantities, functions of “time” which is only a universal parameter and the “fluxions” are the derivatives with respect to “time”; the possibility that Newton allows himself, of changing at need the parameter, is equally present in the method of Barrow, although used by him with less suppleness. ⁶ The language of fluxions, adopted by Newton and imposed by his authority on English mathematicians of the following century, represents thus the last outpost, for the period that has our attention, of the dynamic methods whose true role was already finished.

C) Algebraic Geometry. This is a parasite theme, a stranger to our subject, which is introduced because Descartes, by his systematic approach, claimed to make of algebraic curves

the exclusive object of geometry ([85 a], v. VI, p. 390); also it is a method of algebraic geometry, and not like Fermat a method from the differential calculus, that he gives for the determination of tangents. The results bequeathed by the ancients on the intersection of a straight line and a conic, the thoughts of Descartes himself on the intersection of two conics and the problems that reduce to that, were going to lead him quite naturally to the idea of taking as criterion for contact the coincidence of the two intersections: we know today that in algebraic geometry it is the correct criterion, and of such great generality that it is independent of the concept of limit and of the nature of the “base field”. Descartes applies it first in a manner that is hardly convenient, in seeking to make two intersections of the curve under study and of a circle having its centre on Ox coincide in a given point ([85 a], v. VI, pp. 413-424); his disciples, van Schooten, Hudde, substitute a straight line for the circle, and obtain in the form $-F'_x/F'_y$ the slope of the tangent to the curve $F(x, y) = 0$, the “derived polynomials” F'_x, F'_y being defined by their formal rule of formation ([198 d], v. I, pp. 147-344 and [85 b], pp. 234-237); de Sluse also reaches this result at about the same time ([198 d], pp. 232-234). Of course the clear-cut distinctions that we draw out here, and which alone give meaning to the controversy between Descartes and Fermat, could in no way exist in the minds of mathematicians of the XVIIth century: we have only mentioned them in order to make clearer one of the more curious episodes in the history which concerns us, and to note almost immediately after the complete eclipse of algebraic methods, provisionally absorbed by differential methods.

- D) Classification of problems. This theme, we have seen, seems absent from the work of Archimedes, to whom it appears to be a matter of indifference to solve a problem or to reduce it to a problem already treated. In the XVIIth century, the problems of differentiation appear at first under three distinct aspects: speed, tangents, maxima and minima. As for these last, Kepler [179 a and b] makes the observation (that is already found with Oresmus ([335], p. 141) and which had not even escaped the Babylonian astronomers) that the variation of a function is particularly slow in the neighbourhood of a maximum. Fermat, already before 1630 ([109], v. II, pp. 71 and 113-179), inaugurates in connection with such problems his infinitesimal method, which in modern language reduces in fact

to seeking the first two terms (the constant term and the first order term) in the Taylor expansion, and to writ-

ing that at an extremum the second is zero; he goes from there to extend his method to the determination of tangents, and even applies it to the search for points of inflexion. If one takes account of what has been said above in connection with dynamics, it can be seen that the unification of the three types of problems relative to the first differential was accomplished at a fairly early stage. As for problems relative to the second differential, they only appear quite late, and especially with the works of Huygens on the evolute of a curve (published in 1673 in his *Horologium Oscillatorium* ([169 b], v. XVIII)); at this time, Newton, with his fluxions, was already in possession of all the analytical means necessary for solving such problems; and, in spite of all the geometric talent that Huygens expends on it (and of which later differential geometry would profit in its beginnings), they were hardly used for anything else, during the period with which we are concerned, than to allow the new analysis to become conscious of the power of its means.

As for integration, it had appeared to the Greeks as the calculation of areas, of volumes, of moments, as the calculation of the length of a circle and of areas of spherical segments; to which the XVIIth century adds the rectification of curves, the calculation of the area of surfaces of revolution, and (with the work of Huygens on the compound pendulum ([169 b], v. XVIII)) the calculation of moments of inertia. It was a case first of recognising the link between all these problems. For the areas and the volumes, this first and immense step forward is made by Cavalieri, in his *Geometry of indivisibles* [57 a]. He announces there, and claims to prove, the following principle more or less: if two plane areas are such that every parallel in a given direction intersects them in segments whose lengths are in a constant ratio, then these areas are in the same ratio; an analogous principle is stated for the volumes intersected by planes parallel to a fixed plane where the areas are in a constant ratio. It is likely that these principles were suggested to Cavalieri by theorems such as that of Euclid (or rather Eudoxus) on the ratio of volumes of pyramids with the same height, and that before stating them in a general form, he first verified their validity with a large number of examples taken from Archimedes. He “justifies” them by the use of a language, about whose legitimacy we see him asking Galileo in a letter of 1621,

whereas in 1622 already he uses it without hesitation ([122 b], v. XIII, pp. 81 and 86) and of which here is the essential part. Let there be for example two areas, one $0 \leq x \leq a, 0 \leq y \leq f(x)$, the other $0 \leq x \leq a, 0 \leq y \leq g(x)$; the sums of the ordinates $\sum_{k=0}^{n-1} f(ka/n), \sum_{k=0}^{n-1} g(ka/n)$ are one to the other in a ratio that, for n sufficiently large, is arbitrarily near to the ratio of two areas, and it would even be not difficult to prove it by exhaustion for f and g monotonic; Cavalieri goes to the limit, makes $n = \infty$, and speaks of “the sum of all the ordinates” of the first curve, which is to the analogous sum for the second curve in a ratio rigorously equal to the ratio of the areas; in the same way for the volumes; and this language is afterwards universally adopted, even by authors, like Fermat, who have the clearest realisation of the precise facts that it covers. It is true that subsequently many mathematicians, such as Roberval ([163], pp. 3-67) and Pascal ([244], v. VIII, pp. 334-384), prefer to see, in the ordinates of the curve whose “sum” is being taken, not segments of a straight line like Cavalieri, but rectangles of the same infinitely small height, which is not great progress from the point of view of rigour (whatever is said by Roberval), but perhaps prevents the imagination from going off the rails too easily. In any case, and as it is a case only of ratios, the expression “the sum of all the ordinates” of the curve $y = f(x)$, or in short “all the ordinates” of the curve, is definitively, as it appears clearly for example in the writings of Pascal, the exact equivalent of the $\int y dx$ of Leibniz.

From the language adopted by Cavalieri, the principles stated higher follow inevitably, and imply immediately consequences that we will state in modern notation, it being understood that there $\int f dx$ will only mean the area bounded by Ox and the curve $y = f(x)$. First of all, every plane area, intersected by each straight line $x = \text{constant}$ in segments the sum of whose lengths is $f(x)$, is equal to $\int f dx$; and it is the same with every volume intersected by each plane $x = \text{constant}$ in an area of value $f(x)$. What is more, $\int f dx$ depends linearly on f ; we have $\int (f + g) dx = \int f dx + \int g dx$, $\int cf(x) dx = c \int f dx$. In particular all the problems of area and volume are reduced to quadratures, that is to say to calculations of area of the form $\int f dx$; and, what is perhaps even more novel and important, two problems depending on the same quadrature must be considered equivalent, and there is the means in each case to decide if it is so. The Greek mathematicians never attained (or perhaps never consented to attain) such a degree of “abstraction”. Thus ([57 a], p. 133) Cavalieri “proves” without any trouble that two

similar volumes are in a ratio equal to the cube of the ratio of similarity, whereas Archimedes only states this conclusion for quadrics of revolution and their segments, at the end of his theory of solids ([5 b], v. I, p. 258). But to get there it was necessary to throw Archimedean rigour overboard.

There was thus there the means of classifying problems, at least provisionally, following the degree of real or apparent difficulty that arose with the quadratures related to them. That is where the algebra of the time was of use as a model: for in algebra also, and in the algebraic problems that were arising in geometry, whereas the Greeks were only interested in the solutions, the algebraists of the XVIth and XVIIth centuries began to turn their attention mainly to the classification of problems according to the nature of the means employed for their solution, thus creating a prelude for the modern theory of algebraic extensions; and they had not only proceeded to a first classification of problems according to the degree of the equation on which they depended, but they had already stated for themselves difficult questions about possibilities: the possibility of solving every equation by radicals (in which some of them did not believe), etc.; they were preoccupied also with reducing to a type of geometric form all the problems of given degree. In the same way in the integral Calculus, the principles of Cavalieri put them in a position to recognise immediately that many of the problems solved by Archimedes reduce to quadratures $\int x^n dx$ for $n=1,2,3$; and he invents an ingenious method for carrying out this quadrature for as many values of n as desired (the method reduces to observing that one has $\int_0^{2a} x^n dx = c_n a^{n+1}$ by homogeneity, and to writing

$$\int_0^{2a} x^n dx = \int_{-a}^a (a+x)^n dx = \int_0^a [(a+x)^n + (a-x)^n] dx,$$

whence, by expanding, a recurrence relation for the c_n) ([57 a], p. 159 and [57 b], pp. 269-273). But already Fermat had gone much farther, by proving first (before 1636) that $\int_0^a x^n dx = \frac{a^{n+1}}{n+1}$ for n a positive integer ([109], v. II, p. 83), by means of a formula for the sums of powers of the first N integers (a procedure imitated from the quadrature of the spiral of Archimedes), then in extending the same formula to all rational $n \neq -1$ ([109], v. I, pp. 195-198); from this last result (communicated to Cavalieri in 1644) he only draws up a proof much later, following his reading of the writings of Pascal on integration ⁷ ([109], v. I, pp. 255-288; French translation v.

III, pp. 216-240).

These results joined to geometric considerations that take the place of change of variables and integration by parts, allow already the solution of a large number of problems that reduce to elementary quadratures. Beyond that, is met first the quadrature of the circle and that of the hyperbola: as it is especially “indefinite integrals” with which they were concerned at that time, the solution of these problems in modern terms, is supplied respectively by the inverse circular functions and by the logarithm; these were given geometrically, and we have seen how this last has been introduced little by little into analysis. These quadratures are the subject of much work, by Grégoire de Saint-Vincent [135], Huygens ([169 b], v. XI, pp. 271-337 and v. XII, pp. 91-180), Wallis ([326 a], v. I, pp. 355-478), Gregory [136 a]; the first believes he has succeeded in the quadrature of the circle, the last believes he has proved the transcendence of π ; with the ones and the others procedures of indefinite approximation are developed for the circular and logarithmic functions, the ones with a theoretical tendency, the others oriented towards numerical calculation, which is going to end soon, with Newton ([233 a], v. I, pp. 3-26 and 29-199), Mercator ([220], reproduced in [218], v. I, pp. 167-196), J. Gregory [136 d], then Leibniz [198 d], with general methods of expansion in series. In any case, the conviction comes to light little by little of the “impossibility” of the quadratures in question, that is to say of the non-algebraic character of the functions that they define; and at the same time, it becomes the custom to consider a problem solved as far as its nature allows, when it has been reduced to one of the “impossible” quadratures. It is the case for example with the problems of the cycloid, solved by means of circular functions, and of the rectification of the parabola, reduced to the quadrature of the hyperbola.

The problems of rectification, of which we have just quoted two of the most famous, had a particular importance, as making up a natural geometric transition between differentiation, which they presuppose, and integration which they come under; problems on the area of surfaces of revolution can be associated with them. The ancients had only treated the case of the circle and the sphere. In the XVIIth century, these questions only appear very late; it appears that the difficulty, insurmountable for the time, of the rectification of the ellipse (considered to be the simplest curve after the circle) had discouraged effort. Dynamic methods give some access

to these problems, which allows Roberval ([263], pp. 363-399) and Torricelli ([310], v. III, pp. 103-159), between 1640 and 1645, to obtain results on the arc of spirals; but it is only in the years that precede 1660 that they become the order of the day; the cycloid is rectified by Wren in 1658 ([326 a], v. I, pp. 533-541); soon after the curve $y^3 = ax^2$ is done by several authors ([326 a], v. I, pp. 551-553; [85 b] v. I, pp. 517-520; [109], v. I, pp. 211-255; French translation, v. III, pp. 181-215), and several authors also ([109], v. I, p. 199; [169 b], v. II, p. 334) reduce the rectification of the parabola to the quadrature of the hyperbola (that is to say to an algebraico-logarithmic function). This latter example is the most important, because it is a particular case of the general principle according to which the rectification of a curve $y = f(x)$ is none other than the quadrature of $y = \sqrt{1 + (f'(x))^2}$; and it is from precisely this principle that Heurat deduces it. It is no less interesting to follow the groping of Fermat in his old age, in his work on the curve $y^3 = ax^2$ ([109], v. I, pp. 211-255; French translation v. III, pp. 181-215); to the curve $y = f(x)$ of arc $s = g(x)$, he associates the curve $y = g(x)$, and determines the tangent to this one starting from the tangent to the first one (in modern language, he proves that their slopes $f'(x), g'(x)$ are linked by the relation $(g'(x))^2 = 1 + (f'(x))^2$); one can believe oneself very close to Barrow, and it would only be necessary to combine this result with that of Heurat (which is more or less what Gregory does in 1668 ([136 d], pp. 488-491)) to obtain the relation between tangents and quadratures; but Fermat states only that if, for two curves each with reference to a system of rectangular axes, the tangents at the points with the same abscissa always have the same slope, the curves are equal, or in other words that the knowledge of $f'(x)$ determines $f(x)$ (up to a constant); and he justifies this assertion only by an obscure argument without any convincing value.

Less than ten years later, the *Lectiones Geometricae* of Barrow [16 b] had appeared. From the beginning (Lect. I), he states in principle that, in a rectilinear movement, the spaces are proportional to the areas $\int_0^t v dt$ bounded by the axis of time and the curve of the velocity. One would believe that he is going to deduce from that, and from his dynamic method already quoted concerning the determination of tangents, the link between the differential conceived as the slope of the tangent, and the integral conceived as the area; but there is none of that, and he proves later, in a purely geometric way ([16 b], Lect. X, §11, p. 243) that, if two curves $y = f(x), Y = F(x)$ are

such that the ordinates Y are proportional to the areas $\int_a^x y dx$, that is to say if $c.F(x) = \int_a^x f(x) dx$, then the tangent to $Y = F(x)$ intersects Ox at the point with abscissa $x - T$ determined by $y/Y = c/T$; the proof besides is perfectly precise, starting with the explicit hypothesis that $f(x)$ is monotonic, and it is said that the direction of variation of $f(x)$ determines the direction of concavity of $Y = F(x)$. But it will be noted that this theorem is somewhat lost amongst a crowd of others, many of which are very interesting; the reader who has not been warned is tempted to see there only a means for solving by quadrature the problem $Y/T = f(x)/c$, that is to say a certain problem of the determination of a curve starting from information on its tangent (or, as we would say, a differential equation of a particular type); and that all the more so as the applications of it given by Barrow are concerned above all with problems of the same kind (that is to say with differential equations integrable by “separation of variables”). The geometric language that Barrow imposes on himself is here a cause why the link between differentiation and integration, so clear as long as it was a case of dynamics, is somewhat obscured.

On the other hand, various methods had taken shape, to reduce problems of integration the ones to the others, and to “solve” them or to reduce them to “impossible” problems that had already been classified. In its simplest geometric form, integration by parts consists in writing the area bounded by Ox , Oy and an arc of the monotonic curve $y = f(x)$ joining a point $(a, 0)$ of Ox to a point $(0, b)$ of Oy as $\int_0^a y dx = \int_0^b x dy$; and it is frequently used in an implicit way. With Pascal ([244], v. IX, pp. 17-18) appears the following generalisation, already much better hidden: $f(x)$ being as above, let $g(x)$ be a function ≥ 0 , and let $G(x) = \int_0^x g(x) dx$; then one has $\int_0^a y g(x) dx = \int_0^b G(x) dy$, which he proves ingeniously by evaluating in two ways the volume of the solid $0 \leq x \leq a, 0 \leq y \leq f(x), 0 \leq z \leq g(x)$; the particular case $g(x) = x^n$, $G(x) = \frac{x^{n+1}}{n+1}$ plays an important role, both with Pascal (loc. cit., pp. 19-21) and with Fermat ([109], v. I, p. 271); this latter (whose work carries the significant title of “Transmutation and emendation of equations of curves, and its varied applications to the comparison of curvilinear spaces between them and with rectilinear spaces ...”) does not prove it, no doubt because he judges it useless to repeat what Pascal had just published. These theorems of “transmutation”, where we would see a combination of integration by parts and changes of variables, take the place of this to some extent, which is only introduced very late; it is in effect contrary to the way of thinking, still

too geometric and too little analytic, of the time, to allow the use of variables other than those imposed by the figure, that is to say one or other of the co-ordinates (or sometimes polar co-ordinates) then the arc of the curve. It is thus that we find with Pascal ([244], v. IX, pp. 60-76) results that, in modern notation, are written, putting $x = \cos t$, $y = \sin t$ and, for particular functions $f(x)$:

$$\int_0^1 f(x)dx = \int_0^{\frac{\pi}{2}} f(x)ydt,$$

and, with J. Gregory ([136 d], p. 489), for a curve $y = f(x)$ and its arc s , $\int yds = \int zdx$, with $z = y\sqrt{1 + y'^2}$. It is only in 1669 that we see Barrow in possession of the general theorem for change variables ([16 b], pp. 298-299); its statement, geometric as always, reduces to what follows: let x and y be linked by a monotonic relation, and let p be the slope of the graph of this relation at the point (x, y) ; then, if the functions $f(x), g(y)$ are such that one has $f(x)/g(y) = p$ for every corresponding pair of values (x, y) , the areas $\int f(x)dx, \int g(y)dy$, taken between corresponding limits, are equal; and conversely, if these areas are always equal (f and g being implicitly supposed to have constant sign), we have $p = f(x)/g(y)$; the converse is naturally used to apply the theorem to the solution of differential equations (by “separation of variables”). But the theorem is only inserted by Barrow in an appendix (Lect. XII, app. III, theor. IV), where, by making the observation that several of his preceding results are only particular cases of it, he excuses himself as having discovered it too late to make more use of it.

Thus, around 1670, the situation is as follows. It is known how to treat, by uniform procedures, the problems that arise with the first derivative, and Huygens has tackled geometric questions that arise with the second derivative. It is known how to reduce all problems of integration to quadratures; there are available various techniques, of a geometric aspect, to reduce quadratures the one to the others, in badly classified cases, and it has become customary, from this point of view, to handle circular and logarithmic functions; there is a consciousness of the link between differentiation and integration; the “inverse method of tangents” is beginning to be tackled, a name given at this time to problems that reduce to differential equations of the first order. The sensational discovery of the series $\log(1 + x) = -\sum_1^{\infty} (-x)^n/n$ by Mercator has just opened up quite new perspectives for the application of series, and mainly power se-

ries, to problems called “impossible”. By contrast the ranks of mathematicians are significantly thinned: Barrow has left the chair of professor for that of preacher; Huygens apart (who has almost all his mathematical work behind him, having already obtained all the principal results of the *Horologium Oscillatorium* that he is setting himself to draw up definitively), only Newton at Cambridge, and J. Gregory at Aberdeen are active, whom Leibniz will join soon with the ardour of a neophyte. All three, Newton from 1665 already, J. Gregory from the publication of Mercator in 1668, Leibniz from about 1673, devote themselves principally to the subject in fashion, the study of power series. But, from the point of view of the classification of problems, the principal effect of the new methods seems to be to obliterate all distinctions between them; and in effect Newton, more of an analyst than an algebraist, does not hesitate to announce to Leibniz in 1676 ([198 d], p. 224) that he knows how to solve all differential equations, ⁸ to which Leibniz responds ([198 d], pp. 248-249) that on the contrary it is a case of obtaining the solution in a finite number of terms whenever it is possible “presupposing the quadratures”, and also of knowing if every quadrature can be reduced to those of the circle and the hyperbola as had been noticed in the majority of cases studied so far; he brings out again in this connection that Gregory believed (with reason, as we know today) that the rectification of the ellipse and the hyperbola is irreducible to the quadratures of the circle and the hyperbola; and Leibniz asks to what point the method of series, such as Newton uses it, can give an answer to these questions. Newton, on his side ([198 d], pp. 209-211), declares that he is in possession of criteria, that he does not indicate, for deciding, apparently by the examination of series, about the “possibility” of certain quadratures (in finite terms), and gives an example of a (very interesting) series for the integral $\int x^\alpha (1+x)^\beta dx$.

The immense progress that has been realised in ten years can be seen; questions of classification are stated in these letters in quite modern terms; if one of those that Leibniz raises has been solved in the XIXth century by the theory of abelian integrals, the other, about the possibility of reducing a given differential equation to quadratures, is still open despite important recent work. If it is thus, it is because already Newton and Leibniz, each on his own account, have reduced to an algorithm the fundamental operations of the infinitesimal calculus; it is sufficient to write, in the notations which one or the other uses, a problem on quadratures or differential

equations, in order for its algebraic structure to appear immediately, freed from its geometric crust; the methods of “transmutation” also can be written in simple analytical terms; the problems of classification are stated in a precise way. Mathematically speaking, the XVIIth century has reached its end.

E) Interpolation and the calculus of differences. This theme (from which we will not separate the study of binomial coefficients) appears early and continues throughout the century, for reasons that are both theoretical and practical. One of the big tasks of the time is in fact the calculation of trigonometric, logarithmic, nautical tables made necessary by the rapid progress of geography, navigation, theoretical and practical astronomy, physics, celestial mechanics; and many of the most eminent mathematicians, from Kepler to Huygens and Newton, take part, either directly, or by means of theoretical research on the procedures for the most efficient approximations.

One of the first problems, in the use and even the construction of the tables, is that of interpolation; and, as the precision of the calculations grows, it is noticed in the XVIIth century that the ancient procedure of linear interpolation loses its validity as soon as the first differences (differences between the successive values which occur in the table) cease to be effectively constant; so Briggs for example is seen ([36], chap. 13) making use of differences of higher orders, and even of fairly high order, in the calculation of logarithms. Later, we see Newton ([233 a], v. I, pp. 271-282 and [233 b], book III, lemma 5; see also [114]) and J. Gregory ([136 d], pp. 119-120), each on his own account, pursuing parallel researches on interpolation and on power series; both one and the other end up, by methods which are different besides, on the one hand with the formula for interpolation by polynomials, called “of Newton”, and on the other with the binomial series ([136 d], p. 131; [198 d], p. 180) and with the principal expansions in series of classical analysis ([136 d]; [233 a], v. I, pp. 3-26 and 271-282 and [198 d], pp. 179-192 and 203-225); it is hardly in doubt that these two lines of research reacted one on the other, and were intimately linked in the mind of Newton with the discovery of the principles of the infinitesimal calculus. With Gregory as with Newton a great concern for numerical practice, the construction and use of tables, the numerical calculation of series and integrals comes to light; in particular, although no polished proof of convergence is found in their work, in the way of

that of Lord Brouncker quoted above, both constantly mention convergence of their series from the practical point of view of their aptitude for calculation. It is still thus that we see Newton, in reply to a question set by Collins to a practical end ([262], v. II, pp. 309-310), applying to the approximate calculation of $\sum_{p=1}^N \frac{1}{n+p}$, for large values of N , a particular case of the method of summation of Euler-Maclaurin so-called.

The calculation of the values of a function starting from their differences is also met early on, used as a practical procedure for integration, and even, it might be said, for the integration of a differential equation. Thus Wright, in 1599, having to solve, with nautical tables in view, a problem that we would denote by $\frac{dx}{dt} = \sec t = \frac{1}{\cos t}$, proceeds by adding the values of $\sec t$, by successive intervals of a second of arc ([338]; cf. [230 b], p. 97), naturally obtaining to all intents and purposes a table for the values of $\log \tan(\frac{\pi}{4} + \frac{t}{2})$; and this coincidence, observed from the calculations of the first tables of $\log \tan t$, remained unexplained until the integration of $\sec t$ by Gregory in 1668 ([136 c] and [136 d], pp. 7 and 463).

But these questions also have a purely theoretic and even arithmetic aspect. Let us agree to denote by $\Delta^r x_n$ the sequence of successive differences of a sequence $(x_n)_{n \in \mathbb{N}}$, defined by recurrence by means of $\Delta x_n = x_{n+1} - x_n$, $\Delta^r x_n = \Delta(\Delta^{r-1} x_n)$, and to denote by S^r the inverse operation to Δ and its iterates, therefore stating $y_n = Sx_n$ if $y_0 = 0$, $\Delta y_n = x_n$ and $S^r x_n = S(S^{r-1} x_n)$; we have

$$S^r x_n = \sum_{p=1}^{n-r} \binom{n-p-1}{r-1} x_p;$$

in particular, if $x_n = 1$ for all n , we have $Sx_n = n$, the sequences $S^2 x_n$ and $S^3 x_n$ are those of the “triangular” and “pyramid” numbers already studied by Greek arithmeticians, and in general we have

$$S^r x_n = \binom{n}{r} \text{ for } n \geq r \text{ (and } S^r x_n = 0 \text{ for } n < r);$$

these sequences were introduced, from this point of view, in any case from the XVIth century; they appear by themselves also in combinatorial problems, which, either on their own, or in connection with probabilities, played a fairly big role in the mathematics of the XVIIth century,

for example with Fermat and Pascal, then with Leibniz. They also turn up in the expression for the sum of the m th powers of the first N integers, whose calculation, as we have seen, is at the basis of the integration of $\int x^m dx$ for m an integer, by the first method of Fermat ([109], v. II, p. 83). It is thus that Wallis proceeds in 1655 in his *Arithmetica Infinitorum* ([326 a], v. I, pp. 355-478), without knowing of the work (unpublished) of Fermat, and without knowing either, he says, of the method of indivisibles other than from the lecture of Torricelli; it is true that Wallis, keen to reach the end, does not slow himself down by meticulous research: once the result obtained for the first integer values of m , he states it to be true “by induction” for all integers m , goes on correctly from there to $m = 1/n$ for n an integer, then, by an “induction” even more summary than the first, to m an arbitrary rational. But what is interesting and original in his work, is that he progressively lifts himself from there to the study of the “Eulerian” integral $I(m, n) = \int_0^1 (1 - x^{\frac{1}{m}})^n dx$ (whose value, for m and $n > 0$, is $\Gamma(m+1)\Gamma(n+1)/\Gamma(m+n+1)$) and other such, draws up, for m and n integers, the table of values of $1/I(m, n)$ which is none other than that of the integers $\binom{m+n}{n}$, and, by methods almost identical to those in use today for expounding the theory of the Γ function, ends up with the infinite product for $I\left(\frac{1}{2}, \frac{1}{2}\right) = \frac{\pi}{4} = \left(\Gamma\left(\frac{3}{2}\right)\right)^2$; it is not difficult besides to make the method correct by means of integration by parts and very simple changes of variables, and by considering $I(m, n)$ for all real values of m and n , which he could hardly have thought of, but which the Newtonian analysis would soon make possible. It is in any case the “interpolation” carried out by Wallis of the integers $\binom{m+n}{n}$ to non integral values of m (more precisely to values of the form $n = p/2$, with p an odd integer) which provides a point of departure for Newton when he starts out ([198 d], pp. 204-206), bringing him, first to the study of a particular case $(1 - x^2)^{p/2}$, to the binomial series, from there to the introduction of x^a (denoted thus) for all real a , and to the differentiation of x^a by means of the binomial series; all that without great effort for obtaining rigorous proofs or even definitions; what is more, a remarkable innovation, it is from the knowledge of the derivative of x^a that he deduces $\int x^a dx$ for $a \neq -1$ ([233 a], v. I, pp. 3-26 and [198 d], p.

225). For the remainder, and although he was soon in possession of much more general methods for expansions in power series, such as the method of Newton’s polygon so-called (for algebraic functions) ([198 d], p. 221) and that of indeterminate coefficients, he returns often in

what follows, with a kind of predilection, to the binomial series and to its generalisations; and it is from there, for example, that he seems to have drawn out the expansion of $\int x^\alpha (1+x)^\beta dx$ which was referred to above ([198 d], p. 209).

The evolution of ideas on the continent, meanwhile, is very different, and much more abstract. Pascal had drawn close to Fermat in the study of binomial coefficients (from which he constructs what he calls the “arithmetic triangle”) and their use in the calculation of probabilities and in the calculation of differences; when he tackles integration, he introduces there the same ideas. Like his predecessors, when he uses the language of indivisibles, he conceives of the integral $F(x) = \int_0^x f(x)dx$ as the value of the ratio of the “sum of all the ordinates of the curve”

$$S\left(f\left(\frac{n}{N}\right)\right) = \sum_{0 \leq p < N} f\left(\frac{p}{N}\right),$$

to the “unit” $N = \sum_{0 \leq p < N} 1$ for $N = \infty$ ([244], v. VIII, pp. 352-355) (or, when he abandons this language for the correct language of exhaustion, as a limit of this ratio for N growing indefinitely). But, having problems about moments in view, he observes that, when it is a matter of discrete masses y_i spread out at equidistant intervals, the calculation of the total mass reduces to the operation Sy_n defined above, and the calculation of the moment to the operation S^2y_n ; and, by analogy, he iterates the operation $\int$ to form what he calls the “triangular sum of the ordinates”, thus, in our language, the limits of the sums $N^{-2}S^2\left(f\left(\frac{n}{N}\right)\right)$, that is to say the integrals $F_2(x) = \int_0^x F(x)dx$; a new iteration gives him the “pyramidal sums” $F_3(x) = \int_0^x F_2(x)dx$, limits of $N^{-3}S^3\left(f\left(\frac{n}{N}\right)\right)$; the context besides is sufficient to show that, if he stops there, it is not because of a lack of generality in his thought nor in his language, but solely because he only expects to use these, whose systematic employment is at the basis of a good part of his results, and of which he proves immediately the properties that we would write as

$$F_2(x) = \int_0^x (x-u)f(u)du, F_3 = \frac{1}{2} \int_0^x (x-u)^2 f(u)du$$

([244], v. VIII, pp. 361-367); all that without writing a single formula, but in a language so transparent and so precise that one can immediately transcribe it into formulae as has just been done. With Pascal as with his predecessors, but in a much clearer and more systematic way,

the choice of the independent variable (which is always one of the co-ordinates, or the arc of the curve) is implicit in the convention that fixes the equidistant subdivision points (although “infinitely near”) of the intervals of integration; these points, respectively, are, either on Ox, or on Oy, or on the arc of the curve, and Pascal takes care never to allow any ambiguity about this to remain ([244], v. VIII, pp. 368-369). When he has to change variables, he does it by means of a principle which reduces to saying that the area $\int f(x)dx$ can be written as $S(f(x_i)\Delta x_i)$, for every subdivision of the interval of integration into “infinitely small” intervals Δx_i , equal or not ([244], v. IX, pp. 61-68).

As can be seen, we are already very near Leibniz; and it is, it might be said, a lucky break that this latter, when he wished to be initiated into modern mathematics, met Huygens, who immediately put Pascal’s writings into his hands ([198 d], pp. 407-408); he was particularly well prepared for this by his thoughts on combinatorial analysis, and we know that he made a deep study of it, which is reflected in his work. In 1675, we see him transcribing Pascal’s theorem above in the form $\text{omn}(x\omega) = x.\text{omn}\omega - \text{omn}(\text{omn}\omega)$, where $\text{omn}\omega$ is an abbreviation for the integral of ω taken from 0 to x , for which Leibniz, a few days later, substitutes $\int \omega$ (initial of “summa omnium ω ”) at the same time as he introduces d for the infinitely small “difference”, or, as he will soon say, the differential ([198 d], pp. 147-167). Conceiving of these “differences” as quantities which are comparable amongst themselves but not with finite quantities, he takes besides most often, explicitly or not, the differential dx of the independent variable x as unity, $dx = 1$ (which amounts to identifying the differential dy with the derivative $\frac{dy}{dx}$), and at first omits it from his notation for the integral, which thus appears as $\int y$ rather than $\int ydx$; but he does not delay introducing this, and keeps to it systematically once he realises its invariant character with respect to the choice of independent variable, which dispenses with having this choice constantly in mind; ⁹ and he shows no little satisfaction, when he returns to the study of Barrow that he had neglected up till then, in noting that the general theorem of change of variables, of which Barrow makes such a big thing, follows on immediately from his own notation ([198 d], p. 412). In all this as it happens, he keeps very close to the calculus of differences, from which his differential calculus can be deduced by going to the limit which of course he would be hard put to justify rigorously; and in what followed he voluntarily insists on

the fact that his principles apply without change to both one and the other. He quotes explicitly Pascal, for example, when in his correspondence with Johann Bernoulli ([198 a], v. III, p. 156), referring to his first researches, he gives a formula for the calculation of differences which is a particular case of that of Newton, and deduces from it by “passing to the limit” the formula $y = \sum_1^{\infty} (-1)^n \frac{d^n y}{dx^n} \cdot \frac{x^n}{n!}$ (where y is a function which is zero for $x = 0$, and the $\frac{d^n y}{dx^n}$ are its derivatives for the value x of the variable) a formula equivalent to a similar one that Bernoulli communicated to him ([198 a], v. III, p. 150 and [20 a], v. I, pp. 125-128), and that this latter proves by successive integration by parts. This formula, as can be seen, is very close to Taylor series; and it is even the argument of Leibniz, going to the limit starting with the calculus of differences, that Taylor finds again in 1715 to obtain “his” series [305],¹⁰ without besides making great use of it.

F) One would have already seen, implicit in the evolution described above, the progressive algebraisation of infinitesimal analysis, that is to say its reduction to an operational calculus furnished with a uniform system of notation of an algebraic character. As Leibniz had indicated several times with perfect clarity ([198 a], v. V, pp. 230-233), it was a case of doing for the new analysis what Viète had done for the theory of equations, and Descartes for geometry. In order to understand its necessity, just read a few pages of Barrow; at no time is it possible not to have under one’s eyes a figure, sometimes complicated, described beforehand with meticulous care; no less than 180 figures are needed for the 100 pages (Lect. V-XII) which form the essential part of the work.

It could hardly have been a question of algebraisation, it is true, before some unity had appeared amongst the multiplicity of geometric appearances. Meanwhile Grégoire de St. Vincent [135] already introduces (under the name of “ductus plani in planum”) a kind of law of composition which reduces to the systematic use of integrals $\int_a^b f(x)g(x)dx$ considered as volumes of solids $a \leq x \leq b, 0 \leq y \leq f(x), 0 \leq z \leq g(x)$; but he is far from drawing out the consequences that Pascal later deduces, as has been seen, from study of the same solid. Wallis in 1655, and Pascal in 1658, forge, each for his own use, languages of algebraic character, in which, without writing any formula, they draw up statements that can immediately be transcribed into formulae of the integral calculus as soon as their mechanism has been understood. The language of Pascal

is particularly clear and precise; and, if it can not be understood why he refused to allow himself the use of the algebraic notations, not only of Descartes, but even of Viète, one can only admire the tour de force that he accomplished, and which his mastery of language was the only means of his being capable of doing.

But just let a few years go by, and everything changes. Newton first conceives of the idea of replacing all the operations, of a geometric character, of contemporary infinitesimal analysis, by a unique analytic operation, differentiation, and by the solution of the inverse problem; an operation that it is well understood the method of power series allowed him to carry out with extreme facility. Borrowing his language, we have seen, from the fiction of a universal “temporal” parameter, he qualifies as “fluents” the quantities varying as a function of this parameter, and as “fluxions” their derivatives. He does not seem to have attached particular importance to the notations, and his devoted supporters later even boast that the absence of a systematic notation is an advantage; he nevertheless picks up the habit early on, for his personal use, of denoting fluxion by a point, thus $\frac{dx}{dt}$ by $\dot{x}$, $\frac{d^2x}{dt^2}$ by $\ddot{x}$, etc.. As for integration, it appears that Newton, exactly like Barrow, never saw it except as a problem (find the fluent knowing the fluxion, thus solve $\dot{x} = f(t)$), and not as an operation; also he has no name for the integral, nor, it seems, standard notation (except sometimes a square $\boxed{f(t)}$ or $\square f(t)$ for $\int f(t)dt$). Is it because he dislikes giving a name and a symbol to an object that is not defined in a unique way, but only up to an additive constant?) Lacking a text, the question can only be asked.

As much as Newton is empirical, concrete, circumspect in his greatest boldness, so much is Leibniz systematic, generalisor, adventurous and sometimes rash innovator. From his youth, he had in mind the idea of a “characteristic” or universal symbolic language, which would be to the whole of human thought what algebraic notation was to algebra, where each name or symbol was the key to all the qualities of the object symbolised, and that could not be employed correctly without at the same time reasoning correctly (see p. 6). It is easy to treat such a project as fanciful; it is not by chance though that its author was the man himself who was soon to recognise and isolate the fundamental concepts of the infinitesimal calculus, and endow it with more or less definitive notations. We have already assisted above at the birth of these, and observed the care with which Leibniz, who seems aware of his mission, modifies them progressively until

he ensures the simplicity and above all the invariance that he is seeking ([198 a], v. V, pp. 220-226 and 226-233). What is important to note here, is, as soon as he introduces them (without knowing anything yet of the ideas of Newton), the clear conception of $\int$ and of d , of the integral and the differential, as inverse operators one of the other. It is true that in proceeding in this way he can not avoid the ambiguity inherent in the indefinite integral, which is the weak point of his system, over which he slides adroitly, like his successors. But what is striking, from the first appearance of the new symbols, is to see Leibniz immediately busy with the formulation of rules of use, asking himself if $d(xy) = dx dy$ ([198 d], pp. 165-166), and answering himself in the negative, to arrive progressively at the correct rule ([198 a], v. V, pp. 220-226), which he was later to generalise by his famous rule for $d^n(xy)$ ([198 a], v. III, p. 175). Of course, at the time at which Leibniz is groping thus, Newton knew for ten years already that $z = xy$ implies $\dot{z} = \dot{x}y + x\dot{y}$; but he never takes the trouble to say it, only seeing it as a particular case, not worthy of being named, of his rule for differentiating the relation $F(x, y, z) = 0$ between fluents. On the contrary, the principal concern of Leibniz is not to make his methods useful for the solution of such concrete problems, nor even to deduce them from rigorous and impregnable principles, but above all to put in train an algorithm relying on the formal handling of a few simple rules. It is in this spirit that he improves the algebraic notation by the use of parentheses, that he adopts progressively $\log x$ or lx for the logarithm,¹¹ and that he insists on the “exponential calculus”, that is to say, the systematic consideration of exponentials, a^x, x^x, x^y , where the exponent is a variable. Above all, while Newton does not introduce fluxions of a higher order except strictly in proportion to the necessity in each concrete case, Leibniz turns early towards the creation of an “operational calculus” by the iteration of d and of $\int$; becoming aware little by little of the analogy between the multiplication of numbers and the composition of operators in his calculus, he adopts, by a fortunate boldness, the notation for exponents to write the iterates of d , writing thus d^n for the n -th iterate ([198 d], pp. 595 and 601,¹² and [198 a], v. V, pp. 221 and 378) and even d^{-1}, d^{-n} for $\int$ and its iterates ([198 a], v. III, p. 167); and he seeks even to give a meaning to d^α for α an arbitrary real ([198 a], v. II, pp. 301-302, and v. III, p. 228).

That is not to say that Leibniz is not also interested in the applications of his calculus, knowing well (as Huygens repeats to him often ([198 d], p. 599)) that they are its touch stone; but he

lacks patience to go deeply into them, and seeks there above all the opportunity for new general rules. It is thus that in 1686 ([198 a], v. VII, pp. 326-329) he deals with the curvature of curves, and with the osculating circle, to end up in 1692 ([198 a], v. VII, pp. 331-337) with general principles on the contacts of plane curves; ¹³ in 1692 ([198 a], v. V, pp. 266-269) and in 1694 ([198 a], v. V, pp. 301-306) he sets down the basis for the theory of envelopes; concurrently with Johann Bernoulli, he effects in 1702 and 1703 the integration of rational fractions by decomposition into simple elements, but first of all in a formal manner and without fully realising the circumstances that accompany the presence of complex linear factors in the denominator ([198 a], v. V, pp. 350-366). It is thus again that one day in August 1697, meditating in a carriage on questions of the calculus of variations, he has the idea for the rule for differentiation with respect to a parameter under the sign $\int$; and, enthusiastic, sends it straightaway to Bernoulli ([198 a], v. III, pp. 449-454). But when he gets there, the fundamental principles of his calculus have been secured for a long time, and their use has started to spread: the algebraisation of infinitesimal analysis is an accomplished fact.

- G) The notion of function is introduced and made precise in a multitude of ways during the XVIIth century. All dynamics is based on an intuitive, and in some way experimental, idea, of quantities varying with time, that is to say functions of time, and we have already seen how we end up thus at a function of one parameter, such as it appears with Barrow, and, under the name of fluent, with Newton. The notion of “arbitrary curve” appears often, but is rarely made precise; it may be that often it is thought of in a dynamic form or in any case experimental form, and that without it being judged necessary that a curve should be capable of a geometric or analytic characterisation in order to be of use as an object of reasoning; it is so, in particular (for reasons that we are better able to understand today) when it is a case of integration, for example with Cavalieri, Pascal and Barrow; this latter, reasoning about the curve defined by $x = ct$, $y = f(t)$, with the hypothesis that $\frac{dy}{dt}$ is increasing, even says expressly that “it matters not” that $\frac{dy}{dt}$ increases “regularly following an arbitrary law, or even irregularly” ([16 b], p. 191) that is to say, as we would say, were capable or not of an analytic definition. Unfortunately this clear and fruitful idea, which would, made appropriately precise, reappear in the XIXth century, could not

fight against the confusion created by Descartes, when this latter had, in the first case, banned from “geometry” all curves not capable of a precise analytic definition, and in the second case restricted only to algebraic operations the admissible procedures of formation in such a definition. It is true that, on this last point, he is not followed by the majority of his contemporaries; little by little, and often by very subtle detours, various transcendental operations, the logarithm, the exponential, the trigonometric functions, quadratures, the solution of differential equations, passing to the limit, the summing of series, acquired the right of being quoted, without it being easy in each case to mark the precise moment when the step forward is being taken; and besides the first step forward is often followed by a step back. For the logarithm, for example, one must consider as important stages the apparition of the logarithmic curve ($y = a^x$ or $y = \log x$ according to the choice of axes), of the logarithmic spiral, the quadrature of the hyperbola, the expansion in series of $\log(1 + x)$, and even the adoption of the symbol $\log x$ or lx . In what concerns the trigonometric functions, and even though in a certain sense they go back to antiquity, it is interesting to note that the sinusoidal curve does not appear at first as being defined by an equation $y = \sin x$, but just, with Roberval ([263], pp. 63-65), as a “companion of the roulette” (in fact, it is a case of the curve

$$y = R \left(1 - \cos \frac{x}{R} \right),$$

that is to say as an auxiliary curve whose definition is deduced from that of the cycloid. To meet the general notion of analytic expression, we must go to J. Gregory, who defined it in 1667 ([169 a], p. 413), as a quantity which is obtained starting from other quantities by a succession of algebraic operations “or of every other imaginable operation”; and he tries to make precise this notion in his preface ([169 a], pp. 408-409), by explaining the necessity of adjoining to the five operations of algebra ¹⁴ a sixth operation, which in none other in the end than passing to the limit. But these interesting thoughts are soon forgotten, submerged by the torrent of expansions in series discovered by Gregory himself, Newton, and others; and the prodigious success of this last method created an enduring confusion between functions that were susceptible to analytic definitions, and functions expandable in power series.

As for Leibniz, he seems to keep to the Cartesian point of view, expanded by the explicit adjunction of quadratures, and by the implicit adjunction of the other operations familiar to the analysis of his time, summation of power series, solution of differential equations. In the same way Johann Bernoulli, when he wishes to consider an arbitrary function of x , introduces it as “a quantity made up in any way starting from x and constants” ([198 a], v. III, p. 150), specifying sometimes that it is a matter of a quantity made up “in an algebraic or transcendental manner” ([198 a], v. III, p. 324): and, in 1698, he agrees with Leibniz to give to such a quantity the name “function of x ” ([198 a], v. III, pp. 507-510 and 525-526).¹⁵ Already Leibniz had introduced the words “constant”, “variable”, “parameter”, and specified in connection with envelopes the notion of a family of curves depending on one or more parameters ([198 a], v. V, pp. 266-269). The questions about notation become precise also in correspondence with Johann Bernoulli: this latter readily writes X , or ξ , for an arbitrary function of x ([198 a], v. III, p. 531); Leibniz approves, but also proposes $\bar{x}|_{-1}^1$, $\bar{x}|_{-1}^2$ where we would write $f_1(x)$, $f_2(x)$; and he proposes, for the derivative $\frac{dz}{dx}$ of a function z of x , the notation $\bar{d}z$ (in contrast to dz which is the differential) whereas Bernoulli writes Δz ([198 a], v. III, pp. 537 and 526).

Thus, with this century, the heroic era comes to an end. The new calculus, with its notions and notations, is constituted, in the form that Leibniz gave it. The first disciples, Jakob and Johann Bernoulli, vie in discoveries with the master, exploring freely the rich territories to which he had shown the way. The first treatise on differential and integral calculus was written in 1691 and

1692 by Johann Bernoulli,¹⁶ for the use of a marquis who shows himself a good pupil. It matters little besides that Newton decided finally, in 1693, to publish parcimoniously a short glimpse of his fluxions ([326 a], v. II, pp. 391-396); if his Principia have enough to supply the meditation of more than a century, on the ground of the infinitesimal calculus he has been joined, and on many points overtaken.

The weaknesses of the new system are besides visible, at least to our eyes. Newton and Leibniz, abolishing at a stroke a tradition two millennia old, gave the primary role to differentiation, and reduced integration to being only its inverse; it will take the whole of the XIXth century, and a part of the XXth century, to re-establish a fair equilibrium, by putting integration as the basis

of the general theory of functions of a real variable, and of its modern generalisation (see p. 223). It is from this reversal of points of view also that follows the excessive, and almost exclusive, role, taking place already with Barrow, and especially starting with Newton and Leibniz, played by the indefinite integral at the cost of the definite integral: there also the XIXth century had to put matters back in their place. Finally the properly Leibnizian tendency to a formal handling of symbols was to go on being emphasised throughout the whole of the XVIIIth century, well beyond what could be allowed by the resources of the analysis of the time. In particular, it must be recognised that the Leibnizian notion of a differential has, truth to tell no meaning; at the beginning of the XIXth century, it fell into a disrepute from which it has only recovered little by little; and, if the use of first differentials ended up by being completely legitimised, the differentials of higher order, although so useful, have not yet been really rehabilitated to our day. Whatever the case, the history of the differential and integral calculus, starting with the end of the XVIIth century, is divided into two parts. One is concerned with the applications of this calculus, always richer, more numerous and more varied. To the differential geometry of plane curves, to differential equations, to power series, to the calculus of variations, which have already been referred to above, have been joined the differential geometry of ill-shaped curves, then of surfaces, multiple integrals, partial differential equations, trigonometric series, the study of numerous special functions, and many other types of problems. We only should be concerned here with work which has contributed to clearing up, deepening and consolidating the very principles of the infinitesimal calculus, in what concerns functions of a real variable.

From this point of view, the great treatises of the middle of the XVIIIth century offer few novelties. Maclaurin in England ¹⁷ [214], Euler on the continent ([108 a], (1), vv. X to XIII), remained faithful to the traditions of which each one was the inheritor. It is true that the first tries hard to clarify somewhat the Newtonian concepts ¹⁸ whereas the second, pushing Leibnizian formalism to an extreme, is content, like Leibniz and Taylor, to base the differential calculus on a very obscure passage to the limit starting with the calculus of differences, a calculus of which as it happens he gives a very polished exposition. But above all Euler finishes the work of Leibniz by introducing and ensuring the adoption of the notation still in use today for e , i and the trigonometric functions, and spreading the notation π . On the other hand, and if most of the

time he does not make any distinction between functions and analytic expressions, he insists, in connection with trigonometric series and the problem of vibrating strings, on the need not to limit oneself to the functions thus defined (and that he qualifies as “continuous”) but to consider also, if the need arises, arbitrary, or “discontinuous”, functions given experimentally by one or more arcs of a curve ([108 a], (1), v. XXIII, pp. 74-91). Finally, even though this is somewhat outside our framework, it is not possible not to mention here his extension of the exponential function to the complex domain, from which he obtains the famous formulae linking the exponential to the trigonometric functions, as well as the definition of the logarithm of complex numbers; from that is elucidated definitively the famous analogy between the logarithm and the inverse circular functions, or, in the language of the XVIIth century, between the quadratures of the circle and of the hyperbola, observed already by Grégoire de St. Vincent, made precise by Huygens and above all by J. Gregory, and which, with Leibniz and Bernoulli, appeared in the formal integration of

$$\frac{1}{1+x^2} = \frac{i}{2(x+i)} - \frac{i}{2(x-i)}.$$

Meanwhile d’Alembert, the enemy of all mysticism in mathematics as elsewhere, had, in remarkable articles ([75 a], articles DIFFÉRENTIEL and LIMITE, and [75 b]), defined with the greatest clarity the notions of limit and of derivative, and maintained strongly that in the end that is all the “metaphysics” of the infinitesimal calculus. But this wise advice did not have any immediate effect. The monumental work of Lagrange ([191], vv. IX-X) represents an attempt to base analysis on one of the most arguable of the Newtonian concepts, that which confuses the notions of arbitrary functions and that of functions which can be expanded in a power series, and to draw from that (by consideration of the coefficient of the term of the first order in the series) the notion of differentiation. Of course, a mathematician of the quality of Lagrange could not fail to obtain in this case important and useful results, as for example (and in a way which was in fact independent of the point of departure that we have just indicated) the general proof of the formula of Taylor with the expression for the remainder in the form of an integral, and its evaluation by the theorem of the mean; in any case the work of Lagrange is at the origin of the method of Weierstrass in the theory of functions of a complex variable, as well as the modern algebraic theory of formal series. But, from the point of view of its immediate objective, it

represents a retreat rather than progress.

With the teaching works of Cauchy, on the contrary ([56 a], (2), v. IV) we find ourselves at last on solid ground. He defines a function essentially as we do it today, even though in a language which is still rather vague. The notion of limit, fixed once and for all, is taken as the point of departure; those of continuous function (in the modern meaning) and of derivative are immediately deduced, as well as their principal elementary properties; and the existence of the derivative, instead of being an act of faith, becomes a matter of study by the ordinary methods of analysis. Cauchy truth to tell, is hardly interested in this; and on the other hand, if Bolzano, having reached on his part the same principles, constructed an example of a continuous function not having a derivative at any point [27 a], this example was not published, and the question was only publicly settled by Weierstrass, in a work in 1872 ([329 a], v. II, pp. 71-74).

As far as integration is concerned, the work of Cauchy represents a return to the sane traditions of antiquity and the first part of the XVIIth century, but relying on technical means which are still insufficient. The definite integral, passed over into second place for too long, becomes again the primary notion, for which Cauchy ensures the definitive adoption of the notation $\int_a^b f(x)dx$ proposed by Fourier (instead of the awkward $\int f(x)dx \left[\begin{array}{l} x = a \\ x = b \end{array} \right]$ sometimes used by Euler); and, in order to define it, Cauchy returns to the method of exhaustion, or as we would say, to “Riemann sums” (which would be better called sums of Archimedes, or sums of Eudoxus). It is true that the XVIIth century did not deem it necessary to submit the notion of area to a critical examination, an idea which had appeared to them to be at least as clear as that of incommensurable real number; but the convergence of “Riemann” sums to the area under the curve, so long as it is the case of a monotonic curve or a piecewise monotonic curve, was a familiar idea to all the authors concerned with rigour in the XVIIth century, such as Fermat, Pascal, Barrow; and J. Gregory, particularly well prepared by his thoughts on the passage to the limit and his familiarity with an already very abstract form of the principle of “nested intervals”, had even drawn up, it would appear, a careful proof, which remained unpublished ([136 d], pp. 445-446), which could have been of use to Cauchy almost without any change if he had known it. ¹⁹ Unfortunately for him, Cauchy claimed to prove the existence of the integral, that is to say the convergence of the “Riemann sums”, for an arbitrary continuous function; and his

proof, which would have become correct if it were based on the theorem on uniform continuity of functions continuous in a closed interval, is stripped of all convincing value for lack of this notion. Dirichlet does not seem to have seen either the difficulty at the time when he drew up his famous memoirs on trigonometric series, since he quotes there the theorem in question as “easy to prove” ([92], v. I, p. 136); it is true that he only applies it definitively to bounded piecewise monotonic functions; Riemann, more circumspect, only mentions these latter when it is a matter of using his necessary and sufficient condition for the convergence of “Riemann sums” ([259 a], pp. 227-271). Once the theorem on uniform continuity had been established by Heine (cf. p. 145), the matter did not offer any more difficulties of course; and it is easily dealt with by Darboux in 1875 in his memoir on the integration of discontinuous functions [77], a memoir in which he meets on many points with the important researches of P. du Bois-Reymond, which appeared at about the same time. In the same way for the first time, but this time definitively, the linearity of the integral of continuous functions is proved. On the other hand, the notion of uniform convergence of a sequence or series, introduced among others by Seidel in 1848, and put into prominence in particular by Weierstrass (cf. p. 205) had allowed the construction of a solid basis, under conditions that were a bit too restrictive it is true, for the integration of series term by term, and for differentiation under the $\int$ sign, while awaiting modern theories of which we do not have to speak here, and which would clarify these questions in a manner which is provisionally definitive.

We have thus attained the final stage of the classical infinitesimal calculus, that which is represented by the great Treatises on Analysis of the end of the XIXth century; from the point of view that concerns us, that of Jordan [174c] occupies amongst them a pre-eminent place, for aesthetic reasons on the one hand, but also because, if it constitutes an admirable setting out of the results of classical analysis, it announces in many ways modern analysis and prepares the way for it.

Chapter 18

ASYMPTOTIC EXPANSIONS.

The distinction between the “infinitely small” (or “infinitely great”) of various orders, appears implicitly from the first writings on differential Calculus, and for example in those of Fermat; it becomes fully conscious with Newton and Leibniz, with the theory of “differences of higher order”; and no time is lost in observing that, in the simplest cases, the limit (or “true value”) of an expression of the form $f(x)/g(x)$, at a point where f and g both tend to 0, is given by the Taylor expansion of these functions in a neighbourhood of the point under consideration (“l’Hôpital’s rule”, due probably to Johann Bernoulli).

Apart from this elementary case, the main problem of “asymptotic expansion” which presents itself to mathematicians already at the end of the XVIIIth century is the calculation, exact or approximate, of sums of the form $\sum_{k=1}^n f(k)$, when n is very large; such a calculation is in effect necessary as much for interpolation and the numerical evaluation of the sum of a series, as in the Calculus of probabilities, where the “functions of large numbers” such as $n!$ or $\binom{a}{n}$ play a preponderant role. Already Newton, in order to obtain approximate values of $\sum_{k=1}^n \frac{1}{a+k}$ when n is large, indicates a method that reduces (in this particular case) to calculating the first terms of the Euler-Maclaurin formula ([262], v. II, pp. 309-310). Towards the end of the century, Jakob Bernoulli, in the course of his research on the Calculus of probabilities, proposes determining the sums $S_k(n) = \sum_{p=1}^n p^k$, polynomials ¹ in n for which he discovers the general formation law (without giving a proof of it), introducing thus for the first time, in the expression for the coefficients of these polynomials, the numbers that bear his name, and the recurrence relation that enables their calculation ([19 b], p. 97). In 1730, Stirling obtains

an asymptotic expansion for $\sum_{k=1}^n \log(x + ka)$, n growing indefinitely, by a procedure for the calculation of the coefficients by recurrence.

From 1730 to 1745 is dated the decisive work of Euler on series and the relevant questions. Setting $S(n) = \sum_{k=1}^n f(k)$, he applies Taylor's formula to the function $S(n)$, which gives him

$$f(n) = S(n) - S(n-1) = \frac{dS}{dn} - \frac{1}{2!} \frac{d^2S}{dn^2} + \frac{1}{3!} \frac{d^3S}{dn^3} - \dots,$$

an equation that he "inverts" by the method of indeterminate coefficients, while looking for a solution of the form

$$S(n) = \alpha \int f(n)dn + \beta f(n) + \gamma \frac{df}{dn} + \delta \frac{d^2f}{dn^2} + \dots;$$

he thus obtains step by step

$$S(n) = \int f(n)dn + \frac{f(n)}{2} + \frac{1}{12} \frac{df}{dn} - \frac{1}{720} \frac{d^3f}{dn^3} + \frac{1}{30,240} \frac{d^5f}{dn^5} - \dots$$

without at first being able to determine the formation law for the coefficients ([108 a], (1), v. XIV, pp. 42-72 and 108-123). But around 1735, by analogy with the decomposition of a polynomial into factors of degree one, he does not hesitate to write the formula

$$1 - \frac{\sin s}{\sin \alpha} = \left(1 - \frac{s}{\alpha}\right) \left(1 - \frac{s}{\pi - \alpha}\right) \left(1 - \frac{s}{-\pi - \alpha}\right) \left(1 - \frac{s}{2\pi + \alpha}\right) \left(1 - \frac{s}{-2\pi + \alpha}\right) \dots$$

and in putting the coefficients of the expansions of the two members of the whole series equal, he obtains in particular (for $\alpha = \frac{\pi}{2}$) the famous expressions for the series $\sum_{n=1}^{\infty} \frac{1}{n^{2k}}$ by means of the powers ² of π (loc. cit., pp. 73-86). A few years later, he realises at last that the coefficients of these powers of π are given by the same equations as those of his summation formula, and recognises their link with the numbers introduced by Bernoulli, and with the coefficients of the expansion in series of $z/(e^z - 1)$ (loc. cit., pp. 407-462).

Independently of Euler, Maclaurin had arrived at about the same time at the same summing formula, by a somewhat less hazardous path, close to that followed today: he iterates in effect the "Taylorian" formula which expresses $f(x)$ by means of the differences $f^{(2k+1)}(x+1) - f^{(2k+1)}(x)$,

a formula that he obtains by “inverting” the Taylor expansions of these differences by the method of indeterminate coefficients ([214], v. II, pp. 672-675); he does not notice besides the formation law for the coefficients, discovered by Euler.

But Maclaurin, like Euler and all the mathematicians of his time, presents all his formulae as expansions in series, whose convergence is not even studied. It is not that the notion of convergent series was totally neglected at this time: it is known from Jakob Bernoulli that the harmonic series is divergent, and Euler had himself recorded this result while evaluating the sum of the first n terms of this series by means of his summing formula ([108 a], (1), v. XIV, pp. 87-100 and 108-123); it is also Euler who remarks that the ratio of two consecutive Bernoulli numbers grows indefinitely, and that it follows that an entire series having these numbers as coefficients can not converge (loc. cit., p. 357).³ But the tendency to a formal calculation is the stronger, and the extraordinary intuition of Euler himself does not prevent him from lapsing sometimes into the absurd, when he writes for example $0 = \sum_{n=-\infty}^{+\infty} x^n$ (loc. cit., p. 362).⁴

We have said elsewhere (see p. 153) how the mathematicians of the beginning of the XIXth century, tired of this formalism without restraint and without basis, brought Analysis back into the paths of rigour. Once the notion of a convergent series is made precise, the necessity appeared for simple criteria providing a proof of the convergence of integrals and series by comparison with known integrals or series; Cauchy gives a certain number of criteria in his *Analyse algébrique* ([56 a], (2), v. III), whereas Abel, in a posthumous memoir ([1], v. II, pp. 197-205), obtains logarithmic criteria of convergence. Cauchy, on the other hand ([56 a], (1), v. VIII, pp. 18-25), elucidates the paradox of series such as the Stirling series, obtained by the application of the Euler-Maclaurin formula (and often called “semi-convergent series”): he shows that if (because of the remark of Euler on the Bernoulli numbers) the general term $u_k(n)$ of such a series, for a fixed value of n , grows indefinitely with k , it remains none the less that, for a fixed value of k , the partial sum $s_k(n) = \sum_{h=1}^k u_h(n)$ gives an asymptotic expansion (for n tending to $+\infty$) of the function “represented” by the series, all the more accurate as k is larger.

In the majority of the calculations of classical Analysis, it is possible to obtain a general formation law for asymptotic expansions of a function, having an arbitrarily large number of terms; this fact contributed to creating a lasting confusion (at least in the language) between

series and asymptotic expansions; so much so that H. Poincaré, when he takes the trouble, in 1886 ([251 a], v. I, pp. 290-296), of codifying the elementary rules of asymptotic expansions (following the integral powers of $1/x$ in the neighbourhood of $+\infty$), uses still the vocabulary of the theory of series. It is only with the appearance of asymptotic expansions coming from the analytic theory of numbers that there was a clear distinction operating between the notion of asymptotic expansion and that of series, by reason of the fact that, in the majority of problems that this theory deals with, only a very small number of terms (most often only one) in the sought for expansion can be obtained explicitly.

These problems also made mathematicians familiar with the use of scales of comparison other than those of powers of the variable (real or integral). This extension goes back to the work of P. du Bois-Reymond [94 a and b] who, first, tackled systematically the problems of comparing functions in the neighbourhood of a point, and, in very original works, recognised the “non-Archimedean” character of the scales of comparison, at the same time as he studied in a general way the integration and differentiation of the comparison relations, and drew from this a large number of interesting consequences [94 b]. His proofs however lacked clarity and rigour, and it is to G. H. Hardy [147] that the correct presentation of the results of du Bois-Reymond is due: his main contribution consisted in recognising and proving the existence of a set of “elementary functions”, the functions (H), where the usual operations of Analysis (notably differentiation) are applicable to comparison relations.

Chapter 19

THE GAMMA FUNCTION.

The idea of “interpolating” a sequence (u_n) by the values of an integral dependant on a real parameter λ and equal to u_n for $\lambda = n$, goes back to Wallis (cf. pp. 187 ff.). It is this idea that was the main guide for Euler when, in 1730 ([108 a], (1), v. XIV, pp. 1-24), he has the idea of interpolating the sequence of factorials. He starts by remarking that $n!$ is equal to the infinite product $\prod_{k=1}^{\infty} \left(\frac{k+1}{k} \right)^n \frac{k}{k+n}$, that this product is defined for every value of n (integral or not), and that in particular, for $n = \frac{1}{2}$ it takes the value $\frac{1}{2}\sqrt{\pi}$ from the formula of Wallis. The analogy of this result with those of Wallis leads him then to take up again the integral $\int_0^1 x^e (1-x)^n dx$ (n an integer, e arbitrary), already considered by this latter. Euler obtains its value $\frac{n!}{(e+1)(e+2)\dots(e+n)}$ by the binomial expansion; a change of variable then shows him that $n!$ is the limit, for z tending to 0, of the integral $\int_0^1 \left(\frac{1-x^z}{z} \right)^n dx$, whence the “second Eulerian integral” $n! = \int_0^1 \left(\log \frac{1}{x} \right)^n dx$; by the same method and the use of the formula of Wallis, he obtains the formula $\int_0^1 \sqrt{\log \frac{1}{x}} dx = \frac{1}{2}\sqrt{\pi}$. In his later work, Euler returns frequently to these integrals; he discovers in this way the relation of the complements ([108 a], (1), v. XV, p. 82 and v. XVII, p. 342), the formula $B(p, q) = \Gamma(p)\Gamma(q)/\Gamma(p+q)$ ([108 a], (1), v. XVII, p. 355), and the particular case of the Legendre-Gauss formula corresponding to $x = 1$ ([108 a], (1), v. XIX, p. 483); all of it of course without worrying about questions of convergence.

Gauss continued the study of the Γ function in connection with his research on the hypergeometric function, of which the function Γ is a limit case ([124 a], v. III, pp. 125-162); it is during this research that he obtains the general formula for multiplication (already noted by Legendre not long before for $p = 2$). The subsequent work on Γ has mainly been concerned

with the extension of this function to the complex domain. It is only recently that it has been noticed that the logarithmic convexity property characterised $\Gamma(x)$ (in the real domain) up to a factor amongst all the solutions of the functional equation $f(x+1) = xf(x)$ ([26], pp. 149-164); and Artin has shown [7 d] how all the classical results about $\Gamma(x)$ can be simply linked to this property.

Chapter 20

FUNCTION SPACES.

We know that the notion of an arbitrary function did not emerge much before the beginning of the XIXth century. All the more so the idea of studying in a general way sets of functions, and furnishing them with a topological structure, did not appear before Riemann (see p. 141) and does not begin to be put to use before the end of the XIXth century.

However, the notion of convergence of a sequence of numerical functions was used in a more or less conscious way since the beginnings of the infinitesimal Calculus. But it was only a matter of simple convergence and it could not be otherwise before the notions of convergent series and of continuous functions had been defined accurately by Bolzano and Cauchy. This latter did not see at the first attempt the distinction between simple convergence and uniform convergence, and believed it possible to prove that every convergent series of continuous functions has as sum a continuous function ([56 a], (2), vol. III, p. 120). The error was almost immediately uncovered by Abel, who proved at the same time that every entire series is continuous in the interior of its convergence interval, by means of the argument that has become a classic which uses essentially, in this particular case, the idea of uniform convergence ([1], v. I, pp. 223-224). It only remained to disengage this latter in a general way, which was done independently by Stokes and Seidel in 1847-48, and by Cauchy himself in 1853 ([56 a], (1), vol. XII, p. 30).¹

Under the influence of Weierstrass and of Riemann, the systematic study of the notion of uniform convergence and of connected questions is developed in the last third of the XIXth century by the German school (Hankel, du Bois-Reymond) and especially the Italian school: Dini and Arzelà specify the necessary conditions for the limit of a sequence of continuous functions to be

continuous, whereas Ascoli introduces the fundamental notion of equicontinuity and proves the theorem characterising compact sets of continuous functions [10] (a theorem popularised later by Montel in his theory of “normal families”, which are none other than relatively compact sets of analytic functions).

Weierstrass himself discovered on the other hand ([329 a], v. III, pp. 1-37) the possibility of approximating uniformly by polynomials a continuous numerical function of one or several real variables on a bounded set, a result that aroused immediately a lively interest, and led to numerous “quantitative” studies that remain outside the point of view we follow at present.

The modern contribution to these questions consisted above all in giving them the whole range of which they are capable, by bringing them into play for functions whose set of definition and set of values are no longer restricted to $\mathbb{R}$ or to spaces of finite dimension, and in placing them thus in their natural framework, with the help of general topological concepts. In particular, the theorem of Weierstrass, which had already shown itself to be a tool of the first order in classical Analysis, could, in these latter years, be extended to much more general cases by M. H. Stone: developing an idea introduced by H. Lebesgue (in a proof of the theorem of Weierstrass), he highlighted the important role played in the approximation of continuous numerical functions by lattices (approximation by “lattice polynomials”), and showed on the other hand how the generalised theorem of Weierstrass implies immediately a whole series of analogous approximation theorems, that are thus grouped in a much more coherent way [301 c].

Chapter 21

TOPOLOGICAL VECTOR SPACES.

The general theory of topological vector spaces was founded during the period which goes from 1920 to 1930 approximately. But it had been prepared for a long time before by the study of numerous problems of Functional Analysis; and its history cannot be retraced without indicating, at least summarily, how the study of these problems gradually brought mathematicians (especially from the beginning of the XXth century) to become aware of the common ancestry of the questions under consideration, and of the possibility of formulating them in a much more general way and of applying to them uniform solution procedures.

It can be said that the analogies between Algebra and Analysis, and the idea of considering functional equations (that is to say where the unknown is a function) as “limit cases” of algebraic equations, go back to the beginnings of infinitesimal Calculus, which in a kind of way answers this need to generalise “from the finite to the infinite”. But the direct algebraic ancestor of the infinitesimal Calculus is the calculus of finite differences (cf. pp. 185 ff.), and not the solution of general linear systems; and it is not before the middle of the XVIIIth century that the first analogies between this latter and the problems of the differential Calculus are manifested, in connection with the equation of vibrating cords. We will not go into the detail of the history of this problem here; but we must pick up the appearance of two fundamental ideas, which are found again constantly in what follows, and which both appear to be due to D. Bernoulli. The first consists in considering the oscillation of the cord as a “limit case” of the oscillation of a system of n point masses, as n tends to infinity; it is known that, for n finite, this problem would a little bit later give the first example of the search for eigenvalues of a linear transformation

(cf. p. 88); there correspond to these numbers, in the “passage to the limit” being considered, the frequencies of the “proper oscillations” of the cord, observed experimentally since long before, and whose theoretic existence had been established (notably by Taylor) at the beginning of the century. This formal analogy, although fairly rarely mentioned in what followed ([302 b], p. 390), seems never to have been lost sight of during the XIXth century; but, as we will see later, it did not acquire its full importance until about 1890-1900.

The other idea of D. Bernoulli (perhaps inspired by experimental facts) is the “principle of superposition”, according to which the most general oscillation of the cord must be capable of being “decomposed” into the superposition of “proper oscillations”, which, mathematically speaking, means that the general solution of the equation of vibrating cords must be capable of being expanded as a series $\sum_n c_n \phi_n(x, t)$, where the $\phi_n(x, t)$ represent the proper oscillations. It is known that this principle was to let loose a long quarrel on the possibility of expanding an “arbitrary” function as a trigonometric series, a quarrel that was only settled by the work of Fourier and of Dirichlet in the first third of the XIXth century. But even before this result was reached, other examples had been met of expansion in series of “orthogonal”¹ functions: spherical functions and Legendre polynomials, as well as different systems of the form $(e^{i\lambda_n x})$, where the λ_n are no longer multiples of the same number, and which were introduced from the XVIIIth century in oscillation problems, as well as by Fourier and Poisson during their researches on the theory of heat. Around 1830, all the phenomena observed in these various special cases are systematised, by Sturm [302 a and b] and Liouville [204 a and b] in a general theory of oscillations, for functions of one variable: they consider the differential equation

$$\frac{d}{dx} \left(p(x) \frac{dy}{dx} \right) + \lambda \rho(x) y = 0 (p(x) > 0, \rho(x) > 0) \quad (21.1)$$

with boundary conditions

$$\begin{cases} y'(a) - h_1 y(a) = 0 \\ y'(b) + h_2 y(b) = 0 \end{cases} \quad (h_1 \geq 0, h_2 \geq 0, a < b) \quad (21.2)$$

and prove the following fundamental results:

- 1) the problem only has a solution $\neq 0$ when λ takes one of the values from a sequence (λ_n)

of numbers > 0 , tending to $+\infty$;

- 2) for each λ_n , the solutions are multiples of the same function v_n , that can be assumed to be “normalised” by the condition $\int_a^b \rho v_n^2 dx = 1$, and we have $\int_a^b \rho v_m v_n dx = 0$ for $m \neq n$;
- 3) every function f , twice differentiable in $[a, b]$, and satisfying the boundary conditions (21.2), is expandable as a uniformly convergent series $f(x) = \sum_n c_n v_n(x)$, where $c_n = \int_a^b \rho f v_n dx$;
- 4) we have the equality $\int_a^b \rho f^2 dx = \sum_n c_n^2$ (already proved by Parseval in 1799 — in a purely formal way, besides — for the system of trigonometric functions, and from which flows immediately the “Bessel inequality” stated by this latter (still for trigonometric series) in 1828).

Half a century later, these properties are completed by the work of Gram [133] who, following the researches of Tchebychef, brings to light the relation between the expansions in series of orthogonal functions and the problem of the “best quadratic approximation” (issuing directly from the “method of least squares” of Gauss, in error theory): this latter consists, given a finite sequence of functions $(\psi_i)_{1 \leq i \leq n}$, of finding, for a function f , the linear combination $\sum_i a_i \psi_i$ for which the integral $\int_a^b \rho (f - \sum_i a_i \psi_i)^2 dx$ attains its minimum. In principle it is a case there only of a basic linear algebra problem, but Gram solves it in an original way, by applying to the ψ_i the process of “orthonormalization” generally known by the name of Erhard Schmidt. Going on next to the case of an infinite orthogonal system (ϕ_n) , he sets himself the question of knowing when the “best quadratic approximation” μ_n of a function f by linear combinations of the first n functions of the sequence, tends to 0 when n tends to infinity; ² he is thus led to define the notion of a complete orthonormal system, and recognises that this property is equivalent to the non-existence of functions $\neq 0$ orthogonal to all the ϕ_n . He even seeks to elucidate the concept of “mean quadratic convergence”, but, before the introduction of the fundamental notions of measure theory, he could hardly obtain any very specific results in this direction.

In the second half of the XIXth century, the main effort of the analysts is mainly towards the extension of the Sturm-Liouville theory to functions of several variables, to which led notably

the study of partial differential equations of elliptic type from mathematical Physics, and the boundary problems that are naturally associated with them. Interest is concentrated principally on the “vibrating membranes” equation

$$L_\lambda(u) \equiv \Delta u + \lambda u = 0 \quad (21.3)$$

where the solutions are sought that are zero on the contour for a fairly regular domain G ; and it is only little by little that the considerable analytic difficulties presented by this problem were overcome, a problem to which it was not thought possible to apply the methods that had succeeded for functions of a single variable. Let us remind ourselves of the main stages towards the solution: the introduction of the “Green’s function” of G , whose existence was proved by Schwarz; the proof, also due to Schwarz, of the existence of the smallest eigenvalue; finally, in 1894, in a famous memoir ([251 a], v. IX, pp. 123-196), H. Poincaré succeeded in proving the existence and essential properties of all the eigenvalues, by considering, for a given “second member” f , the solution u_λ of the equation $L_\lambda(u) = f$ which is zero on the contour, and in proving, by a clever generalisation of the method of Schwarz, that u_λ is a meromorphic function of the complex variable λ , having only real simple poles λ_n , which are precisely, the sought for eigenvalues.

This research is closely linked to the beginnings of the theory of linear integral equations, which was to contribute no doubt most to the coming of modern ideas. We will limit ourselves here to giving a few short indications on the development of this theory. This type of functional equation, having first appeared sporadically in the first half of the XIXth century (Abel, Liouville), had acquired importance since Beer and C. Neumann had reduced the solution of “Dirichlet’s problem” for a fairly regular domain G , to the solution of an “integral equation of the second type”

$$u(x) + \int_a^b K(x, y)u(y)dy = f(x) \quad (21.4)$$

for an unknown function u ; an equation that C. Neumann had succeeded in solving by means of a procedure of “successive approximations” in 1877. Driven no doubt as much by the alge-

braic analogies already mentioned as by the results that he had just obtained on the vibrating membranes equation, H. Poincaré, in 1896 ([251 a], v. IX, pp. 202-272), had the idea of introducing a variable parameter λ in front of the integral in the preceding equation, and affirms that, as for the vibrating membranes equation, the solution is then a meromorphic function of λ ; but he does not succeed in proving this result, which was only established (for a continuous “kernel” K and a finite interval $[a, b]$) by I. Fredholm four years later [116]. This latter, more consciously perhaps even than his predecessors, lets himself be completely guided by the analogy of (21.4) with the linear system

$$\sum_{q=1}^n \left(\delta_{pq} + \frac{1}{n} a_{pq} \right) x_q = b_p \quad (1 \leq p \leq n) \quad (21.5)$$

in order to obtain the solution of (21.4) as a quotient of two expressions constructed on the model of determinants that occur in the formulae of Cramer. It was not besides a new idea: from the beginning of the XIXth century, the method of “indeterminate coefficients” (consisting of obtaining an unknown function assumed expandable in series $\sum_n c_n \phi_n$, where the ϕ_n are known functions, by calculating the coefficients c_n) had led to “linear systems in an infinity of unknowns”

$$\sum_{j=1}^{\infty} a_{ij} x_j = b_i \quad (i = 1, 2, \dots). \quad (21.6)$$

Fourier, who comes across such a system, “solves” it still like a mathematician of the XVIIIth century: he suppresses all the terms having an index i or j greater than n , solves explicitly the finite system thus obtained, by means of Cramer’s rule, then “passes to the limit” by making n tend to $+\infty$ in the solution! When later it was no longer sufficient to play such lazy games of pass the parcel, it is still by means of the theory of determinants that the first attempts to attack the problem were made; starting in 1886 (following the work of Hill), H. Poincaré, then H. von Koch, had built a theory of “infinite determinants” which allows the solution of certain types of systems (21.6) following the classical model; and if these results were not directly applicable to the problem Fredholm aimed at, at least it is certain that the theory of von Koch, in particular, was of use as a model for the formation of his “determinants”.

It is at this moment that Hilbert enters the scene and gives a new impulsion to the theory [163 b]. He starts by completing the work of Fredholm in achieving effectively the passage to the limit that leads from the solution of (21.5) to that of (21.4); but he adds to it immediately the corresponding passage to the limit for the theory of real quadratic forms, to which the types of integral equations with symmetric kernel (that is to say such that $K(y, x) = K(x, y)$) were leading naturally, by far the most frequent in mathematical Physics. He reaches thus the fundamental formula that generalises directly the reduction of a quadratic form to its axes

$$\int_a^b \int_a^b K(s, t) x(s) x(t) ds dt = \sum_{n=1}^{\infty} \frac{1}{\lambda_n} \left(\int_a^b \phi_n(s) x(s) ds \right)^2, \quad (21.7)$$

the λ_n being the eigenvalues (necessarily real) of the kernel K , the ϕ_n forming the orthogonal system of corresponding eigenfunctions, and the second member of the formula (21.7) being a convergent series for $\int_a^b x^2(s) ds \leq 1$. He shows also how every function “representable” in the form $f(x) = \int_a^b K(x, y) g(y) dy$ allows the “expansion” $\sum_{n=1}^{\infty} \phi_n \int_a^b \phi_n(y) f(y) dy$, and carrying on the analogy with the classical theory of quadratic forms, he indicates a procedure for the determination of the λ_n by a variational method, which is none other than the extension of the well-known extremal properties of the axes of a quadric ([163 b], pp. 1-38).

These first results of Hilbert were almost immediately taken up by E. Schmidt, in a simpler and more general form, avoiding the introduction of the “Fredholm determinants” as well as the passage from finite to infinite, and already very near an abstract statement, the fundamental properties of linearity and of positivity of the integral being visibly used only in the proofs [274 a]. But already Hilbert had reached conceptions which are much more general still. All the preceding work brought out the importance of functions that are square integrable, and the Parseval formula established a tight link between these functions and the sequences (c_n) such that $\sum_n c_n^2 < +\infty$. It is no doubt this idea that guides Hilbert in his memoirs of 1906 ([163 b], chap. XI-XIII) where, taking up the old method of “indeterminate coefficients”, he shows that the solution of the integral equation (21.4) is equivalent to a system of an infinity of linear equations

$$x_p + \sum_{q=1}^{\infty} k_{pq} x_q = b_p (p = 1, 2, \dots) \quad (21.8)$$

for the “Fourier coefficients” $x_p = \int_a^b u(t) \omega_p(t) dt$ of the unknown function u with respect to a given complete orthonormal system (ω_n) (with $b_p = \int_a^b f(t) \omega_p(t) dt$ and $k_{pq} = \int_a^b \int_a^b K(s, t) \omega_p(s) \omega_q(t) ds dt$). As well, the only solutions of (21.8) to be considered from this point of view are those for which $\sum_n x_n^2 < +\infty$; also it is to this type of solution that Hilbert systematically limits himself; but over against that he relaxes the conditions imposed on the “infinite matrix” (k_{pq}) (which, in (21.8), is such that $\sum_{p,q} k_{pq}^2 < +\infty$). From that moment, it is clear that the “Hilbert space” of sequences $x = (x_n)$ of real numbers such that $\sum_n x_n^2 < +\infty$, although not explicitly introduced, underlies all the theory, and appears as a “passage to the limit” starting from Euclidean space of finite dimension. Over and above that, what is particularly important for subsequent developments, Hilbert is led to introduce in this space, not only one, but two distinct notions of convergence (corresponding to what has been called the weak topology and the strong topology³), as well as a “principle of choice” which is none other than the property of weak compactness of the unit ball. The new linear algebra that he develops in connection with the solution of the systems (21.8) rests entirely on these topological notions: linear maps, linear forms and bilinear forms (associated with linear maps) are classified and studied according to their “continuity”⁴ properties. In particular, Hilbert discovers that the success of Fredholm’s method rests on the notion of “complete continuity”, that he brings out by formulating it for bilinear forms⁵ and studying it deeply; we can not give here more details about this important notion, nor on the admirable and deep work where Hilbert inaugurates the spectral theory of symmetric bilinear forms (bounded or not).

The language of Hilbert still remains classical, and, throughout the “Grundzüge”, he never ceases to have in view the applications of the theory, of which he develops numerous examples (taking up about half the volume). The following generation will already adopt a more abstract point of view. Under the influence of ideas of Fréchet and of F. Riesz on general topology (see p. 142), E. Schmidt [274 b] and Fréchet himself introduce deliberately, in 1907-1908, the language of Euclidean geometry into “Hilbert space” (real or complex); it is in these works that is found the first mention of the norm (with the actual notation $\|x\|$), the triangle inequality that

it satisfies, the fact that Hilbert space is “separable” and complete; moreover, E. Schmidt proves the existence of the orthogonal projection on a closed linear variety, which allows him to give a simpler and more general form to Hilbert’s theory of linear systems. In 1907 also, Fréchet and F. Riesz remark that the space of square summable series has a fully analogous “geometry”; an analogy that is explained perfectly when, some months later, F. Riesz and E. Fischer prove that this space is complete and isomorphic to “Hilbert space”, at the same time bringing out in a startling way the value of the tool newly created by Lebesgue. From that moment the essential points of the theory of Hilbert spaces can be considered as acquired; among more recent progress, mention should most notably be made of the axiomatic presentation of the theory given around 1930 by M. H. Stone and J. von Neumann, as well as the casting off of the restrictions of “separability”, which is effected around 1934, in the works of Rellich, Löwig, and F. Riesz [260 g].

However other streams of ideas were coming, in the first years of the XXth century, to reinforce the tendency that was leading to the theory of normed spaces. The general idea of a “functional” (that is to say a function with numerical values defined on a set whose elements are themselves numerical functions of one or several real variables) was brought out in the last decades of the XIXth century, linked to the calculus of variations, on the one hand, the theory of integral equations on the other. But if it is mainly to the Italian school, around Pincherle and especially Volterra, that is due the bringing to light of this notion, as well as the more general idea of “operator”, the work of this school remained often of a passably formal nature and attached to problems of a particular type, lacking a sufficiently deep analysis of the underlying topological concepts. In 1903, Hadamard inaugurates the modern theory of “topological” duality, by seeking the most general continuous linear “functionals” on the space $\mathcal{C}(I)$ of numerical functions on a compact interval I (a space furnished with a topology of uniform convergence), and in characterising them as limits of sequences of integrals $x \mapsto \int_I k_n(t)x(t)dt$. In 1907, Fréchet and F. Riesz show in the same way that, on a Hilbert space, continuous linear forms are the “bounded” forms introduced by Hilbert; then, in 1909, F. Riesz puts into definitive form Hadamard’s theorem, by expressing all continuous linear functionals on $\mathcal{C}(I)$ by a Stieltjes integral, a theorem that would later become the point of departure for the modern theory of integration (see p. 226).

The following year, it is still F. Riesz [260 c] who makes new and important progress in the theory by introducing and studying (copied from the theory of Hilbert space) the spaces $L^p(I)$ of functions that are p -power summable in an interval I (for an exponent p such that $1 < p < +\infty$), a study that he follows three years later [260 e] with analogous work on the spaces of sequences $L^p(\mathbf{N})$; these researches, as we will see later, were greatly to contribute to clarifying the ideas on duality, because of the fact that here were met for the first time two spaces which were dual and not naturally isomorphic. ⁶

From that moment, F. Riesz was thinking about an axiomatic study encompassing all these results ([260 c], p. 452); and it seems that only the scruples of an analyst concerned not to distance himself too much from classical mathematics kept him from writing his famous memoir of 1918 on Fredholm theory in this form [260 f]. He considers there in principle the space $\mathcal{C}(I)$ of functions continuous on a compact interval; but, after having defined the norm of this space, and having remarked that $\mathcal{C}(I)$, together with this norm, is complete, he never uses again in his arguments any thing other than the axioms of normed complete spaces. ⁷ Without going into a detailed examination of this work here, let us mention that it is there that is to be found for the first time defined in a general way the notion of a completely continuous linear map (by the property of transforming a neighbourhood into a relatively compact set); ⁸ and, by a chef-d'oeuvre of axiomatic analysis, the whole of Fredholm's theory (in its qualitative aspect) is reduced to a single fundamental theorem, namely that every normed locally compact space is finite dimensional.

The general definition of normed spaces was given in 1920-22 by S. Banach, H. Hahn and E. Helly (this latter only considering spaces of sequences of real or complex numbers). In the ten years that followed, the theory of spaces is developed mainly around two questions of fundamental importance for applications: the theory of duality and the theorems that are attached to the notion of Baire "category".

We have seen that the idea of duality (in the topological sense) goes back to the beginning of the XXth century; it underlies Hilbert's theory and occupies a central place in the work of F. Riesz. This latter observes, for example, already in 1911 ([260 d], pp. 41-42), that the relation $|f(x)| \leq M\|x\|$ (taken as the definition of "bounded" linear functionals in Hilbert space) is

equivalent to the continuity of f when one is in the space $\mathcal{C}(I)$, and that by an argument of fully general character. In connection with the characterisation of continuous linear functionals on $\mathcal{C}(I)$, he remarks also that the condition for a set A to be dense in $\mathcal{C}(I)$ is that there does not exist any Stieltjes measure $\mu \neq 0$ on I that is “orthogonal” to every function of A (generalising thus the Gram condition for complete orthonormal systems); he notes finally, in the same work, that the dual of the space L^∞ is “bigger” than the space of Stieltjes measures ([260 d], p. 62).

On the other hand, in his work on the spaces $L^p(I)$ and $L^p(\mathbb{N})$, F. Riesz succeeds in modifying the method for the solution of linear systems in Hilbert space, given by E. Schmidt [274 b], in such a way as to make it applicable to more general cases. The idea of E. Schmidt consisted in determining an “extremal” solution of (21.6) by seeking the point of the closed linear variety represented by the equations (21.6), whose distance to the origin is minimal. By using the same idea, F. Riesz shows that a necessary and sufficient condition that there exists a function $x \in L^p(a, b)$ satisfying the equations

$$\int_a^b \alpha_i(t)x(t)dt = b_i (i = 1, 2, \dots) \quad (21.9)$$

(where the α_i belong to L^q (with $\frac{1}{p} + \frac{1}{q} = 1$), and such that as well $\int_a^b |x(t)|^p dt \leq M^p$, is that, for all finite sequences $(\lambda_i)_{1 \leq i \leq n}$ of real numbers, one has

$$\left| \sum_{i=1}^n \lambda_i b_i \right| \leq M \cdot \left(\int_a^b \sum_{i=1}^n |\lambda_i \alpha_i(t)|^q dt \right)^{1/q}. \quad (21.10)$$

In 1911 [260 d], he treats in an analogous way the “problem of generalised moments”, consisting of solving the system

$$\int_a^b \alpha_i(t)d\xi(t) = b_i (i = 1, 2, \dots) \quad (21.11)$$

where the α_i are continuous and the unknown is a Stieltjes measure ξ ;⁹ and it is visible here that the problem can be stated by saying that it is a case of determining a continuous linear functional on $\mathcal{C}(I)$ by its values on a sequence of given points in this space. It is also in this form that Helly treats the problem in 1912, — obtaining the conditions of F. Riesz by a fairly different method and of wider applicability¹⁰ — and that he takes it up again in 1921, under

much more general conditions. Introducing the notion of norm (on spaces of sequences) as we have seen above, he remarks that this notion generalises that of the “gauge” of a convex body in space of n dimensions, used by Minkowski in his famous work on the “geometry of numbers” [221 a and b]. In the course of this work, Minkowski had also defined (in R^n) the notions of support hyperplane and of “support function” [221 b], and proved the existence of a support hyperplane at all boundary points of a convex body ([221 a], pp. 33-35). Helly extends these notions to a space of sequences E , furnished with an arbitrary norm; he establishes a duality between E and the space E' of sequences $u = (u_n)$ such that, for all $x = (x_n) \in E$, the series $(u_n x_n)$ should be convergent; $\langle u, x \rangle$ designating the sum of this series, he defines in E' a norm by the formula $\sup_{x \neq 0} |\langle u, x \rangle| / \|x\|$, which gives the support function in finite dimensional spaces.

¹¹ The solution of a system (21.6) in E , where each of the sequences $u_i = (a_{ij})_{j \geq 1}$ is assumed to belong to E' , reduces, as Helly shows then, to solving successively the following two problems: 1° find a continuous linear form L on the normed space E' , such that $L(u_i) = b_i$ for every index i , which, as he indicates, leads to conditions of type (21.10); 2° seek if such a linear form can be written $u \mapsto \langle u, x \rangle$ for an $x \in E$. This last problem, as Helly observes, does not necessarily have a solution even when L exists, and he limits himself to giving some sufficient conditions that imply the existence of the solution $x \in E$ in certain particular cases [156].

These ideas acquire their definitive form in 1927, in a fundamental memoir of H. Hahn [142] whose results are found again (independently) by S. Banach two years later [14 c]. The procedure of Minkowski-Helly is applied by Hahn to an arbitrary normed space, and therefore gives to the dual the structure of a (complete) normed space; which allows Hahn immediately to consider successive duals of a normed space, and to state in a general way the problem of reflexive spaces, foreseen by Helly. But above all the crowning problem of the extension of a continuous linear functional while preserving its norm is definitively solved by Hahn in a completely general way, by a transfinite induction argument on the dimension — thus giving one of the first examples of an important application of the axiom of choice to Functional Analysis.

¹² To these results, Banach adds a thorough study of the relations between a continuous linear map and its transpose, extending to general normed spaces results known until then only for the spaces L^p [260 c], by means of a deep theorem on the weakly closed subsets of a dual; these

results are expressed besides in a more striking way by using the notion of quotient space of a normed space, introduced a few years later by Hausdorff and Banach himself. Finally, it is again Banach who discovers the link between the weak compactness of the unit ball (observed in many particular cases, as we have pointed out above) and reflexivity, at least for spaces of countable type ([14 a], p. 189). The theory of the duality of normed spaces can, from this moment, be considered as settled in its major aspects.

At the same time some theorems of paradoxical aspect were clarified, the first examples of which go back to the neighbourhood of 1910. Hellinger and Toeplitz had in effect proved in substance that year, that a sequence of bounded bilinear forms $B_n(x, y)$ on a Hilbert space, whose values $B_n(a, b)$ for every given pair (a, b) are bounded (by a number dependant a priori on a and b), is in fact uniformly bounded on every ball. Their proof proceeds by contradiction and consists in constructing a particular pair (a, b) violating the hypothesis, by an induction method known since by the name of “method of the sliding hump”, and which is still in use in many analogous questions. Already in 1905, Lebesgue had besides used an analogous procedure to prove the existence of continuous functions whose Fourier series diverges at certain points; and, the same year as Hellinger and Toeplitz, he applied the same method to prove that a weakly convergent series in L^1 is bounded in the norm.¹³ These examples are multiplied in the years that follow, but without the introduction of any new ideas until 1927, the date when Banach and Steinhaus (with the partial collaboration of S. Saks) link these phenomena to the notion of thin sets and to the theorem of Baire on complete metric spaces, obtaining a general statement that covers all the previous particular cases [15]. The study of questions of “category” in normed complete spaces besides leads Banach, at the same time, to numerous other results on continuous linear maps; the most remarkable and no doubt the deepest is the theorem of the “closed graph” which, like the Banach-Steinhaus theorem, showed itself to be a tool of the first order in modern Functional Analysis [14 c].

The publication of the treatise of Banach on “Linear Operators” [14 a] marks, one could say, the beginning of the adult age for the theory of normed spaces. All the results of which we have been speaking, as well as many others, are set out in this volume, in a way that is still a bit disorganised, but accompanied with many striking examples taken from varied domains of

Analysis, and which seemed to forecast a brilliant future for the theory. As it happened, the work had considerable success, and one of its most immediate effects was the quasi-universal adoption of the language and notations used by Banach. But, in spite of a large amount of research undertaken for 40 years on Banach spaces ([203 bis]), if one makes an exception of the theory of Banach algebras and its applications to commutative and non-commutative harmonic analysis, the almost total absence of new applications of the theory to the big problems of classical Analysis has somewhat disappointed the hopes based upon it.

It is rather in the sense of an expanding and a more thrusting axiomatic analysis of the concepts relative to normed spaces that the most fruitful developments have occurred. Although the functional spaces met since the beginning of the XXth century arose for the most part furnished with a “natural” norm, it had not happened without some exceptions being noticed. Around 1910, E. H. Moore had proposed generalising the notion of uniform convergence by replacing it with the notion of “relative uniform convergence”, where a neighbourhood of 0 is made up of the functions f satisfying a relation $|f(t)| \leq \epsilon g(t)$, g being a function everywhere > 0 , which could vary with the neighbourhood. It had on the other hand been observed, before 1930, that notions such as simple convergence, convergence in measure for measurable functions, or compact convergence for entire functions, are not capable of being defined by means of a norm; and in 1926, Fréchet had noted that vector spaces of this nature can be metrisable and complete. But the theory of these more general spaces was only to develop in a fruitful way in combination with the idea of convexity. This latter (that we saw appearing with Helly) was an object of study for Banach and his pupils, who recognised the possibility of interpreting thus in a more geometric way numerous statements of the theory of normed spaces, preparing the way for the general definition of locally convex spaces, given by J. von Neumann in 1935. The theory of these spaces, and notably the questions dealing with duality, have been especially developed during the last ten years. What must be noted in this context, is on the one hand, the progress in simplicity and in generality made possible by the setting up of the fundamental notions of general Topology, carried out between 1930 and 1940; on the other hand, the importance taken by the notion of bounded set, introduced by Kolmogoroff and von Neumann in 1935, and whose fundamental role in duality theory has been brought to light by the work of Mackey [213 a and b] and of

Grothendieck [138 b and c]. Finally and especially, it is certain that the main impulsion that motivated this research came from new possibilities for applications to Analysis, in domains where Banach theory was inoperative: the theory of sequence spaces must be mentioned in this context, developed by Köthe, Toeplitz and their pupils since 1934 in a series of memoirs [184], the recent setting up of the theory of “analytic functionals” of Fantappi , and above all the theory of distributions of L. Schwartz [280], where the modern theory of locally convex spaces found a field of applications that is without doubt a long way from being exhausted.

Chapter 22

INTEGRATION IN LOCALLY COMPACT SPACES.

The development of the modern notion of integral is tightly linked to the evolution of the idea of function, and to a deep study of the numerical functions of real variables, which has been taking place since the beginning of the XIXth century. It is known that Euler already conceived of the notion of function in a fairly general way, since for him a given “arbitrary” curve intersected in a single point by every parallel to the axis Oy defined a function $y = f(x)$ (cf. p. 196); but, like the majority of his contemporaries, he refused to allow that such functions could be expressed “analytically”. This point of view was to change hardly at all until the work of Fourier; but the discovery, by this latter, of the possibility of representing discontinuous functions as sums of trigonometric series, ¹ was to exercise a decisive influence on subsequent generations. Truth to say, the proofs of Fourier totally lacked in rigour, and their domain of validity did not appear clearly; however the integral formulae

$$a_n = \frac{1}{\pi} \int_{-\pi}^{+\pi} \phi(x) \cos nxdx, b_n = \frac{1}{\pi} \int_{-\pi}^{+\pi} \phi(x) \sin nxdx (n \geq 1) \quad (22.1)$$

giving the coefficients in the expansion of ϕ as a Fourier series, had an intuitively obvious meaning as soon as ϕ is assumed to be continuous and piecewise monotonic. ² Also it is at first to these functions that Dirichlet limits himself, in the famous memoir ([92], v. I, pp. 117-132) where he established the convergence of Fourier series; but already, at the end of his work, he

is concerned with the extension of his results to wider classes of functions. It is known that it is on this occasion that Dirichlet, making the ideas of Fourier precise, defines the general notion of function such as we understand it today; the first point to clarify was naturally to know in which cases it was still possible to attach a meaning to the formulae (22.1). “When the solutions for the continuity [of ϕ] are infinite in number ...” says Dirichlet (*loc. cit.*, pp. 131-132), “it is necessary that then the function $\phi(x)$ should be such that, if one denotes by a and b two arbitrary quantities between $-\pi$ and $+\pi$, it is always possible to put between a and b other quantities r and s close enough so that the function remains continuous in the interval r to s . The necessity for this restriction will be felt by considering that the different terms of the series [of Fourier] are definite integrals and in going back to the fundamental notion of integrals. It will be seen then that the integral of a function does not mean anything except in as much as the function satisfies the condition previously stated.”

In modern terms, Dirichlet seems to believe that integrability is equivalent to the fact that the points of discontinuity make up a sparse set; he points out besides, a few lines later, the famous example of the function equal to c for x rational, and to a different value d for x irrational, and affirms that this function “could not be substituted” in the integral. He announced besides subsequent work on this subject, but this work was never published,³ and for 25 years, no one seems to have sought to advance along this path, perhaps because the consideration of such “pathological” functions seemed at the time completely lacking in interest; in any case, when Riemann, in 1854 ([259 a], pp. 227-264), takes up the question (always in connection with trigonometric series⁴) he experiences the need to justify his work: “Whatever our ignorance concerning the way in which the forces and the states of matter vary with time and place in the infinitely small, we can nonetheless take it as certain that the functions to which the researches of Dirichlet do not apply, do not occur in natural phenomena. However”, he continues, “it appears that these cases not treated by Dirichlet deserve attention for two reasons. Firstly, as Dirichlet himself remarks at the end of his work, this subject is in a very close relation to the principles of the infinitesimal calculus, and can be of use to bring more clarity and certainty to these principles. From this point of view, his study has an immediate interest. In the second place, the application of Fourier series is not limited to research in physics; they are at present applied also with success

in a domain of pure mathematics, number theory, and there it seems that they are precisely the functions whose expansion in trigonometric series has not been studied by Dirichlet, that are of importance" ([259 a], pp. 237-238).

The idea of Riemann is to start from the approximation procedure for the integral, brought back to respectability by Cauchy, and to determine when the "Riemann sums" of a function f , in a bounded interval $[a, b]$, tend to a limit (the maximal length of the intervals in the subdivision tending to 0); a problem of which he obtains without trouble the solution in the following form: for every $\alpha > 0$, there is a subdivision of $[a, b]$ into partial intervals of sufficiently small maximal length so that the sum of the lengths of the intervals of this subdivision, where the oscillation of f is $> \alpha$ is arbitrarily small. He shows as well that this condition is satisfied, not only for continuous and piecewise monotonic functions, but also for functions that may have an everywhere dense set of points of discontinuity.⁵

The memoir of Riemann was not published until after his death, in 1867. But this time, the times were more favourable for this kind of research, and the "Riemann integral" took its place naturally in the stream of ideas that was leading then to a sustained study of the "continuum" and of functions of real variables (Weierstrass, Du Bois-Reymond, Hankel, Dini) and was to end up, with Cantor, at the blossoming of set theory. The form given by Riemann to the integrability condition suggested the idea of "measure" for the set of points of discontinuity of a function in an interval; but nearly 30 years were to pass by before a fruitful and convenient definition of this notion would be given.

The first attempts in this direction are due to Stolz, Harnack and Cantor (1884-1885); the first two, in order to define the "measure" of a bounded subset E of $\mathbb{R}$, consider sets $F \supset E$ which are finite unions of intervals, take for each F the sum of the lengths of the corresponding intervals, and say that the "measure" of E is the lower bound of these numbers; whereas Cantor, starting from the beginning in the space $\mathbb{R}^n$, considers for a bounded set E and for $\rho > 0$ the neighbourhood $V(\rho)$ of E constructed of points whose distance to E is $\leq \rho$, and takes the lower bound of the "volume" of $V(\rho)$.⁶ With this definition, the "measure" of a set is equal to that of its closure, whence in particular it is possible that the "measure" of the union of two sets with no point in common can be strictly less than the sum of the "measures" of these two sets. No

doubt in order to alleviate this last difficulty, Peano [246 a] and Jordan ([174 c], v. I, pp. 28-31), a few years later, introduce, alongside the “measure” of Cantor $\mu(A)$ of a set A contained in a block I , its “interior measure” $\mu(I) - \mu(I - A)$, and call “measurable” the sets A (now called “quarriable”) for which these two numbers coincide. The union of two quarriable sets A, B with no point in common is then quarriable and has as “measure” the sum of the “measures” of A and B ; but an open bounded set is not necessarily quarriable, and neither is the set of rational numbers contained in a bounded interval, which removed much of the interest in the notion of Peano-Jordan.

It is to E. Borel [32 a] that the credit for having known how to discern the faults of the previous definitions and for having seen how it was possible to remedy them is due. It was known since Cantor that every open set U in $\mathbf{R}$ is the union of a countable family of its “components”, pairwise disjoint open intervals; instead of seeking to approach U “from the outside” by enclosing it in a finite sequence of intervals, Borel, relying on the preceding result, proposes taking as the measure of U (when U is bounded) the sum of the lengths of its components. Then he describes very summarily ⁷ the class of sets (also called “Borel”) that can be obtained, starting from the open sets, by iterating indefinitely the operations of countable union and of “difference” $A - B$, and indicates that, for these sets, a measure can be defined that possesses the fundamental property of complete additivity: if a sequence (A_n) is made up of pairwise disjoint Borel sets, the measure of their union (assumed bounded) is equal to the sum of their measures.

This definition was to inaugurate a new era in Analysis: on the one hand, in tandem with the contemporary work of Baire, it became the point of departure for a whole series of researches of a topological nature on the classification of sets of points (see p. 165); and above all, it was to be of use as the basis of the extension of the notion of integral, created by Lebesgue in the first years of the XXth century.

In his thesis [196 a], Lebesgue starts by making precise and developing the succinct indications of E. Borel; imitating the method of Peano-Jordan, the “exterior measure” of a bounded set $A \subset \mathbf{R}$ is defined as the lower bound of the measures of the open sets containing A ; then, if I is an interval containing A , the “interior measure” of A is the difference between the exterior measures of I and of $I - A$; thus is obtained a notion of a “measurable set” which only differs

from the initial “constructive” definition of Borel by the adjunction of a subset of a set of measure zero in the sense of Borel. This definition extended immediately to the spaces $\mathbf{R}^n$; the old conception of the definite integral $\int_a^b f(t)dt$ of a bounded function ≥ 0 as the “area” bounded by the curve $y = f(x)$, the straight lines $x = a, x = b$ and $y = 0$, supplied thus an immediate extension of the Riemann integral to all the functions f for which the measure of the preceding set was defined. But the originality of

Lebesgue does not reside so much in the idea of this extension,⁸ than in his discovery of the fundamental theorem on the passage to the limit in the integral thus conceived, a theorem which appears in his work as a consequence of the complete additivity of the measure;⁹ he realises immediately all its importance, and makes it the key stone of the didactic exposé of his theory that he gives, already in 1904, in the famous “Lessons on integration and the quest for primitive functions” [196 c].¹⁰

We cannot describe here in detail the innumerable advances that the results of Lebesgue were to bring about in the study of the classical problems of infinitesimal Calculus. Lebesgue himself had already, in his thesis, applied his theory to the extension of the classical notions of length and area to more general sets than the usual curves and surfaces; concerning the considerable development of this theory over half a century, we send the reader back to the recent exposé by L. Cesari [59]. Let us mention also the applications to trigonometric series, developed by Lebesgue almost immediately after his thesis [196 b], and which were to open new horizons for this theory, whose exploration is far from being over in our day (see [345]). Finally and especially, the definition of the spaces L^p and the Fischer-Riesz theorem ([111], [260 a and c]; cf. p. 213) brought to light the role that could be played in Functional Analysis by the new notion of integral; a role that could only grow with the subsequent generalisations of this notion, of which we shall speak in a moment.

Beforehand, we will stop a little longer with one of the problems to which Lebesgue devoted the most effort, the link between the notions of integral and of primitive. With the generalisation of the integral introduced by Riemann the question that arose naturally was to know if the classical correspondence between integral and primitive, valid for continuous functions, still subsisted in the more general case. But it is easy to give examples of functions f , integrable in

the Riemann sense, and such that $\int_a^x f(t)dt$ does not have a derivative (nor even a derivative on the right or on the left) at certain points;

conversely, Volterra had shown, in 1881, that a function $F(x)$ may have a bounded derivative in an interval I , but be non integrable (in the Riemann sense) in I . By an analysis of great subtlety (where the theorem on the passage to the limit in the integral is far from being sufficient), Lebesgue succeeds in showing that, if f is integrable (in his sense) in $[a, b]$, $F(x) = \int_a^x f(t)dt$ has almost everywhere a derivative equal to $f(x)$ [196 c]. Conversely, if a function g is differentiable in $[a, b]$ and if its derivative $g' = f$ is bounded, f is integrable and we have the formula $g(x) - g(a) = \int_a^x f(t)dt$. But Lebesgue notes that the problem is much more complex when g' is not bounded; g' is not necessarily integrable in this case, and the first problem is therefore to characterise the continuous functions g for which g' exists almost everywhere and is integrable. By limiting himself to the case where one of the derived numbers¹¹ of g is everywhere finite, Lebesgue showed that g is necessarily a function with bounded variation.¹² Finally he establishes the converse of this last result: a function with bounded variation g has almost everywhere a derivative, and g' is integrable; but it is no longer necessarily true that

$$g(x) - g(a) = \int_a^x g'(t)dt; \quad (22.2)$$

the difference between the two parts of this relation is a function of bounded variation which is not constant and with zero derivative almost everywhere (a “singular” function). It remained to characterise the functions with bounded variation g such that the relation (22.2) held. Lebesgue established that these functions (called “absolutely continuous” by Vitali, who made a detailed study of them) are those that have the following property: the total variation of g in an open set U (sum of the total variations of g in each of the connected components of U) tends to 0 with the measure of U .

We will see below how, in a weakened form, these results would later acquire a much more general scope. In their initial form, their field of application remained fairly restricted, and did not go beyond the framework of the “fine” theory of functions of a real variable, of which we will not study here the subsequent development; we will be content with mentioning the deep work of Denjoy and his followers and successors (Perron, de la Vallée-Poussin, Khintchine, Lusin,

Banach, etc.); the reader will find a detailed exposé in the book of S. Saks [268].

One of the essential advances brought about by the Lebesgue theory concerns multiple integrals. This notion was introduced about the middle of the XVIIIth century, and first in the form of an “indefinite integral” (by analogy with the theory of the integral of functions of a single variable, $\int \int f(x, y) dx dy$ denotes a solution of the equation $\frac{\partial^2 z}{\partial x \partial y} = f(x, y)$); but already from 1770, Euler had a very clear conception of the double integral extended to a bounded domain (bounded by arcs of analytic curves), and wrote down correctly the formula for evaluating such an integral by means of two simple integrals in succession ([108 a], (1), v. XVII, pp. 289-315). It was not difficult to justify this formula starting from “Riemann sums”, so long as the function being integrated was continuous, and the domain of integration not too complicated; but as soon as one wanted to tackle more general cases, the procedure of Riemann ran into serious difficulties ($f(x, y)$ may be integrable in the Riemann sense, without $\int dx \int f(x, y) dy$ having a meaning when the simple integrals are taken in the Riemann sense). These difficulties vanish when we pass to the definition of Lebesgue; already this latter had showed in his thesis that, when $f(x, y)$ is a bounded “Baire function”, it is the same for the functions $y \mapsto f(x, y)$ (for all x) and $x \mapsto \int f(x, y) dy$, and we have the formula

$$\int \int f(x, y) dx dy = \int dx \int f(x, y) dy \quad (22.3)$$

(integral taken in the rectangle). A little later, Fubini [121] brought an important complement to this result by proving that, if it is only assumed that f is integrable, then the set of x such that $y \mapsto f(x, y)$ is not integrable is of zero measure, which allowed the extension of the formula (22.3) to this case.

Finally, in 1910 [157 e], Lebesgue tackles the extension to multiple integrals of his results on the derivatives of simple integrals. He is thus brought to associate with a function f , integrable on all compact subsets of $\mathbf{R}^n$, the function of a set $F(E) = \int_E f(\mathbf{x}) dx$, defined for all integrable subsets E of $\mathbf{R}^n$, which generalises the concept of “indefinite integral”; and he observes on this occasion that this function possesses the following two properties: 1° it is completely additive; 2° it is “absolutely continuous” in the sense that $F(E)$ tends to 0 with the measure of E . The essential part of the memoir of Lebesgue consists in proving the converse of this proposition. ¹³

But he does not stop there, and in the same direction, signals the possibility of generalising the notion of function with bounded variation, by considering functions of a measurable set $F(E)$, completely additive and such that $\sum_n |F(E_n)|$ remains bounded for every countable partition of E into measurable subsets E_n . And, if he limits himself in fact to not considering such functions except on the set of blocks of $\mathbf{R}^n$, it is quite clear that there only remained one step to take to end up with the general notion of measure that J. Radon is going to define in 1913, gathering into the same synthesis the Lebesgue integral and the Stieltjes integral, of which we must now speak.

In 1894, T. Stieltjes published, under the title “Research on continuous fractions” [297], a very original memoir where, starting from a question which is very restricted in appearance, were to be found stated and solved, with a rare elegance, problems of a quite new nature in the theory of analytic functions and that of functions of a real variable.¹⁴ In order to represent the limit of a certain sequence of analytic functions, Stieltjes had been brought, among others, to introduce on the straight line, the concept of a positive “mass distribution”, a familiar notion for a long time in the physical sciences, but which until then had never been considered in mathematics except under restrictive hypotheses (in general, the existence of a “density” at every point, varying in a continuous way); he remarks that giving such a distribution is equivalent to that of an increasing function $\phi(x)$ which gives the total mass contained in the interval with ends 0 and x for $x > 0$, and this mass having changed sign for $x < 0$, the discontinuities of ϕ corresponding to masses “concentrated at a point”.¹⁵ Stieltjes then forms, for such a mass distribution in an interval $[a, b]$, the “Riemann sums” $\sum_i f(\xi_i)(\phi(x_{i+1}) - \phi(x_i))$ and shows that, when f is continuous in $[a, b]$, these sums tend to a limit that he denotes $\int_a^b f(x)d\phi(x)$. Only needing to integrate continuous functions (and even differentiable functions) Stieltjes did not push further forward with the study of this integral¹⁶ and for about ten years this notion does not seem to have attracted attention.¹⁷ But in 1909, F. Riesz [260 d], solving a problem stated a few years earlier by Hadamard (cf. p. 213), proved that the Stieltjes integrals $f \mapsto \int_a^b f d\phi$ are the most general continuous linear functionals on the space $\mathcal{C}(I)$ of numerical functions continuous on $I = [a, b]$ ($\mathcal{C}(I)$) being furnished with the uniform convergence topology¹⁸; and the elegance and simplicity of this result caused almost immediately several generalisations of it. The most successful was that

of J. Radon, in 1913 [256]: combining the ideas of F. Riesz and of Lebesgue, he showed how an integral could be defined by the procedures of Lebesgue, starting with an arbitrary “completely additive function of a set” (defined on measurable sets for the Lebesgue measure) instead of starting with the Lebesgue measure. In the notion of “Radon measure” on R^n , thus defined, that of function with “bounded variation” is found to be absorbed: the decomposition of such a function as a difference of two increasing functions is a particular case of the decomposition of a measure as a difference of two positive measures; in the same way, the “measure with base μ ” corresponds to the notion of an “absolutely continuous” function, and the decomposition of an arbitrary measure into a measure with base μ and a measure foreign to μ , to the Lebesgue decomposition of a function with bounded variation as the sum of an absolutely continuous function and a “singular” function. As well, Radon showed that the “density” with respect to μ of a measure with base μ exists still when μ is a measure having as base the Lebesgue measure, by using a previous idea of F. Riesz (taken up again and popularised later by J. von Neumann among others), which consists in constructing an image of the measure μ by a map θ of R^n into R , chosen so that $\theta(\mu)$ is the Lebesgue measure on R .

Almost immediately after the appearance of the memoir of Radon, Fréchet remarked that nearly all the results of this work could be extended to the case where the “completely additive functions of a set”, instead of being defined on the measurable sets of R^n , are defined for certain subsets of an arbitrary set E (these subsets being such that the operations of countable union and of “difference” still give sets for which the function is defined). However, the expression of a measure with base μ in the form $g \cdot \mu$ relied, with Lebesgue and Radon, on arguments using in an essential way the topology of R^n (and we have seen that the proof of Radon only applies if μ is a measure with base the Lebesgue measure); it is only in 1930 that O. Nikodym [235] obtained this theorem in its general form, by a direct argument (notably simplified a few years later by J. von Neumann, thanks to the use of the properties of the spaces L^2 ([324 g], pp. 127-130)).

With the memoir of Radon, the general theory of integration could be considered as completed in its main lines; as subsequent substantial acquisitions, it is only possible to mention the definition of the infinite product of measures, due to Daniell [76 b], and that of the integral of a function with values in a Banach space, given by Bochner in 1933 [24 a], and which was a

prelude to the more general study of the “weak integral” developed a few years later by Gelfand [125 a], Dunford and Pettis ([95] and [96]). But it remained to popularise the new theory, and to make of it a mathematical instrument in general use, while the majority of mathematicians, around 1910, still saw in the “Lebesgue integral” only an instrument of great precision, delicate handling, destined only for researches of extreme subtlety and of an extreme abstraction. That was the work of Carathéodory, in a book which remained a classic for a long time [49] and which enriched Radon’s theory with numerous original remarks.

But it is with this book also that the notion of integral, which had been in the front line of the concerns of Lebesgue (as was sufficiently shown by the titles of his thesis [196 a] and of his main work on these questions [196 c]) yields the field for the first time to that of measure, which had been for Lebesgue (as before him with Jordan) an auxiliary technical means. This change in point of view was due no doubt, with Carathéodory, to the excessive importance that he seems to have attached to “p-dimensional measures”.¹⁹ Since then, the authors that have treated integration are divided between these two points of view, not without involving themselves in debates which have caused a lot of ink to flow if not a lot of blood. One group have followed Carathéodory; in their exposés ever more abstract and more axiomatised, measure, with all the technical refinements to which it lends itself, not only plays the dominant role, but also tends to lose contact with the topological structures to which it is in fact linked in the majority of problems where it is involved. Other exposés follow more or less closely a method already indicated in 1911 by W. H. Young, in a memoir unfortunately little noticed [339 b], and developed since by Daniell. The first, dealing with the Lebesgue integral, parted company with the “Cauchy integral” of continuous functions with compact support, assumed known, in order to define successively the upper integral of lower semi-continuous functions, then of arbitrary numerical functions, whence a definition of integrable functions, modelled on that of Lebesgue for sets, by purely “functional” means. Daniell, in 1918 ([76 a]; cf. [207]) extended this exposé, with some variants, to functions defined on an arbitrary set; in the same stream of ideas (and in close link with the methods used in spectral theory before Gelfand), we must also point out the memoir of F. Riesz [260 h] which puts into an elegant and concise form the few results of the theory of ordered spaces that play a role in the theory of Integration.

Rather than in works of exposition, more or less pleasant to read, but whose substantial content could not vary much, it is on the side of applications that progress arising from the theory of Integration since 1920 must be sought: the theory of probability (at other times the pretext for guessing and paradoxes, and become a branch of the theory of Integration since its

axiomatisation by Kolmogoroff [183], but an autonomous branch with its own methods and problems); ergodic theory; spectral theory and harmonic analysis, since the discovery by Haar of the measure that bears his name, and the movement of ideas provoked by this discovery, have made of the integral one of the most important tools in the theory of groups.

Chapter 23

HAAR MEASURE. CONVOLUTION.

The notions of length, area and volume among the Greeks are essentially based on their invariance under displacements: “Things which coincide (ἐφαρμόζοντα) are equal” (Eucl. El., Book I, “Common notions” 4); and it is by an ingenious use of this principle that all the formulae are obtained giving the areas or volumes of the classical “figures” (polygons, conics, polyhedra, spheres, etc.), sometimes by a procedure of finite decomposition, sometimes by “exhaustion”.¹ In modern language, it can be said that what was done by the Greek geometers, is to prove the existence of “functions of sets” which are additive and invariant under displacements, but defined only for sets of a very particular type. The integral calculus can be considered as replying to the need for enlarging the domain of definition of these functions of sets, and, from Cavalieri to H. Lebesgue, it is this preoccupation that will be in the forefront of the research of the analysts; as for the property of invariance under displacements, it becomes a secondary consideration, having become a trivial consequence of the general formula for change of variables in double or triple integrals and because of the fact that an orthogonal transformation has determinant equal to ± 1 . Even in the non-Euclidean geometries (where however the group of displacements is different), the point of view remains the same: in a general way, Riemann defines the infinitesimal elements of area or volume (or their analogues for dimensions ≥ 3) starting with a ds^2 by means of the classical Euclidean formulae, and their invariance under transformations that leave invariant the ds^2 is thus almost a tautology.

It is only around 1890 that other less immediate extensions of the notion of measure invariant under a group appear, with the development of the theory of the integral invariants, notably by

H. Poincaré and E. Cartan;

H. Poincaré only foresees groups with one parameter operating on a part of space, whereas E. Cartan is mainly interested in groups of displacements, but operating in other spaces than those in which they are defined. For example, he determines thus among others ([52 a], v. II₁, pp. 265-302) the invariant (under a group of displacements) measure in the space of straight lines in $\mathbb{R}^2$ or in $\mathbb{R}^3$; ² further, he points out that in general the integral invariants for a Lie group are none other than particular differential invariants, and that it is thus possible to determine all of them by the methods of Lie. It seems however that no one thought to consider or to use a measure invariant on the group itself before the fundamental work of A. Hurwitz in 1897 [168]. Seeking to construct polynomials (on $\mathbb{R}^n$) invariant under the orthogonal group, Hurwitz starts from the remark that, for a finite group of linear transformations, the problem is solved immediately by taking the mean of the transforms s.P of an arbitrary polynomial P by all the elements s of the group; which gives him the idea of replacing, for the orthogonal group, the mean by an integral relative to an invariant measure; he gives explicitly the expression for this latter by means of the parametric representation by the angles of Euler, but observes immediately (independently of E. Cartan) that the methods of Lie supply the existence of an invariant measure for all Lie groups. Perhaps because of the decline of the theory of invariants at the beginning of the XXth century, the ideas of Hurwitz had hardly any immediate reaction, and would not be considered valuable until after 1924, with the extension to compact Lie groups, by I. Schur and H. Weyl, of the classical theory of Frobenius (p. 121) on the linear representations of finite groups. The first limits himself to the case of the orthogonal group, and shows how the method of Hurwitz allows the extension of the classical relations of orthogonality of characters; an idea that H. Weyl combines with the work of E. Cartan on semisimple Lie algebras to obtain the explicit expression for the characters of the irreducible representations of compact Lie groups and the complete reducibility theorem [331 b], then, by a bold extension of the notion of “regular representation”, the famous theorem of Peter-Weyl, a perfect analogy of the decomposition of the regular representation into its irreducible components in the theory of finite groups [332].

One year previously, O. Schreier had founded the general theory of topological groups [276 a], and from that moment it was clear that the arguments of the memoir of Peter-Weyl would

remain valid as such for every topological group on which could be defined an “invariant measure”. Truth to tell, the general notions of topology and measure were still at this time in full formation, and neither the category of topological groups on which one could hope to define an invariant measure, nor the sets for which this “measure” would be defined, seemed clearly delimited. The only evident point was that one could not hope to extend to the general case the infinitesimal methods proving the existence of an invariant measure on a Lie group. But, another stream of ideas, issuing from work on the Lebesgue measure, was leading precisely to more direct methods of attack. Hausdorff had proved, in 1914, that there does not exist an additive function on sets not identically zero defined for all subsets of $\mathbf{R}^3$ and invariant under displacements, and it was natural to find out if this result was still valid for $\mathbf{R}$ and $\mathbf{R}^2$: a problem that was solved by S. Banach in 1923 in a surprising way, by showing on the contrary that in these two cases such a “measure” well and truly existed [14 b]; his procedure, very ingenious, relied already on a construction by transfinite induction and on the consideration of “means” $\frac{1}{n} \sum_{k=1}^n f(x + \alpha_k)$ of translates of a function by elements of the group. ³ It is analogous ideas that allowed A. Haar, in 1933 [139], to make the decisive step by proving the existence of an invariant measure for locally compact groups with a countable base of open sets: inspired by the method for the approximation of a volume, in classical integral Calculus, by a juxtaposition of congruent cubes with arbitrarily small sides, he obtains, with the help of the diagonal procedure, the invariant measure as the limit of a sequence of “approximating measures”, a procedure that still is essentially the one used today. This discovery had a very big repercussion, in particular because it allowed J. von Neumann to solve immediately, for compact groups, Hilbert’s famous “5th problem” on the characterisation of Lie groups by purely topological properties (to the exclusion of all previously given differential structure). But it was noticed immediately that, in order to be able to use the invariant measure in an efficacious way, it was not enough only to know of its existence, but also to know that it was unique up to a factor; this point was first proved by J. von Neumann for compact groups, by using a method for the definition of the Haar measure by “means” of continuous functions, analogous to those of Banach [324 e]; then J. von Neumann [324 f] and A. Weil [330 c], by different methods, obtained simultaneously the uniqueness in the case of locally compact groups, A. Weil indicating at the same time how

the procedure of Haar could be extended to general locally compact groups. It is also A. Weil (loc. cit.) who obtained the condition for the existence of a relatively invariant measure on a homogeneous space, and showed finally that the existence of a “measure” (endowed with reasonable properties) on a separated topological group, implied ipso facto that the group is locally precompact. These works completed essentially the general theory of Haar measure; the only more recent addition that needs quoting is the notion of quasi-invariant measure, which hardly emerged until around 1950, linked with the theory of representations of locally compact groups in Hilbert spaces.

The history of the convolution product is more complex. Already from the beginning of the XIXth century, it is noticed that if, for example, $F(x, t)$ is an integral of a partial differential equation in x and t , linear and with constant coefficients, then

$$\int_{-\infty}^{+\infty} F(x-s, t) f(s) ds$$

is also an integral of the same equation; Poisson, among others, already before 1820, uses this idea to write the integrals of the heat equation in the form

$$\int_{-\infty}^{+\infty} \exp\left(-\frac{(x-s)^2}{4t}\right) f(s) ds. \quad (23.1)$$

A little later, the expression

$$\frac{1}{2\pi} \int_{-\pi}^{+\pi} \frac{\sin \frac{2n+1}{2}(x-t)}{\sin \frac{x-t}{2}} f(t) dt \quad (23.2)$$

for the partial sum of a Fourier series and the study, made by Dirichlet, of the limit of this integral when n tends to $+\infty$, gives the first example of “regularisation” $f \mapsto \rho_n * f$ on the torus T (in actual fact by a sequence of non-positive “kernels” ρ_n , which complicates considerably its study); under the name of “singular integrals”, the analogous integral expressions will be a favourite subject for the analysts of the end of the XIXth century and the beginning of the XXth century, from P. du Bois-Reymond to H. Lebesgue. On $\mathbb{R}$, Weierstrass uses the integral (23.1) for a proof of his theorem on approximation by polynomials, and gives in this connection the general principle of regularisation by a sequence of positive “kernels” ρ_n of the form $x \mapsto c_n \rho(nx)$. On

T, the most famous example of a regularisation by positive kernels is given a little later by Fejér, and from that moment, it is the standard procedure which will be at the basis of the majority of “summation methods” of series of functions.

However, this work, because of the lack of symmetry of the roles that are played by the “kernel” and the regularised function, hardly made apparent the algebraic properties of the convolution product. It is to Volterra especially that is owed the stressing of this point. He studies in a general way the “composition” $F * G$ of two functions of two variables

$$(F * G)(x, y) = \int_x^y F(x, t)G(t, y)dt$$

that he sees as a generalisation, by “passing from the finite to the infinite”, of the product of two matrices [322 a]. He distinguishes very early on the case (“of the closed cycle” so called because of its interpretation in the theory of heredity) where F and G depend only on $y - x$; it is then the same with $H = F * G$, and if we set $F(x, y) = f(y - x)$, $G(x, y) = g(y - x)$, we have

$$H(x, y) = h(y - x),$$

with

$$h(t) = \int_0^t f(t - s)g(s)ds$$

so that, for $t \geq 0$, h coincides with the convolution of functions f_1, g_1 equal respectively to f and g for $t \geq 0$, and to 0 for $t < 0$.

Meanwhile, the algebraic formalism developed by Volterra did not bring out the links with the group structure of $\mathbb{R}$ and the Fourier transform. We do not need to write the history of this latter here; but it is appropriate to note that starting with Cauchy the analysts who deal with the Fourier integral are mainly concerned with finding wider and wider conditions for the validity of the different formulae of “inversion”, and neglect somewhat its algebraic properties. It is certainly not possible to say as much of the work of Fourier himself (or of that of Laplace on the analogous integral $\int_0^{+\infty} e^{-st}f(t)dt$); but these transformations had been introduced essentially in connection with linear problems, and it is therefore not very surprising that for a long

time it had not been thought worth considering the product of two Fourier transforms (with the exception of the products of trigonometric series or of entire series, but the link with the convolution of discrete measures could obviously not have been noticed in the XIXth century). The first mention of this product and of the convolution on $\mathbb{R}$ is probably to be found in a memoir of Tchebychef [306], in connection with questions about the calculation of probabilities. Indeed, in this theory, the convolution $\mu * \nu$ of two “probability laws” on $\mathbb{R}$ (positive measures of total mass 1) is none other than the “product” probability law of μ and ν (for the sum of the corresponding “random variables”). Of course, with Tchebychef, it is still only a matter of the convolution of probability laws having a density (with respect to the Lebesgue measure), hence of the convolution of functions; besides it only occurs with him in an episodic way, and it will be thus in the few rare works where it appears before the period 1920-1930. In 1920, P.-J. Daniell, in a note rarely consulted [76 c], defines the convolution of two arbitrary measures on $\mathbb{R}$ and the Fourier transform of such a measure, and observes explicitly that the Fourier transform takes the convolution to the ordinary product; a formalism that, starting in 1925, is going to be intensively used by the probabilists, following P. Lévy especially. But the fundamental importance of the convolution in the theory of groups is only fully recognised by H. Weyl in 1927; he realises that, for a compact group, the convolution of functions plays the role of multiplication in the group algebra of a finite group, and allows him as a consequence to define the “regular representation”; at the same time, he finds in the regularisation the equivalent of the unit element of the algebra of a finite group. It remained to make the synthesis of all these points of view, which is accomplished in the book of A. Weil [330 c], a prelude to subsequent generalisations that will constitute, on the one hand the normed algebras of I. Gelfand, and on the other the convolution of distributions.

The Haar measure and convolution have rapidly become essential tools in the tendency to algebraisation that distinguishes so strongly modern Analysis; we will have to mention numerous applications of it in subsequent notes. We will limit ourselves here to pointing out that one which concerns the “variation” of closed subgroups (and notably discrete subgroups) of a locally compact group. This theory, starting with a result of K. Mahler in the Geometry of numbers, was inaugurated in 1950 by C. Chabauty, and has just been considerably developed and deepened

by Macbeath and Swierczkowski [212].

Chapter 24

INTEGRATION IN NON LOCALLY COMPACT SPACES.

If the study of the links between topology and measure theory goes back to the beginnings of the modern theory of functions of real variables, it is only very recently that integration in discrete topological spaces has been set up in a general way. Before giving the history of the work which preceded the actual synthesis, we will recollect some of the steps in the evolution of the ideas concerning the relations between topology and measure.

For Lebesgue, it is only a question of integrating the functions of one or several real variables. In 1913, Radon defined general measures on $\mathbf{R}^n$ and the corresponding integrals; this theory is expounded in detail in the work [82] of Ch. de la Vallée Poussin and relies in a constant way on the topological properties of Euclidean spaces. A bit later, in 1915, Fréchet defines in [115 c] “abstract” measures on a set furnished with a σ -algebra and the integrals with respect to these measures; he notes that it is possible to establish thus the main results of Lebesgue’s theory without using any topological means. He justifies his undertaking with the following words, taken from the introduction to [115 d], first part: “That for example in a space with an infinity of co-ordinates where various applications of Analysis had led to various non equivalent definitions of a convergent sequence, one of these definitions is replaced by another, nothing will be changed in the properties of the families and additive functions of sets in these spaces”. The research of Fréchet is completed by Carathéodory, to whom is due the important theorem on the

extension of a function of sets into a measure. The beginning of the book of Saks [268] offers a condensed exposition of this point of view.

The discovery of Haar measure on locally compact groups (pp. 231 ff.) and the numerous applications that it receives immediately, then the work of Weil and Gelfand in Harmonic Analysis, lead in 1940 to a deep modification of this point of view: in this kind of question, it is most convenient to consider a measure to be a linear form on a space of continuous functions. This method forces a restriction to compact or locally compact spaces, but it is not a hindrance for almost the totality of the applications; even better, the introduction of Harmonic Analysis on the p -adic groups and the adèle groups by J. Tate and A. Weil allowed a spectacular renewing of analytic Number theory.

It is from quite another direction that the necessity comes to enlarge this point of view by the consideration of measures on topological spaces which are not locally compact: little by little, Probability Theory leads to the study of such spaces and supplies numerous non trivial examples. Perhaps the reason for the late influence of these developments on measure theory must be sought in the relative isolation of Probability Theory, which remained on the margin of traditional mathematical disciplines until recent times.

MEASURES ON SEQUENCE SPACES.

One of the most developed branches of classical Probability Theory is that of limit theorems (the law of large numbers, tendency towards the Gauss-Laplace law, ...); it is a case of a deepening of the notion of statistical regularity manifested by phenomena bringing into play very large populations. The correct mathematical formulation of these problems requires the introduction of measures on the space of sequences; these spaces, which constitute the most obvious generalisation of spaces of finite dimension, are the favourite subject of the researches on “General Analysis” undertaken around 1920 by Fréchet, Lévy, Lusin, ... It is in any case not fortuitous that Khintchine and Kolmogoroff, the creators of the new methods in Probability Theory, are both disciples of Lusin, and that Lévy very soon turned to probabilistic problems: these constituted the touch stone of the new methods.

The first implicit occurrence of a measure on a sequence space appears in the work devoted

by E. Borel in 1909 to countable probabilities [32 b]. A very original idea of Borel consisted of the application of probabilistic results that he had just obtained to the proof of properties possessed by the decimal expansion of nearly every real number contained between 0 and 1. This application rests on the following fundamental remark: let us define every real number contained between 0 and 1 by the sequence of digits in its expansion in a given base q ($q \geq 2$); if one takes at random successively the different digits from a number x , independently the ones from the others and with an equal probability $1/q$ for $0, 1, \dots, q-1$, the probability that x is to be found in an interval of $[0,1]$ is equal to the length of this interval.

In 1923, Steinhaus [293] establishes these results rigorously and describes the precise mathematical model of the unbounded sequence of selections of the kind considered by Borel: let us take $q = 2$ to simplify matters and denote by I the set with two elements $\{0, 1\}$; we furnish I with the measure μ defined by $\mu(0) = \mu(1) = \frac{1}{2}$; the elements of the product space $I^{\mathbb{N}}$ are the sequences $\varepsilon = (\varepsilon(n))_{n \in \mathbb{N}}$ of numbers equal to 0 or 1 and the map $\phi : \varepsilon \mapsto \sum_{n \geq 0} \varepsilon(n) \cdot 2^{-n-1}$ is, up to a countable set, a bijection of $I^{\mathbb{N}}$ on the interval $[0, 1]$; what is more, ϕ^{-1} transforms the Lebesgue measure on $[0, 1]$ into the measure P on $I^{\mathbb{N}}$ the product of the measures μ on each of the factors. In fact Steinhaus does not have at his disposal a construction for product measures;

he uses the existence of the quasi-bijection ϕ to construct the measure P on $I^{\mathbb{N}}$ starting from the Lebesgue measure on $[0,1]$, then he gives an axiomatic characterisation of P . The isomorphism thus obtained allows the translation of the language of probability into that of measure and to apply the known theorems about the Lebesgue integral.

In the same work, Steinhaus considers the random series $\sum_{n \geq 0} \sigma_n \cdot a_n$, where the signs $\sigma_n = \pm 1$ are chosen at random independently the ones from the others and with the same probability $\frac{1}{2}$; between 1928 and 1935, he studies numerous other random series. On their side, Paley, Wiener and Zygmund consider random Fourier series¹ of the form $\sum_{-\infty}^{\infty} a_n \exp(2\pi i(nt + \Phi_n))$; the “amplitudes” a_n are fixed, and the “phases” Φ_n are independent random variables uniformly spread over $[0, 1]$. If the analytic difficulties vary enormously from problem to problem, the translation in terms of measure theory is the same in all cases and represents an extension of the case treated by Borel and Steinhaus; it is a matter of constructing a measure on $\mathbf{R}^{\mathbb{N}}$, the product of a family of measures all identical to one particular positive measure μ of mass 1 on $\mathbf{R}$; for

example, the preceding random Fourier series correspond to the case where μ is the Lebesgue measure on $[0, 1]$.

To construct such product measures, two methods can be used. The first is a direct method, set up for the first time by Daniell [76 b] in 1918; it is found again in 1934 by Jessen [173] who will make a detailed study of the case where μ is the Lebesgue measure on $[0, 1]$. The second method is the search for tricks analogous to those of Steinhaus in order to reduce to the case of the Lebesgue measure on $[0, 1]$; this way of proceeding had the advantage of convenience as long as there is not at our disposal a complete account of the general theory of measure, for it allows the use without new proofs of the Lebesgue theorems. ²

THE THEORY OF BROWNIAN MOVEMENT.

This theory occupies an exceptional place in contemporary scientific development because of the constant and fruitful exchange which it witnesses between physical problems and “pure” mathematics. The study of brownian movement, discovered in 1829 by the botanist Brown, has been carried on intensively in the XIXth century by numerous physicists, ³ but the first satisfactory mathematical model was invented by Einstein only in 1905. In the simple case of a particle moving along a straight line, the fundamental hypotheses of Einstein are formulated thus: if $x(t)$ is the abscissa of the particle at time t , and if $t_0 < t_1 < \dots < t_{n-1} < t_n$, the successive displacements $x(t_i) - x(t_{i-1})$ (for $1 \leq i \leq n$) are independent gaussian random variables. It is not the place to evoke here in detail the important experimental work of J. Perrin which motivated Einstein’s theory: for our purposes, it is sufficient to remember only a remark of Perrin, according to which the observation of the trajectories of brownian movement suggested irresistibly to him “the functions without derivative of the mathematicians”. This remark was the initial spark for Wiener.

Quite another stream of ideas draws its origins from the kinetic theory of gases, developed between 1870 and 1900 by Boltzmann and Gibbs. Let us consider a gas made up of N molecules of mass m at (absolute) temperature T and let us denote by $v_1, \dots, v_N$ the velocities of the N molecules of the gas; the kinetic energy of the system is equal to

$$\frac{m}{2}(\mathbf{v}_1^2 + \dots + \mathbf{v}_N^2) = 3NkT \quad (24.1)$$

where k is Boltzmann's constant. According to the ideas of Gibbs, the multitude of collisions between molecules does not allow the precise determination of the velocities of the molecules, and it is convenient to introduce a probability law P on the sphere S of the space with $3N$ dimensions defined by the equation (24.1). The “microcanonical” hypothesis consists in assuming that P is the measure of mass 1 invariant under rotation on the sphere S . Besides, Maxwell's law of velocities states that the law of probability of the component of the velocity of a molecule has a Gaussian measure of variance $2kT/m$. E. Borel seems to have been the first to remark in 1914 that Maxwell's law is a consequence of the hypotheses of Gibbs and of properties of the sphere when the number of molecules is very large. He considers a sphere S in Euclidean space of large dimension and the measure P of mass 1 invariant under rotation on S ; using the classical methods of approximation based on the Stirling formula, he shows that the projection of P on a co-ordinate axis is approximately Gaussian. These results are made precise a little later by Gâteaux and Lévy [201 a]. Given an integer $m \geq 1$ and a number $r > 0$, let us denote by $S_{m,r}$ the set of sequences of the form $(x_1, \dots, x_m, 0, 0, \dots)$ with $x_1^2 + \dots + x_m^2 = r^2$; let us denote also by $\sigma_{m,r}$ the measure of mass 1 invariant under rotation on $S_{m,r}$. Stated in modern language, the result of Gâteaux and Lévy is the following: the sequence of measures $\sigma_{m,1}$ tends strictly to the unit mass at the origin $(0, 0, \dots)$ and the sequence of measures $\sigma_{m,\sqrt{m}}$ tends strictly to a measure Γ of the form

$$d\Gamma(x_1, x_2, \dots) = \prod_{n=1}^{\infty} d\gamma(x_n)$$

(γ is the Gaussian measure with variance 1 on $\mathbb{R}$).

The preceding measure Γ plays the role of a Gaussian measure in infinite dimensions. It appears strongly that Lévy had confusedly hoped to define in an intrinsic manner a Gaussian measure on all Hilbert spaces of infinite dimension. In fact, as was shown by Lévy and Wiener, the measure Γ is invariant in a certain sense ⁴ under the automorphisms of l^2 ; unfortunately, the set l^2 of square summable sequences $(x_1, x_2, \dots, x_n, \dots)$ is of measure zero for Γ . It is

known today that we must be content with a Gaussian premeasure on a Hilbert space of infinite dimension. ⁵

To Wiener is owed the essential progress: if there is no reasonable Gaussian measure on a Hilbert space of infinite dimension, it is possible to construct by means of the primitive operation a measure w on a space of continuous functions starting with a Gaussian premeasure. We shall explain succinctly the initial construction of w by Wiener [336]; it is directly influenced by the relation $\Gamma = \lim_{m \rightarrow \infty} \sigma_{m, \sqrt{m}}$ of Gâteaux and Lévy. For every integer $m \geq 1$, let us denote by H_m the set of functions on $T =]0, 1]$ which are constant on each of the intervals $\left[\frac{k-1}{m}, \frac{k}{m}\right]$ (for $k = 1, 2, \dots, m$), and π_m the measure of mass 1 invariant under rotation on the Euclidean sphere of radius 1 in R^m . Let f_m be the isomorphism of H_m on R^m which associates with each function taking the value a_k on the interval $\left[\frac{k-1}{m}, \frac{k}{m}\right]$ the vector $(a_1, a_2 - a_1, \dots, a_m - a_{m-1})$ (whence the name of “differential space” dear to Wiener); let us denote by w_m the measure on H_m image of π_m under f_m^{-1} . Wiener defines the measure w being sought as the limit of the measures w_m . Precisely, let us denote by H the set of regular functions on T , with the uniform convergence topology (we have $H_m \subset H$ for every integer $m \geq 1$); for every uniformly continuous function F bounded on H , the limit $A\{F\} = \lim_{m \rightarrow \infty} \int_{H_m} F(x) dw_m(x)$ exists; Wiener then obtains certain majorations by a subtle analysis of the fluctuations of the items in play, and, taking up again the arguments of compactness brought out by Daniell, he shows that we are in a situation for the application of Daniell’s extension theorem. We conclude the existence of a measure w carried by $C(T)$ and such that $A\{F\} = \int_{C(T)} F(x) dw(x)$. Wiener can then show that the measure w corresponds to the hypotheses of Einstein, ⁶ and his estimates allowed him to give a precise meaning to the remark of Perrin on functions without a derivative: the set of functions satisfying a Lipschitz condition of order $\frac{1}{2}$, is negligible for w (on the contrary, for every a with $0 < a < \frac{1}{2}$, nearly every function satisfies a Lipschitz condition of order a).

We know today numerous constructions for the Wiener measure. Thus, Paley and Wiener use random Fourier series ([243], chap. IX): to every real sequence $\mathbf{a} = (a_n)_{n \geq 1}$ and every integer $m \geq 0$, let us define the function $f_{m, \mathbf{a}}$ on $]0, 1]$ by

$$f_{m, \mathbf{a}}(t) = a_1 t + 2 \sum_{k=2}^{2^{m+1}} \frac{1}{\pi k} a_{k-1} \sin \pi k t;$$

it can be shown that, for Γ -almost all sequences $\mathbf{a}$, the sequence of functions $f_{m,\mathbf{a}}$ tends to a continuous function $f_{\mathbf{a}}$ and that w is the image of Γ by the map (defined almost everywhere) $\mathbf{a} \mapsto f_{\mathbf{a}}$. Later, Lévy gave in [201 b and c] another construction. Finally Kac, Donsker and Erdős show around 1950 how to replace in Wiener’s initial construction the spherical measures π_m on $\mathbf{R}^m$ by more general measures. Their results establish a solid link between Wiener’s measure and the limit theorems of Probability Theory; they will be completed and systematised by Prokhorov [254] in work to which we will return later.

This is not the place to analyse the numerous and important probabilistic works caused by Wiener’s discovery; today, the brownian movement appears only as one of the most important examples of a Markov process. We will only mention the application made by Kac of Wiener’s measure to the solution of certain parabolic partial differential equations; it is a matter there of an adaptation of the ideas of Feynman in quantum field theory — another example of this reciprocal influence between mathematics and the problems of physics.

PROJECTIVE LIMITS OF MEASURES.

This is the case of a theory that was developed especially in view of the needs of Probability Theory. The problems concerning a finite sequence of random variables $X_1, \dots, X_n$ are solved in principle when the law P_X of this sequence is known: it is a positive measure of mass 1 on $\mathbf{R}^n$, such that the probability of obtaining simultaneously the inequalities $a_1 \leq X_1 \leq b_1, \dots, a_n \leq X_n \leq b_n$ is equal to $P_X(C)$ where C is the closed box $[a_1, b_1] \times \dots \times [a_n, b_n]$ of $\mathbf{R}^n$.

In practice, the measure P_X has discrete support or else admits a density with respect to Lebesgue measure. When it is the case of an infinite sequence $(X_n)_{n \geq 1}$ of random variables, the law P_n of the partial sequence $(X_1, \dots, X_n)$ is known in general for every integer $n \geq 1$; these given laws satisfy a compatibility condition that expresses the fact that the sequence $(P_n)_{n \geq 1}$ is a projective system of measures. Until around 1920, one defined in a more or less implicit manner the probabilities of events linked to the infinite sequence by a “natural” passing to the limit starting with probabilities in the finite case; thus it will be assumed that the probability that a game stops is the limit, for n tending to infinity, of the probability that it stops in at most n steps. Naturally, such a theory is hardly coherent, and nothing excludes the presence of “paradoxes”, a

same probability receiving two distinct estimations according to whether it is evaluated by one or other of two procedures each as “natural” as the other.

Steinhaus [293] seems to have been the first to feel the necessity of considering (for the case of the toss of a coin) not only the projective system $(P_n)_{n \geq 1}$ but its limit. Shortly before, in 1919, Daniell [76 d] had proved the existence of such projective limits in general,⁷ but this result seems to have remained unknown in Europe. It is found again in 1933 by Kolmogoroff in the work [183] where this author formulates the axiomatic conception of Probability Theory. The proofs of Daniell and Kolmogoroff use a compactness argument that rests on Dini’s theorem.

The Daniell-Kolmogoroff theorem leaves nothing to be desired in the case of random sequences $(X_n)_{n \geq 1}$, but the study of random functions undertaken starting in 1935 by Kolmogoroff, Feller and Doob conceals difficulties of quite another order. Let us consider for example an interval T of $\mathbb{R}$, which represents the set of instants of observations of a “stochastic process”; the set of possible trajectories is the product space R^T , considered as a projective limit of partial products R^H , where H runs through the set of finite subsets of T ; one obtains in general a projective system of measures (μ_H) . Kolmogoroff’s theorem does indeed provide a measure on R^T , but it is defined on a family that is notably smaller than the Borel family.⁸ A variant of Kolmogoroff’s construction, which provides a measure on a topological space, is due to Kakutani [177], and has been rediscovered many times since: one considers μ_H as a measure on $\bar{R}^H$ carried by R^H ,⁹ the compact space $E = \bar{R}^T$ is a projective limit of the finite products $\bar{R}^H$ and a measure μ on E can be defined as a projective limit of the μ_H . But this procedure has a severe inconvenience; the elements of $\bar{R}^T$ do not possess any regularity property allowing the continuation of the probabilistic study of the process — or even simply the elimination of the parasitic values $\pm\infty$ introduced by the compactification $\bar{R}$ of $\mathbb{R}$. It can be remedied by restricting the measure μ of $\bar{R}^T$ to a particular subspace (for example $C(T)$ in the case of brownian movement); the fundamental difficulty arises from the fact that a functional space, even of the usual type, is not necessarily μ -measurable in $\bar{R}^T$, and even the choice of the functional space can be in question.¹⁰

A decisive step was accomplished in 1956 by Prokhorov in a work [254] which exercised a determining influence on the theory of stochastic processes. By putting into a convenient ax-

in the same form the methods used by Wiener in the article analysed above, he establishes a general existence theorem for projective limits of measures on functional spaces.

A more restricted class of projective systems was introduced by Bochner [24 b] in 1947; it is a case of projective systems made up of real vector spaces of finite dimension and surjective linear transformations. The projective limit of such a system is identified in a natural way with the algebraic dual E^* of a real vector space E supplied with a weak topology $\sigma(E^*, E)$; a corresponding projective system of measures has a limit which is a measure μ defined for a family which is notably smaller than the Borel family of E^* . Bochner characterised completely such “premeasures” by their Fourier transform, which is a function on E . But this result is hardly usable in the absence of a topology on E , in which case the possibility of considering μ as a measure on the topological dual E' of E must be examined. In an independent way, R. Fortet and E. Mourier, while seeking to generalise to random variables with values in a Banach space certain classical results of Probability Theory (the law of large numbers, the central limit theorem) brought out also the fundamental role played by the Fourier transform in these questions. But substantial progress was only made in 1956 when Gelfand [125 c] suggested that the natural framework for the Fourier transform is not that of Banach spaces or Hilbert spaces, but that of nuclear Fréchet spaces. He conjectured that every continuous function of positive type on such a space is the Fourier transform of a measure on its dual, a result established soon after by Minlos [222]. Its importance comes above all from the fact that it is applicable to distribution spaces, and that the quasi-totality of functional spaces are Borel subspaces of the space of distributions (which thus constitutes a much better setting than $\mathbf{R}^T$).¹¹ The theory of random distributions is an area in full expansion, and we will be content with referring the reader to the work of Gelfand and Vilenkin [126].

The results that we have just mentioned on projective limits use the existence of topologies on the base spaces. It can be asked if there exists an analogous theory in the case of “abstract” measures. Von Neumann proved already in 1935 the existence of product measures in all cases, but the discovery of a counterexample by Jessen and Andersen [290] ruined the hope that every projective system of measures admits a limit. Two palliatives were discovered: in 1949, C. Ionescu-Tulcea established the existence of countable projective limits, granted the existence of

suitable decompositions, ¹² a very interesting result for the study of Markov processes; besides, it became clear that the topology of the spaces does not come in except by means of the set of compact subsets. It was therefore natural to seek to axiomatise this situation inside an abstract theory, by means of the notion of a compact class of subsets of a set. This work was done in 1953 by Marczewski (who establishes by this means a theorem on abstract projective limits) and Ryll-Nardzewski (who worked on the decomposition of measures). ¹³

MEASURES ON GENERAL TOPOLOGICAL SPACES AND STRICT CONVERGENCE.

The study of the links between topology and measure theory was especially thought of as the study of regularity properties of measures, and in particular that of “exterior” regularity and of “interior” regularity; ¹⁴ interior regularity is equivalent to exterior regularity on a locally compact space countable at infinity. The construction that Lebesgue gives of the measure of sets on the straight line brings out these two kinds of regularity, and the property of exterior regularity of measures on a Polish space seems to have had public notoriety around 1935. But it is only in 1940, in an article whose diffusion was slowed down by the war, that A. D. Alexandroff [3] brought out the role of interior regularity and showed that this is possessed by the measures on a Polish space; this result was found again later by Prokhorov [254] and is often attributed mistakenly to this author. It only appeared very recently that this property extended to sublinian spaces; because of this, the importance of these spaces increased greatly, all the more so that it became known that their theory could be done without a metrisability hypothesis, and that the quasi-totality of functional spaces were sublinian (and even most often lusinian). ¹⁵

The definition of a mode of convergence (weak or strict) for measures is done in the most convenient way by putting in duality the space of measures with a space of continuous functions. Generalising an ancient result of F. Riesz, A. A. Markhoff established in 1938 a bijective correspondence between the positive functionals on $\mathcal{C}(X)$ and the regular measures on a compact space X . In the work [3] already quoted, A. D. Alexandroff extends these results to the case of a completely regular space: he introduces a hierarchy on the set of positive linear forms on the

space $\mathcal{C}^b(X)$ of bounded continuous functions on a completely regular space X ,¹⁶ he defines strict convergence of bounded measures and proves among others the following two theorems:

- a) if X is Polish, the set of linear forms on $\mathcal{C}^b(X)$ corresponding to measures is closed in the weak convergence of sequences;
- b) if a sequence of bounded measures has a strict limit, “there is no mass fleeing to infinity” (it is a weak form of the reciprocal of the theorem of strict convergence of Prokhorov).

From this fusion of notions and theorems, Prokhorov will know how to extract important results for the theory of stochastic processes, and to present them in a simple and striking form. In his great work of 1956 already quoted [254], an important part is devoted to bounded positive measures on a Polish space; in generalising a construction of Lévy, he defines on the set of positive measures of mass 1 a distance which makes it into a Polish space, then he establishes an important compactness criterion for strict convergence. Independently of Prokhorov, Le Cam [197] has obtained a certain number of compactness results for the strict convergence of measures; he puts no metrisability hypotheses on the spaces that he considers, and his results reduce to previous theorems of Dieudonné in the locally compact case.

Chapter 25

LIE GROUPS AND LIE ALGEBRAS.

I. GENESIS.

The theory, which has been called for almost a century the “theory of Lie groups”, was essentially set up by one mathematician: Sophus Lie.

Before starting on its history, we will summarise briefly various earlier researches that prepared its development.

a) Transformations (Klein-Lie, 1869-1872).

Around 1860, the theory of permutation groups of a finite set is developing and begins to be used (Serret, Kronecker, Mathieu, Jordan). On the other hand, the theory of invariants, then in full flight, familiarises mathematicians with certain infinite sets of geometric transformations closed under composition (notably linear or projective transformations). But, before the work of 1868 of Jordan [174 b] on the “movement groups” (closed subgroups of the groups of displacements in Euclidean space of 3 dimensions), it does not seem as if any conscious link had been established between these two streams of ideas.

In 1869, the young Felix Klein (1849-1925), a pupil of Plücker, becomes the friend at Berlin of the Norwegian Sophus Lie (1842-1899), a few years older, being brought together by their common interest in the “geometry of straight lines” of Plücker and notably the theory of complexes of straight lines (p. 133). It is around this period that Lie conceives of one of his most

original ideas, the introduction of the notion of invariant in Analysis and in differential geometry; one of the sources of this is his observation that the classical methods of integration “by quadratures” of differential equations all rely on the fact that the equation is invariant for a “continuous” family of transformations. It is from 1869 that is dated the first work (drawn up by Klein) where Lie uses this idea; he studies there the “Reye complex” (a set of straight lines intersecting the faces of a tetrahedron in 4 points having a given cross-ratio) and the curves and surfaces having as tangents straight lines of this complex ([202], vol. I, Abh. V, pp. 68-77): his method relies on the invariance of the Reye complex under the commutative group with 3 parameters (a maximal torus of $\text{PGL}(4, \mathbb{C})$) leaving invariant the corners of the tetrahedron. This same idea dominates the work written together by Klein and Lie when they found themselves in Paris in the spring of 1870 ([182], v. I, pp. 416-420); they essentially determine there the connected commutative subgroups of the projective group of the plane $\text{PGL}(3, \mathbb{C})$, and study the geometric properties of their orbits (under the name of V curves or surfaces); that gives them, by a uniform procedure, properties of varied curves, algebraic or transcendental, such as $y = cx^m$ or the logarithmic spirals. Their testimony agrees in underlining the profound impression that was produced on them by the theories of Galois and of Jordan (the commentary of Jordan on Galois had appeared in *Math. Annalen* in 1869; in any case, Lie had heard of the theory of Galois already in 1863). Klein, who in 1871, begins to be interested in non-Euclidean geometries, sees there the beginning of his research for a classification principle for all known geometries, research that was to lead him in 1872 to the “Erlangen programme”. On his side, Lie in a letter of 1873 to A. Mayer ([202], vol. 5, p. 584), dates from his stay in Paris the origin of his ideas on transformations groups, and in a work of 1871 ([202], vol. I, Abh. XII, pp. 153-214), he uses already the term “transformation group” and states explicitly the problem of the determination of all the subgroups (“continuous or discontinuous”) of $\text{GL}(n, \mathbb{C})$. Truth to tell, Klein and Lie must both have had difficulty in penetrating this new mathematical universe, and Klein speaks of the “Treatise” of Jordan, newly appeared, as a “book sealed with seven seals” ([182], v. I, p. 51); he writes elsewhere in connection with ([182], v. I, pp. 424-459): “It is to Lie that belongs all that is concerned with the heuristic idea of a continuous group of operators, in particular all that touches on the integration of differential equations or partial differential equations. All the notions that

he developed later in his theory of continuous groups were already to be found in embryo with him, but so little elaborated however, that I had to convince him of many details, for example at the beginning even the existence of the curves V, during the course of long conversations” ([182], v. I, p. 415).

b) Infinitesimal transformations.

The conception of an “infinitely small” transformation goes back to the beginnings of the infinitesimal Calculus; it is known that Descartes discovers the instantaneous centre of rotation by assuming that “in the infinitely small” every plane movement can be assimilated into a rotation; the elaboration of analytic Mechanics, in the XVIIIth century, is entirely founded on similar ideas. In 1851, Sylvester, seeking to form invariants of the linear group

$GL(3, C)$ or of certain of its subgroups, gives to the parameters z_j occurring in these matrices “infinitely small” increments of the form $\alpha_j dt$, and expresses that a function $f((z_j))$ is invariant by writing down the equation $f((z_j + \alpha_j dt)) = f((z_j))$; this gives him a linear equation for f with partial derivatives $Xf = 0$, where

$$Xf = \sum_j \alpha_j \frac{\partial f}{\partial z_j}, \quad (25.1)$$

X being thus a differential operator, “a derivative in the direction of directed parameters α_j ” ([304], vol. 3, pp. 326 and 327); Sylvester seems to feel that there is there a general principle of fairly wide scope, but does not seem to have returned to the question. A little later, Cayley ([58], v. II, pp. 164-178) proceeds in the same way for invariants of $SL(2, C)$ in certain representations of this group and shows that it is the solution of two partial differential equations of the first order $Xf = 0$, $Yf = 0$, where X and Y are obtained as above starting from “infinitely small” transformations

$$\begin{pmatrix} 0 & 0 \\ dt & 0 \end{pmatrix} \text{ and } \begin{pmatrix} 0 & dt \\ 0 & 0 \end{pmatrix}.$$

In modern terms, this is explained by the fact that X and Y generate the Lie algebra $\mathfrak{sl}(2, C)$; besides Cayley calculates explicitly the bracket $XY - YX$ and shows that it also comes from an

“infinitely small” transformation.

In his memoir of 1868 on groups of movements [174 b], Jordan uses from beginning to end the concept of “infinitely small transformation”, but exclusively from a geometric point of view. There is no doubt that it is with him that the idea of a group with one parameter “generated” by an infinitely small transformation appears: for Jordan, it is the set of transformations obtained by “repeating suitably” the infinitely small transformation (loc. cit. p. 243). Klein and Lie, in their memoir of 1871, use the same expression “repeated infinitely small transformation” ([182], v. I, pp. 424-459), but the context shows that they mean by that an integral of a differential system. If the group with one parameter that they consider is made up of transformations $x' = f(x, y, t)$, $y' = g(x, y, t)$, the corresponding “infinitely small transformation” is given by

$$dx = p(x, y)dt, dy = q(x, y)dt$$

where $p(x, y) = \frac{\partial f}{\partial t}(x, y, t_0)$, $q(x, y) = \frac{\partial g}{\partial t}(x, y, t_0)$, where t_0 corresponds to the identical transformation of the group. As Klein and Lie know the functions f and g explicitly, they have no problem in checking that the functions

$$t \mapsto f(x, y, t) \text{ and } t \mapsto g(x, y, t)$$

give in a parametric form the integral curve of the differential equation

$$q(\xi, \eta)d\xi = p(\xi, \eta)d\eta$$

going through the point (x, y) , but do not give any general reason for that; besides they do not use this fact again in the rest of their memoir.

c) Contact transformations.

In the following two years, Lie seems to abandon the theory of transformation groups (although he remains in very close contact with Klein, who publishes in 1872 his “Programme”) in order to study contact transformations, the integration of partial differential equations of the first order and the relations between these two theories. We do not have to spell out the history of these

questions here, and we will limit ourselves to mentioning a few points that seem to have played an important role in the genesis of the theory of transformation groups.

The notion of contact transformation generalises at the same time the point transformations and the transformations by reciprocal polars. Grosso modo, a contact transformation ¹ in C^n is an isomorphism of an open set Ω of the variety $T'(C^n)$ of cotangent vectors to C^n onto another open set Ω' of $T'(C^n)$ transforming the canonical 1-form of Ω into that of Ω' . In other words, if $(x_1, \dots, x_n, p_1, \dots, p_n)$ describes the canonical co-ordinates of $T'(C^n)$, a contact transformation is an isomorphism $(x_i, p_i) \mapsto (X_i, P_i)$ satisfying the relation $\sum_{i=1}^n P_i dX_i = \sum_{i=1}^n p_i dx_i$. Such transformations occur in the study of partial differential equations of the form

$$F\left(x_1, x_2, \dots, x_n, \frac{\partial z}{\partial x_1}, \dots, \frac{\partial z}{\partial x_n}\right) = 0 \quad (25.2)$$

Lie becomes familiar during the course of his research on these questions with the handling of Poisson parentheses

$$(f, g) = \sum_{i=1}^n \left(\frac{\partial f}{\partial x_i} \frac{\partial g}{\partial p_i} - \frac{\partial g}{\partial x_i} \frac{\partial f}{\partial p_i} \right) \quad (25.3)$$

and brackets ² $[X, Y] = XY - YX$ of differential operators of type (25.1); he interprets the Poisson parenthesis (25.3) as the effect on f of a transformation of type (25.1) associated with g , and observes on this occasion that the Jacobi identity for Poisson parentheses means that the bracket of differential operators corresponding to g and h is associated with the parenthesis (g, h) . The search for functions g such that $(F, g) = 0$, which occurs in the method of Jacobi for integrating the partial differential equation (25.2), becomes for Lie that for an infinitesimal contact transformation leaving the given equation invariant. Finally, Lie is led to study the set of functions $(u_j)_{1 \leq j \leq m}$ of the x_i and the p_i such that the parentheses (u_j, u_k) are functions of the u_h , and calls these sets “groups” (already considered in substance by Jacobi).

II. CONTINUOUS GROUPS AND INFINITESIMAL TRANSFORMATIONS.

Suddenly, in the autumn of 1873, Lie takes up again the study of transformation groups and obtains decisive results. For as much as it is possible to follow the unravelling of his thoughts in a few letters to A. Mayer from the years 1873-1874 ([202], vol. 5, pp. 585-608), he starts from a “continuous group” of transformations on n variables

$$x'_i = f_i(x_1, \dots, x_n, a_1, \dots, a_r) \quad (1 \leq i \leq n) \quad (25.4)$$

depending effectively ³ on r parameters $a_1, \dots, a_r$; he observes that if the transformation (25.4) is the identity for the values $a_1^0, \dots, a_r^0$ of the parameters, ⁴ then the Taylor expansions of the x_i , limited to the first order:

$$f_i(x_1, \dots, x_n, a_1^0 + z_1, \dots, a_r^0 + z_r) = x_i + \sum_{k=1}^r z_k X_{ki}(x_1, \dots, x_n) + \dots \quad (1 \leq i \leq n) \quad (25.5)$$

give a “generic” infinitely small transformation depending linearly on the r parameters z_j

$$dx_i = \left(\sum_{k=1}^r z_k X_{ki}(x_1, \dots, x_n) \right) dt \quad (1 \leq i \leq n). \quad (25.6)$$

Proceeding as in his memoir with Klein, Lie integrates the differential system

$$\frac{d\xi_1}{\sum_k z_k X_{k1}(\xi_1, \dots, \xi_n)} = \dots = \frac{d\xi_n}{\sum_k z_k X_{kn}(\xi_1, \dots, \xi_n)} = dt, \quad (25.7)$$

which gives him, for each point $(z_1, \dots, z_r)$, a group with one parameter

$$t \mapsto x'_i = g_i(x_1, \dots, x_n, z_1, \dots, z_r, t) \quad (1 \leq i \leq n) \quad (25.8)$$

such that $g_i(x_1, \dots, x_n, z_1, \dots, z_r, 0) = x_i$ for every i . He shows in an ingenious way, using the fact that the transformations (25.4) make up a set closed under composition, that the group with one parameter (25.8) is a subgroup of the given group ([202], vol. 5, Abh. VII, pp. 32-63). The

new idea, key to the whole theory, is to proceed on to the second order in the Taylor expansions of the functions (25.4). The thread of his argument is fairly confused and heuristic ([202], vol. 5, Abh. VII, pp. 32-63 and [202], vol. 5, pp. 600-601); it can be set out as follows. For the z_j that are fairly small, one can put $t=1$ in (25.8), and thus are obtained new parameters $z_1, \dots, z_r$ for the transformations of the group (it is in fact the first appearance of the “canonical parameters”). By definition, we have with regard to (25.7)

$$\frac{\partial g_i}{\partial t} = \sum_k z_k X_{ki}(x'_1, \dots, x'_n),$$

from where

$$\begin{aligned} \frac{\partial^2 g_i}{\partial t^2} &= \sum_{k,j} z_k \frac{\partial X_{ki}}{\partial x_j}(x'_1, \dots, x'_n) \frac{\partial x'_j}{\partial t} = \\ &= \sum_{k,j} z_k \frac{\partial X_{ki}}{\partial x_j}(x'_1, \dots, x'_n) \left(\sum_h z_h X_{hj}(x'_1, \dots, x'_n) \right) \end{aligned}$$

which gives

$$\begin{aligned} x'_i &= x_i + \left(\sum_k z_k X_{ki}(x_1, \dots, x_n) \right) t \\ &+ \frac{1}{2} \left(\sum_{k,h,j} z_k z_h \frac{\partial X_{ki}}{\partial x_j}(x_1, \dots, x_n) X_{hj}(x_1, \dots, x_n) \right) t^2 + \dots, \end{aligned}$$

from where, for $t = 1$, the Taylor expansions with respect to the parameters z_j

$$x'_i = x_i + \left(\sum_k z_k X_{ki} \right) + \frac{1}{2} \left(\sum_{k,h,j} z_k z_h X_{hj} \frac{\partial X_{ki}}{\partial x_j} \right) + \dots \quad (1 \leq i \leq n). \quad (25.9)$$

Let us write these relations in brief as $x' = G(x, z)$ between vectors

$$x = (x_1, \dots, x_n), x' = (x'_1, \dots, x'_n), z = (z_1, \dots, z_r);$$

the fundamental property of closure of the sets of these transformations under composition is written as

$$G(G(x, u), v) = G(x, H(u, v)) \quad (25.10)$$

where $H = (H_1, \dots, H_r)$ is independent of x ; it is immediate that $H(u, 0) = u, H(0, v) = v$,
whence the expansions

$$H_i(u, v) = u_i + v_i + \frac{1}{2} \sum_{h,k} c_{ikh} u_h v_k + \dots, \quad (25.11)$$

the terms not written down being non linear either in u or in v . Transforming (25.10) with the help of (25.9) and (25.11), then comparing the terms in $u_h v_k$ of the two members, Lie obtains the relations

$$\sum_{j=1}^n \left(X_{hj} \frac{\partial X_{ki}}{\partial x_j} - X_{kj} \frac{\partial X_{hi}}{\partial x_j} \right) = \sum_{l=1}^r c_{lkh} X_{li} \quad (1 \leq h, k \leq r, 1 \leq i \leq n). \quad (25.12)$$

His expertise with the theory of partial differential equations leads him to write these conditions in a simpler form: following the model of (25.1), he associates with each of these r infinitely small transformations obtained by putting $z_k = 1$, $z_h = 0$ for $h \neq k$ in (25.6), the differential operator

$$A_k(f) = \sum_{i=1}^n X_{ki} \frac{\partial f}{\partial x_i}, \quad (25.13)$$

and rewrites the conditions (25.12) in the form

$$[A_h, A_k] = \sum_l c_{lkh} A_l, \quad (25.14)$$

a keystone of his theory. Until then, he had used indiscriminately the terms “infinitely small transformation” and “infinitesimal transformation” (e. g. [202], vol. 5, Abh. I, pp. 1-4); the simplicity of the relations (25.14) leads him to call the operator (25.13) the “symbol” of the infinitesimal transformation $dx_i = X_{ki} dt$ ($1 \leq i \leq n$) ([202], vol. 5, Abh. III, pp. 42-75) and very rapidly, it is the operator (25.13) itself that he will call “infinitesimal transformation” ([202], vol. 5, Abh. III, pp. 42-75 and [202], vol. 5, p. 589).

He then becomes aware of the tight links which unite the theory of “continuous groups” with his previous research on contact transformations and partial differential equations. This link fills him with enthusiasm: “My old work was so to speak all ready in advance to found the

new theory of groups of transformations” he wrote to Mayer in 1874 ([202], vol. 5, p. 586).

In the following years, Lie pursued the study of transformation groups. Apart from the general theorems summarised here (§ III), he obtains a certain number of more particular results: the determination of transformation groups of the straight line and of the plane, subgroups of small codimension in projective groups, groups with at most 6 parameters, etc.. For all that he does not abandon differential equations. In fact, it even appears that for him, the theory of transformation groups was to be an instrument for integrating differential equations, where the transformation group would play a role analogous to that of the Galois group of an algebraic equation. ⁵ Let us note that this research leads him likewise to introducing certain sets of transformations with an infinity of parameters, which he calls “infinite and continuous groups”; ⁶ he keeps the name “finite and continuous groups” for the transformation groups with a finite number of parameters of type (25.4) above.

III. THE “DICTIONARY” LIE GROUPS — LIE ALGEBRAS. THE “DICTIONARY” LIE GROUPS — LIE ALGEBRAS.

The theory of “finite and continuous” groups, developed by Lie in numerous memoirs starting in 1874, is set out systematically in the imposing treatise “Theorie der Transformationsgruppen” ([203], 1888-1893), written in collaboration with F. Engel; ⁷ it is the object of the first volume and of the last five chapters of the third, the second being consecrated to contact transformations.

As the title indicates, there was never any question in this work of anything other than transformation groups, in the meaning of the equations (25.4), where the space of “variables” x_i and the space of “parameters” a_j initially both play equally important roles. Besides the concept of “abstract” group is not clearly expressed at this time; when in 1883 ([202], vol. 5, Abh. XII, pp. 311-313) Lie remarks that with the notation of (25.10), the equation $w = H(u, v)$, which gives the parameters of the product of two transformations of the group, defines a new group, it is as a transformation group on the space of parameters that he is considering it, obtaining thus

what he calls the “group of parameters” (he even obtains two of them, which are none other than the group of left translations and the group of right translations).⁸

The variables x_i and the parameters a_j in the equations (25.4) are assumed complex in principle (except in chaps. XIX-XXIV of v. 3), and the functions f_i analytic; Lie and Engel are of course conscious of the fact that these functions are not in general defined for all complex values of the x_i and the a_j and that, in consequence, the composition of such transformations raises serious difficulties ([203], v. I, pp. 15-17, pp. 33-40 and *passim*); and although, later, they express themselves almost always as if composition of the transformations that they study were always possible without restriction, it is no doubt only for the convenience of the statements, and they re-establish explicitly the “local” point of view each time that it is necessary (cf. *loc. cit.*, pp. 168 or 189 for example or *ibid.*, v. 3, p. 2, note at the bottom of the page); in other words, the mathematical object that they are studying is near to what we would now call a partial operation law. They do not fail to consider, on occasion, global groups, for example the 4 series of classical groups ([203], v. 3, p. 682), but do not seem to have set themselves the question of what in general a “global group” can be; it suffices for them to be able to obtain, for the “parameters” of the classical groups (the “variables” of these groups do not introduce any difficulty, since it is a case of linear transformations of $\mathbf{C}^n$), systems of “local” parameters in the neighbourhood of the identical transformation, without their being concerned about the domain of validity of the formulae that they write down. They set themselves however one problem that stands out clearly from the local theory:⁹ the study of “mixed” groups, that is to say groups having a finite number of connected components, such as the orthogonal group ([203], v. I, p. 7). They present this study as that of a set of transformations closed under composition and taking inverses, which is the union of sets H_j such that each is described by systems of functions $f_i^{(j)}$ as in (25.4); the number of (essential) parameters of each H_j is even assumed a priori to depend on j , but they show in fact that this number is the same for all the H_j . Their main result is then the existence of a finite continuous group G such that $H_j = G \circ h_j$ for an $h_j \in H_j$ and for all j ; they establish also that G is normal in the mixed group and remark that the determination of the invariants of this latter reduces to that of the invariants of G and of a discontinuous group ([203], v. I, chap. 18).

The general theory developed in [203] finishes up (without that being stated in a very systematic way by the authors) by forging a “dictionary” making the translation from properties of “finite continuous” groups to those of the set of their infinitesimal transformations. It is based on the “three theorems of Lie”, each one of which is made up of an assertion and its converse.

The first theorem ([203], v. I, pp. 33 and 72 and v. 3, p. 563) affirms in the first instance that if in (25.4) the parameters are effective, the functions f_i satisfy a system of partial differential equations of the form

$$\frac{\partial f_i}{\partial a_j} = \sum_{k=1}^r \xi_{ki}(f(x, a)) \psi_{kj}(a) \quad (1 \leq i \leq n) \quad (25.15)$$

where the matrix (ξ_{kl}) is of maximum rank and $\det(\psi_{kj}) \neq 0$; reciprocally, if the functions f_i have this property, the formulae (25.4) define a groupuscule of transformations.

The second theorem ([203], v. I, pp. 149 and 158, and v. 3, p. 590) gives relations between the ξ_{ki} on the one hand, the ψ_{ij} on the other: the conditions on the ξ_{ki} can be written in the form

$$\sum_{k=1}^n \left(\xi_{ik} \frac{\partial \xi_{jl}}{\partial x_k} - \xi_{jk} \frac{\partial \xi_{il}}{\partial x_k} \right) = \sum_{k=1}^r c_{ij}^k \xi_{kl} \quad (1 \leq i, j \leq r, 1 \leq l \leq n) \quad (25.16)$$

where the c_{ij}^k are constants $(1 \leq i, j, k \leq r)$ anti-symmetric in i, j . The conditions on the ψ_{ij} in the form given by Maurer, are:

$$\frac{\partial \psi_{kl}}{\partial a_m} - \frac{\partial \psi_{km}}{\partial a_l} = \frac{1}{2} \sum_{1 \leq i, j \leq r} c_{ij}^k (\psi_{il} \psi_{jm} - \psi_{jl} \psi_{im}) \quad (1 \leq k, l, m \leq r). \quad (25.17)$$

By introducing the matrix (α_{ij}) contragredient to (ψ_{ij}) and the infinitesimal transformations

$$X_k = \sum_{i=1}^n \xi_{ki} \frac{\partial}{\partial x_i}, A_k = \sum_{j=1}^r \alpha_{kj} \frac{\partial}{\partial a_j} \quad (1 \leq k \leq r), \quad (25.18)$$

(25.16) and (25.17) can be written respectively:

$$[X_i, X_j] = \sum_{k=1}^r c_{ij}^k X_k \quad (1 \leq i, j \leq r) \quad (25.19)$$

$$[A_i, A_j] = \sum_{k=1}^r c_{ij}^k A_k. \quad (25.20)$$

Conversely, if given r infinitesimal transformations $X_k (1 \leq k \leq r)$, linearly independent and satisfying conditions (25.19), the one parameter subgroups generated by these transformations generate a group of transformations with r essential parameters.

Finally, the third theorem ([203], v. I, pp. 170 and 297 and v. 3, p. 597) reduces the determination of systems of infinitesimal transformations $(X_k)_{1 \leq k \leq r}$ satisfying (25.19) to a purely algebraic problem: we must have

$$c_{ij}^k + c_{ji}^k = 0 \quad (25.21)$$

$$\sum_{l=1}^r (c_{il}^m c_{jk}^l + c_{kl}^m c_{ij}^l + c_{jl}^m c_{ki}^l) = 0 (1 \leq i, j, k, m \leq r). \quad (25.22)$$

Conversely, ¹⁰ if (25.21) and (25.22) are satisfied, there exists a system of infinitesimal transformations satisfying relations (25.19), from whence a group of transformations with r parameters (in other words, the linear combinations with constant coefficients of the X_k make up a Lie algebra, and conversely every Lie algebra of finite dimension can be obtained in this way).

These results are completed by the study of isomorphism questions. Two groups of transformations are said to be similar if it is possible to pass from one to the other by an invertible transformation of co-ordinates on the variables and an invertible transformation of co-ordinates on the parameters: from the beginning of his research, Lie had met this notion in a natural way in connection with the definition of the “canonical parameters”. He shows that two groups are similar if, by means of a transformation on the “variables”, one can bring the infinitesimal transformations of the one onto those of the other ([203], v. I, p. 329). A necessary condition for this to be so is that the Lie algebras of the two groups be isomorphic, which Lie expresses by saying that the groups are “gleichzusammengesetzt”; but this condition is not sufficient, and a whole chapter ([203], v. I, chap. 19) is devoted to obtaining supplementary conditions insuring that the groups should be “similar”. On the other hand the theory of permutation groups supplied the notion of the “holoedric isomorphism” of two such groups (isomorphism of the underlying “abstract” groups); Lie transposes this notion to groups of transformations, and shows that two such groups are “holoedric isomorphic” if and only if their Lie algebras are isomorphic ([203],

v. I, p. 418). In particular, every group of transformations is holodetically isomorphic to each of its groups of parameters, and that shows that, when one wishes to study the structure of the group, the “variables” on which they operate do not matter much and that in fact everything reduces to the Lie algebra. ¹¹

Always by analogy with the theory of permutation groups. Lie introduces the notions of subgroups, normal subgroups, “meriedric isomorphisms” (surjective homomorphisms), and shows that they correspond to that of subalgebras, ideals and surjective homomorphisms of Lie algebras; besides he had met very early on a particularly important example of “meriedric isomorphism”, the adjoint representation, and had recognised its links with the centre of the group ([202], vol. 5, Abh. III, pp. 42-75). For these results, as for the fundamental theorems, the essential tool is the theorem of Jacobi-Clebsch giving the complete integrability of a differential system (one of the forms of the theorem called “of Frobenius”); he gives a new proof of it, as it happens, using groups with one parameter ([203], v. I, chap. 6).

The notions of transitivity and of primitivity, so important for groups of permutations, arise also naturally for the “finite and continuous” groups of transformations, and the treatise of Lie-Engel makes a detailed study of them ([203], v. I, chap. 13 and passim); the relations with the subgroups stabilising a point and the notion of homogeneous space are observed (for as much as they could be without putting oneself in a global point of view) ([203], v. I, p. 425).

Finally, the “dictionary” is completed, in [203], by the introduction of the notions of derived group and soluble group (called “integrable group” by Lie; this terminology, suggested by the theory of differential equations, will remain in use until the work of H. Weyl) ([203], v. I, p. 261 and v. 3, pp. 678-679); the relation between commutators and brackets had besides been observed by Lie already in 1883 ([202], vol. 5, p. 358).

Other proofs of fundamental theorems.

In [278 b], F. Schur shows that in canonical co-ordinates the ψ_{ik} of (15) satisfy the differential equations

$$\frac{d}{dt}(t\psi_{ik}(ta)) = \delta_{ik} + \sum_{j,l} c_{jl}^k t a_l \psi_{ji}(ta). \quad (25.23)$$

These integrate and give a formula equivalent to the formula

$$\varpi(X) = \sum_{n \geq 0} \frac{1}{(n+1)!} (\text{ad}(X))^n \quad (25.24)$$

for the right differential $\varpi(X)$ of the exponential map at the point X ; in particular, in canonical co-ordinates, the ψ_{ij} are extended into entire functions of the a_k . F. Schur deduces from this a result making precise an earlier remark of Lie: if, in the definition (25.4) of groups of transformations, it is only assumed that the f_i are in the class C^2 , then the group is holoedrically isomorphic to an analytic group. ¹² Following his research on the integration of differential systems, E. Cartan ([52 a], v. II₂, p. 371) introduces in 1904 the Pfaffian forms

$$\omega_k = \sum_{i=1}^r \psi_{ki} da_i \quad (1 \leq i \leq r) \quad (25.25)$$

(with the notation of (25.15)), later called Maurer-Cartan forms. The conditions (25.17) of Maurer can then be written

$$d\omega_k = -\frac{1}{2} \sum_{i,j} c_{ij}^k \omega_i \wedge \omega_j;$$

E. Cartan shows that it is possible to develop the theory of finite continuous groups starting with the ω_k and establishes the equivalence of this point of view and that of Lie. But, for him, the interest in this method is especially that it adapts itself to “infinite and continuous” groups for which he pushes the theory much further than had been done by Lie, and that it allows the setting up of his theory of the generalised “mobile marker”.

IV. THE THEORY OF LIE ALGEBRAS.

Once the correspondence between groups of transformations and Lie algebras is acquired, the theory is going to take a clearly more algebraic turn and will be centred on a deep study of Lie algebras. ¹³

A first and short period, from 1888 to 1894, marked by the work of Engel, of his pupil Umlauf and especially Killing and E. Cartan, ended up with a series of spectacular results on complex

Lie algebras. We have seen above that the notion of soluble Lie algebra was due to Lie himself, who had proved (in the complex case) the reduction theorem for soluble linear Lie algebras to triangular form ([203], v. I, p. 270).¹⁴ Killing observes [180] that there exists in a Lie algebra a biggest soluble ideal (that is called the radical today), and that the quotient of the Lie algebra by its radical has null radical; he calls the Lie algebras with null radical semisimple, and proves that they are the product of simple algebras (this last notion had already been introduced by

Lie, who had proved the simplicity of the “classical” Lie algebras ([203], v. 3, p. 682)).

On the other hand, Killing introduces, in a Lie algebra, the characteristic equation $\det(\text{ad}(x) - \omega \cdot 1) = 0$, already met by Lie in studying the Lie subalgebras of dimension 2 containing a given element of a Lie algebra. We refer to other historical Notes for the analysis of the methods whereby Killing, in studying in a penetrating manner the properties of the roots of the “generic” characteristic equation for a semisimple algebra, ends up with the most remarkable of his results, the complete determination of the simple (complex) Lie algebras.¹⁵

Killing proves that the derived algebra of a soluble algebra is “of rank 0” (which means that $\text{ad } x$ is nilpotent for every element x of the algebra). Soon after, Engel proves that the algebras “of rank 0” are soluble (this statement is in substance what is called today Engel’s theorem). In his thesis, E. Cartan introduces on the other hand what is now called the “Killing form”, and establishes the two fundamental criteria that characterise by means of this form the soluble Lie algebras and the semisimple Lie algebras.

Killing had affirmed ([180], IV) that the derived algebra of a Lie algebra is the sum of a semisimple algebra and of its radical, which is nilpotent, but his proof was incomplete. A little later, E. Cartan announced without proof ([52 a], v. I, p. 104) that more generally every Lie algebra is the sum of its radical and of a semisimple subalgebra; the only result in this direction that is indisputably established at this time is a theorem of Engel affirming the existence, in every non-soluble Lie algebra, of a simple Lie subalgebra of dimension 3. The first published proof (for complex Lie algebras) of the statement of Cartan is due to E. E. Levi [200]; another proof (equally valid for the real case) was given by J. H. C. Whitehead in 1936 [334 a]. In 1942, A. Malcev completed this result by the uniqueness theorem on the “Levi sections” up to conjugation.

Already from his first work, Lie had set himself the problem of the isomorphism of every Lie algebra with a linear Lie algebra. He had thought that he had solved it affirmatively by considering the adjoint representation (and so deducing a proof of his “third theorem”) ([202], vol. 5, Abh. III, pp. 42-75); he recognised rapidly that his proof was only correct for Lie algebras with null centre; after him, the question remained open for a long time, and was solved affirmatively by Ado in 1935 [2 a]. On the other hand, Lie had substantially set himself the problem of determining the linear representations of minimal dimension of the simple Lie algebras, and had solved it for the classical algebras; in his Thesis, Cartan had also resolved this problem for the exceptional simple algebras; ¹⁶ the methods that he employs in this context will be generalised by him twenty years later in order to obtain all the irreducible representations of simple real or complex Lie algebras.

The property of complete reducibility of a linear representation seems to have been met for the first time (in a geometric form) by Study. In an unpublished manuscript, but quoted in ([203], v. 3, pp. 785-788) he proves this property for the linear representation of the Lie algebra of $SL(2, \mathbb{C})$, and obtains partial results for $SL(3, \mathbb{C})$ and $SL(4, \mathbb{C})$. Lie and Engel conjecture on this occasion that the complete reducibility theorem is valid for $SL(n, \mathbb{C})$ whatever n may be. The complete reducibility of the linear representations of semisimple Lie algebras was established by H. Weyl in 1925 ¹⁷ by an argument of a global nature (see later). The first algebraic proof was obtained in 1935 by Casimir and van der Waerden [55]; other algebraic proofs were given afterwards by R. Brauer [34 b] and J. H. C. Whitehead [334 b].

Finally, during his research on the exponential map (cf. *infra*). H. Poincaré ([251], v. III) considers the associative algebra of differential operators of all orders, generated by the operators of a Lie algebra; he shows, substantially that, if $(X_i)_{1 \leq i \leq n}$ is a basis of a Lie algebra, the associative algebra generated by the X_i has as basis certain symmetric functions of the X_i (sums of the non-commutative “monomials” obtained from a given monomial by all permutations of its factors). The essential part of his proof is algebraic in nature, and allows him to obtain the structure of the enveloping algebra. Analogous proofs have been given in 1937 by G. Birkhoff [22 b] and E. Witt [337 b]. ¹⁸

The majority of the work quoted above is limited to real or complex Lie algebras, which are

the only ones corresponding to Lie groups in the usual sense. The study of Lie algebras over a field other than $\mathbb{R}$ or $\mathbb{C}$ is tackled by Jacobson [172 a] who shows that the greater part of the classical results remain valid over a field of characteristic zero.

V. THE EXPONENTIAL AND THE FORMULA OF HAUSDORFF.

The first research concerning the exponential map is due to E. Study and F. Engel; Engel [104 b] remarks that the exponential is not surjective for $\mathbf{SL}(2, \mathbb{C})$ (for example $\begin{pmatrix} -1 & a \\ 0 & -1 \end{pmatrix}$ is not an exponential if $a \neq 0$), but that it is for $\mathbf{GL}(n, \mathbb{C})$, thus also for $\mathbf{PGL}(n, \mathbb{C})$ (this latter property had already been noted by Study for $n = 2$); thus $\mathbf{SL}(2, \mathbb{C})$ and $\mathbf{PGL}(2, \mathbb{C})$ give an example of two groups that are locally isomorphic, but which are nonetheless very different from a global point of view. Engel shows also that the exponential is surjective in the other classical groups, augmented by homotheties; this work is taken up again and continued by Maurer, Study and others, without bringing in substantial new results.

In 1899, H. Poincaré ([251], v. III, pp. 169-172 and 173-212), tackles the study of the exponential map from a different point of view. His memoirs seem to have been hurriedly written, for in several places he affirms that every element of a connected group is an exponential, whereas he gives examples to the contrary elsewhere. His results refer mainly to the adjoint group: he shows that a semisimple element of such a group G can be the exponential of an infinity of elements of the Lie algebra $L(G)$, whereas a non-semisimple element may be not an exponential. If $\text{ad}(X)$ does not have an eigenvalue which is a non-zero multiple of $2\pi i$, then $\exp$ is étale in X . He proves also that, if U and V describe loops in $L(G)$, and if W is defined by continuity such that $e^U \cdot e^V = e^W$, we do not return necessarily to the initial determination of W . He uses a formula for residues that reduces essentially to

$$\Phi(\text{ad } X) = \frac{1}{2\pi i} \int \frac{\Phi(\xi) d\xi}{\xi - \text{ad } X}$$

where $\text{ad}(X)$ is a semisimple element whose non-zero eigenvalues have multiplicity 1, Φ is

an entire series with a sufficiently large radius of convergence, the integral being taken over a loop enclosing the eigenvalues of $\text{ad } X$; he studies also what happens when X tends towards a transformation having multiple eigenvalues.

The search for expressions for W as a function of U and V in the formula $e^U \cdot e^V = e^W$ had already, a little before the work of Poincaré, been the subject of two memoirs of Campbell [46 a and b]. As is written a little later by Baker “...the theory of Lie suggests in an obvious way that the product $e^U e^V$ is of the form e^W where W is an alternating series in U and V ...”. The subsequent work on this subject aimed to make this assertion precise and to give an explicit formula (or a method of construction) for W (“Hausdorff formula”). After Campbell and Poincaré, Pascal, Baker [13] and Hausdorff [152 a] return to the question; each considers that the proofs of his predecessor are not convincing; the main difficulty resides in what is meant by “alternating”: is it a case of elements of the particular Lie algebra under consideration, or of universal “symbolic” expressions? Neither Campbell, nor Poincaré, nor Baker express themselves clearly on this point. The memoir of Hausdorff, on the contrary, is perfectly precise; he works at first in the algebra of formal associative (non commutative) series in a finite number of indeterminates and considers U, V, W as elements of this algebra. He proves the existence of W by an argument on differential equations analogous to those of his predecessors. The same argument is used by him to prove the convergence of the series when the indeterminates in it are replaced by elements of a Lie algebra of finite dimension. As Baker had remarked, and Poincaré independently, this result can be used to give a proof of the third theorem of Lie; he clarifies the correspondence between groups and Lie algebras, for example as far as the group of commutators is concerned.

In 1947, Dynkin [98 a] takes up the question again, and obtains the explicit coefficients of the Hausdorff formula, by considering from the beginning a normed Lie algebra (of finite dimension or not, over $\mathbb{R}$, $\mathbb{C}$ or an ultrametric field).¹⁹

VI. LINEAR REPRESENTATIONS AND GLOBAL LIE GROUPS.

None of the work of which we have been speaking tackled frankly the problem of the definition and the study of global Lie groups. The first steps along this path go back to H. Weyl. He is inspired by two theories, which until then had developed independently: that of linear representations of semisimple complex Lie algebras, due to E. Cartan, and that of linear representations of finite groups, due to Frobenius and which had just been transposed to the orthogonal group by I. Schur using an idea of Hurwitz. This latter had shown [168] how invariants can be constructed for the orthogonal groups or the unitary groups by replacing the operation of the mean on a finite group by integration relative to an invariant measure. He had also remarked that in applying this method to the unitary group, invariants for the general linear group are obtained, a first example of the “unitarian trick”. In 1924, I. Schur [279 e] used this procedure to show the complete reducibility of the representations of the orthogonal group $O(n)$ and of the unitary group $U(n)$, by the construction of a positive non-degenerate invariant hermitian form; he deduces from this, by the “unitarian trick”, the complete reducibility of the holomorphic representations of $O(n, C)$ and of $SL(n, C)$, establishes orthogonality relations for the characters of $O(n)$ and of $U(n)$ and determines the characters of $O(n)$. H. Weyl also extends this method to complex semisimple Lie algebras [331 b]. Given such an algebra $\mathfrak{g}$, he shows that it possesses a “real compact form” (which amounts to saying that it comes from an algebra $\mathfrak{g}_0$ over R , whose adjoint group G_0 is compact, by extending the scalars from R to C). Furthermore he shows that the fundamental group of G_0 is finite, thus that the universal cover ²⁰ of G_0 is compact. He deduces from this, by an appropriate adaptation of the procedure of Schur, the complete reducibility of the representations of $\mathfrak{g}$, and gives also, by global means, the determination of the characters of the representations of $\mathfrak{g}$. In a letter to I. Schur [331 a], H. Weyl summarises the results of Cartan, that Schur did not know (cf. [279 e], p. 299, note at the foot of the page) and compares the two points of view: the method of Cartan supplies all the holomorphic representations of the simply connected group of the Lie algebra $\mathfrak{g}$; in the case of the orthogonal group, the representations of a cover in two sheets (later called the spinor group), which had escaped Schur’s notice, are also

obtained; from another side, Schur's method has the advantage of proving complete reducibility and of giving the characters explicitly.

After the work of H. Weyl, E. Cartan adopts an openly global point of view in his research on symmetric spaces and Lie groups. It is this point of view that is at the basis of his exposé of 1930 ([52 a], v. I₂, pp. 1165-1225) of the theory of "finite and continuous" groups. There is found in particular the first proof of the global variant of the third fundamental theorem (existence of a Lie group with given Lie algebra); Cartan shows also that every closed subgroup of a real Lie group is a Lie group, which generalises a result of J. von Neumann on the closed subgroups of the linear group [324 b]. In this Memoir, von Neumann showed also that every continuous representation of a complex semisimple group is real analytic.

After this work, the theory of Lie groups in the "classical" sense (that is to say of finite dimension over $\mathbb{R}$ or $\mathbb{C}$) is more or less fixed in its main lines. The first detailed exposé of it is given by Pontrjagin in his book on topological groups [253]; he keeps there to a point of view still fairly close to that of Lie, but in distinguishing carefully the local from the global. It is followed by the book of Chevalley [62 d] which contains also the first systematic discussion of the theory of analytic varieties and of the exterior differential calculus; the "infinitesimal transformations" of Lie appear there as fields of vectors and the Lie algebra of a Lie group G is identified with the space of fields of vectors invariant on the left under G . He leaves to one side the "groupuscul" aspect and the "groups of transformations" aspect.

VII. EXTENSIONS OF THE NOTION OF A LIE GROUP.

In our day, the vitality of Lie theory is manifested by the diversity of its applications (in topology, differential geometry, arithmetic, etc.) as well as by the creation of parallel theories where the structure of the underlying differential variety is replaced by a neighbouring structure (p -adic, algebraic variety, schema, formal schema, ...). We do not need to provide here the history of all these developments, and we will limit ourselves to two of them: Banach Lie groups and p -adic Lie groups.

a) Banach Lie groups.

It is a case of Lie groups “of infinite dimension”. From the local point of view, a neighbourhood of 0 in an Euclidean space is replaced by a neighbourhood of 0 in a Banach space. It is what is done by G. Birkhoff in 1936 [22 a], ending up thus with the notion of a complete normed Lie algebra and with its correspondence with a “groupuscule” defined on an open set of a Banach space. Around 1950, Dynkin completes these results by an extension to this case of the Hausdorff formula (cf. *supra*).

The definitions and results of Birkhoff and Dynkin are local. Until recent days, it does not seem that an attempt has been made to make explicit the corresponding global theory, no doubt because of the lack of applications.

b) p-adic Lie groups.

Such groups are met for the first time in 1907 in the work of Hensel [157 e] on the p-adic analytic functions (defined by expansions in entire series). He studies notably the exponential and the logarithm; in spite of the a priori surprising behaviour of the series that define them (for example the exponential series does not converge everywhere), their functional properties remain valid, which supplies a local isomorphism between the additive group and the multiplicative group of $\mathbb{Q}_p$ (or, more generally, of every complete ultrametric field of characteristic zero).

It is equally about commutative groups (but not linear this time) that the work of A. Weil [330 a] and E. Lutz [210] on the elliptic p-adic curves (1936) is concerned. Other than the arithmetic applications, there can be found the construction of a local isomorphism of the group with the additive group, based on the integration of an invariant differential form. This method can be equally applied to abelian varieties, as is remarked shortly afterwards by C. Chabauty who uses it without any more explanation in order to prove a particular case of the “Mordell conjecture” [61].

From this moment, it was clear that the local theory of Lie groups can be applied almost without change to the p-adic case. The fundamental theorems of the “dictionary” Lie groups-Lie algebras are established in 1942 in the thesis of R. Hooke [166], a pupil of Chevalley; this work contains also the p-adic analogue of the theorem of E. Cartan on the closed subgroups of

real Lie groups.

More recently, M. Lazard [195 b] develops a more precise form of the “dictionary” for compact analytic groups over $\mathbf{Q}_p$. He shows that the existence of an analytic p -adic structure on a compact group G is tightly linked to that of certain filtrations on G , and gives various applications of it (for example to the cohomology of G). One of Lazard’s tools is an improvement of Dynkin’s results on the convergence of the p -adic Hausdorff series [195 a].

VIII. FREE LIE ALGEBRAS.

It remains for us to speak of a series of works on Lie algebras where the link with the theory of Lie groups is very tenuous; these researches have on the other hand important applications in the theory of “abstract” groups and more especially nilpotent groups.

The origin of this is the work of P. Hall [144], which appeared in 1932. There is however there no question of Lie algebras: P. Hall has in view the study of a certain class of p -groups, those that he calls “regular”. But that leads him to examine in detail iterated commutators and the lower central series of a group; he establishes on this occasion a variant of the Jacobi identity, as well as the “Hall formula”

$$(xy)^n = x^n y^n (x, y)^{n(n-1)/2} \dots$$

Soon after (in 1935-1937) the fundamental works of W. Magnus [215 a and b] and E. Witt [337 b] appear. In [215 a] Magnus uses the same algebra of formal series $\hat{A}$ as Hausdorff (since called the “Magnus algebra”); he embeds the free group F in it and uses the natural filtration of $\hat{A}$ in order to obtain a decreasing sequence (F_n) of subgroups of F ; it is one of the first examples of filtration. He conjectures that the F_n coincide with the terms of the lower central series of F . This conjecture is proved in his second memoir [215 b]; it is also there that he explicitly makes the link between his ideas and those of P. Hall, and that he defines the free Lie algebra L (as a sub algebra of A) of which he shows in substance that it is identified with the graded F . In [337 b], Witt completes this result on several points. He shows notably that the enveloping algebra of L is a free associative algebra and deduces from that immediately the rank of the homogeneous

components of L (“Witt formulae”).

As for the determination of the basis of L known by the name of “Hall basis”, it seems that it only appears for the first time in 1950, in a note of M. Hall [143], although it is implicit in the works of P. Hall and W. Magnus quoted above.

Chapter 26

GROUPS GENERATED BY REFLECTIONS; ROOT SYSTEMS.

The groups considered in this Note appeared in connection with various questions of Geometry, Analysis and the Theory of Lie groups, sometimes in the form of permutation groups, sometimes in the form of groups of displacements in Euclidean or hyperbolic geometry, and these various points of view have only been co-ordinated in recent times.

Historically, the beginnings of the theory are a good deal older than the introduction of the concept of a group: it takes its source indeed in the studies on the “regularity” or the “symmetries” of geometrical figures, and notably in the determination of regular polygons and polyhedra (going back no doubt to the Pythagoreans), which constitute the crowning achievement of the Elements of Euclid, and one of the most admirable creations of the Greek genius. Later, notably with the Arab authors of the high Middle Ages, then with Kepler, the beginnings of a mathematical theory of regular “tilings” of the plane or the sphere by polygons congruent in pairs (but not necessarily regular) appear, no doubt linked to the origin of the different types of ornament thought up by the ancient and Arab civilisations (that can be considered with good cause as an authentic part of the mathematics developed by these civilisations [291]).

Towards 1830-1840, studies in crystallography (Hessel, Bravais, Möbius) lead to considering a problem that is exactly that of the determination of the finite groups of displacements in Euclidean space of 3 dimensions, although the authors quoted above do not use yet the language

of the theory of groups; this latter hardly comes into use until around 1860, and it is in the context of the classification of groups that Jordan, in 1869 [174 b], determines the discrete subgroups of displacements in $\mathbf{R}^3$ which preserve orientation (and more generally, all the closed subgroups of the groups of displacements preserving orientation).

Until the last years of the XIXth century, this stream of ideas is developed in many directions, of which the most marked are the following:

1. Conforming to a tendency that appears very early on in the theory of finite groups, “presentations” of finite groups of displacements by generators and relations of a simple type are sought. It is thus that Hamilton, already in 1856 [145 b], proves that the finite groups of rotations in Euclidean space

R^3 are generated by two generators S, T linked by the relations $S^p = T^q = (ST)^3 = 1$ for appropriate values of p and q .

2. Discrete groups of displacements may or may not contain reflections. Already in 1852, Möbius determines substantially the finite groups of displacements in spherical geometry generated by reflections (which is equivalent to the same problem for finite groups of Euclidean displacements in $\mathbf{R}^3$); he finds that, with the exception of the cyclic groups, such a group has as fundamental domain a spherical triangle having angles of the form $\pi/p, \pi/q, \pi/r$, where p, q, r are three integers > 1 such that $\frac{1}{p} + \frac{1}{q} + \frac{1}{r} > 1$ ([223], v. II, pp. 349-360 and pp. 561-708). He also notices that these groups contain all the finite groups of displacements as subgroups.
3. This last stream of ideas finds a new amplification when following work of Riemann and Schwarz on hypergeometric functions and conformal representations, the study of “tilings” of the complex plane or of the half-plane by means of figures bounded by arcs of a circle is started; Klein and Poincaré make it the foundation of the theory of “automorphic functions” and recognise in it (for the case of arcs of a circle orthogonal to a fixed straight line) a problem equivalent to that of the search for the discrete subgroups of the group of displacements of the non-Euclidean hyperbolic plane (identified with the “Poincaré half-plane”) [118].

4. The notions of regular polyhedron and of the tiling of R^3 by such polyhedra are extended to all the Euclidean spaces R^n by Schläfli, in work that goes back to the neighbourhood of 1850, but was only published much later and remained ignored for a long time ([273], v. I, pp. 167-387); he determines completely the regular “polytopes” in each R^n , the group of displacements leaving invariant such a polytope, and a fundamental domain of this group, which, as in the case $n = 3$ studied by Möbius, is a “room” whose trace on the sphere S_{n-1} is a spherical simplex. However, he does not tackle the inverse problem of the search for the finite displacement groups generated by reflections in R^n ; this problem will only be solved much later, by Goursat [132] for $n = 4$, and, for arbitrary n , its solution will have to wait for the work of E. Cartan ([52 a], v. I₂, pp. 1003-1020) and of Coxeter [70 c], to which we will return later on.

Around 1890, with the first work of Killing and of E. Cartan on Lie groups, begins a new stream of ideas that for a long time will develop without links to the preceding work. Killing [180] and Cartan ([52 a], v. I₁, pp. 137-287), in their study of the structure of complex semisimple Lie algebras, will plan straightaway a primordial role for certain linear forms ω_α on a “Cartan sub algebra” $\mathfrak{h}$ of such a Lie algebra $\mathfrak{g}$; they are the “roots” relative to $\mathfrak{h}$, so called because with Killing they appear as the roots of the characteristic equation $\det(\text{ad}_{\mathfrak{g}}(x) - T) = 0$, considered as functions of $x \in \mathfrak{h}$. The properties of these “roots” established by Killing and Cartan reduce to affirming that, in the actual geometric language, they form a “reduced system of roots”;

they show then that the classification of the complex semisimple Lie algebras is reduced to that of associated “systems of roots”, which itself reduces to the determination of certain matrices with integer coefficients (later called “Cartan matrices”). Killing and Cartan also bring out, for every root ω_α , the existence of an involutive permutation S_α of the set of roots; they use in an essential way the transformation $C = S_{\alpha_1} S_{\alpha_2} \dots S_{\alpha_l}$, the product of permutations associated with l roots making up a fundamental system (a transformation today called a “Coxeter transformation”); they even extend this permutation into a linear transformation of the vector space generated by the fundamental roots $\omega_{\alpha_i} (1 \leq i \leq l)$, and study its eigenvalues ([180], II, p. 20; [52 a], v. I₁, p. 58). But neither Killing, nor at first Cartan, seem to have thought of considering the group G' generated by the S_α ; and when Cartan, a little later ([52 a], v. I₁, pp. 293-353),

determines the Galois group G of the characteristic equation

$$\det(\text{ad}_{\mathfrak{g}}(x) - T) = 0$$

of a “general element” $x \in \mathfrak{h}$, he studies it at first without bringing in the S_{α} ; thirty years later, already under the influence of the work of H. Weyl, he proves ([52 a], v. I₁, pp. 555-568) that G has as a normal subgroup the group G' and determines in all cases the structure of the quotient group G/G' which (for a simple algebra $\mathfrak{g}$) is of order 1 or 2 except for the type D_4 where it is isomorphic to S_3 ; it is also on this occasion that he interprets G' as the group induced by the inner automorphisms of a complex semisimple Lie algebra, leaving fixed a Cartan subalgebra.

The work of H. Weyl, to which we have just alluded, is that which inaugurated the geometric interpretation of the group G' (since called the “Weyl group” of $\mathfrak{g}$); in the same way as Killing and Cartan had done it for the transformation C , he had the idea of considering the S_{α} as reflections in the vector space of linear forms on $\mathfrak{h}$. It is also in the memoir of H. Weyl [331 b] that the fundamental domain of the “affine Weyl group” is seen to appear (without besides the link with the “Weyl group” G' being clearly indicated); Weyl uses it in order to prove that the fundamental group of a compact semisimple group is finite, a crucial point in his proof of the complete reducibility of the linear representations of a complex semisimple Lie algebra. Soon after, E. Cartan brings to fruition the synthesis of the global points of view of H. Weyl, of his own theory of real or complex semisimple Lie algebras, and of the theory of symmetric Riemann spaces that he was building at this time. In the memoir ([52 a], v. I₂, pp. 793-840), he completes the determination of the fundamental polytopes of the Weyl group and of the affine Weyl group, and introduces the weight systems and radical weight systems; he extends this discussion to symmetric spaces, and meets thus notably the first examples of non reduced root systems ([52 a], v. I₂, pp. 1003-1010). Finally the article ([52 a], v. I₂, pp. 867-989) gives the first proof of the fact that every finite group generated by reflections in R^n and irreducible has a fundamental domain having as trace on S_{n-1} a spherical simplex; it is also in this work that he proves the uniqueness of the largest root (under an arbitrary lexicographic ordering on the system of roots) by geometrical considerations.

A little later, van der Waerden [317 b], relying on the memoir of H. Weyl, shows that the classification of the complex semisimple Lie algebras is equivalent to that of the reduced root systems, which he carries out by elementary geometric considerations (whereas, with Killing and Cartan, this classification is a result of complicated calculations with determinants). At about the same time, Coxeter determines explicitly all the finite irreducible groups of Euclidean displacements which are generated by reflections [70 c]; he completes thus the results of the memoir ([52 a], v. I₂, pp. 793-840) of E. Cartan, who had only determined the “crystallographic” groups (i. e. associated to a root system, or also susceptible of being embedded in an infinite discrete group of displacements). The following year [70 d], Coxeter shows that the finite groups generated by reflections are the only finite groups (up to isomorphism) admitting a presentation by involutive generators R_i subject to relations of the form $(R_i R_j)^{m_{ij}} = 1$ (m_{ij} integers), whence the name of “Coxeter groups” given since to groups (finite or not) admitting such a presentation.

The first link between the two streams of research that we have described above seems to have been established by Coxeter [70 e], then by Witt [337 c]. They notice that the infinite irreducible groups of Euclidean displacements generated by reflections correspond bijectively (up to isomorphism) to complex simple Lie algebras. Witt gives a new determination of discrete groups of this type, and extends also the theorem of Coxeter [70 d] recalled above by characterising as well the Coxeter groups isomorphic to infinite discrete groups of Euclidean displacements. This result, and the fact that the analogous groups in hyperbolic geometry are also Coxeter groups,¹ have led to the open tackling of the study of these latter, first by putting the emphasis on a geometric realisation ([71], [308 a]), then, following J. Tits [308 b and c] in a purely algebraic framework.

Starting with the work of Witt, the theory of semisimple Lie groups and that of discrete groups generated by reflections will not cease from reacting extremely fruitfully the one on the other. Already from 1941, Stiefel [296] remarks that the Weyl groups are exactly the finite groups generated by reflections, and that leave a network invariant. Chevalley [62 e] and Harish-Chandra [149 a] give in 1948-51 proofs a priori of the bijective correspondence between “crystallographic” groups and complex semisimple Lie algebras; it was not possible until then to verify separately this correspondence for each type of simple Lie algebra.

Around 1950, it is noticed on the other hand that the polynomials invariant under the Weyl group play an important role in the theory of linear representations of infinite dimension [149 a] and in the topology of Lie groups.

On his side, Coxeter [70 g], taking up again the study of the transformation C , product of the fundamental reflections of a finite group W generated by reflections, notices (by means of a separate examination of each type) that the algebra of polynomials invariant under W is generated by algebraically independent elements, whose degrees are linked in a simple way to the eigenvalues of C . Proofs a priori of these results were subsequently given by Chevalley [62 f] for the first, and by Coleman [68] and Steinberg [292] for the second.

With the work of A. Borel on linear algebraic groups [30] new developments in the theory of Lie groups start which were to lead to a notable enlargement of this latter. A. Borel brings out the importance of the maximal connected soluble subgroups (since called “Borel subgroups”) in a Lie group, and makes of them the principal tool for transposing a large part of the classical theory to algebraic groups over an algebraically closed field (without however obtaining yet a classification of the simple algebraic groups²). The Borel subgroups (in the case of real or complex classical groups) had already turned up some years previously in the work of Gelfand and Neumark on the representations of infinite dimension; and in 1954, F. Bruhat had discovered the remarkable fact that, for the classical simple groups, the decomposition of the group into double cosets over a Borel group is indexed in a canonical way by the Weyl group [40]. This result was subsequently extended to all real and complex semisimple groups by Harish-Chandra [149 b]. On the other hand, in 1955, Chevalley [62 g] had succeeded in associating with every complex semisimple Lie algebra $\mathfrak{g}$ and with every commutative field k , a group of matrices with coefficients in k , possessing a Bruhat decomposition; and he used this last fact to show that, with a small number of exceptions, the group thus defined was simple (in the sense of the theory of abstract groups). He “explained” thus the coincidence, already noticed since Jordan and Lie, between the simple Lie groups (in the sense of the theory of Lie groups) of types A, B, C, D and the classical simple groups defined in a purely algebraic way over an arbitrary field (a coincidence that until then could not be extended except to the exceptional types G_2 and E_6 by Dickson [88c and d]). In particular, by taking a finite field k , the construction of Chevalley supplied, for each

type of complex simple Lie algebra, a family of finite simple groups, containing a large part of the then known finite simple groups, as well as three new series (corresponding to the types of simple Lie algebras F_4, E_7 and E_8). Soon after, by various procedures, using modifications of the methods of Chevalley, several authors (Hertzig, Suzuki, Ree, Steinberg and Tits) on the one hand showed that one can obtain in an analogous way the other finite simple groups known at that time, with the exception of the alternating groups and the Mathieu groups, and on the other hand constructed other series of new finite simple groups (cf. [54]).

At almost the same time, Chevalley [62 h], still using the technique of Bruhat decompositions, joined to a key result on the normaliser of a Borel subgroup, took up again the study of linear algebraic groups and reached the result that over an algebraically closed field k of arbitrary characteristic, the theory of semisimple linear algebraic groups ³ leads essentially to the same types as in the Killing-Cartan classification for $k = \mathbb{C}$. Following on, J. Tits [308 a and b], analysing the methods of Chevalley, reached an axiomatised version (the “BN-pairs”) of the Bruhat decompositions, in a remarkably flexible form, only bringing in the structure of the group; it is this notion that is at present known under the name “Tits system”. All the simple groups (in the various meanings of the word) of which we have been speaking above are canonically supplied with Tits systems, and Tits himself [308 c] has proved that the existence of such a system in an abstract group G , linked with a few supplementary properties from pure group theory, allows the proof that G is simple, a theorem which covers the majority of the proofs of simplicity given until then for these groups. In collaboration with A. Borel, he has on the other hand generalised the results of Chevalley of [62 h], by showing the existence of Tits systems in the group of rational points of a semisimple linear algebraic group over an arbitrary field [31].

All the Tits systems occurring in these questions have a finite Weyl group. Another category of examples was discovered by Iwahori and Matsumoto [170]; they have shown that if, in Chevalley’s construction of [62 g], k is a p -adic field, then the group obtained has a Tits system whose Weyl group is the affine Weyl group of the complex semisimple Lie algebra with which one started. This result has just been extended by Bruhat and Tits [41] to all semisimple algebraic groups over a local field.

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